Training dynamics of tree tensor networks: saddle-to-saddle, lazy and rich regimes

TTN regimes project*

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Abstract

How do gradient methods train tree tensor networks? We study the model $f(x) = \langle W, \phi_0(x_0) \otimes \cdots \otimes \phi_{N-1}(x_{N-1}) \rangle$, with W contracted from K cores on a tree, trained on the exact population loss of a teacher. From small initialization (scale σ), gradient flow is saddle-to-saddle: plateaus sit at critical points of the problem with fixed bond ranks, and each escape raises the bond ranks along a route, a connected set of cores that must move together; on large trees a route often dies first, and the sequence ends early. The first plateau lasts $\propto \sigma^{-(K-2)}$, as in other homogeneous models, with a prefactor computed per initialization. Later, the tree decides: the smallest route, of k^* cores, gives no plateau ($k^* = 1$), a logarithmic one ($k^* = 2$) or one $\propto \sigma^{-(k^*-2)}$ (derived; confirmed up to $k^* = 5$). At a saddle with one learned component and a spare bond channel along the route, we prove that k^* is the saddle order (the power of the distance at which the loss first decreases) under a spectral condition, which every two-component teacher learned in weight order satisfies, and that the order can be lower from $k^* = 5$ on. With non-orthogonal teachers, the saddle after the first escape is strict on matrix product states, not on binary or hierarchical Tucker trees whose internal cores carry no site. The learning order is a race set by the initialization unless competing components differ on only two cores. With the initial function removed, larger σ leads through a rich regime to a lazy one, which exists only when every bond dimension reaches the rank of the matricization it cuts.

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Conventions. Every claim carries a status tag: proven (proof given or cited), derived (derivation under stated assumptions), empirical (supported by experiments only) or conjecture (stated without sufficient derivation or evidence). Results we believe to be new are marked new, known results known with a reference; Section 1.3 discusses both. Unmarked statements are definitions or standard background. Sites are numbered from 0, and a pattern is a digit string with site 0 first (0110: basis vector 1 at sites 1 and 2, basis vector 0 elsewhere). Notation is defined in Section 2 and collected in Table 3.

1 Introduction and summary of findings

1.1 Question and motivation

Tensor networks whose contraction pattern is a tree (*tree tensor networks*, TTNs), among them matrix product states (MPS, or tensor trains), hierarchical Tucker (HT) formats and binary trees, are used as machine-learning models for classification and generative modelling (Stoudenmire and Schwab, 2016; Novikov et al., 2018; Han et al., 2018; Stoudenmire, 2018; Cheng et al., 2019; Liu et al., 2019; Glasser et al., 2019), and their expressive power has been related to that of deep

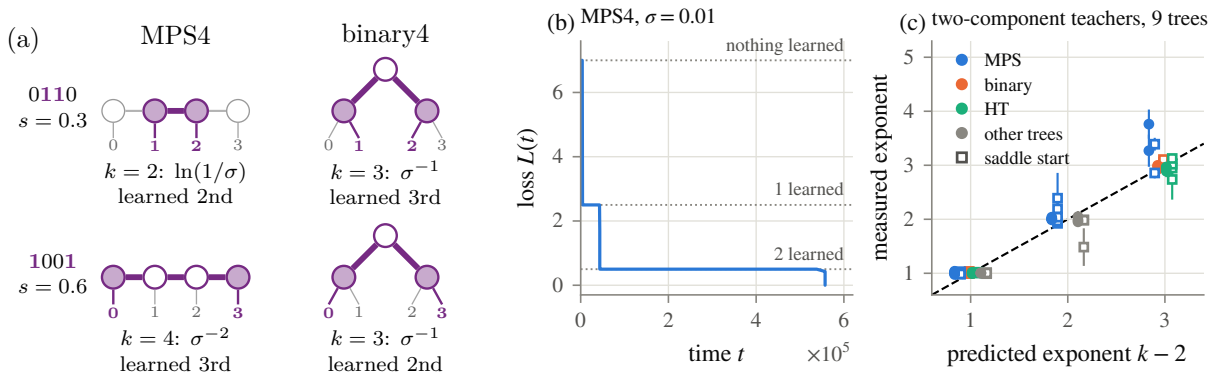


Figure 1: The tree decides the later plateaus. (a) Teacher B4: patterns 0000 (weight 1), 0110 (0.3) and 1001 (0.6), with local dimension $d = 2$ and bond dimension $\chi = 3$, on MPS4 (a chain of four cores, core i carrying site i) and binary4 (two cores carrying sites $0, 1$ and $2, 3$, joined by a root without sites). Both trees learn 0000 first. A remaining component then needs the cores that carry the sites where it differs from 0000 (filled) and every core on the paths between them: its route, the Steiner subtree of those cores, with k cores (outlined; its bonds in bold). The next plateau lasts $\propto \ln(1/\sigma)$ for $k = 2$ and $\propto \sigma^{-(k-2)}$ for $k \geq 3$, where σ is the initialization scale. On MPS4 the smaller component 0110 ($k = 2$) comes before 1001 ($k = 4$): locality beats weight. On binary4 both have $k = 3$ and the larger comes first. Each order holds in every run (60 on MPS4, 48 on binary4); measured exponents 2.02 ± 0.01 on MPS4 and 1.05 ± 0.01, 1.03 ± 0.01 on binary4 (mean $\pm$ s.e. over 5 and 8 seeds). (b) Loss of a run on MPS4 with an orthogonal teacher of weights $3, 2, 1$ ($d = 4$, $\chi = 3$, $\sigma = 0.01$; top left of Figure 4): it rests at the partial-fit levels (dotted: the loss with 0, 1 or 2 components learned) and drops by one component per escape. (c) Plateau exponent after the first component against the prediction $k - 2$: two-component teachers on nine trees, every case with $k = 3$–5 (mean $\pm$ s.e. over up to eight seeds; filled: small initialization, open: started at the saddle). Larger k and the slopes for $k = 2$: Figure 6. (source: App. B, Tables 12 and 13)

networks (Cohen et al., 2016; Levine et al., 2018). How they train is less understood. Work on their trainability has focused on barren plateaus, which occur for global but not for local losses of unitary-embedded MPS (Liu et al., 2022) and are absent from the energy optimization of isometric MPS, TTNS and MERA (Miao and Barthel, 2024; Barthel and Miao, 2025). Which dynamics gradient methods produce on a TTN, from which initializations, and why, has not been characterized (Section 1.3).

A TTN regression model, $f(x) = \langle W, \phi_0(x) \otimes \cdots \otimes \phi_{N-1}(x) \rangle$ with W contracted from K cores (Section 2.1), is multilinear in the cores and homogeneous of degree K in them, like a deep linear network. There, small initialization gives saddle-to-saddle (incremental) learning, in which the components of the target are learned one at a time after plateaus that lengthen as the initialization shrinks (Saxe et al., 2014; Jacot et al., 2022), and large initialization gives lazy training, in which the model stays close to its linearization (Jacot et al., 2018; Chizat et al., 2019). What does the tree add? Four things that a product of matrices lacks. (i) A component of the target can be *local* on the tree: it can differ from what has been learned on a few input positions (*sites*) only, and then only the cores that hold those sites, and the cores between them, need to change (Figure 1a). (ii) The number of cores K and the depth of the tree are different quantities: an MPS with N sites has depth $N - 1$, a binary tree $\log_2 N - 1$, and the binary tree with four sites (binary4) has depth one but is a product of three matrices, a depth-3 linear network. (iii) The bond dimension χ plays the role of width, but what a bond can carry is capped by its *cut dimension*, the largest rank that the matricization of W across that bond can have. (iv) Teachers can be genuinely tensorial, so that the correlation $\langle W, W^* \rangle$ has many local maxima on the sphere.

We ask, for gradient flow on the exact population loss of a teacher (Section 2), which saddles

are visited from small initialization (Q1); how the plateaus scale with the initialization scale σ, the number of cores K and the bond dimension χ (Q2); how the position of a component on the tree sets its escape (Q3, *locality*); which component is learned first (Q4); which regimes occur and where their boundaries lie (Q5); and what an input covariance (Q6), the structure of the teacher (Q7), discrete steps, noise and finite samples (Q8), the optimizers used for tensor networks (Q9: Adam, the sweeps of alternating least squares (ALS) and of the density-matrix renormalization group (DMRG), Riemannian methods) and real data (Q10) change. The model is kept minimal (whitened Gaussian inputs unless stated, realizable teachers, double precision; Section 2.1 says why), so that theory can be compared with experiment number by number. The main focus is the saddle-to-saddle regime (Section 4).

1.2 Summary of findings

The answer in six points.

- From small initialization, gradient flow learns the teacher one component at a time. Each plateau sits near a critical point with all bond ranks fixed, and each escape raises the bond ranks along a *route*, a connected set of cores whose new entries must grow together (Figure 1; definitions in Section 2.7).
- The first plateau lasts $\propto \sigma^{-(K-2)}$, with a prefactor that one integration computes for each initialization before training. The estimate from the norm of the initialization alone is only a lower bound, off by factors of up to $3 \cdot 10^7$.
- The tree sets the later plateaus. The next component needs the k^* cores of the smallest subtree that contains the cores carrying the sites where it differs (a *Steiner subtree*), and the next plateau is absent ($k^* = 1$), $\propto \ln(1/\sigma)$ ($k^* = 2$) or $\propto \sigma^{-(k^*-2)}$ ($k^* \geq 3$). Hence, as $\sigma \to 0$, locality beats weight, and the same teacher has different plateau laws on an MPS and on a binary tree. At a saddle with one learned component and a spare bond channel along the route, k^* is exactly the order of the saddle under a spectral condition; from $k^* = 5$ on the order can be lower.
- Which component comes first is a race set by the initialization when two components differ on sites carried by three or more cores; when they differ on two cores, learning becomes greedy as $\sigma \to 0$. In every case tested, teachers whose components are not orthogonal make the saddle after the first escape strict on an MPS, but not on binary and HT trees.
- At larger σ, with the initial function removed, training becomes rich and then lazy. A lazy regime exists only when every bond reaches its cut dimension; below it training stalls on a loss floor.
- Gradient descent (GD) with steps below a sharp stability limit, $2/(K s_1^{2-2/K})$ with s_1 the largest teacher weight, keeps the saddle sequence on a shifted clock; gradient noise (SGD) lengthens later plateaus on average and can trap; Adam keeps the first component but leaves the origin in a number of steps that does not depend on σ; ALS, DMRG and Riemannian gradient descent remove the plateaus. What decides is invariance: steps proportional to the gradient vanish near the origin, steps invariant under rescaling the gradient do not, and methods equivariant under the gauge group of the network see the initialization only through its frames. In short, the tree, the teacher and the input covariance shape the landscape; σ and the step size mostly set the clock.

Table 1: Findings by research question (sections in brackets). σ: initialization scale; K: number of cores; k^*: number of cores of the cheapest route (Figure 1a); χ: bond dimension. Orthogonal teacher: rank-one terms with orthonormal factors at every site; binary4: the binary tree with four sites, a depth-3 linear network; full bonds: every bond at its cut dimension. Tags as in the conventions; "new": no prior statement found (Section 1.3).

question	finding	status; new or known
Q1 sequence (4.1)	Orthogonal teachers are learned one component at a time; plateaus at the critical points with all bond ranks fixed; each escape raises every bond rank by one	derived, empirical; known (binary4), new (other trees)
Q2 scaling (4.3)	First plateau $\propto \sigma^{-(K-2)}$, with a prefactor computed for each initialization; the bound from the initialization's norm is low by up to $3 \cdot 10^7$	proven (scaling), derived (offset), empirical; known (exponent), new (TTN form, tests)
Q3 locality (4.4)	Next plateau none, $\propto \ln(1/\sigma)$ or $\propto \sigma^{-(k^*-2)}$ for $k^* = 1, 2, \geq 3$; confirmed for $k^* \leq 5$ on nine trees, with named exceptions; k^* is the saddle order under a spectral condition (one learned component)	proven (route, order), derived, empirical; known (mechanism), new (route, theorem)
Q4 order (4.7)	Differing on three or more cores: a race set by the initialization, independent of σ; on two cores: greedy as $\sigma \to 0$, the wrong order dying out like a power of σ	proven (race), derived, conjecture (rate), empirical; known (binary4), new
Q5 regimes (3, 5.1–5.3)	Initial function removed: saddle-to-saddle, rich, then lazy as σ grows; lazy only with full bonds; both boundaries $\propto \chi^{-1/2}$ on binary4	empirical, derived (lazy), conjecture (large K); known (other models), new (TTNs)
Q6 covariance (6.1–6.3)	A product covariance preconditions a whitened problem; an aligned one reweights race and landscape; misalignment and correlations make the MPS escape logarithmic	proven (reduction), derived, empirical; new (TTNs; two-layer analogue known)
Q7 teachers (6.2–6.6)	Plateaus at fixed-rank critical points; strict escapes on MPS, higher-order on binary and HT trees; stars forget a larger learned component (started at it)	proven (Hessians), derived, empirical; known (critical points), new (the rest)
Q8 steps, noise (5.4)	Gradient descent keeps the sequence below a sharp step-size limit; SGD lengthens later plateaus on average and can trap; small initialization recovers the teacher from finite samples	proven (limit), derived, empirical, conjecture (SGD law); known (analogues), new (TTN forms)
Q9 optimizers (5.5)	Adam: same first component, first escape independent of σ until its regularizer ϵ restores the law; ALS, DMRG, Riemannian GD: no plateaus	derived, proven (sweeps), empirical; new (TTNs; Adam's ϵ law concurrent)
Q10 real data (6.7)	MNIST: the first route after the escape on a binary tree is a sibling pair ($k = 3$); the final loss across pixel orders correlates with a data-only share of route couplings	empirical (pre-registered), conjecture (mechanism); new

Main findings. Each item states a claim, the scale of its evidence and its status; numbers and scope are in the sections cited, and novelty is discussed in Section 1.3.

1. **Saddle-to-saddle learning on tensor networks** (Section 4.1). From small initialization, gradient flow learns an orthogonal teacher one component at a time on binary4, MPS4, binary8 and HT4. Each plateau is the critical point with all bond ranks fixed, each escape raises every bond rank by one (497 of 531 runs), and the plateaus grow as $\sigma^{-(K-2)}$ at higher-order saddles (over less than a decade of σ at $K = 7, 8$) empirical (predicted derived).

2. **The first escape: a prefactor for each initialization** (Section 4.3). Near the origin the norm of the parameters blows up at $\sigma^{-(K-2)}\mathcal{T}(Z)$, an integral along the spherical flow of the correlation $g = \langle W, W^* \rangle$ from the initialization direction Z proven, plus an $O(1)$ offset derived. Computed before training, it predicts the first escape (231 pairs) and its component (1,847 of 1,848 runs) empirical; the norm-based time is a lower bound proven, low by up to $3 \cdot 10^7$ empirical.

3. **Later escapes follow the tree: the route rule** (Section 4.4). With one component learned,

the cheapest route to the next one is the Steiner subtree of the cores that carry the sites where the two differ, with k^* cores proven (orthogonal pattern teachers), and the next plateau is absent, logarithmic or $\propto \sigma^{-(k^*-2)}$ for $k^* = 1,\ 2,\ \geq 3$ derived. In about 11,600 runs on nine trees the exponent is $k-2$ for $k \leq 5$, with named exceptions; with three or four components the rule predicts the plateau levels, with one exception empirical (as an algorithm: conjecture).

4. **When k^* is the saddle order** (Section A.4). At a saddle with one learned component j and a spare bond channel along the route, the saddle order is exactly k^* unless a "piggyback" coupling of the residual on small sets of cores exceeds the learned weight s_j, which never happens for two-component teachers learned in weight order proven. From $k^* = 5$ on it can: on MPS6 a teacher with $k^* = 5$ has order 4 proven, and gradient flow escapes in $\propto \sigma^{-2}$ instead of σ^{-3} derived. Several learned components are not covered.

   [Pending (saddle-dyn): one sentence on the pre-registered test of the theorem's dynamics.]

5. **Strict or higher-order: the tree decides** (Sections 4.4 and 6.2). With one learned component of a pattern teacher the saddle is strict iff a two-core route exists or the piggyback coupling exceeds the learned weight proven. Non-orthogonal teachers open two-core routes wherever two adjacent cores carry sites: on an MPS the escape from the rank-one saddle becomes strict and logarithmic, at the predicted rate from the saddle, while binary and HT trees, whose internal cores carry no site, keep a higher-order saddle (13 generic teachers, $N = 4, 8$) derived empirical.

6. **Order of learning: a race or greedy** (Section 4.7). When two components differ on sites carried by three or more cores, the first one learned is decided by a race set by the initialization, independent of σ (exact at leading order on an MPS proven; 67,200 runs empirical); when they differ on two cores, learning becomes greedy as $\sigma \to 0$ derived, the wrong order dying out like a power of σ conjecture empirical. As $\sigma \to 0$ a smaller component with a shorter route is learned first (locality beats weight) derived empirical (teacher B4, Figure 1a).

7. **Failure modes: dead routes** (Section 4.8). A sequence can end early when a route dies (an end core decays to roundoff); a conservation-law mechanism explains the decay in a reduced model derived, but its criterion is neither necessary nor sufficient empirical. Most runs with $K = 7, 8$ and the bond dimension equal to the teacher rank never learn the last component, and 398 of 5,682 pattern-teacher runs stall, all with $k \geq 4$; balanced initialization removes the dead routes in the runs checked, but not all stalls empirical.

8. **The map of regimes** (Section 3; Sections 5.1 to 5.3). With the initial function removed, raising σ moves training from saddle-to-saddle through a rich window to lazy training empirical, which needs full bonds derived empirical; on binary4 both boundaries scale as $\chi^{-1/2}$, the lazy one as predicted derived empirical. The saddle-to-saddle boundary rises with K and, with the initial function kept and spare channels, does not tend to 1: at $K = 8$–15 all 97 per-seed lower bounds exceed both candidate laws, and 76 of 96 seeds stay saddle-to-saddle up to where the initial function overlaps the teacher, about $1.15\sqrt{d}$ empirical. At large χ a μP-type parameterization keeps the kernel moving derived empirical.

9. **Steps, noise and practical optimizers** (Sections 5.4 and 5.5). Gradient descent is stable at the balanced global minimum only below $\eta_c = 2/(K s_1^{2-2/K})$ proven, and below it keeps gradient flow's sequence on a clock shifted by $O(\eta)$ derived empirical. SGD lengthens later plateaus on average and can trap empirical (law: conjecture). Adam keeps the first component, and mostly the order, with a σ-independent first escape until ϵ restores the law derived empirical; ALS, DMRG and Riemannian GD have no plateaus derived, and ALS and DMRG do not see σ proven.

10. **Covariance, teacher structure and real data** (Section 6). A product covariance is a whitened problem with a preconditioned flow proven; an aligned one reweights the race by the input variance along each component and the landscape by its square root derived empirical. Random TTN teachers plateau at fixed-rank critical points, not at truncated singular value decompositions; binary8 students learn bottom-up; started at a larger learned component, star teachers (smaller components sharing a core) make the flow forget it empirical. On MNIST a data-only share of route couplings correlates with the final loss across pixel orders (pre-registered) empirical; why is a conjecture.

1.3 What is known and what is new

A systematic literature search (about 270 queries, arXiv listings up to October 2026, and a check of every citation and of prior work for every new result; Section C) found no paper on the saddle-to-saddle dynamics of gradient flow on TTNs with dense cores and $K \geq 3$; it is weakest for 2025–2026 work on training MPS and tensor trains. This is the only place where novelty is discussed; elsewhere the markers known and new carry it. Findings are numbered as in Section 1.2.

Deep linear networks. They have exact solutions from decoupled initialization (Saxe et al., 2014, 2019), incremental learning (Gidel et al., 2019; Gissin et al., 2020; Arora et al., 2019b; Li et al., 2021), and saddle-to-saddle dynamics from small initialization, with the first escape proven and the later saddles conjectured (Jacot et al., 2022); the full sequence is proven at depth two (Jiang et al., 2023), the TTN with two cores, and for decoupled or identity-type deeper factorizations (Chou et al., 2024). With three or more matrices the saddles visited from an exactly balanced initialization are non-strict (Achour et al., 2024, Thm 7 and the remark after Prop. 8); balance is conserved by gradient flow (Arora et al., 2018; Du et al., 2018; Kunin et al., 2021; Marcotte et al., 2023). Binary4 is such a network (Remark A.23), so its saddle sequence (finding 1), greedy limit (Jacot et al., 2022, Thm 2) and strict wrong-order saddles (Achour et al., 2024, Thm 7), (Jacot et al., 2022, Lemma 15) (finding 6) are known, as are the decoupled modes of aligned initialization, our case $k = K$, and, for two layers, the reweighting by a commuting input covariance (Gidel et al., 2019, Thm 2) and early alignment (Atanasov et al., 2022) (finding 10). New: the sequence on genuine tensor networks (MPS4, binary8, HT4) and the preconditioned form of a product covariance on trees.

Homogeneous models near the origin. Their weights align with a KKT point of the correlation with the target, selected by the initialization (Kumar and Haupt, 2025a, Thm 1), and escape after a time that grows as a negative power of the initialization (Kumar and Haupt, 2025a,b); dormant weights of the lowest homogeneity degree stay small for a time of the same kind (Kumar and Haupt, 2026, Thm 8); escapes that need 1, 2 or $r \geq 3$ small layers take a time of order one, logarithmic, or $\propto \epsilon^{-(r-2)}$, with ϵ their scale (Rawal and DeWeese, 2026, Thm 4.2); alternating gradient flows carry dormant amplitudes across phases (Kunin et al., 2025); weaker directions get a head start in two-layer linear networks (Zhang et al., 2026, Thm 4); and the leap of Abbe et al. (2023) counts the coordinates that must be learned at once. Hence known in our setting: the exponent $K-2$ of the first escape (Jacot et al., 2022; Kumar and Haupt, 2025a), its per-initialization blow-up identity (the activation law of Kunin et al., 2025, Eqs. 2–3), the norm-based time as the escape time from a maximizer (Kumar and Haupt, 2025b), the balance of the escape direction (linear chains: Kumar and Haupt, 2025a, Lemma 5; it also follows from the proof of Furman et al., 2026, Prop. F.10) and the head start of two-core routes. New: the evaluation on trees, the per-run tests and the gap to the norm-based time (finding 2); and which cores are dormant later: the Steiner

route of the residual's sites, its coupling and rate, and the closed-form head start for $3 \leq k < K$ (finding 3).

The landscape of our model. Furman et al. (2026) study it for exactly our model and loss: minimum-norm local minima are global; at a minimum-norm critical point every connected component of rank-deficient bonds gives a descent curve whose order is its number of cores, and with several components "the smallest core delivers the least suppressed escape" (Furman et al., 2026, Lemmas D.8–D.9, Remark D.10); one parity saddle has a lower bound on its order; points off the minimum-norm set can be spurious minima (Furman et al., 2026, Thm G.1). For two cores their curve has negative curvature, so the saddle is strict (a one-line consequence, not stated there). They do not treat dynamics. New relative to this: a route can be a proper subtree of one rank-deficient component, so k^* can lie far below its number of cores (K at a rank-one saddle with $\chi \geq 2$; finding 3); the matching lower bound on the saddle order with one learned component, and its failure from $k^* = 5$ on (finding 4; lower bounds were known for deep linear networks, order two or more, Achour et al., 2024, Thm 7, and for one parity saddle, Furman et al., 2026, Prop. 5.3); the piggyback criterion for strict learned saddles with $k^* \geq 3$, and the forgetting by stars (findings 5, 10; in deep linear networks the piggyback graph is one edge, Achour et al., 2024, Thm 7); and the tree contrast for generic teachers, whose rank-one saddle on an MPS has order 2 where the landscape bound allows K (finding 5). Locality beats weight has only analogues: the smallest component escaping first, and the lowest leap first (Abbe et al., 2023). Dead routes (finding 7) relate to known shrinkage of dormant weights (Jacot et al., 2022, App. C.1), (Kumar and Haupt, 2026), (Kunin et al., 2025, Lemma B.1) and to the failure of exactly balanced initialization in four-layer matrix factorization (Luo et al., 2025); new are the route-end condition and the measurements.

Tensor factorization. Small initialization biases CP factorization towards low rank (Razin et al., 2021) and hierarchical tensor factorization, whose internal nodes are local CP factorizations rather than dense cores, towards low hierarchical rank (Razin et al., 2022); over-parameterized orthogonal tensor decomposition learns the largest component first (Ge et al., 2021); lazy training of a tensor decomposition needs many components (Wang et al., 2020); growth exponents depend on depth in deep CP factorization (Hariz et al., 2022); the Tucker landscape has descents of order 2, 3 and 4 (Frandsen and Ge, 2022); and tensor-train completion shows a plateau, then abrupt rank growth (Milanesi et al., 2021). In multi-spike tensor PCA the recovery order is a greedy selection on $(\text{weight}) \times (\text{initial correlation})^{p-2}$ for $p \geq 3$ and follows the signal-to-noise ratios for $p = 2$ (Ben Arous et al., 2025b, Thm 1.5), (Ben Arous et al., 2025a, gradient flow, Thm 1.3), the dichotomy we find between pairs that differ on three or more cores and on two; related orderings hold for the tensor power method (Anandkumar et al., 2014, Lemma 5.1) and for units quadratic in their weights (Zhang et al., 2026, Prop. 5). New for TTNs (finding 6): the classification by the number of cores, the exact σ-independent race on MPS (the TTN form of the selection rule), the order statistics, the finite-σ law of the greedy case, the branches after a wrong first escape beyond binary4, and three components. The critical points of the rank-constrained problem are known in substance (Eldén and Savas, 2011; Holtz et al., 2012a); new are that generic teachers plateau there and not at truncated SVDs, the tree contrast and the bottom-up order (findings 5, 10).

Lazy and rich training. Lazy training and the tangent kernel (Jacot et al., 2018; Chizat et al., 2019; Lee et al., 2019), its failure when the tangent space misses the target (Ghorbani et al., 2019), the crossover read from the kernel (Geiger et al., 2020; Bordelon and Pehlevan, 2022), the kernel-to-rich transitions of diagonal networks (Woodworth et al., 2020; Pesme and Flammarion, 2023;

Berthier, 2023), imbalance (Kunin et al., 2024; Dominé et al., 2025), mixed dynamics (Tu et al., 2024; Lyu et al., 2024) and parameterizations of wide networks (Yang and Hu, 2021; Yang et al., 2024) are known; for tensor networks, kernel limits of MPS and tensor trains (Guo and Draper, 2021; Wesel and Batselier, 2025) and factor-wise scaling rules (Qiu et al., 2024; Potapczynski et al., 2024). Known in our setting: maps of regimes as such, the effect of a light layer (Kunin et al., 2024; Dominé et al., 2025), and the floor below full bonds as a linearized model that fits only its tangent space (Chizat et al., 2019; Ghorbani et al., 2019). New for TTNs (finding 8): the map, measured on one tree and as lower bounds beyond seven cores, the full-bond condition, the lazy constants, and a feature-learning (μP-type) parameterization.

Tensor networks, steps and optimizers. TTNs rest on the numerical analysis of hierarchical formats and their sweeping algorithms (Hackbusch and Kühn, 2009; Grasedyck, 2010; Oseledets, 2011; Holtz et al., 2012b; Uschmajew and Vandereycken, 2013; Holtz et al., 2012a; White, 1992), which also underlie Riemannian training (Novikov et al., 2018; Steinlechner, 2016); their barren plateaus (Liu et al., 2022; Miao and Barthel, 2024; Barthel and Miao, 2025) are of another kind: there the gradient concentrates over random initializations, here it is small because the initialization is. Known for steps and noise: the edge of stability and catapults (Cohen et al., 2021; Lewkowycz et al., 2020), stability limits and two-cycles in deep linear and scalar networks (Mulayoff and Michaeli, 2020; Kreisler et al., 2023; Ghosh et al., 2025; MacDonald et al., 2025), the modified flow behind our event shift (Barrett and Dherin, 2021), SGD stability and stochastic collapse (Li et al., 2017; Mulayoff and Michaeli, 2024; Chen et al., 2023), SGD experiments in deep linear networks (Corlouer et al., 2026), and plateaus at singular points, which the natural gradient avoids (Amari et al., 2006); concurrently with this work, Adam's bounded first escape and its ϵ law in deep matrix factorization (Sabanayagam and Lucey, 2026). New for TTNs (finding 9): the stability limit on general trees, the event shift, the delay and trap laws of SGD, the first component kept by Adam, and the σ-independence of sweeps.

1.4 Outline

Section 2 sets up the model and defines routes, stalls and event times (notation: Table 3); Section 3 maps the regimes; Section 4, the main section, treats the saddle-to-saddle regime; Section 5 the other regimes and practical training (steps, noise, samples, optimizers); Section 6 input covariance, teacher structure and real data; and Section 7 summarizes how each parameter acts and lists open questions. Sections A to D give the derivations, the settings of every experiment, how every figure is regenerated, and supplementary results.

2 Setup: model, data, teacher, training and diagnostics

This section fixes the model, the measurements used to classify training regimes, and the terms the later sections use: routes, saddle order and stalls (Section 2.7), events and regime labels (Section 2.8). Every experiment uses these definitions; deviations are stated where they occur. Table 3 collects the notation.

2.1 Tree tensor network regression

Inputs. An input x has N *sites* (input positions), numbered $i = 0, \dots, N-1$ throughout; site i carries a local feature vector $\phi_i(x) \in \mathbb{R}^{d_i}$ (default $d_i = d$). Unless stated otherwise $\phi_i(x) = x^{(i)} \sim$

$\mathcal{N}(0, \Sigma_i)$, independent across sites, and the inputs are *whitened*, $\Sigma_i = I$. We write

$$\Phi(x) = \phi_0(x) \otimes \cdots \otimes \phi_{N-1}(x) \in \mathbb{R}^{d_0 \times \cdots \times d_{N-1}}, \qquad \Sigma = \Sigma_0 \otimes \cdots \otimes \Sigma_{N-1}.$$

Trees and cores. A tree tensor network (TTN) is specified by a rooted tree $\mathcal{T}$ with node set $\mathcal{V}$, $|\mathcal{V}| = K$, and an assignment of each site to one node. For a node v, S_v is its (possibly empty) set of sites, $\mathrm{ch}(v)$ its children and $p(v)$ its parent. Every non-root node v is joined to its parent by the edge (bond) e_v of *bond dimension* χ_v; $\chi_{\mathrm{root}} = 1$. Node v carries a *core* G_v with one *physical leg* of dimension d_i per $i \in S_v$, one leg of dimension χ_c per child c, and a parent leg of dimension χ_v (at the root, the scalar output). The dimensions of the other (*child*) legs multiply to the *fan-in*

$$F_v = \prod_{i \in S_v} d_i \prod_{c \in \mathrm{ch}(v)} \chi_c,$$

so G_v is an $F_v \times \chi_v$ matrix; the parameters are $\theta = (G_v)_{v \in \mathcal{V}}$.

Model. The tensor $W(\theta) \in \mathbb{R}^{d_0 \times \cdots \times d_{N-1}}$ is the contraction of all cores over all bonds, and the model is

$$f_\theta(x) = \langle W(\theta),\, \Phi(x) \rangle. \tag{1}$$

Equivalently, f_θ is computed by passing messages $h_v(x) \in \mathbb{R}^{\chi_v}$ from the leaves to the root,

$$h_v(x)_a = \sum_{(j_i),(b_c)} G_v[(j_i)_{i \in S_v}, (b_c)_{c \in \mathrm{ch}(v)}, a] \prod_{i \in S_v} \phi_i(x)_{j_i} \prod_{c \in \mathrm{ch}(v)} h_c(x)_{b_c}, \qquad f_\theta(x) = h_{\mathrm{root}}(x). \tag{2}$$

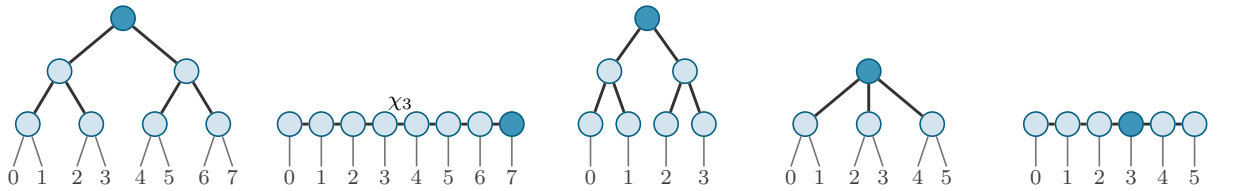
The model is multilinear in the site features, and W is multilinear in the cores (each appears once), so $\theta \mapsto W(\theta)$ is homogeneous of degree K: $G_v \mapsto c_v G_v$ rescales W by $\prod_v c_v$.

Bipartitions and bond spectra. Let A_v be the sites in the subtree rooted at v; the edge e_v separates them from the rest, and $W^{\langle e_v \rangle}$ is the matricization of W with the sites of A_v as rows. Contracting the subtree below v and the rest separately gives $W^{\langle e_v \rangle} = T_v C_v^\top$ with $T_v \in \mathbb{R}^{(\prod_{i \in A_v} d_i) \times \chi_v}$, so the *bond rank* $\operatorname{rank} W^{\langle e_v \rangle}$ is at most χ_v (Hackbusch and Kühn, 2009; Grasedyck, 2010). A bipartition is written by its two site sets as digit strings, e.g. 01|23 for $\{0,1\}$ against $\{2,3\}$ (ranges: 0–3|4–7). The *bond spectrum* of $e = e_v$ is the list $s_1^e \geq \cdots \geq s_{\chi_v}^e \geq 0$ of the χ_v largest singular values of $W^{\langle e \rangle}$ (the Schmidt values of tensor-network practice; we say singular values). They include all nonzero ones, so the bond rank is the number of nonzero s_j^e, and $\sum_j (s_j^e)^2 = \|W\|_F^2$ on every edge. With anisotropic inputs we use the function-space spectra of $(\Sigma^{1/2} W)^{\langle e \rangle}$, $\Sigma^{1/2} = \bigotimes_i \Sigma_i^{1/2}$. QR sweeps over the cores give them to about 10^{-15} relative accuracy without forming W.

Why this model. It is the simplest genuine TTN. (i) With at least three sites and an internal bond, W is a genuine tensor; a deep linear network is a chain with physical legs only at its two ends. (ii) Whitened inputs reduce the population loss to a Frobenius distance (Section 2.3), the setting in which Saxe et al. (2014) solved deep linear networks. (iii) The matrix product state (MPS) classifiers of Stoudenmire and Schwab (2016) compute, for each class y, $f^y(x) = \langle W^y, \bigotimes_i \phi_i(x) \rangle$ (the class index sits on one core), with non-Gaussian local features $(\cos(\pi x^{(i)}/2), \sin(\pi x^{(i)}/2))$ that are correlated across sites. (iv) Training is then a fully observed tensor factorization. On HT trees it has the mode tree of the hierarchical tensor factorization of Razin et al. (2022), whose internal nodes are local CP factorizations (their Eq. 3) with one scalar invariant per local component; our dense cores carry the matrix gauge symmetries of Section 2.6 instead.

Table 2: Tree families for N sites (K cores; depth: edges on the longest root-to-leaf path), and the two further trees of the locality experiments. Binary-tree depths are for $N = 2^m$; otherwise the sites are split recursively into halves ($\lceil N/2 \rceil$, $\lfloor N/2 \rfloor$), a pair of sites becomes a bottom core and a single site joins the core above, so $K = N - 1$.

name	structure	K	depth
binary	balanced binary tree; bottom cores carry two adjacent sites, internal cores none	$N-1$	$\log_2 N - 1$
MPS	chain (caterpillar): core i carries site i, its parent is core $i+1$	N	$N-1$
HT	hierarchical Tucker: balanced binary tree over N leaf cores, one site each; internal cores none	$2N-1$	$\lceil \log_2 N \rceil$
star3	a root without sites and three children carrying sites 01, 23, 45	4	1
midroot6	the chain of MPS6 rooted at the core of site 3	6	3



(a) binary8, $K = 7$ **(b)** MPS8, $K = 8$ **(c)** HT4, $K = 7$ **(d)** star3, $K = 4$ **(e)** midroot6, $K = 6$

Figure 2: The trees of Table 2. Circles are cores G_v; the shaded core is the root, whose parent leg (dimension one, the scalar output) is not drawn. Thick lines are bonds: the edge e_v between core v and its parent has dimension χ_v (one label shown in (b), where core i carries site i). Thin dangling lines are physical legs, labelled by their site i (dimension d_i). (d) and (e) are the two further trees of the locality experiments (Section 4.4).

2.2 Tree topologies

Table 2 lists the tree families and Figure 2 draws them; binary$N$, MPSN and HTN denote the member with N sites (binary4, MPS8, ...). Binary trees and MPS with the same N are the extremes of tree shape at nearly equal K, the default pair for saddle-to-saddle studies. The hierarchical Tucker (HT) format adds one leaf core per site; with leaf bond dimension d this adds no expressivity but changes K and the number of cores that must move to learn a component. Tree depth is not network depth: binary4 (tree depth one) computes the three-factor product $W^{\langle 01|23\rangle} = ABC^\top$ of its bottom cores A, C (as $d^2 \times \chi$ matrices) and its root B, a three-layer deep linear network on a $d^2 \times d^2$ matrix. Results on binary4 can therefore be compared with deep-linear-network theory; tensor-specific behaviour starts with MPS4, binary8 and HT4. Unless stated otherwise every edge has the same nominal bond dimension χ, capped at the largest rank its bipartition allows,

$$\chi_v = \min\Big(\chi,\ \prod_{i \in A_v} d_i,\ \prod_{i \notin A_v} d_i\Big). \tag{3}$$

The cap $\min(\prod_{i\in A_v} d_i, \prod_{i\notin A_v} d_i)$, the largest rank $W^{\langle e_v\rangle}$ can have, is the *cut dimension* of e_v, and a bond is *full* when χ_v reaches it (binary4 with $d = 3$: both bonds have cut dimension 9). A bond below its cut dimension limits the tensors the network can represent; a lazy regime needs every bond full (Section 3).

2.3 Teacher, data and loss

Teachers. The target is realizable, $f^*(x) = \langle W^*, \Phi(x)\rangle$. A *CP teacher* has

$$W^* = \sum_{r=1}^{R} s_r U_r, \qquad U_r = u_r^{(0)} \otimes \cdots \otimes u_r^{(N-1)}, \qquad \|u_r^{(i)}\| = 1, \qquad s_1 \geq \cdots \geq s_R > 0. \tag{4}$$

Three kinds are used:

- *Odeco* (orthogonally decomposable): $\langle u_r^{(i)}, u_{r'}^{(i)}\rangle = \delta_{rr'}$ at every site (so $R \leq \min_i d_i$); by default the factors at each site are columns of an independent Haar-random orthogonal matrix.
- *Pattern teachers*: site i has an orthonormal basis $(\beta_j^{(i)})_{j=0}^{d-1}$ and component r uses $u_r^{(i)} = \beta_{\pi_r(i)}^{(i)}$, where the *pattern* π_r maps each site to a basis index in $\{0, \ldots, d-1\}$; we write it as the digit string $\pi_r(0)\pi_r(1)\cdots\pi_r(N-1)$; powers mark repeats ($0^3 1^2 0^3 = 00011000$). Components with distinct patterns are orthonormal. Two components *differ* on the site set $D = \{i : \pi_r(i) \neq \pi_{r'}(i)\}$: 0000 and 1100 differ on $D = \{0, 1\}$, so the second is a *local* modification of the first. We write V_D for the set of cores that carry a site of D, $m = |V_D|$, and $\mathrm{St}(V_D)$ for their *Steiner subtree*, the smallest subtree of $\mathcal{T}$ that contains them (V_D and every core on a path between two of its cores).
- *Random CP*: $u_r^{(i)} = g/\|g\|$ with independent $g \sim \mathcal{N}(0, I)$; the components are generically non-orthogonal.

A *TTN teacher* gives W^* as a TTN on the student's tree or on another tree; a random TTN teacher is a fan-in initialization (Section 2.4) with $\sigma = 1$, rescaled to $\|W^*\|_F = 1$. The overlap of the student with component r of a CP teacher,

$$a_r = \langle W, U_r\rangle_\Sigma / \langle W^*, U_r\rangle_\Sigma, \tag{5}$$

is 0 at $W = 0$ and 1 at $W = W^*$; for an odeco teacher with whitened inputs, $a_r = \langle W, U_r\rangle / s_r$.

Proposition 2.1 (Teacher bond spectra). proven *Let e be an edge whose two sides both carry sites (every edge of the trees of Table 2). (a) For an odeco teacher the bond spectrum of W^* on e is $(s_1, \ldots, s_R)$, for every tree. (b) For a two-component pattern teacher whose components differ exactly on the site set D, the bond rank of W^* on e is 2 if e is an edge of the Steiner subtree $\mathrm{St}(V_D)$, and 1 otherwise.*

Proof. With A the sites on one side of e, $x_r = \bigotimes_{i\in A} u_r^{(i)}$ and $y_r = \bigotimes_{i\notin A} u_r^{(i)}$, $(W^*)^{\langle e\rangle} = \sum_r s_r x_r y_r^\top$. (a) Both families are orthonormal, so this is a singular value decomposition. (b) If $D \subseteq A$ or $D \cap A = \emptyset$, then $y_1 = y_2$ or $x_1 = x_2$: rank one. Otherwise the components differ (by orthogonal basis vectors) on both sides, so $x_1 \perp x_2$, $y_1 \perp y_2$: rank two. And e separates two sites of D exactly when it lies on the tree path between their nodes. □

Its number of cores, $k = |\mathrm{St}(V_D)|$, is the size of the route along which the second component is learned (Section 2.7).

Population loss. Training minimizes the population square loss

$$L(\theta) = \tfrac{1}{2}\,\mathbb{E}_x\big[(f_\theta(x) - f^*(x))^2\big] = \tfrac{1}{2}\|W(\theta) - W^*\|_\Sigma^2, \qquad \|A\|_\Sigma^2 = \langle A, \Sigma A\rangle. \tag{6}$$

The second equality holds because, by independence across sites, $\mathbb{E}[\Phi\otimes\Phi] = \bigotimes_i \mathbb{E}[\phi_i\phi_i^\top] = \Sigma$. With whitened inputs L is half the squared Frobenius distance. Only second moments enter: every local feature map with $\mathbb{E}[\phi_i\phi_i^\top] = I$ gives the same dynamics, and a non-orthonormal one is equivalent to $\Sigma_i = \mathbb{E}[\phi_i\phi_i^\top] \neq I$.

Closed form. Expanding (6),

$$L(\theta) = \tfrac{1}{2}\langle W, W\rangle_\Sigma - \langle W, W^*\rangle_\Sigma + \tfrac{1}{2}\langle W^*, W^*\rangle_\Sigma, \tag{7}$$

and every term is an exact contraction. Two TTNs on the same tree (bond dimensions may differ) are contracted from the leaves to the root at a cost linear in K; for a CP teacher, $\langle W, U_r\rangle_\Sigma$ is the student (2) evaluated at the site vectors $\Sigma_i u_r^{(i)}$; a TTN teacher on another tree is glued to the student along the physical legs and the loopy network is contracted exactly along an optimized path. For an odeco teacher with whitened inputs, (7) reads $L = \frac{1}{2}\|W\|_F^2 - \sum_r s_r^2 a_r + \frac{1}{2}\sum_r s_r^2$. For orthonormal components (odeco or pattern teachers) and whitened inputs, the *partial fit* $W_S = \sum_{r\in S} s_r U_r$ of a set S of components has loss

$$L_S = \tfrac{1}{2}\sum_{r\notin S} s_r^2, \tag{8}$$

the candidate plateau levels; learning in decreasing order of s_r passes through $S = \emptyset, \{1\}, \{1,2\}, \ldots$.

Anisotropic and correlated inputs. Anisotropic product covariances are handled exactly by the same contractions; the default family has power-law site spectra, $\Sigma_i = c_\beta\,\mathrm{diag}(1^{-\beta}, 2^{-\beta}, \ldots, d^{-\beta})$ with c_β fixed by $\mathrm{tr}\,\Sigma_i = d$, aligned with the teacher's factors or rotated away from them. Cross-site correlations break the product structure of Σ; for small N the loss is then computed with the dense second-moment matrix (Section 6).

Finite samples. A training set $(x_j, y_j)_{j=1}^n$ has $x_j \sim \mathcal{N}(0,\Sigma)$ i.i.d. and $y_j = f^*(x_j) + \varepsilon_j$ with optional label noise $\varepsilon_j \sim \mathcal{N}(0,\varsigma^2)$. The empirical loss is $\hat{L}_n(\theta) = \frac{1}{2n}\sum_j (f_\theta(x_j) - y_j)^2$, and the test error is the population loss (6).

2.4 Initialization and its variants

Fan-in initialization. The default initialization is

$$G_v = \frac{\sigma_v}{\sqrt{F_v}}\,Z_v, \qquad Z_v \text{ with i.i.d.} \mathcal{N}(0,1) \text{ entries}, \tag{9}$$

with a single per-core scale $\sigma_v = \sigma$ unless stated otherwise. Each core is then approximately σ_v times an isometry (when $F_v \gg \chi_v$), so with $\sigma_v = 1$ the messages (2) keep their size up the tree.

Proposition 2.2 (Initial scales). proven *Under* (9), $\mathbb{E}\|W(\theta_0)\|_F^2 = \prod_v \sigma_v^2$ *exactly,* $\mathbb{E}\|G_v\|_F^2 = \sigma_v^2\chi_v$, *and hence* $\mathbb{E}\|\theta_0\|^2 = \sum_v \sigma_v^2\chi_v$.

Proof. Let T_v be the contraction of the subtree below v (χ_v columns). Given the children's (independent) subtrees, the entries of G_v, of variance σ_v^2/F_v, give $\mathbb{E}[T_v^\top T_v] = \frac{\sigma_v^2}{F_v}\prod_{i\in S_v} d_i \prod_{c\in\mathrm{ch}(v)} \mathrm{tr}(T_c^\top T_c)\, I$, so by induction from the leaves $\mathbb{E}[T_v^\top T_v] = (\prod_u \sigma_u^2) I$ over the subtree of v; T_{root} is W flattened. Finally, $\mathbb{E}\|G_v\|_F^2 = F_v\chi_v\,\sigma_v^2/F_v$. □

We report both σ and the function-space scale $\alpha = \prod_v \sigma_v$ ($= \sigma^K$ with one scale), the root-mean-square norm of $W(\theta_0)$: small σ makes $W(\theta_0)$ exponentially small in K, while $\|\theta_0\|^2 \approx \sigma^2\sum_v \chi_v$ grows with the bond dimensions.

Variants.

(a) *Imbalance*: per-core scales $\sigma_v = \sigma c_v$ with a fixed profile (c_v) (for example root versus rest), compared at fixed $\prod_v \sigma_v$.

(b) *Centred model.* The *plain model* (1) trained on W^* starts, at large σ, from a large random function f_{θ_0}. The centred model removes it: the target is $W^* + W(\theta_0)$, that is, $f_\theta - f_{\theta_0}$ is trained on f^* (Chizat et al., 2019, Sec. 3.2), so that σ changes the dynamics but not the target, and σ alone moves the model from small to large initialization. Overlaps and bond spectra refer to $W - W(\theta_0)$. This is the main construction of the regime map (Section 3).

(c) *Two-copy model* (Chizat et al., 2019, Sec. 3.1; Woodworth et al., 2020, Sec. 4): $W(\theta^+) - W(\theta^-)$ with $\theta^+(0) = \theta^-(0) = \theta_0$, zero at initialization for every σ; a cross-check of the centred model. It is trained as one TTN with doubled bonds and block-diagonal cores, which needs a gradient mask (Remark 2.3; implementation in Section B.1).

(d) *Aligned (decoupled) initialization* for CP teachers (Proposition 2.4): cores diagonal in the teacher's basis, giving exactly solvable Saxe-type dynamics.

(e) *Per-core learning-rate multipliers* η_v (13). A fixed forward multiplier c_v ($G_v = c_v \tilde{G}_v$, $\tilde{G}_v$ trained) is equivalent to the initial scale $c_v \sigma_v$ with the multiplier $c_v^2 \eta_v$; this expresses the NTK and μP parameterizations of wide networks (Yang and Hu, 2021), transposed to large bond dimension (Section 5).

Remark 2.3 (The two-copy model needs a gradient mask). proven Gradient flow on the doubled TTN does not keep its block-diagonal form: at a root with children c_1, c_2 and no sites, the entry that pairs index b_1 of the $+$ half of e_{c_1} with index b_2 of the $-$ half of e_{c_2} has, at initialization, the gradient $-\langle W^*, T^+_{c_1}[:, b_1] \otimes T^-_{c_2}[:, b_2] \rangle_\Sigma$ (up to the order of sites), generically nonzero. The two-copy dynamics is therefore the flow (13) with multipliers set to zero off the diagonal blocks; it keeps the blocks and equals gradient flow of $\frac{1}{2}\|W(\theta^+) - W(\theta^-) - W^*\|^2_\Sigma$ in (θ^+, θ^-).

Proposition 2.4 (Aligned initialization). proven known (deep linear networks: Saxe et al., 2014) *Let W^* be odeco, inputs whitened, $\chi_v \geq R$ on every edge, and let every subtree and its complement carry a site (true for the trees of Table 2). Initialize*

$$G_v = \sum_{r=1}^{R} q_{v,r} \Big(\bigotimes_{i \in S_v} u_r^{(i)} \Big) \otimes \Big(\bigotimes_{c \in \mathrm{ch}(v)} \varepsilon_r \Big) \otimes \varepsilon_{r(v)}, \tag{10}$$

with ε_r the r-th standard basis vector of the bond in question, $r(v) = r$ for $v \neq$ root and $r(\text{root}) = 1$. Under gradient flow with multipliers η_v the cores keep this form, $W = \sum_r \Pi_r U_r$ with $\Pi_r = \prod_v q_{v,r}$, $a_r = \Pi_r / s_r$, $L = \frac{1}{2} \sum_r (s_r - \Pi_r)^2$, and

$$\dot{q}_{v,r} = \eta_v \, (s_r - \Pi_r) \prod_{u \neq v} q_{u,r}. \tag{11}$$

With $\eta_v = 1$ the differences $q^2_{v,r} - q^2_{u,r}$ are conserved; a balanced start $q_{v,r}(0) = q_r(0)$ stays balanced, and $\dot{q}_r = q_r^{K-1}(s_r - q_r^K)$.

Proof. On the family (10), $W = \sum_r \Pi_r U_r$ with orthonormal U_r. The derivative of W with respect to an entry of G_v is a product of physical basis vectors, child-subtree columns and the environment column above v; on the family these are multiples of products of teacher factors, so their inner product with $W - W^* = \sum_r (\Pi_r - s_r) U_r$ vanishes unless all bond indices equal one r ($a = 1$ at the root). The gradient thus stays in the family, with component $(\Pi_r - s_r) \prod_{u \neq v} q_{u,r}$ along $q_{v,r}$; with $\eta_v = 1$, $\frac{\mathrm{d}}{\mathrm{d}t} q^2_{v,r} = 2(s_r - \Pi_r)\Pi_r$ for every v. □

For $K \geq 3$ a balanced mode started at $q_r(0) \ll s_r^{1/K}$ reaches $a_r = \frac{1}{2}$ at

$$t_r = \frac{q_r(0)^{-(K-2)}}{(K-2)\, s_r} + O\big(s_r^{-(2-2/K)}\big), \tag{12}$$

and for $K = 2$ at $t_r = \log(s_r/q_r(0)^2)/(2s_r) + O(q_r(0)^2/s_r^2)$, by direct integration of the balanced ODE proven. With $q_r(0) \propto \sigma$ the first plateau grows like $\sigma^{-(K-2)}$.

2.5 Training dynamics

Gradient flow is the reference dynamics,

$$\dot{\theta} = -\boldsymbol{\eta} \odot \nabla L(\theta), \tag{13}$$

where the diagonal $\boldsymbol{\eta}$ holds, for each parameter of core v, the learning-rate multiplier η_v (default 1), set to zero where a gradient mask applies (Remark 2.3). Here t denotes gradient-flow time. It is integrated by the adaptive Dormand–Prince 5(4) pair (Dormand and Prince, 1980) with standard step-size control (Hairer et al., 1993, Sec. II.4) (relative and absolute tolerances 10^{-8} and 10^{-12}) and, past the first escape, where the flow is stiff, mostly by LSODA with the exact Hessian; steps follow the local time scale, so long plateaus are cheap (choices and tolerances per experiment: Section B.1). Diagnostics are recorded on a log-spaced time grid plus *fine rows*, kept whenever θ has moved by 1% of its norm, or $\ln L$ by 0.01, since the last kept row, so that every escape is sampled about 20 times or more. A run stops at $t_{\max}$, below a prescribed loss, or at a step or wall-clock limit; together these are its *budget*. Accuracy is checked with the conservation laws (Section 2.6) and the aligned solutions (Proposition 2.4). All computations are in double precision: with $\sigma = 10^{-2}$ and $K = 7$, $\|W(\theta_0)\|_F \approx 10^{-14}$. A dormant component whose escape signal falls below about $10^{-15}\|W\|_F$ is seeded by roundoff, not by the initialization; such runs are flagged.

Gradient descent with learning rate η iterates $\theta \leftarrow \theta - \eta\, \boldsymbol{\eta} \odot \nabla L(\theta)$, and each step advances t by η. A run has *diverged* when L becomes non-finite or exceeds $10^6 L(\theta_0)$.

Stochastic gradient descent uses the minibatch loss $\hat{L}_B(\theta) = \frac{1}{2B}\sum_{b=1}^{B}(f_\theta(x_b) - y_b)^2$ in the same update. Online SGD draws fresh inputs $x_b \sim \mathcal{N}(0, \Sigma)$ with $y_b = f^*(x_b)$ at every step; with a fixed training set, each epoch uses a new random permutation. Optional label noise is drawn afresh each step.

2.6 Symmetries and conservation laws

For a bond $e = e_v$ joining core v to its parent $p = p(v)$, let $G_{u,(e)}$ denote core $u \in \{v, p\}$ matricized with its e-leg as rows (a $\chi_v \times F_v$ matrix for $u = v$). For every invertible $\Omega \in \mathrm{GL}(\chi_v)$ the *gauge transformation*

$$G_{v,(e)} \mapsto \Omega\, G_{v,(e)}, \qquad G_{p,(e)} \mapsto \Omega^{-\top} G_{p,(e)} \tag{14}$$

leaves W, and hence any loss that depends on θ only through W, unchanged.

Proposition 2.5 (Conservation laws). proven known (deep linear networks: Arora et al., 2018; Du et al., 2018) *Under gradient flow $\dot{G}_u = -\eta_u \nabla_{G_u} L$ of any loss that depends on θ only through W (population or empirical), the $\chi_v \times \chi_v$ matrix*

$$Q_e = \frac{1}{\eta_v} G_{v,(e)} G_{v,(e)}^\top - \frac{1}{\eta_p} G_{p,(e)} G_{p,(e)}^\top \tag{15}$$

is constant in time, for every bond $e = e_v$ with $p = p(v)$.

Proof. With $X = G_{v,(e)}$, $Y = G_{p,(e)}$ and $\Omega = I + \epsilon M$ in (14), $X \mapsto X + \epsilon MX$ and $Y \mapsto Y - \epsilon M^\top Y + O(\epsilon^2)$; first-order invariance of L for every M gives $C := \nabla_X L\, X^\top = Y(\nabla_Y L)^\top$, so $\frac{\mathrm{d}}{\mathrm{d}t}(XX^\top)/\eta_v = -(C + C^\top) = \frac{\mathrm{d}}{\mathrm{d}t}(YY^\top)/\eta_p$. □

This is the tree version of the balancedness of deep linear networks, an instance of symmetry-induced conservation laws (Kunin et al., 2021). Consequences:

- *Norm balance*: $\operatorname{tr} Q_e = \|G_v\|_F^2/\eta_v - \|G_p\|_F^2/\eta_p$; with equal multipliers all differences $\|G_u\|_F^2 - \|G_{u'}\|_F^2$ are conserved, so all cores grow together.
- *Balancedness at small initialization*: under (9), $\mathbb{E}\, Q_e(0) = (\sigma_v^2/\eta_v - \sigma_p^2\chi_p/(\eta_p\chi_v))I$, so at small σ the cores stay balanced up to $O(\sigma^2)$ while they grow to $O(1)$. Imbalanced initializations or multipliers set Q_e to other values, a second knob for the regime map (Kunin et al., 2024).
- The masked two-copy flow conserves the Q_e of each copy; gradient descent and SGD conserve Q_e only up to $O(\eta^2)$ per step.
- *Integrator check*: the conservation drift

$$\mathrm{drift}(t) = \frac{\max_e \|Q_e(t) - Q_e(0)\|_{\max}}{\max_e \|Q_e(0)\|_{\max} + \max_u \|G_u(t)\|_F^2/\eta_u}, \tag{16}$$

  with $\|\cdot\|_{\max}$ the largest absolute entry, measures the integration error relative to the terms that cancel in Q_e; it is required to stay below 10^{-6} under gradient flow (exceptions are reported in Section B).

2.7 Saddles, routes and saddle order

The terms below are used from Section 1 on; Section 4 states the results and Sections A.3 and A.4 derive them.

Learned saddles and routes. From small initialization gradient flow rests, between escapes, near a *learned saddle* θ_S: a critical point that represents a partial fit W_S (for orthonormal teachers the sum of the learned components, (8); in general a best approximation of W^* with the current bond ranks). Each bond uses only its first $\mathrm{rank}(W_S)^{\langle e\rangle}$ (*learned*) channels; the others are zero. A new channel opened in one core meets zeros in its neighbours and leaves W unchanged: new channels move W only when every core of a connected set C opens one together, and then by a product of $|C|$ small entries, a small TTN on C attached to the learned network (Lemma A.16). Such a set is a *route*. It lowers the loss at order $|C|$ in the new entries, while the quadratic (Gauss–Newton) term $\frac{1}{2}\|\Delta W\|_\Sigma^2$ is of order $2|C|$, so near the saddle it behaves like a fresh $|C|$-core TTN started near zero (Proposition A.18). The *ends* of a route are its leaves as a subtree; the *slice* of an end is that core's part on the route's new channel (for a leaf core, one column). The learned channels can also rotate; when such moves descend faster than the cheapest route is decided in Section A.4.

Route size k and k^*. For an orthonormal pattern teacher with one learned component, the route that adds component r is the Steiner subtree $\mathrm{St}(V_D)$ of the cores that carry the sites D where r differs from the learned component, and the bonds whose rank must grow are exactly its edges (Propositions A.19 and 2.1) proven. Its size $k = |\mathrm{St}(V_D)|$ is the number of cores that must move together, and k^* is the smallest k among the components not yet learned (in general, the smallest route with a nonzero local residual, Proposition A.18). If D lies on one core ($k = 1$), that core

fits the component through its learned channel and the flow does not stop. Otherwise, when the saddle order is k^* (below), the next plateau follows the first-escape law with K replaced by k^*: it lasts $\propto \ln(1/\sigma)$ for $k^* = 2$ and $\propto \sigma^{-(k^*-2)}$ for $k^* \geq 3$ **derived** (Section 4.4). Example: 0000 and 0110 differ on sites 1 and 2; on MPS4 these sit on adjacent cores ($k = 2$), on binary4 on the two bottom cores, joined through the root ($k = 3$).

Saddle order. The *order* of a saddle is the power of the distance at which the loss first decreases: the largest exponent ord with $L(\theta) - L_S \geq -C\|\theta - \theta_S\|^{\mathrm{ord}}$ near θ_S (Section A.4). A *strict* saddle has order 2 (a negative Hessian eigenvalue); at a *higher-order* saddle (order at least 3) every descent direction is flat to second order. A route of k^* cores descends at order k^*, so the order is at most k^* **proven** (Proposition A.19); when it equals k^* is stated in Section 4 and proven in Section A.4.

2.8 Regime diagnostics

Regimes are assigned from measured quantities, never by eye; Section B gives the settings of each experiment.

Recorded quantities. Along every run we record the loss, the bond spectra of W, the overlaps, the relative parameter change $\delta_\theta = \|\theta(t) - \theta(0)\|/\|\theta(0)\|$ (also per core), the relative kernel change δ_Θ and the conservation drift (16), and at plateaus the Hessian H and its gauge projection $H_\perp$ (Table 3). The *effective bond rank* rk_e counts the s_j^e above 0.1 times the smallest nonzero teacher singular value on e (threshold-free: the entropy rank $\mathrm{rk}_e^H = \exp(-\sum_j \hat{s}_j^2 \ln \hat{s}_j^2)$, $\hat{s}_j = s_j^e/\|W\|_F$). The tangent kernel $\Theta = J\,\mathrm{diag}(\boldsymbol{\eta})\,J^\top$, $J = \partial W/\partial\theta$, is the operator on $\mathbb{R}^{d^N}$ that drives the function-space dynamics of (13), and $\delta_\Theta = \|\Theta_t - \Theta_0\|_F/\|\Theta_0\|_F$; it is computed where $d^N \dim\theta \leq 2 \cdot 10^7$ (binary4 at every bond dimension used), and Section A.8 derives its constants. The *probe kernel* restricts Θ to $P = 128$ fixed inputs $x_p \sim \mathcal{N}(0, \Sigma)$ (for whitened inputs, $\Phi(x_p)^\top \Theta\, \Phi(x_{p'})$); where it is used instead, as in the regime labels of Section 3, the change is written $\delta_\Theta^{\mathrm{probe}}$.

Learning events. For CP teachers with orthogonal components (in the metric Σ), *event* r is the time t_r at which the overlap a_r rises through $\frac{1}{2}$, and its *width* w_r is the duration of that final rise, from $a_r = 0.1$ to 0.9. Experiments date the crossing by slightly different rules: the last upward crossing before a_r first reaches 0.9 (recommended), the last upward crossing after which a_r stays above $\frac{1}{2}$, or the first crossing. The rules agree unless a learned component later dips below $\frac{1}{2}$ (star teachers, gradient noise, rotations to a better component; the second rule then dates the relearning, or gives none) or a transient crosses $\frac{1}{2}$ while another component is learned (the third rule then dates the transient). For other teachers one escape can raise several overlaps, and events are read from the bond spectra instead (for example the onset, when s_j^e first reaches 0.05 times the teacher's value); on real data from the loss rate. Table 11 gives each experiment's rule. Crossings are interpolated on the recorded rows or found by root finding on the integrator's dense output; an event narrower than three sample intervals is *unresolved*, and separation ratios involving it are lower bounds.

Plateaus and the saddle-to-saddle signature. Number the events in time order, $t_1 < t_2 < \cdots$, with widths w_i; the *separation ratios* are

$$\rho_i = \frac{t_{i+1} - t_i}{\max(w_i, w_{i+1})}. \tag{17}$$

Table 3: Notation and where it is defined (§: section; (n): equation). Sites and cores are numbered from 0, teacher components from 1. Abbreviations: GF gradient flow; GD, SGD (stochastic) gradient descent; S2S saddle-to-saddle; LO leading order; odeco orthogonally decomposable; MPS matrix product state; HT hierarchical Tucker; SP, NTK, μP parameterizations (Section 5.3); ALS, DMRG sweep methods (Section 5.5).

symbol	meaning	defined in
N, i; d_i, d	number of sites, site index $i = 0, \dots, N-1$; local feature dimension (default d)	§2.1
ϕ_i, Φ; Σ_i, Σ	site features, their tensor product; site and product input covariance	§2.1
$\mathcal{T}$, $\mathcal{V}$, K; $p(v)$, $\mathrm{ch}(v)$	rooted tree, its cores, their number; parent and children of core v	§2.1
S_v, A_v; e_v, χ_v, χ	sites of core v, sites below v; bond to the parent, its dimension; nominal value	§2.1, (3)
G_v, F_v, θ	core of v, its fan-in; all parameters	§2.1
W, W^*, W_0	tensor of the student, of the teacher, at initialization	§2.1, §2.3
$W^{\langle e \rangle}$, 01|23	matricization across e; a bipartition written by its site sets	§2.1
s_j^e, $s_j^{e,*}$; rk_e	j-th singular value across e of W, of W^* (their nonzero count: the bond rank); effective bond rank	§2.1, §2.8
cut dimension, full	largest rank across e_v; a bond is full when χ_v reaches it	§2.2
binary$N$, MPSN, HTN	trees with N sites; also star3, midroot6	Table 2
R, s_r	number of teacher components; weights $s_1 \geq \cdots \geq s_R$	(4)
$u_r^{(i)}$, U_r	unit factor of component r at site i; unit rank-one term	(4)
π_r	pattern of component r, a digit string, site 0 first (0110, $0^3 1^2 0^3$)	§2.3
a_r	overlap with component r (0 at $W = 0$, 1 at $W = W^*$)	(5)
D; V_D, m	sites where a component differs from the learned one; cores carrying them, $m = \lvert V_D \rvert$	§2.3
$\mathrm{St}(\cdot)$	Steiner subtree: the smallest subtree that contains a set of cores	§2.3
L, ℓ; W_S, L_S	population loss, $L/L(0)$; partial fit of the components in S, its loss	(6), (8)
n, $\hat{L}_n$; B	training-set size, empirical loss; minibatch size	§2.3, §2.5
c; Λ_r	cross-site input correlation; aligned covariance: per-component product of eigenvalues	§A.6; §6.1
σ, σ_v, α	initialization scale; per-core scale; function-space scale $\prod_v \sigma_v$	(9), §2.4
$q_{v,r}$, Π_r	aligned-initialization amplitudes; their product	Prop. 2.4
Z; ε	initialization direction, $\theta_0 = \sigma Z$; perturbation of a saddle start	§4.3; §B.1
t, $t_{\max}$; T	gradient-flow time, time budget; end time of a run	§2.5; Table 4
η; η_v, $\boldsymbol{\eta}$	learning rate (step size); per-core multiplier, the diagonal preconditioner	(13)
η_c	$2/(K s_1^{2-2/K})$, the step size up to which GD keeps the saddle sequence	(23)
Q_e	conserved quantity of bond e	(15)
Θ, J	tangent kernel $J\,\mathrm{diag}(\boldsymbol{\eta})J^\top$; $J = \partial W/\partial\theta$	§2.8
δ_Θ, $\delta_\Theta^{\mathrm{probe}}$; P	relative kernel change (tensor, probe kernel); probe-set size	§2.8
δ_θ; drift	relative parameter change; conservation drift	§2.8, (16)
H, $H_\perp$, $\lambda_{\min}$	Hessian, its gauge projection, its smallest eigenvalue	§2.8
t_r, w_r	time and width of event r	§2.8
ρ_i, t_i^{mid}	separation ratio of consecutive events; plateau midpoint	(17)
θ_S; $\mathcal{R}$	learned saddle; residual $W^* - W_S$	§2.7; §A.3
route, end, slice	cores that open a channel together; its leaves; an end's part on the channel	§2.7
k, k^*	route size $\lvert \mathrm{St}(V_D) \rvert$; smallest route with a nonzero residual	§2.7
saddle order	power of the distance at which the loss first decreases (2: strict)	§2.7; §A.4
Δt_2, λ_2	time between the first two events; $k = 2$ escape rate $s_2 s_1^{(K-2)/K}$	(21)
head start	factor $1 - s_2/s_1$: the route grew during the previous plateau	§4.4
c_r; $T(Z)$	coupling of the route of component r; first-escape prefactor of Z	(38); §4.3
Γ	piggyback graph; the saddle is strict if $\lambda_{\max}(\Gamma) > s_j$	Prop. A.22
$Q_{\nu\nu}$	conserved quantity on the new channel ν of a route end	Prop. A.28
w; γ	weight ratio s_2/s_1; exponent in P(smaller first) $\propto \sigma^\gamma$	§4.7
σ_c, σ_{lazy}; λ	S2S and lazy boundaries in σ; root scale at fixed α	§3, §3.4
r_S; $r(\cdot)$	Spearman correlation; spectral radius	§6.7; §A.4

Table 4: Regime of one run trained up to time T (S2S signature: Section 2.8). Threshold sensitivity: the kernel threshold 0.1 is varied over $\{0.05, 0.1, 0.2\}$, the separation threshold 3 over $\{2, 3, 5\}$.

regime	rule
lazy	the loss converges and $\delta_\Theta(T) < 0.1$ (so δ_θ is small as well)
saddle-to-saddle	$\delta_\Theta(T) \geq 0.1$ and the S2S signature (i)–(iii)
rich, non-incremental	$\delta_\Theta(T) \geq 0.1$ and the loss converges, but some $\rho_i \leq 3$ (overlapping transitions)
stalled	$L(T) \geq 10^{-3} L(0)$ when the budget ends; subclasses dead route, slow and floor (see the text)
diverged	L non-finite or above $10^6 L(0)$ (GD and SGD; for gradient flow, an integrator failure)

Between two events with $\rho_i > 3$ the loss sits on a plateau; its midpoint t_i^{mid} is the log time of minimal speed $\|\dot\theta\|$ between them. A run shows the *saddle-to-saddle (S2S) signature* when (i) there are at least two events, (ii) every $\rho_i > 3$, and (iii) across a sweep, plateau durations grow without bound as $\sigma \to 0$ (like $\log(1/\sigma)$ or a power of $1/\sigma$). Plateaus are judged in linear time: an escape that needs two cores takes a time of order $\log(1/\sigma)$, invisible on a log-time axis after a long first plateau.

Regime assignment. Each run, trained up to time T, receives one label (Table 4).

Remark 2.6 (Reading of the rules). "Converges" means $L(T) < 10^{-3} L(0)$; labels are checked in the order diverged, stalled, then the converged regimes. Condition (iii) belongs to a sweep: a run is labelled S2S on (i)–(ii), a region of a phase diagram only if (iii) holds there. A converged run with $\delta_\Theta(T) \geq 0.1$ and a single (merged) event is rich, non-incremental. The regime map measures convergence against $\frac{1}{2}\|W^*\|_\Sigma^2$, which equals $L(0)$ when the fitted tensor vanishes at $t = 0$ (centred and two-copy models).

Stalled runs. *Stalled* is an umbrella label (Table 4); its four causes fall into three subclasses. *Dead route* (trap): an end of the route that the next component needs has its slice at roundoff. The route's drive is a product over its cores, so it cannot escape, and the run stays at a partial fit. The proposed mechanism is the conservation-law trap (Proposition A.28, derived in a reduced model): on a long plateau the conservation laws can make an end's slice decay at a rate $\propto \sigma^2$ (tests and open conflicts: Section 4.8). *Slow*: every route is live, but the budget ends first, on a plateau longer than the time budget (a route of k cores needs a time $\propto \sigma^{-(k-2)}$) or at the step limit (unfinished lazy fits at full bonds, some deep-tree runs). *Floor*: in the centred model with a bond below its cut dimension, the target $W^* + W(\theta_0)$ is not representable and the loss stops at the best fit with the available bond ranks, above the threshold at large σ (Section 5.1); only this cause is specific to the centred model. Experiments test the subclass in their own ways (a route-end slice or a bond singular value at roundoff, a gradient below a threshold, a continued run; Section B); runs whose subclass is not determined are reported as stalled.

Saddle type at a plateau. At the plateau midpoint we compute the Hessian H and its restriction $H_\perp$ to the orthogonal complement of the well-conditioned gauge directions, which are the tangent vectors $(MG_{v,(e)}, -M^\top G_{p,(e)})$ of the orbits (14) (cutoff: Section B.1); the raw $\lambda_{\min}(H)$ is reported too. The type is read across a sweep from how $\lambda_{\min}(H_\perp)$ at the midpoint scales with σ

(Proposition A.26): it tends to a negative constant at a strict saddle and vanishes like σ^{k-2} at a higher-order saddle reached through a route of $k \geq 3$ cores (the origin: $k = K$). A fixed threshold at one σ, strict if $\lambda_{\min}(H_\perp) < -10^{-3} \max_j |\lambda_j(H_\perp)|$, cannot separate the two: it calls $k = 3$ plateaus strict for $\sigma \gtrsim 0.01$–0.04.

Reporting rules. Scaling fits use at least five initialization scales over at least one decade and at least five seeds per point, and report mean ± standard error, a fitted exponent with its standard error, and the theoretical prediction. Fits that fall short of this rule are reported but marked *limited range*, with their range of σ and their seed count. Every regime boundary comes with the threshold sensitivity check of Table 4, and every number states the dynamics that produced it (integrator tolerances, step size, batch size).

2.9 What the results may depend on

Plateaus and saddle sequences are properties of population gradient flow; implicit bias and generalization need finite samples. The features are multilinear, with no pointwise nonlinearity, and the population loss is a tree contraction only for a product input measure, which cross-site correlations break. Binary trees and HT differ in K and in how many cores must change to learn a component, which sets plateau-time exponents (Section 4.4). Fan-in initialization fixes the relative scales of the cores; exponents should not depend on it, but prefactors and the learning order might, so imbalanced, balanced ($Q_e = 0$) and aligned initializations are also studied.

3 Map of regimes

Which regime of Table 4 does a run fall into, and which parameters decide it? This section maps the regimes of gradient flow on the centred model over the per-core initialization scale σ, the bond dimension χ, the number of cores K and the imbalance between the root and the other cores (Figure 3); Table 5 at its end lists which parameter moves which boundary. On the binary tree with $N = 4$, training is saddle-to-saddle below $\sigma_c \approx 0.3/\sqrt{\chi}$, rich in a window of about 20 in σ, and lazy above $\sigma_{\text{lazy}} \approx 6.5/\sqrt{\chi}$ (values reached at $\chi \geq 32$), but lazy only when every bond reaches its cut dimension; below it the runs end on a floor. With more cores the saddle-to-saddle boundary rises (in the plain model above $\sigma = 1$ for most seeds at $K \geq 8$) and stalled runs invade its side, and a light root makes training lazy at an up to 100 times smaller function-space scale. Section 5 gives the detail of each regime other than saddle-to-saddle, and Section 5.4 adds discrete steps and gradient noise. new (for TTNs) known (maps for linear and diagonal networks: Woodworth et al., 2020; Tu et al., 2024; Kunin et al., 2024; Dominé et al., 2025)

3.1 Protocol

The main family is the *centred* model (Section 2.4, variant (b)): its target $W^* + W_0$ removes the initial function at every σ, so σ alone moves a run from small to large initialization; the *plain* and *two-copy* models are cross-checks. Odeco teachers with weights $(3, 2, 1)$ at $d = 3$ or $(3, 2)$ at $d = 2$, fan-in initialization with one scale σ for every core unless stated. On the binary tree with $N = 4$ ($K = 3$, *binary4*) the bonds are not capped, and $\chi = 9 = d^2$ is the cut dimension. Labels follow Table 4, with the tensor kernel (δ_Θ) on binary4 and the probe kernel ($\delta_\Theta^{\text{probe}}$) on the larger trees (Section 2.8). Boundaries are found per seed (six seeds; saddle-to-saddle where $\min_i \rho_i$ crosses 3, lazy where the kernel change crosses 0.1) and reported as medians, or as geometric means ×/ their

geometric standard error f (the interval from the mean divided by f to the mean times f). Settings: Section B.5; supplementary results: Section D.2.

3.2 Initialization scale and bond dimension

Setup. Binary4, $d = 3$, teacher $(3, 2, 1)$: $\chi = 3$–36 with 19 values of $\sigma \in [0.01, 40]$, and $\chi = 9$–128 with 22 values of $\sigma \in [0.004, 4]$ (Figure 3(a)).

Three regimes, ordered in σ. empirical At every χ the runs are saddle-to-saddle at small σ and rich (converged, $\delta_\Theta \geq 0.1$, overlapping events) at intermediate σ; at large σ they are lazy if $\chi \geq 9$ and stalled on a floor if $\chi < 9$. The saddle-to-saddle boundary falls from 0.24 at $\chi = 3$ to 0.136 at $\chi = 9$ and 0.043 at $\chi = 36$ (not monotone at small χ: 0.099 at $\chi = 4$ and 0.075 at $\chi = 12$, where per-seed values spread by up to a factor 16; Section 5.2). Below it the smallest separation ratio grows like σ^{-1} per seed (fitted exponents 1.015–1.048 in 96 of 101 fits, against $K - 2 = 1$, derived for the first gap, Section A.8): the gaps grow while the event widths stay at the values of aligned initialization (Proposition A.48). The χ-dependence of σ_c is not derived. Per-χ values and fits: Section D.2.

The lazy boundary. empirical confirming derived Proposition A.47 With the tensor kernel and $\chi = 32$–128 the lazy boundary lies inside the predicted brackets, 0.07% below the lower edge at $\chi = 128$: $\sigma_{\text{lazy}}\sqrt{\chi} = 6.40$–$6.56$ against the asymptote 6.50 of (59). At $\chi = 9$–16, where the isotropic constants are not expected to hold, it lies 1.5–6% above the bracket ($\sigma_{\text{lazy}}\sqrt{\chi} = 8.96$ at $\chi = 9$). At large χ a coherent second-order kernel change sets the boundary (Section 5.1). known (the criterion, a threshold on the kernel change: Geiger et al., 2020; a coherent second-order kernel change in wide networks: Bordelon and Pehlevan, 2022) new (the TTN constants)

No lazy regime below the cut dimension. derived empirical With $\chi < 9$ the centred target has more bond rank than the student can carry: from $\sigma = 1.0, 1.0, 1.6$ at $\chi = 3, 4, 6$ every seed stalls on the floor, the best approximation of $W^* + W_0$ across the cut $01|23$, and no run is lazy (Section 5.1).

Both boundaries scale like $\chi^{-1/2}$. empirical With full bonds, fits to the per-seed boundaries give the slope -0.53 ± 0.05 against χ for the saddle-to-saddle boundary ($\chi = 16$–128) and -0.504 ± 0.011 for the lazy boundary ($\chi = 32$–128). At $\chi = 32$–128, $\sigma_c\sqrt{\chi} = 0.27$–$0.33$, and the rich window $\sigma_{\text{lazy}}/\sigma_c$ spans a factor 19–24 in σ (18–23 with the probe kernel), about 10^4 in the function-space scale $\alpha = \sigma^3$. At large χ the map thus depends on $\sigma\sqrt{\chi}$, or on the per-bond Gram scale $\chi\sigma^2/d^2$ (the eigenvalue scale of the subtree Grams $T_v^\top T_v$): saddle-to-saddle below about 0.01, lazy above about 4.7. Neither σ^K (slope 0) nor $\sigma^3\chi/d^2$, the amplitude per path times the number of paths (slope $-\frac{1}{3}$, 2.7–2.9 standard errors from the saddle-to-saddle slope), collapses the data. Scope: one tree (a three-factor matrix product), fan-in initialization, less than one decade of χ. The saddle-to-saddle scaling is not derived; heuristically, the leaf Grams grow like $\chi\sigma^2/d^2$ while the root is light by $\sqrt{\chi}$, so the effective per-core amplitude of the escape grows like $\sigma\sqrt{\chi}$ conjecture. known (as a scaling: the width-dependent scales $\sigma^2 w$ of Tu et al., 2024 and $\alpha\sqrt{h}$ of Geiger et al., 2020)

Seven cores. empirical On the binary tree with $N = 8$ ($K = 7$) the regimes come in the same order. At $d = 3$ no centred run with $\chi \leq 24$ is lazy (the top bonds have cut dimension 81; floor from $\sigma = 0.91$–1.34). At $d = 2$ (cut dimensions 4 and 16) the saddle-to-saddle boundary is 0.83

$[0.72, 0.97]$ for $\chi = 4$, 8, 16 and 0.53 $[0.46, 0.61]$ for full bonds over-parameterized twice (medians of six seeds and of their brackets), and only the latter reach a lazy regime within the step budget, from $\sigma \approx 1.4$: the rich window spans a factor 2.7 in σ, against about 20 at $K = 3$ (with another d and teacher). At $\chi = 16$, where the bonds are exactly full, the runs above $\sigma = 2$ stop at the step limit unconverged (Section 5.1).

3.3 Number of cores

Setup. $d = 2$, teacher $(3, 2)$, $\chi = 4$ (twice the teacher rank) on every tree, so the bonds are full for $K \leq 5$ and deficient from $K = 6$ on: binary trees with $N = 4$, 8, 16 ($K = 3$, 7, 15) and MPS with $N = 4$, 5, 6, 8, 10, 12; 11–14 values of σ from $\max(10^{-8/K}, 0.02)$ to 2 (Figure 3(b)).

Resolved up to seven cores. empirical new (for TTNs) The saddle-to-saddle boundary rises with K: medians of the per-seed boundaries 0.17, 0.49, 0.61, 0.68 and 0.75 at $K = 3$–7 (quartiles in Section D.2), with all six seeds resolved except at $K = 6$ (four). It stays above the aligned-initialization boundary $a_{0c}(K) = 0.066$–0.49 of Proposition A.48. The plain model gives the same boundaries within one grid step up to $K = 7$; beyond, it must first remove W_0.

Beyond seven cores. empirical At $K = 8$, 10, 12, 15 at most one of six seeds is resolved: the run after the last saddle-to-saddle run stalls on the floor of the unrealizable target, so the label rule gives only brackets near 1 (median brackets between 0.63 and 1.36), lower bounds rather than crossings (Section D.2).

The limit at large K, in realizable problems. empirical new Two laws were candidates: a fixed σ_c^K, which gives $\sigma_c \to \lim_K a_{0c}(K) = 1$ like the per-layer threshold of aligned initialization known (Gissin et al., 2020, Thm 2), and a fixed per-component amplitude $\sigma_c^K/\sqrt{d^N}$ (Proposition A.46(b)), which gives $\sigma_c \to \sqrt{d}$. A pre-registered second sweep with realizable targets (binary trees with $N = 4$–16, teacher $(3, 2)$, $d = 2$ and 3, six seeds per cell) tests them. In the plain model with spare channels ($\chi = 4$), 85 of the 96 seeds at $K \geq 8$ find no crossing: at every sampled σ they stay saddle-to-saddle up to the scale at which the initial function itself overlaps a teacher component, where the event rule stops applying, or to the end of the budget. Their lower bounds on σ_c, 1.07–2.03, all exceed 1 and both laws fitted on the resolved cells, so in the plain model σ_c does not tend to 1. A proposed mechanism, consistent with the data but not tested causally, is that the plain model unlearns W_0 while it waits on the plateau, which lowers the effective amplitude conjecture. The centred model with full bonds has no W_0 to unlearn but is affordable only up to $K = 8$ at $d = 2$ and $K = 4$ at $d = 3$: there σ_c rises from 0.16 to 0.83 over $K = 3$–8, and the slope of $\ln(\sigma_c^K/a_{0c}^K)$ against $\ln\sqrt{d^N}$ (0 for the first law, 1 for the second) is -0.07 ± 0.51, which resolves neither law. Whether $\sigma_c \to \sqrt{d}$ remains a conjecture (details: Section D.2).

Stalls on the saddle-to-saddle side. empirical Below the boundary stalled runs take a growing part of the map (plain grey in Figure 3(b)): at the smallest σ, from none of six seeds on binary4 to 6 of 6 on MPS12 and 5 of 5 on binary16, each at a partial fit with one component learned. A stalling MPS seed stalls up to a seed-dependent σ, then turns saddle-to-saddle. Such stalls are dead routes or slow plateaus (the centred runs did not record the spectra that tell the two apart), the failure modes of Section 4.8, which also covers the more frequent stalls with χ equal to the teacher rank.

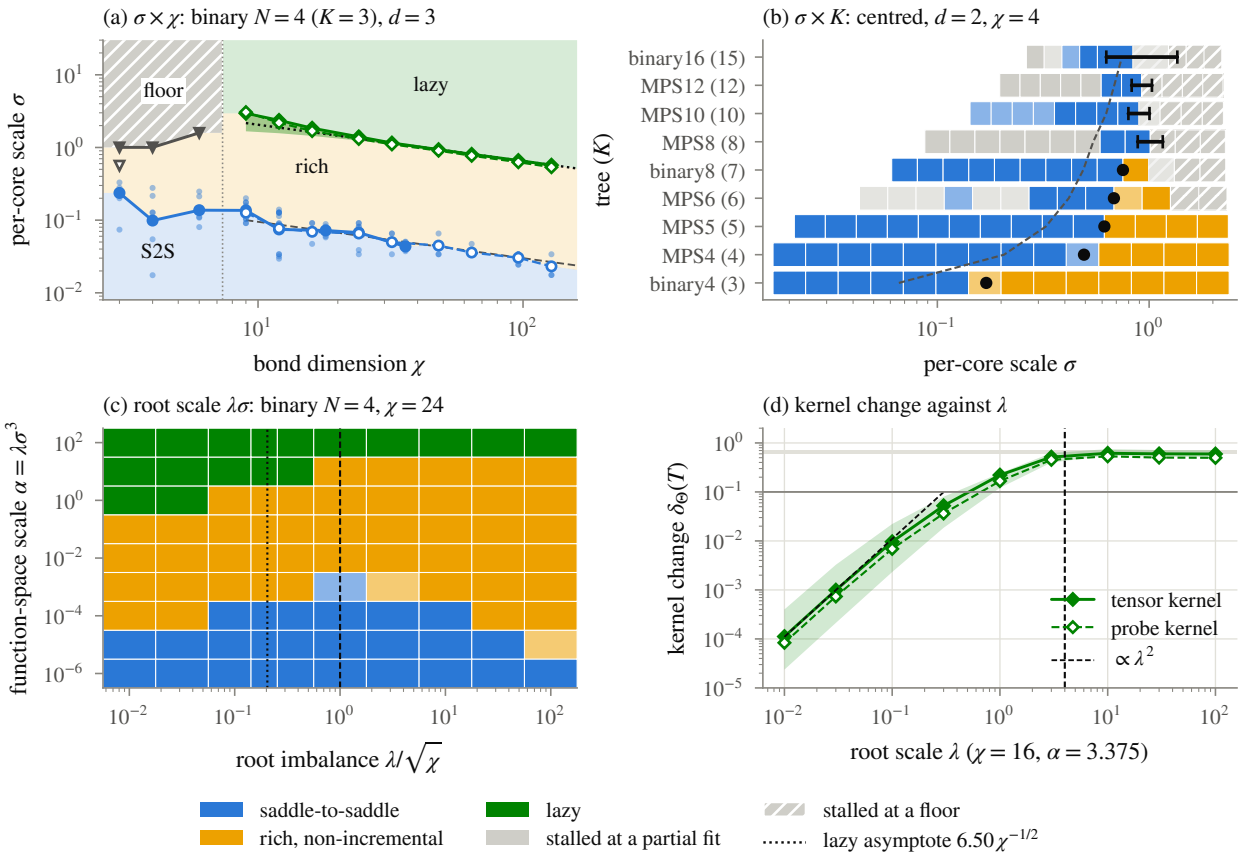


Figure 3: Map of regimes: gradient flow, centred model, fan-in initialization; S2S: saddle-to-saddle. Regions in (a), majority labels of six seeds in (b), (c) (pale: at most three runs agree; binary16 has 2–5 runs in 7 of its cells). (a) Binary $N = 4$, $d = 3$, teacher $(3, 2, 1)$; per-seed boundaries (dots, six seeds) and geometric means. Blue: saddle-to-saddle boundary from the $\sigma \times \chi$ sweep (filled, $\min_i \rho_i = 3$) and the χ sweep (open, label bracket); dashed: $0.30\,\chi^{-1/2}$ (a guide). Green: lazy boundary (kernel change 0.1; filled: tensor kernel, open: probe kernel), bracket of Proposition A.47 (band) and its asymptote (dotted). Triangles: σ from which every seed stalls on the floor (filled) and, at $\chi = 3$ only, where the small-σ floor (60) would reach the convergence threshold (open); bonds are full right of the dotted vertical. (b) $d = 2$, teacher $(3, 2)$, $\chi = 4$ (full bonds only for $K \leq 5$). Dots: median per-seed boundary; bars: median bracket from the last saddle-to-saddle run to the next, stalled one; dashed: the aligned-initialization boundary $a_{0c}(K)$ (Proposition A.48). Hatched: stalled above the saddle-to-saddle runs, on the floor of the unrealizable target; plain grey: stalled below them, at a partial fit. (c) Root at scale $\lambda\sigma$, other cores at σ, fixed $\alpha = \lambda\sigma^3$, $\chi = 24$; dashed: balance ($\mathbb{E}Q_e = 0$ on the root bonds); dotted: fan-in default $\lambda = 1$. (d) δ_Θ against λ at $\chi = 16$, $\alpha = 3.375$: medians and range of six seeds; dashed: $\propto \lambda^2$; grey band: predicted saturation; grey line: the lazy threshold $\delta_\Theta = 0.1$; vertical: balance. Takeaway: on binary4 both boundaries scale like $\chi^{-1/2}$ and only full bonds give a lazy regime; with more cores the saddle-to-saddle boundary rises and stalls invade its side; a light root makes training lazy. (source: Section B.5, Table 15)

3.4 Imbalance between cores

Setup. Binary4, $d = 3$, teacher $(3, 2, 1)$. The root starts at scale $\lambda\sigma$, the other cores at σ, so $\alpha = \lambda\sigma^3$ is fixed while λ sets the conserved imbalance of the root bonds, $\mathbb{E}Q_e = \sigma^2(1 - \lambda^2/\chi)I$ (Section 2.6): balance at $\lambda = \sqrt{\chi}$; the fan-in default $\lambda = 1$ is root-light. Grids: $\chi = 24$ (full bonds) and $\chi = 3$, nine values of α from 10^{-6} to 10^2 times ten of $\lambda/\sqrt{\chi}$ from 0.01 to 100; and $\chi = 16$, $\alpha = 3.375$ ($\xi = \|W^*\|_F/\alpha = 1.11$), $\lambda = 0.01$–100 (Figure 3(c), (d)).

A light root makes training lazy; a heavy one does not. empirical new (for TTNs) At $\chi = 24$ every seed is lazy from $\alpha = 1$ for a light root ($\lambda/\sqrt{\chi} = 0.01, 0.03$), from $\alpha = 10$ for $\lambda/\sqrt{\chi} = 0.1$–$0.32$ (the fan-in default is 0.2), and only from $\alpha = 100$ at balance (2 of 6 lazy at $\alpha = 10$) and for a heavy root ($\lambda/\sqrt{\chi} \geq 3$, none at $\alpha = 10$): a light root lowers the lazy boundary by a factor 100 in α, a heavy one does not move it. The saddle-to-saddle majority reaches $\alpha = 10^{-4}$ for $0.1 \leq \lambda/\sqrt{\chi} \leq 10$ and 10^{-5} or 10^{-6} outside; on this one-decade grid no widest point near balance can be read off. With $\chi = 3$ every run stalls on the floor from $\alpha = 1$.

Kernel change at fixed α. empirical confirming derived (Section A.8) At $\chi = 16$, $\alpha = 3.375$, δ_Θ grows like λ^2 from small λ (local slope 1.85 ± 0.06 between $\lambda = 0.03$ and 0.1, predicted 1.8–1.9) and saturates at 0.59–0.61 for $\lambda \geq 10$ (medians; at the lower edge of the predicted 0.6–0.7); the runs are lazy up to $\lambda = 0.3$ and rich from $\lambda = 1$. In the aligned mode a small root is the only core that moves, so the model is linear in it, while a large root freezes and leaves a degree-$(K-1)$ model derived. Two-layer linear networks share the first effect, a lazy phase that turns rich when the features are incomplete, but there a large output weight also gives lazy training known (Kunin et al., 2024; Dominé et al., 2025); for $K \geq 3$ the model left by a frozen root is not linear.

4 The saddle-to-saddle regime

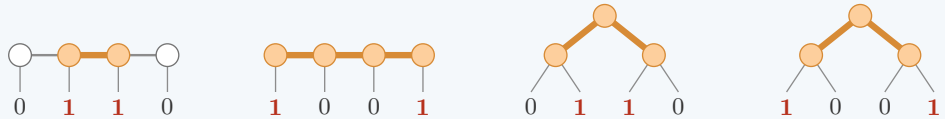
This section treats gradient flow from small initialization, where a run passes from plateau to plateau (Table 4). It answers questions Q1–Q4 of Section 1.1: which saddles are visited, how long the flow stays near each, which component comes next, and how all of this depends on the tree. The organizing idea is the *route* (Section 2.7; box below): near a saddle, new parameters lower the loss at no Gauss–Newton cost only through routes, and, when the spectral condition of Section A.4 holds, the number of cores of the cheapest route sets the type of the saddle and the length of the plateau derived (Section A.3). Unless stated otherwise, inputs are whitened, the loss is the exact population loss, the initialization is fan-in (Section 2.4), and events are dated as in Section 2.8, by the rule of each experiment (Section B).

Section 4.1 shows complete sequences for odeco teachers, Section 4.2 the exactly solvable aligned case, and Section 4.3 predicts the first escape of each run from its initialization. Section 4.4 tests the route rule for the later escapes on two-component teachers, Section 4.5 says when the route gives the saddle order, and Section 4.6 treats teachers with more components. Section 4.7 asks which component is learned first, and Section 4.8 how sequences end early. Supplementary tables, secondary checks and the provenance of the numbers are in Section D.1.

Table 5: Which parameter moves which boundary. Gradient flow, centred model, fan-in initialization, odeco teachers, six seeds per point; binary4: binary tree with $N = 4$, $d = 3$, teacher $(3, 2, 1)$. Lazy boundaries with the tensor kernel; predictions (in brackets) from Section A.8. Scope: trees with at most 16 sites; the χ scalings rest on one tree and less than one decade.

parameter	effect	measured (predicted)	status
scale σ	saddle-to-saddle, rich, then lazy or a floor	binary4: $\sigma_c = 0.24$ to 0.023 ($\chi = 3$–128), $\sigma_{\text{lazy}} = 3.0$ to 0.57 ($\chi = 9$–128)	empirical
bond dimension χ, full bonds	both boundaries $\propto \chi^{-1/2}$	slopes -0.53 ± 0.05 and -0.504 ± 0.011 ($-\frac{1}{2}$ for the lazy one); $\sigma_{\text{lazy}}\sqrt{\chi} = 6.40$–$6.56$ (6.50); rich window 19–24	empirical; derived (lazy constants)
χ below the cut dimension	no lazy regime, a floor	all seeds stalled from $\sigma = 1.0, 1.0, 1.6$ at $\chi = 3, 4, 6$	derived empirical
cores K ($d = 2$, $\chi = 4$)	σ_c rises; stalls below it	$\sigma_c = 0.17, 0.49, 0.61, 0.68, 0.75$ for $K = 3$–7; brackets near 1 for $K = 8$–15; realizable, plain model ($d = 2$, 3): no crossing in 85 of 96 seeds at $K \geq 8$, lower bounds 1.07–2.03	empirical; conjecture (limit $\sqrt{d}$)
root imbalance λ at fixed α	light root: lazy at a 100× smaller α	$\delta_\Theta \propto \lambda^2$ (slope 1.85 ± 0.06; 1.8–1.9), saturation 0.59–0.61 (0.6–0.7)	empirical derived
parameterization, large χ	lazy (SP, NTK) or not (μP-TTN)	$\delta_\Theta \propto \chi^{-3.02 \pm 0.06}$ (-3); NTK: local slope -1.01 between $\chi = 64$ and 128 (-1; full-range fit -0.79 ± 0.04); μP-TTN: $\delta_\Theta \to 6.2$ (6.2–6.3); Section 5.3	empirical derived
thresholds δ_*, ρ	boundaries move, labels robust	×1.30–1.35 for δ_*: $0.05 \to 0.2$ (1.27–1.36); ×2.14–2.34 for ρ: $2 \to 5$ (2.5); at most 7.7% of labels change, each to a neighbouring regime; Section D.2	empirical

How to read every plateau law. *Toy model.* A product of k scalars, each started at σ, with $\dot{a} = a^{k-1}(s - a^k)$ (the balanced mode of (11)), reaches $O(1)$ in a time of order one for $k = 1$, in $\ln(1/\sigma)/s$ for $k = 2$, and in $\sigma^{-(k-2)}/((k-2)s)$ for $k \geq 3$. If the factors start unequal, those that start smallest set the time: k of them at scale σ, far below the others, give the same three laws derived (Proposition A.13) known (Rawal and DeWeese, 2026, Thm 4.2). Every plateau law of this section has this form, with k the number of cores that must move together; prefactors and the head start of $k = 2$ need the full analysis.
Why routes. A route (Section 2.7) moves W only through all of its cores at once, by the product of their small new entries, so its gain comes at no Gauss–Newton cost of the same order (Lemma A.16): near the saddle it fits the residual like a fresh k-core TTN started near zero, and its escape is the first escape (Section 4.3) with K replaced by k, the toy model with k factors (Proposition A.18). The next plateau is therefore set by k^*: it is absent ($k^* = 1$), lasts $\propto \ln(1/\sigma)$ ($k^* = 2$), or lasts $\propto \sigma^{-(k^*-2)}$ ($k^* \geq 3$). For a component that differs from the learned one on the sites D, the route is the Steiner subtree of the $m = |V_D|$ cores that carry D, so it includes the cores on the paths between them proven (Proposition A.19).
Example (teacher B4, as in the introduction: patterns 0000, 0110, 1001 with weights 1, 0.3, 0.6; 0000 is learned first on both trees). On MPS4 the 0.3 component (0110) differs on sites 1 and 2, on adjacent cores: $k = 2$, a logarithmic plateau. The 0.6 component (1001) differs on sites 0 and 3, at the two ends of the chain: $k = 4$, a σ^{-2} plateau. So the smaller component is learned first (*locality beats weight*): 0110 before 1001 in all 60 runs. On binary4 both components need the two leaf cores and the root: $k = 3$ for both, each comes after a σ^{-1} plateau, and they are learned in the order of their weights, 1001 before 0110, in all 48 runs (Section 4.6).



MPS4, weight 0.3, $k = 2$: $\ln(1/\sigma)$ | MPS4, weight 0.6, $k = 4$: σ^{-2} | binary4, weight 0.3, $k = 3$: σ^{-1} | binary4, weight 0.6, $k = 3$: σ^{-1}

Routes of teacher B4 after its first component (0000): pattern of the next component under the sites (in red where it differs from 0000); orange: the cores and bonds of the Steiner subtree.

4.1 Phenomenology

Gradient flow from a small fan-in initialization learns an odeco teacher one component at a time. This subsection shows such runs and which saddles the plateaus are.

Setup (details in Section B.2). Odeco teachers (Section 2.3) with $R = 3$, $d = 4$, Haar-random site bases and weights $(3, 2, 1)$ (*separated*) or $(1, 0.85, 0.7)$ (*close*); trees binary4 ($K = 3$), MPS4 ($K = 4$), binary8 and HT4 ($K = 7$), MPS8 ($K = 8$); $\chi = 3 = R$ or $\chi = 6$, capped by (3). Fan-in initialization with seeds 0–9 at eight values of $\sigma \in [0.003, 0.1]$ ($K = 3, 4$) or five or six in $[0.07, 0.4]$ ($K = 7, 8$, whose later plateaus grow like σ^{-5} and σ^{-6}); also the same W_0 re-gauged to $Q_e = 0$ (*balanced*). A run is *complete* if it learns all three components; Section 4.8 treats the others.

Prediction. derived Two odeco components differ at every site, so the route to a new component is the whole tree at every saddle, $k = K$ (Proposition A.19). Hence (i) the plateaus sit at the partial fits $W_S = \sum_{r \in S} s_r U_r$, S the components learned so far, at the loss

$$L_S = \tfrac{1}{2} \sum_{r \notin S} s_r^2, \tag{18}$$

with bond spectrum $(s_r)_{r \in S}$ on every edge (Proposition 2.1(a)); (ii) each event raises the rank of every edge by one; (iii) every plateau is the first-escape problem of a K-core route (Proposition A.18) and lasts $\propto \sigma^{-(K-2)}$ for a fixed initialization direction; (iv) the plateaus are higher-order saddles, $\lambda_{\min}(H_\perp) \propto \sigma^{K-2}$ (Proposition A.26). The exception to (iv) is binary4, where only two cores carry sites: a learned set that misses a larger component is a strict saddle, $\lambda_{\min} = -(s_r - s_j)\, s_j^{(K-2)/K}$ (j learned, r larger and not learned; Proposition A.22), reached only after an escape out of weight order (in our runs the second escape: order 1-3-2). Binary4 is a depth-3 deep linear network (Remark A.23), for which all of this is known known (Saxe et al., 2014; Jacot et al., 2022; Achour et al., 2024); the other four trees are genuine tensor networks.

Plateaus at the rank-constrained partial fits, one rank step per event. empirical Complete sequences were measured on binary4 and MPS4 over 1.5 decades of σ, on binary8 and HT4 at $\chi = 6$ over 0.6 decade, and on MPS8 up to its second component (run counts in Section D.1.1). At every plateau between resolved events ($\rho_i > 3$) the loss equals (18) for the components learned so far, in whatever order: to a relative $2 \cdot 10^{-2}$ at worst (binary4, close teacher), and to $7.4 \cdot 10^{-7}$ in the other cells at $\sigma = 0.003$. The top $|S|$ singular values of every edge equal the learned weights (to a relative $7.7 \cdot 10^{-3}$ at worst), and the next one is below 0.07 times the smallest weight: the plateaus are the critical points W_S of the problem with all bond ranks fixed at $|S|$. In 497 of the 531 complete runs that recorded spectra in their event windows, every event raised by exactly one, on every edge, the number of singular values above about half the weight being learned (the exceptions: overlapping events on binary4 with the close teacher at $\sigma \geq 0.05$, and 6 HT4 runs). new (MPS4, binary8, HT4; known for binary4)

Plateaus grow as $\sigma^{-(K-2)}$. empirical Per-seed fits of $\ln(t_{i+1} - t_i)$ against $\ln \sigma$ give $K - 2$ to within 0.1 per seed on binary4 and 0.04 on MPS4, over 1.5 decades (limited range for some binary4 close-teacher seeds: after the cut $\rho_i > 3$ they keep four or five values of σ, about 0.8 decade). At $K = 7, 8$ the means agree with $K - 2$ (binary8 5.02 ± 0.01 and 4.97 ± 0.02, MPS8 5.95 ± 0.05 for its only gap), but over 0.6 decade (limited range); HT4 is not a clean confirmation (per seed 3.93–5.78). All fits: Table 21. new (MPS4, binary8, HT4, MPS8; known for binary4)

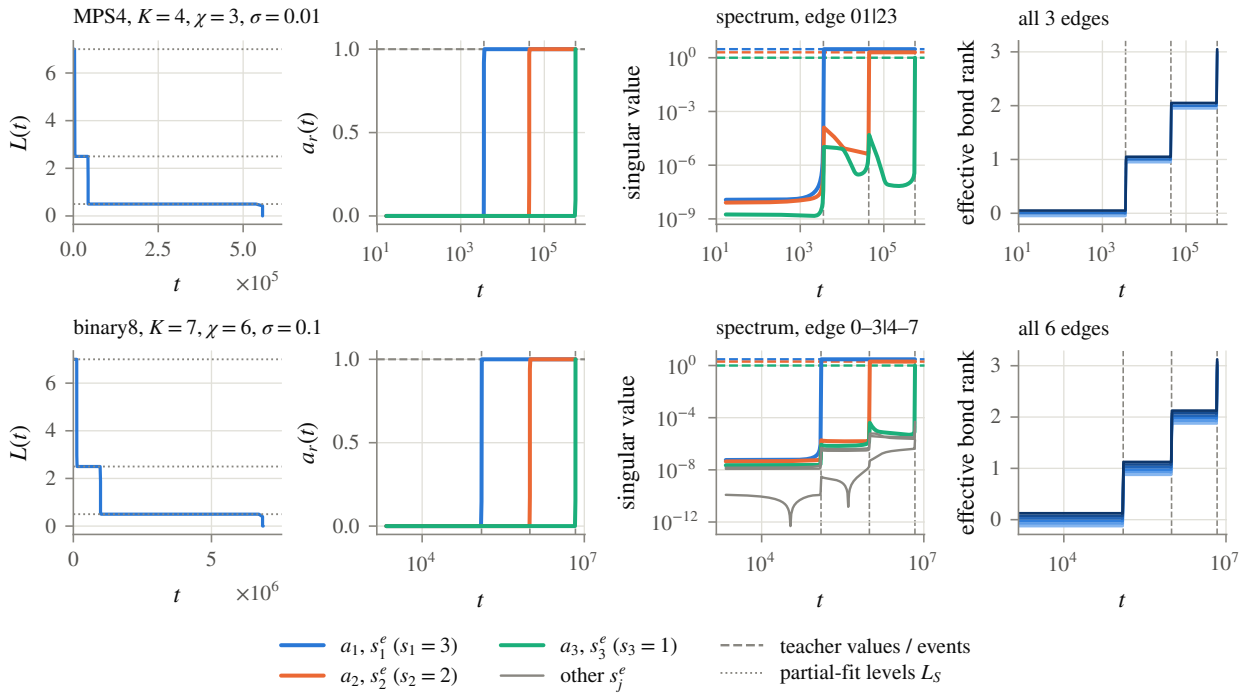


Figure 4: Two saddle-to-saddle runs: separated teacher $(3, 2, 1)$, fan-in initialization, seed 0; top: MPS4, $\chi = 3$, $\sigma = 0.01$; bottom: binary8, $\chi = 6$, $\sigma = 0.1$. Columns: loss in linear time with the partial-fit levels (18) (dotted); overlaps a_r; singular values s_j^e of the middle edge 01|23 (MPS4) and the root edge 0–3|4–7 (binary8), teacher values dashed (edges are named by the sites on either side, numbered from 0); effective bond rank of every edge, offset slightly per edge. Dashed verticals: events. Takeaway: one component per event, in the order of the weights; every edge gains one singular value at each event, all at the same time, and the next ones stay far below the teacher's in between (source: Section B, Table 12).

Saddle type. empirical At the plateau midpoints $\lambda_{\min}(H_\perp)$ vanishes like σ^{K-2}: the cell means of the per-seed slopes are 1.02–1.18 (binary4), 1.99–2.02 (MPS4), 4.88–5.13 ($K = 7$, $\chi = 6$) and 5.99 (MPS8). In the medians of the fan-in cells the magnitude is 0.87–1.15 times the balanced-route value of (45) at $K = 3, 4$ (1.19 for balanced binary4) and 1.36–3.9 times it at $K = 7, 8$, where the routes are not balanced. The strict exception appears as predicted: on binary4 with the close teacher, the runs that learn in the order 1-3-2 rest at $\lambda_{\min}$ between -0.11 and -0.15, flat in σ, against $-(0.85 - 0.7)\, 0.7^{1/3} = -0.133$, and leave within about one event width; the wrong order, and with it the strict saddle, becomes rare as $\sigma \to 0$ (Proposition A.15; seeds in Section D.1.1). Otherwise the full order on binary4 is greedy (all 160 complete runs with the separated teacher; 89% and 95% at $\chi = 3$, 6 with the close one); on the other trees it varies from seed to seed (Section 4.7). new (MPS4, binary8, HT4, MPS8) known (binary4 and its strict saddle, deep linear networks: Achour et al., 2024, Thm 7; Jacot et al., 2022, Lemma 15)

4.2 An exactly solvable case: aligned initialization

If every core starts diagonal in the teacher's basis (Proposition 2.4), the components decouple exactly, a balanced start stays balanced, and each component follows the mode equation of deep linear networks, here on any tree proven known (Saxe et al., 2014, eq. 15). By (12), with one initial amplitude for all components they are learned in the order of their weights, at times in the ratio $1/s_1 : 1/s_2 : 1/s_3$, and every gap grows like $\sigma^{-(K-2)}$ known (from eq. 15 of Saxe et al., 2014; symmetric

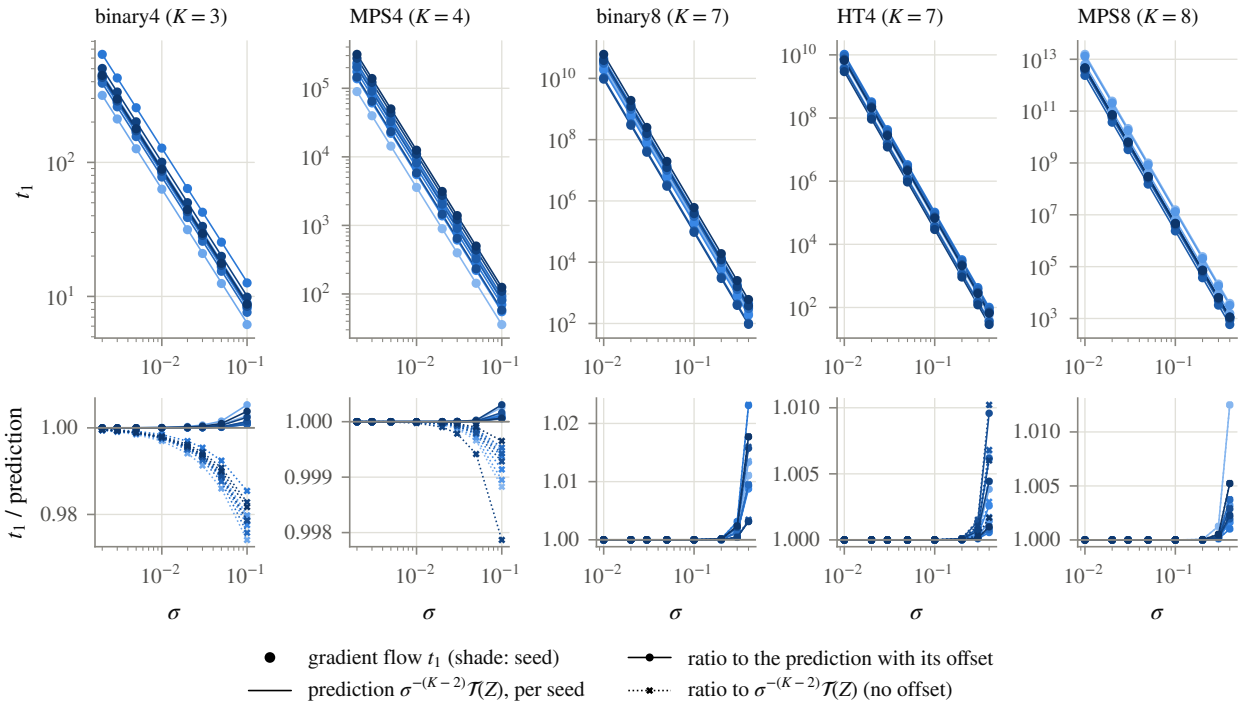


Figure 5: First escape against σ: $\chi = 3$, separated teacher, fan-in initialization, ten seeds (one shade of blue per seed). Top: t_1 from gradient flow (dots) and each seed's prediction $\sigma^{-(K-2)}\mathcal{T}(Z)$ (lines), with $\mathcal{T}(Z)$ computed from that seed's initialization before training. Bottom: t_1 over the prediction (19), with the offset $\delta_K s_{r_1}^{-(2-2/K)}$ (solid) and without it (dotted). Takeaway: one number per initialization, computed before training, predicts the first escape of every run, to within 10^{-5} at the smallest σ in every panel; the deviations at large σ are the offset and, at $K = 7, 8$, a direction that has not converged (source: Section B, Table 12).

matrix factorization from a multiple of the identity: Li et al., 2021, Thm 6.2). This is the case $k = K$ of Proposition A.25: a route through every core never sees the components already learned. Random initialization is unbalanced, so its escape times are not those of a balanced start of the same norm (the toy model of the box; Section 4.3). The exact solution also checks the integrators: gradient flow on the full TTN follows it on all five trees, every overlap to within $3 \cdot 10^{-9}$ empirical (Figure 19 in Section D.1.2).

4.3 Escape from the origin

How long does the first plateau last? The answer is a blow-up time computed for each initialization, with no free parameter; the norm of the initialization gives only a lower bound, low by factors from 3 to $3 \cdot 10^7$.

Prediction: one blow-up time per initialization. On the first plateau $\|W\|_F$ is far below the teacher's weights, and gradient flow is to leading order (LO) the flow $\dot{\theta} = \nabla g$ of the correlation $g = \langle W, W^* \rangle$ (Section A.2). In the time $\mathrm{d}\tau = \|\theta\|^{K-2}\mathrm{d}t$ the direction $\hat{\theta}$ follows the spherical gradient flow of g, independent of σ, and $\mathrm{d}\ln\|\theta\|/\mathrm{d}\tau = K\, g(\hat{\theta})$ (Proposition A.4). Hence $\|\theta\|$ blows up at

$$t^* = \|\theta_0\|^{-(K-2)} \int_0^\infty e^{-(K-2)KG(\tau)}\, \mathrm{d}\tau, \qquad G(\tau) = \int_0^\tau g(\hat{\theta})\, \mathrm{d}\tau',$$

which for $\theta_0 = \sigma Z$ (Z the initialization at $\sigma = 1$) is $\sigma^{-(K-2)}\mathcal{T}(Z)$, with $\mathcal{T}(Z)$ independent of σ (Corollary A.5). The direction tends to the balanced rank-one configuration of one component r_1, which then escapes as a balanced mode (Proposition A.6):

$$t_1 = \sigma^{-(K-2)}\,\mathcal{T}(Z) + \delta_K\, s_{r_1}^{-(2-2/K)} + o(1), \tag{19}$$

with δ_K from (29) ($\delta_3 = -0.850$, $\delta_4 = -0.266$, $\delta_7 = 0.148$, $\delta_8 = 0.201$). $\mathcal{T}(Z)$ and r_1 come from integrating the LO flow from Z (Remark A.7), so (19) predicts the exponent, the prefactor and the first component of every run. Status: the scaling and the blow-up identity are proven (at LO); the offset is derived (the direction must have converged to r_1 by the time $\|W\|_F$ is comparable to s_{r_1}), and so is the first component, which needs the KKT point reached to be a component configuration (Proposition A.10; not for stars, Remark A.27). The escape direction is exactly balanced, with core norms $1/K$ (Proposition A.9) proven; at the end of the LO flow every core holds $1/K$ of $\|\theta\|^2$ to within 1.3% in all 1,200 initializations of the d sweep empirical. known (the exponent: deep linear networks, Jacot et al., 2022; homogeneous models, Kumar and Haupt, 2025a, Thm 1; the identity is the activation law of Kunin et al., 2025, Eqs. 2–3; balance: Kumar and Haupt, 2025a, Lemma 5, and the proof of Furman et al., 2026, Prop. F.10) new (TTN form: its evaluation on trees and the tests)

Test, seed by seed (Figure 5). empirical For 231 pairs of initialization and teacher on the five trees (1,848 runs, eight values of σ each), the prediction matches every run at the smallest σ to a relative $4 \cdot 10^{-8}$ to $2.2 \cdot 10^{-3}$ (the largest for binary4 with the close teacher, whose direction has not converged at $\sigma = 0.002$). With the offset removed, the per-seed exponents are $K - 2$ to within 0.004, except in that cell (0.990–0.996). Details: Section D.1.3.

The size of the initialization gives only a lower bound. From an initialization along a maximizer of g the escape takes $t_{\mathrm{KH}} = K^{K/2-1}\|\theta_0\|^{-(K-2)}/((K-2)s_1)$ known (Kumar and Haupt, 2025b, text before Thm 11), and every initialization of the same norm takes longer (Corollary A.12) proven. The median $\mathcal{T}/t_{\mathrm{KH}}$ over the cells of the first-escape sweep is 3.1–4.8 on binary4, 7.9–13.7 on MPS4, 95–210 on HT4, 184–2650 on binary8 and 1460 on MPS8, and it reaches $2.9 \cdot 10^7$ on binary16 empirical new (the bound on trees and the ratios). Most of $\|\theta_0\|$ sits in core slices orthogonal to the teacher's factors, which do not move at leading order (Proposition A.11), and the moving slices form an unbalanced product path (Proposition A.13; the toy model of the box): the norm law is a bound, not a prediction.

Dependence on d, χ and K. empirical The prefactor $\mathcal{T} = \lim_{\sigma\to 0}\sigma^{K-2}t_1$ falls slowly with χ (log-log slopes -0.26 to -0.76, against -0.47 to -2.5 for t_{KH}): χ acts only through the moving slices. At full bond $\chi = d^2$ it does not depend on d (binary4 medians $\approx \pi/2$ for $d = 2, 3, 4$, the closed form of Proposition A.13, derived given end directions that do not rotate). Its median ranges from 0.42 (HT3, $K = 5$) to 814 (binary16, $K = 15$), growing with K within each tree family, and a heuristic with no free parameter tracks it within a factor 0.87–2.13 (conjecture, heuristic). Re-gauging to $Q_e = 0$ changes $\mathcal{T}$ per seed by a factor 0.88–1.92 and leaves the exponent unchanged. Details and Figure 20: Section D.1.3. new

4.4 Locality of later escapes: two components

After the first escape, gradient flow rests near a saddle that represents what has been learned. Which component comes next, and how long is the next plateau? In a TTN the answer depends on where the next component sits in the tree (box at the start of this section). We state the route rule, derived in Section A.3, and test it on two-component teachers.

Prediction: the route rule. Take a two-component pattern teacher (Section 2.3), $W^* = s_1U_1 + s_2U_2$ with $s_1 > s_2$, whose components differ exactly on the site set D (sites numbered from 0, as in Table 10). As in Section 2.7, the route size is

$$k = |\mathrm{St}(V_D)|, \tag{20}$$

the number of cores of the Steiner subtree of the cores V_D that carry D, and the bonds whose teacher rank is two are exactly the bonds of $\mathrm{St}(V_D)$ (Proposition 2.1(b)). Once U_1 is learned, the flow sits near the balanced saddle θ_1 that represents s_1U_1, and new parameters can lower the loss at no Gauss–Newton cost only through routes (Lemma A.16); for two-component teachers in the correct order no other curve descends faster (Corollary A.32). The cheapest route to U_2 is $\mathrm{St}(V_D)$, with coupling $c_2 = s_2 s_1^{(K-k)/K}$ (Proposition A.19), and the escape is the first-escape problem of a k-core TTN (Proposition A.18). With $\Delta t_2 = t_2 - t_1$ the time between the two learning events,

$$\Delta t_2 \simeq \begin{cases} O(1) \text{ (no plateau)}, & k = 1, \\ \dfrac{1 - s_2/s_1}{\lambda_2} \ln \dfrac{1}{\sigma}, \qquad \lambda_2 = s_2\, s_1^{(K-2)/K}, & k = 2, \\ A\, \sigma^{-(k-2)}, & k \geq 3, \end{cases} \tag{21}$$

(Proposition A.25). For $k = 2$ the saddle is strict, with $\lambda_{\min}(H_\perp) = -\lambda_2$ (Proposition A.21); for $k \geq 3$ its Hessian has no negative eigenvalue (a higher-order saddle), and on the plateau $\lambda_{\min} \propto \sigma^{k-2} \to 0$ (Proposition A.26). The factor $1 - s_2/s_1$ is a *head start*: the route grows during the first plateau, so from the saddle itself the $k = 2$ slope is $1/\lambda_2$. The prefactor A is set by the route amplitudes when the plateau starts and is not predicted from σ alone for random initialization. Status: the Steiner identity, the upper bound k on the saddle order and, for two-component teachers learned in this (greedy) order, the matching lower bound, so that the order is exactly k on every tree (Corollary A.32), are proven; (21) is derived (leading order, small σ). known (mechanism and trichotomy: Kumar and Haupt, 2026, Thm 8; Rawal and DeWeese, 2026, Thm 4.2; descent along a whole connected set of rank-deficient bonds: Furman et al., 2026, Lemma D.8; head start for $k = 2$: Zhang et al., 2026, Thm 4, Kunin et al., 2025, Sec. 2.2) new (the route is the Steiner subtree, which can be a proper subtree of such a set; the coupling and rate; the head start for $k \geq 3$)

Setup (details in Section B.3). Orthogonal pattern teachers with $d = 2$ and weights $(1, 0.5)$ unless stated, $\chi = 2$, whitened inputs, gradient flow on the exact population loss. Nine trees: MPS with $N = 4, 6, 8$, binary4, binary8, HT4, HT8, and two trees with $N = 6$, *star3* (a root without sites, three children with two sites each) and *midroot6* (the MPS6 chain rooted at a middle core). Protocols: *small* (fan-in initialization at scale σ, first plateau included), *saddle* (the exact balanced saddle s_1U_1 plus a fan-in perturbation of scale σ; $\Delta t_2 = t_2$ times the escape alone) and *balanced* (the fan-in W_0 re-gauged to $Q_e = 0$). Per seed (seeds 0–7) we fit $\ln \Delta t_2$ against $\ln(1/\sigma)$ (exponent p), or Δt_2 against $\ln(1/\sigma)$ (the $k = 2$ slope), over the values of σ at which the seed escapes (at least three; fewer than the row's six or seven for seeds that stall at some σ, Section 4.8), and report mean ± s.e. over seeds (median). Log and power laws are compared by the residual of $\ln \Delta t_2$ on the same data (*fair comparison*). Fits use the runs that learn U_1 first, without rotations (an overlap crossing $\frac{1}{2}$ more than once). About 11,600 runs in all.

Two components: the plateau exponent is $k - 2$. empirical (Figure 6; every row in Table 22) There is no plateau for $k = 1$, a logarithmic one for $k = 2$, and the exponent is $k - 2$ for $k = 3$ (all nine trees), 4 and 5, on trees with K from 3 to 15: it follows the Steiner count, not K or the

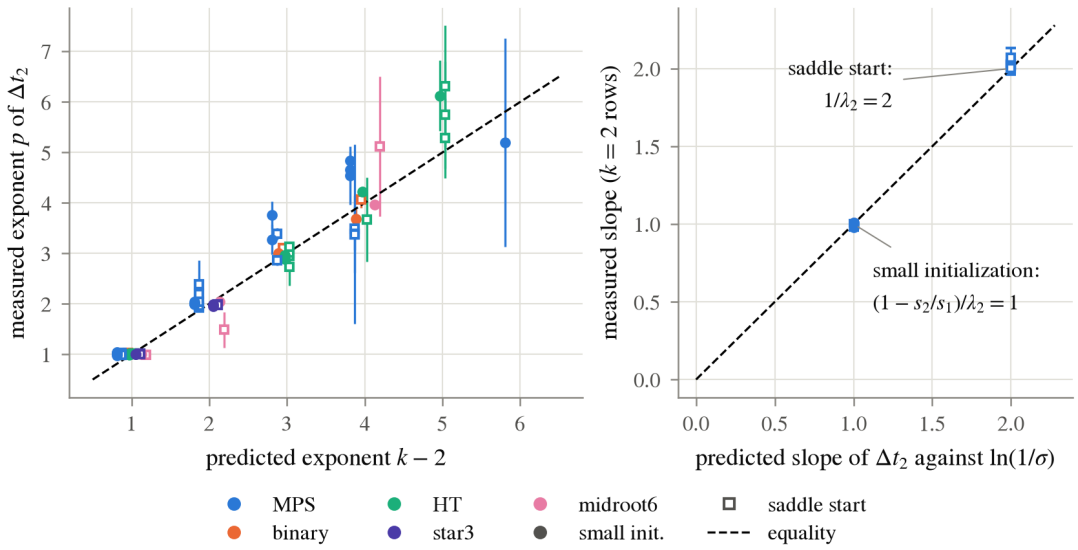


Figure 6: Plateau law after the first component: two-component pattern teachers, $d = \chi = 2$, weights $(1, 0.5)$, all rows of the main sweep. Left: per-seed exponent p of $\Delta t_2 \propto \sigma^{-p}$ (mean $\pm$ s.e. over the fitted seeds, at most eight) against the prediction $k-2$, $k = |\mathrm{St}(V_D)|$, for every row with $k \geq 3$; filled circles: small initialization, open squares: saddle start; colour: tree family (MPS blue, binary orange, HT green); points are offset horizontally by family and protocol. Right: slope of Δt_2 against $\ln(1/\sigma)$ for the $k = 2$ rows, against $(1 - s_2/s_1)/\lambda_2 = 1$ (small initialization) and $1/\lambda_2 = 2$ (saddle start). Dashed: equality. The points at $k - 2 \geq 4$ have limited range (0.5–0.8 decades of σ, 1–5 fitted seeds) and are not measurements of the exponent (one-decade values in Table 22). Takeaway: up to $k = 5$ the exponent is set by the number of cores of the Steiner subtree, on every tree; for $k = 2$ the head start halves the slope (source: Section B, Table 13).

tree family. Named exceptions: the saddle-start $k = 4$ rows and HT8 $\{1, 2\}$ are carried by seeds that stall at some σ (Section 4.8); the MPS rows with $k = 5$ exceed 3 for $\sigma \in [0.03, 0.3]$ but give 2.97–3.15 at $\sigma \leq 0.03$ or on never-stalling seeds, except MPS8 $\{1, 5\}$ from the saddle, 3.67 ± 0.20 (limited range: 4 seeds, 0.4 decade). For $k = 6$, 7 the exponent is $k - 2$ on binary8 and HT4 *with balanced initialization only*, over one decade; MPS8 with $k = 6$ gives 4.51 ± 0.19, and the other $k \geq 6$ rows (limited range) show only that the exponent grows with k. $k = 8$ is not measured. Coverage is uneven: $k = 2$ occurs only on chains (MPS, midroot6), $k = 4$ on chains and star3, $k = 7$ only on HT trees. new

$k = 2$: a logarithmic plateau with a head start. empirical The fair comparison prefers the logarithmic law in all 14 rows with $k = 2$ (1.1–3 decades of σ). Its slope carries the head start: 0.980–1.009 from small initialization against $(1 - s_2/s_1)/\lambda_2 = 1$, and 2.002–2.071 from the saddle against $1/\lambda_2 = 2$ (Figure 6, right); without a head start both would be 2. Other weights and trees agree as well, including the dependence of λ_2 on K: with weights $(2, 1)$ from the saddle the slopes on MPS4, MPS6 and MPS8 are within 3% of $1/\lambda_2 = 0.707$, 0.630, 0.595 (Section D.1.4). known (head start: two-layer linear networks, Zhang et al., 2026, Thm 4; dormant norms carried across phases, Kunin et al., 2025, Sec. 2.2) new (the TTN rate λ_2 and its dependence on K)

Saddle type: strict for $k = 2$, higher order for $k \geq 3$. empirical At the plateau midpoint we compute $\lambda_{\min}$ of $H_\perp$ and of the raw Hessian and read the type from its scaling with σ (Proposition A.26). For $k = 2$, $\lambda_{\min} = -0.5000 = -\lambda_2$ from the saddle at every σ, with one negative eigenvalue per active route (also at $\chi = 4$, 8, and for other weights); from small initialization the

raw value is -0.498 to -0.499 for $\sigma \le 10^{-3}$. For $k \ge 3$, $\lambda_{\min} \propto \sigma^x \to 0$, with x within 0.2 of $k-2$ in 126 of 135 rows with $k = 3$–5, and the raw x is within 0.2 of the row's duration exponent p in 95% of 169 rows with $k \ge 3$: where the plateau law deviates, the curvature deviates the same way. No plateau with $k \ge 3$ has σ-independent negative curvature, as predicted for saddles visited in the greedy order; wrong-order saddles (Sections 4.1 and 4.7) and star saddles (Section 6.6) were not part of this sweep. Per-row values and Figure 21: Section D.1.4. new (for TTNs) known (deep linear networks: Achour et al., 2024, Thm 7)

One teacher, different trees. empirical Since k depends on the tree, one teacher has different plateau laws on different trees. For $D = \{3, 4\}$ and $N = 8$ the route has $k = 2$ on MPS8, 5 on binary8 and 7 on HT8 (Table 10). MPS8 shows a logarithmic plateau (saddle slope 2.002 ± 0.010 against 2; residuals 0.003 for the log law, 0.064 for a power law), binary8 a σ^{-3} plateau (2.998 ± 0.078 small, 3.106 ± 0.057 saddle), and HT8 $p = 5.74 \pm 1.06$ (median 5.01; saddle start, limited range: 0.6 decade, one seed with a pinned route), consistent with 5 but unable to separate $k = 6$, 7, 8. The order of learning changes too (teacher B4, box and Section 4.6). new

Bond spectra: the new component enters through the Steiner edges only. empirical The second component raises the teacher's bond rank exactly on the edges of $\mathrm{St}(V_D)$ (Proposition 2.1(b)), so the bond spectra show which cores the escape uses (Figure 7). Through all 5,131 two-component events the second singular value of every edge outside the Steiner subtree stays at the initialization-noise level, from the plateau midpoint to the end of the run: $s_2^e \le 4.3\,\sigma^2 s_2$, scaling as $\sigma^{2.07}$. This *leakage bound* is the substantive result: the new component enters through the Steiner edges only, up to $O(\sigma^2)$. Given it, a check of which edges gain effective rank tests that every Steiner edge rises, and that they rise together: the edges whose effective rank is higher at $a_2 = 0.9$ than at the plateau midpoint are the Steiner edges in 99.75% of the 4,448 events with a resolved plateau, and their crossing times differ by a median $5 \cdot 10^{-5}$ event widths. new

4.5 When is k^* the saddle order?

The route gives a descent of order k^* at every learned saddle, so the saddle order (Section 2.7) is at most k^*. Can anything descend faster?

The theorem, informally. proven (Theorems A.30 and A.31 and Corollary A.32) At a saddle with one learned component j, and with a spare bond channel along the route, the saddle order is exactly k^* unless a piggyback coupling of the residual on small sets of cores exceeds the learned weight s_j. This never happens for $k^* = 2$, nor for two-component teachers learned in weight order, on any tree. For $k^* = 3$ or 4 it happens exactly when the saddle is strict ($\lambda_{\max}(\Gamma) > s_j$ below; equality at $k^* = 4$ is open). From $k^* = 5$ on it can happen while the Hessian is positive semidefinite, and the order then drops below k^*. Saddles with two or more learned components are not covered. new

The piggyback idea. Besides opening new channels, the flow can rotate the site vectors of single learned cores, piggybacking on the learned channel. Two such rotations together produce a component that differs from the learned one on exactly those two cores, at second order and without new channels. The *piggyback graph* (Proposition A.22) has one node per pair (core, rotated pattern) and one edge of weight s_r per residual component r that differs from the learned one on exactly two cores; if its largest eigenvalue $\lambda_{\max}(\Gamma)$ exceeds s_j, the learned saddle is strict. This happens when

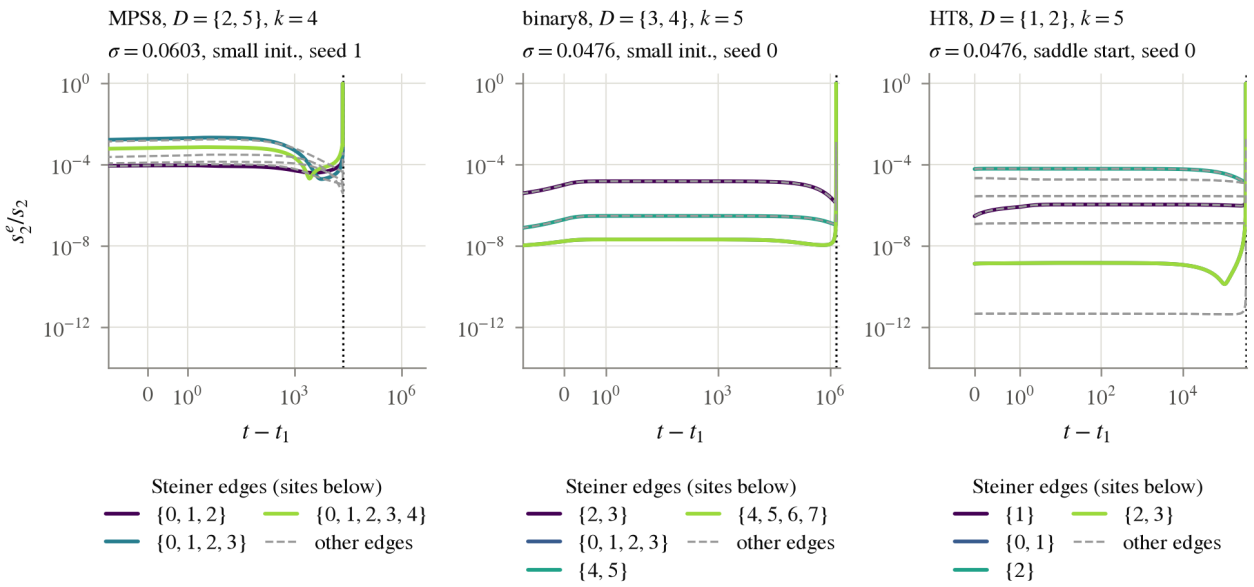


Figure 7: Bond spectra through the second event, two-component teachers, $d = \chi = 2$, weights $(1, 0.5)$. Second singular value s_2^e of every edge, relative to s_2, against the time since the first event (symmetric-log axis); coloured: the edges of the Steiner subtree $\mathrm{St}(V_D)$ (legend: the sites below each edge, numbered from 0), grey dashed: all other edges, drawn on top (edges with equal values, such as the two root edges of a binary tree, coincide); dotted vertical line: the second event. Left: MPS8, $D = \{2, 5\}$, $k = 4$, small initialization; middle: binary8, $D = \{3, 4\}$, $k = 5$, small initialization; right: HT8, $D = \{1, 2\}$, $k = 5$, saddle start (one run each). Before the event every edge sits at the initialization-noise level, Steiner edges included; at the event exactly the Steiner edges rise, together. Takeaway: the escape uses the cores of the Steiner subtree and no others (source: Section B, Table 13).

a larger component is still waiting (a wrong order, Section 4.7) or when several components share a core (a *star*, Section 6.6). From $k^* = 5$ on, small clusters of cores can piggyback in the same way: new entries on a cluster whose Steiner subtree has fewer than $k^*/2$ cores, together with rotations of other cores, couple through the residual, and a generalized piggyback operator (Definition A.29) decides whether they descend before the route.

A counterexample. On MPS6 the teacher 000000 (weight 1), 110001 and 110010 (0.8 each) has $k^* = 5$ and a positive semidefinite Hessian at the saddle of its first component, yet new entries on cores 0 and 1 aimed at the non-residual pattern 110000, together with rotations of cores 4 and 5, descend at order 4 proven (Section A.4). Gradient flow from this saddle escapes in $\propto \sigma^{-2}$ instead of σ^{-3} derived (checked on this teacher: 5 seeds from the saddle, and the seeds that visit it from small initialization), and then forgets the first component empirical (seen from small initialization). For the matched control with weights 0.6, which meets the condition and has order 5, the small-initialization exponent is about 3 in some seeds and about 4 in others, unexplained (Section 4.8).
[Pending (saddle-dyn): the dynamics test of the theorem; one sentence here when it lands.]

Which experiments satisfy the condition. Every two-component fit of Section 4.4 (the runs that learn the larger component first), and every saddle with one learned component of the multi-component teachers of Section 4.6 (there $\lambda_{\max}(\Gamma) \le 0.7 < s_j$). The condition fails for the HT4 runs with $D = \{0, 1, 2, 3\}$ that learn the smaller component first (order 6, not $k = 7$; excluded from the fits): in 6 of the 7 such runs ($\chi = 2$, $\sigma \ge 0.4$) the smaller component is forgotten when the larger

one is learned, and then relearned empirical. A check of all rank-one saddles that our runs can reach finds 30 such order-6 saddles, all after a wrong first escape on binary and HT trees (Section 4.6).

4.6 Several components

Prediction: the route rule at rank-constrained optima. With several learned components "the sites where the next component differs" is ambiguous, and saddles need not be partial sums of components. The *route rule* (Remark A.27) works at the saddle actually reached, the best approximation of W^* with the current bond-rank profile: if the learned structure is stable (no wrong order, no star, and from $k^* = 5$ on no generalized star; Section 4.5), the next escape takes the smallest route, of size k^*, with a nonzero local residual (Definition A.17), raises the bond rank on its internal edges, and the flow moves to the best approximation with the new profile. We evaluate it numerically (rank-constrained optima by L-BFGS with restarts); for two components $k^* = k$. Status: derived at leading order for each step, conjecture as an algorithm.

Test. empirical (predictions derived at leading order) Seven teachers (patterns as site values, weights in brackets): TA 0000 (1), 1100 (0.6), 1110 (0.35); TB as TA plus 1111 (0.2); N4 0000 (1), 1100 (0.6), 1111 (0.4); B4 0000 (1), 0110 (0.3), 1001 (0.6); E8 0^8 (1), $0^3 1^2 0^3$ (0.4), $010^4 10$ (0.7); TC 0^8 (1), $1^2 0^6$ (0.6), $1^4 0^4$ (0.35), 1^8 (0.2); TD 0^8 (1), $0^3 1^2 0^3$ (0.6), $1^2 0^6$ (0.35); $\chi = 2$ (3 for B4, N4, E8; Section B.3). We detect the plateaus of $L(t)$ (separation ratio above 3, linear time) and match them to the levels predicted by the route rule (within 5%). Every predicted level with $k^* \geq 3$ is found as a plateau (88–100% of the runs; 35–65% for a few $k^* = 3$ levels within 10% of the next one), with one exception below, and its duration exponent is within 0.2 of $k^* - 2$ (1.02–1.19, 2.02–2.08, 2.97–3.17, 4.05 for $k^* = 3, 4, 5, 6$). The exception is B4's second level on HT4 with fan-in initialization, 4.73 ± 0.90 (limited range: 4 fitted seeds, 23 of 48 runs stalled; 3.17 ± 0.10 with balanced initialization). The $k^* = 2$ levels are logarithmic where measured over three decades from the saddle (four levels of TC, the first of TD); elsewhere (TD's second level, the first levels on MPS8 from small initialization) a power law below 1 fits better and the law is not determined. At the predicted saddles the measured bond-rank profile equals the predicted one in 93.9% of 3,114 resolved (run, level) pairs. new

Locality beats weight. empirical On MPS4 all 60 runs learn B4's 0.3 component before its 0.6 component, after a $k^* = 2$ level too short to be detected as a plateau; the 0.6 component follows after a σ^{-2} plateau ($p = 2.015 \pm 0.013$, 5 seeds). On binary4 all 48 runs learn them in weight order, after two σ^{-1} plateaus (1.051 ± 0.008 and 1.028 ± 0.005, 8 seeds). As $\sigma \to 0$ a logarithmic route beats a power-law route whatever the weights (Remark A.27) derived; at finite σ weights and prefactors compete. new

The route rule, not the literal rule. empirical Three teachers discriminate between rules. The literal rule (the Steiner count of the sites where the next component differs from the closest learned one) predicts no plateau before the third component of TA on any tree; there is a σ^{-1} plateau on binary4 (1.023 ± 0.005, 8 seeds) and a σ^{-2} one on HT4 (2.084 ± 0.040, 7 seeds), as the route rule predicts (Figure 8). For TD on MPS8 (from the saddle) the last plateau has $k = 2$ by the literal rule, $k = 4$ by the exact bond-rank rule (cores whose bond rank must grow to add the component exactly, Lemma A.20) and $k^* = 3$ by the route rule, which first learns part of the component through a two-core route: measured 1.012 ± 0.010 (8 seeds). For TC on MPS8 (from the saddle) the route rule predicts a $k^* = 4$ plateau where the bond-rank rule predicts $k = 5$: measured 2.018 ± 0.017. new

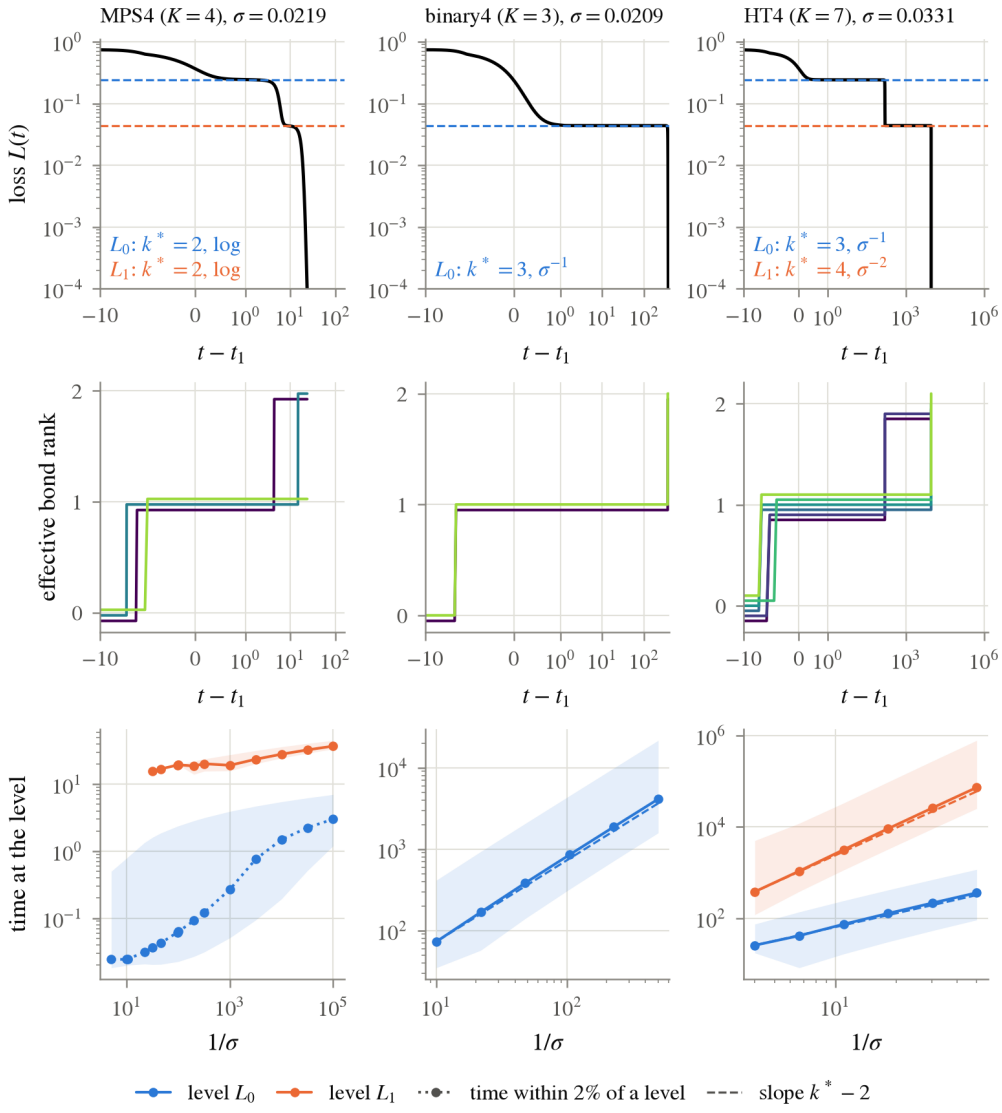


Figure 8: One teacher on three trees: teacher TA (patterns 0000, 1100, 1110, weights 1, 0.6, 0.35), $d=\chi=2$, small fan-in initialization. Columns: MPS4 ($K=4$), binary4 ($K=3$), HT4 ($K=7$). Top: loss of one run (seed 0, σ in the panel title) against time since the first event (symmetric-log axis), with the plateau levels predicted by the route rule (dashed; labels: k^* and the predicted law). Middle: effective bond rank of every edge in that run, one colour per edge (threshold 0.1 times the smallest nonzero teacher singular value of the edge; edges offset vertically by 0.05). Bottom: duration of each predicted level against $1/\sigma$, median over 8 seeds with the range shaded; solid: detected plateaus, dotted: time spent within 2% of a level too short to be detected; dashed: slope k^*-2. Measured: MPS4 two short levels (the second logarithmic in σ; the law of the first is undetermined); binary4 one σ^{-1} plateau, where the first escape already fits U_1, U_2 and part of U_3; HT4 σ^{-1} and σ^{-2} plateaus (1.09, 2.08). Takeaway: the tree, not the teacher alone, sets the number of plateaus and their laws (source: Section B, Table 13).

Whole sequences: a saddle hopper. empirical (bookkeeping of known theory) The sequence rules of this section (first escape, route rule, branches after a wrong first escape, race of logarithmic routes) were implemented as a predictor that hops from saddle to saddle, and scored on all committed runs without tuning (Section D.1.5). Where its prediction is informative (one predicted order, run not stalled) it matches 97.0% of the event sequences, against 95.8% for the baseline "learn the components in weight order"; it beats the baseline only on the 107 runs where the route rule overrides the weight order (B4 on MPS4, E8 on MPS8: 88.8% against 0%). Its race across several saddles of the nested teacher TC (MPS8, from the saddle) predicts five plateau slopes within 3.5% empirical new. Checked at the 587 rank-one saddles that the scored runs can reach, the condition of Section 4.5 fails at 30, where the order drops from 7 to 6 proven (the theorems applied by exact computation); all are saddles reached by a wrong first escape on binary and HT trees, and in one HT4 seed the escape has the predicted signature of the clusters (a ratio of loss decrease to overlap of 0.2998, against 0.300; about 1 for a route) empirical new.

4.7 Order of learning

Which component is learned first, and how random is the answer? With aligned initialization every tree learns the components in the order of their weights (Section 4.2), and a deep linear network of depth three leaves the origin along its top singular pair as $\sigma \to 0$ known (Jacot et al., 2022, Thm 2; with probability one if the gradient at the origin has full rank). In a TTN with random initialization the answer depends on $m = |V_D|$, the number of cores that carry a site on which the two components differ (for odeco teachers $m = N$ on an MPS, 4 on binary8 and 2 on binary4): the order is a race set by the initialization when $m \geq 3$ and greedy as $\sigma \to 0$ when $m = 2$.

Prediction. As $\sigma \to 0$ the first component learned is the local maximum of $g = \langle W, W^* \rangle$ on the sphere whose basin contains the initial direction, since the spherical flow does not depend on σ (Section 4.3) derived known (convergence to a KKT point selected by the initialization: Kumar and Haupt, 2025a, Thm 1). For $m \geq 3$ every component is such a maximum (*tensor-like* pairs); for $m = 2$ only the larger one, the smaller having n unstable directions, with $n = \chi_e$ if its two cores share the bond e and $n = 1$ otherwise (*matrix-like* pairs; Proposition A.10) derived. With weights $(1, w)$:

(i) *Race.* On an MPS the components of an odeco teacher evolve independently at leading order (Proposition A.11); component r blows up at $\sigma^{-(K-2)}\hat{t}_{1,r}/s_r$ with $\hat{t}_{1,r}$ i.i.d., so

$$P(\text{larger first}) = P(\hat{t}_{1,2}/\hat{t}_{1,1} > w), \tag{22}$$

independent of σ (Proposition A.14) proven (at LO). Cores without sites (binary, HT trees) couple the components and make the race approximate.

(ii) *Matrix-like pairs.* $P(\text{smaller first}) \to 0$ derived. To end near the smaller component, all n unstable deviations must start below $\sigma^{(1-w)/w}$ (Proposition A.15(b)), which gives $P(\text{smaller first}) \propto \sigma^{\gamma}$ with $\gamma = n(1-w)/w$: for $n = 1$ empirical (heuristic mechanism; the data below support it), for $n = \chi_e$ a conjecture. When a wrong order does happen, the deviation is therefore of order one at the start of the wrong plateau, and the correction in (iii) takes a time $O(1/|\lambda_{\min}|)$, not $\ln(1/\sigma)/|\lambda_{\min}|$ empirical (heuristic mechanism).

(iii) *After a wrong first escape* (Remark A.24) derived, with the Hessian at the saddle proven (Propositions A.21 and A.22) and k the Steiner count (20): *rotation* for $m = 2$, $k \geq 3$: the saddle is strict along the piggyback direction, $\lambda_{\min} = -(1-w)w^{(K-2)/K}$; the learned bond index turns to U_1, and U_2 is forgotten (the loss rests at $\frac{1}{2}w^2$) and relearned after a plateau $\propto \sigma^{-(k-2)}$.

Adding for $m = 2$, $k = 2$: a two-core route with $\lambda_{\min} = -w^{(K-2)/K}$ (multiplicity $\chi_e - 1$) puts U_1 on a spare channel and nothing is forgotten. *Kept* for $m \geq 3$: no negative Hessian eigenvalue; U_1 is added after a plateau $\propto \sigma^{-(k-2)}$ and the wrong order stays.

A correction through a strict saddle thus exists only where the wrong order becomes rare as $\sigma \to 0$.

Setup (Section B.4). Two-component teachers $W^* = U_1 + wU_2$: odeco on five trees, or pattern teachers on MPS4 and MPS6 whose components differ on the sites D (numbered from 0; on an MPS core i carries site i), in the standard site basis, which the rotation invariance of the initialization makes general; $d = 2$ unless stated, fan-in initialization, exact population loss. The component learned first is the one whose overlap a_r first crosses $\frac{1}{2}$; probabilities are frequencies over seeds 0–399 (0–999 in the σ scans of binary4 and MPS4), the same seeds for every w and σ, with Wilson 95% intervals or binomial s.e. (≤ 0.025).

The first component of the odeco runs. empirical In the first-escape test of Section 4.3 the first component learned is the LO flow's r_1 in 1,847 of 1,848 runs (the exception: a wrong first escape on binary4 at $\sigma = 0.1$). On binary4, a matrix-like tree, it is the largest in 639 of 640 runs known (deep linear networks: Jacot et al., 2022, Thm 2 and Lemma 15); on the tensor-like trees it varies from seed to seed but not with σ, and the race of independent per-component flows names it in every MPS run, where the race is exact at leading order, in 480 of 488 binary8 and HT4 runs, where cores without sites couple the components, and in only 537 of 640 binary4 runs, where that coupling decides (Section D.1.6).

Tensor-like pairs: a race set by the initialization (Figure 9(a,b)). empirical In the main sweep (five trees, 28 settings of (d, χ), six values of w, $\sigma = 0.01$; 67,200 runs) P(larger first) is 0.60–0.88 at $w = 0.5$ and 0.51–0.65 at $w = 0.8$ on the 23 tensor-like settings, and every one of the 40,800 MPS runs learns first the winner of its own LO race, $\arg\min_r \hat{t}_{1,r}/s_r$ (96–99.5% of the runs on binary8, whose internal cores are shared). The order does not depend on σ (MPS4, $\chi = 4$: P(smaller first) $= 0.185, 0.395, 0.490$ at $w = 0.5, 0.8, 0.95$ for every $\sigma \in [0.003, 0.1]$, with the same winner per seed; likewise on binary8 and MPS6), and it becomes more random as m grows (MPS6, $w = 0.8$: 0.17, 0.32, 0.41, 0.46 for $m = 2, 3, 4, 6$). new (related: units ordered by their initialization, Zhang et al., 2026, Prop. 5)

Number of cores, bond and physical dimension; the selection rule (Figure 9(c)). empirical Through the spread of the race, the order becomes more random with K (P(larger first) $= 0.79, 0.69, 0.63$ at $w = 0.5$ on MPS with $K = 4, 6, 8$) and slowly greedier with uncapped χ (at $\chi = 32$, $w = 0.8$, from 0.86 on MPS4 to 0.72 on binary8, confirmed by gradient flow); d enters only through the bond caps (derived; no effect at equal bonds, $|\Delta P| \leq 0.021$). The winner follows the TTN form of the selection rule of tensor PCA: "the winner is $\arg\max_r s_r X_r^{K-2}$", X_r the projection of the initialization onto the orbit of component r's balanced rank-one configuration, names 76–92% of the winners on the tensor-like settings, also out of sample (Section D.1.6). new known (the (weight)$\times$(overlap)$^{p-2}$ rule of multi-spike tensor PCA, Ben Arous et al., 2025b, Thm 1.5, and for gradient flow Ben Arous et al., 2025a, Thm 1.3; the tensor power method, Anandkumar et al., 2014, Lemma 5.1; over-parameterized orthogonal tensor decomposition learns the largest component first, Ge et al., 2021)

Matrix-like pairs: greedy as $\sigma \to 0$, slowly (Figure 9(b)). empirical (heuristic mechanism) For the three pairs on non-adjacent cores ($n = 1$: odeco binary4, $D = \{0, 5\}$ on MPS6, $D = \{0, 2\}$ on MPS4) P(smaller first) is consistent with $C\sigma^\gamma$, $\gamma = (1-w)/w$, for $w = 0.7$–0.95: fits over the three

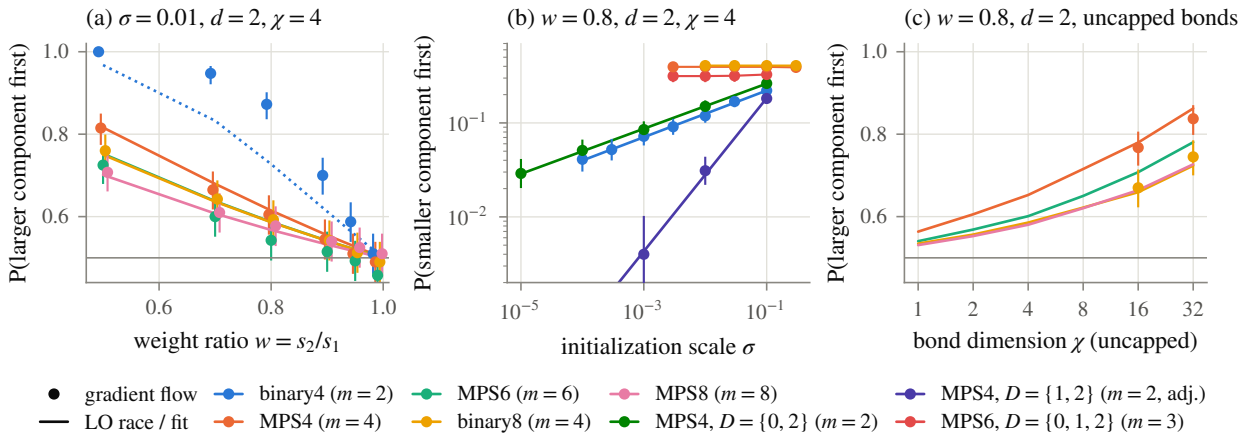


Figure 9: Which component is learned first: weights $(1, w)$, $d = 2$; dots: gradient flow, Wilson 95% intervals (400 seeds; 1,000 for binary4 and MPS4 in (b)). (a) $\sigma = 0.01$, $\chi = 4$, odeco teachers; lines: the leading-order (LO) race (22) of the same initializations (dotted for binary4, where it is not the dynamics). (b) $w = 0.8$; lines: fitted $C\sigma^{\gamma}$ for $m = 2$ (Table 23); zero counts not drawn. (c) Uncapped bonds, $w = 0.8$: race (lines, 500 initializations), gradient flow at $\chi = 16$, 32. Legend: m, the number of cores that carry a site where the two components differ; D, the differing sites of pattern teachers; "adj.", a pair whose two cores share a bond; "LO race / fit", the race in (a) and (c) and the fitted power law in (b). Takeaway: for $m \geq 3$ a race set by the initialization, independent of σ and slowly greedier with χ; for $m = 2$ the wrong order dies out as a power of σ (source: Section B, Table 14).

smallest values of σ agree within 1.1 s.e. in all 13 sets (limited range), while fits over the whole range lie up to 24% below, a pre-asymptotic deficit (Table 23). The wrong-order seeds are nested (a seed wrong at some σ is wrong at every larger σ) in all 18 matrix-like sets, as the threshold picture of (ii) predicts. Adjacent pairs ($n = \chi_e = 4$) are 2.2–3.4 times steeper, consistent with $n(1-w)/w$ only at the small-σ end of MPS4 (0.40 ± 0.05 against 0.444 at $w = 0.9$), so $n = \chi_e$ remains a conjecture. new (the power law in σ) known (the limit for binary4: Jacot et al., 2022, Thm 2)

After a wrong first escape (Figure 10). empirical From the exact wrong saddle plus a fan-in perturbation of scale σ ($d = 2$, $\chi = 4$; for $m = 2$, 8 seeds at $\sigma = 10^{-7}$–10^{-2}) every run follows the predicted branch: binary4 and MPS4 $D = \{0, 2\}$ rotate, MPS4 $D = \{1, 2\}$ adds, the odeco MPS4, MPS6 and binary8 keep the wrong order. On the strict branches $\lambda_{\min}$ matches the prediction to 0.05% at every σ, and the escape takes $\ln(1/\sigma)/|\lambda_{\min}|$ plus a constant (slopes within 1.5%); on the kept branch $|\lambda_{\min}| \propto \sigma^{k-2}$ (MPS4 $\sigma^{1.999 \pm 0.014}$; MPS6 $\sigma^{4.22 \pm 0.09}$ and binary8 $\sigma^{5.2 \pm 0.3}$, limited range: 5 and 6 seeds over factors 4 and 2.3 in σ). Natural runs at $\sigma = 10^{-4}$–10^{-2} that learned the smaller component first show no $\ln(1/\sigma)$ part, as (ii) predicts: 296 of 301 binary4 and 237 of 240 MPS4 $D = \{0, 2\}$ runs rotate, within 0.98–1.38 times $1/|\lambda_{\min}|$, flat in σ; at $\sigma = 0.1$, where the wrong saddle is barely reached, only 30 of 120 binary4 runs rotate (Section D.1.6). new (beyond binary4) known (binary4: Achour et al., 2024, Thm 7; Jacot et al., 2022, Lemma 15)

Three and four components. empirical With three odeco components (weights $(1, w, w^2)$, $w = 0.8$, 0.9, $d = \chi = 4$, 200 seeds, one σ per tree), binary4 ($m = 2$) is far greedier than a race of independent clocks (greedy in 0.705 ± 0.032 and 0.490 ± 0.035 of runs, against 0.518 and 0.319). For $m \geq 3$ the order is a random permutation favouring the larger components, consistent with that race ($|z| \leq 1.6$) and no less greedy than each run's own LO race (McNemar $p \geq 0.45$), which names the whole order in 70–89% of runs. With four components most runs stop before the last

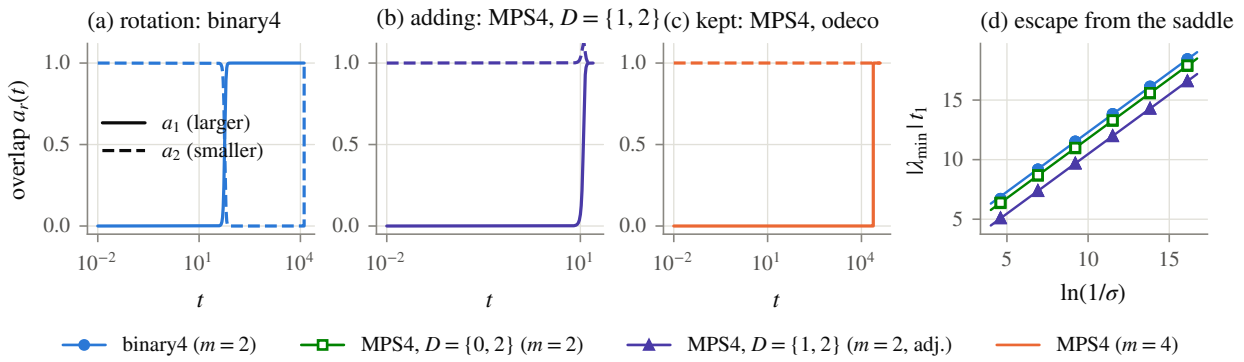


Figure 10: After a wrong first escape, from the wrong saddle (U_2 learned, $w = 0.8$) plus a fan-in perturbation of scale σ; $d = 2$, $\chi = 4$. (a)–(c) Overlaps of one run per branch (seed 0): (a) binary4 ($\sigma = 10^{-4}$): the learned bond index turns to U_1 at $t \approx 57$, U_2 is forgotten and relearned at $t \approx 1.3 \cdot 10^4$; (b) MPS4, $D = \{1, 2\}$ ($\sigma = 10^{-4}$): U_1 is added at $t \approx 11$, a_2 stays at 1; (c) MPS4 odeco ($\sigma = 0.01$): U_1 is added after a σ^{-2} plateau ($t \approx 2.4 \cdot 10^4$). (d) Escape time in units of the predicted $1/|\lambda_{\min}|$, mean over 8 seeds for each w (binary4: 0.5, 0.7, 0.8, 0.9; MPS4: 0.5, 0.8, 0.9; they nearly coincide); lines: slope 1; legend as in Figure 9. Takeaway: $m = 2$ wrong-order saddles are strict and are left at the predicted rate, by rotation or by adding; for $m \geq 3$ nothing corrects the wrong order (source: Section B, Table 14).

one (MPS4 289 of 400; binary4 none): the stalls of Section 4.8, here also at $K = 4$. Teachers with stars forget a learned larger component and relearn it last (Section 6.6). Details: Section D.1.6. new

Status. The order is set by the basin of the initial direction for tensor-like pairs (proven at LO on MPS, derived elsewhere; empirical on four trees), not for matrix-like pairs, which become greedy as $\sigma \to 0$. A wrong order is corrected through a strict saddle in a time $\propto \ln(1/\sigma)$ only from the saddle and only for $m = 2$ pairs on non-adjacent cores; natural runs correct in $O(1/|\lambda_{\min}|)$, adjacent pairs add the larger component, and tensor-like pairs keep the wrong order. Open: the prefactor of the $m = 2$ law, the exponent of adjacent pairs, and a closed-form race when cores without sites couple the components.

4.8 Failure modes

A saddle-to-saddle run can end *stalled* (Section 2.8): with a dead route (trap), whose route ends have decayed to roundoff; on a slow route, live but with a plateau longer than the budget; or at the floor of an unrealizable centred target (Section 5.1). This subsection treats dead and slow routes in the odeco runs of Section 4.1, whose routes span the tree, in the pattern-teacher runs of Section 4.4, and in the regime maps of Section 3 (ends and slices of a route: Section 2.7).

Mechanism of dead routes. derived (reduced model, Proposition A.28) The conservation laws (Proposition 2.5) tie the new entries of each route end to the imbalance of its bond, of order σ^2 under fan-in initialization. On a plateau the Gauss–Newton term makes the new entries of an end that the imbalance does not favour decay at a rate of order σ^2. Decay in a time $\propto \sigma^{-2}$ competes with an escape in $\propto \sigma^{-(k-2)}$: for $k = 3$ the route escapes first, for $k = 4$ both are of the same order, and for $k \geq 5$ the ends are decided long before the route could escape. Exactly balanced initialization ($Q_e = 0$) is expected to protect the ends conjecture (for ends that are tree leaves see below; in four-layer matrix factorization exactly balanced initialization can fail, Luo et al., 2025).

known (related shrinkage of dormant weights: Jacot et al., 2022, App. C.1; Kumar and Haupt, 2025b; Kunin et al., 2025, Lemma B.1)

A criterion from the initialization (pending verification). derived (one learned component, leading order in the small entries, $\sigma \to 0$) On the plateau only the new entries normal to the learned manifold pay a Gauss–Newton cost. For a route end u with route neighbour w this conserves $C_u = Q_{\nu\nu} + \|Y_{\mathrm{nv}}\|^2 - \|X_{\mathrm{nv}}\|^2$, the Gram of the end's new entries minus its partner's ($Q_{\nu\nu}$ the imbalance of their bond on the route's channel; X_{nv}, Y_{nv} the new entries of u and w that pair with nothing at second order), and the end dies iff $C_u < 0$ and the route is long enough ($k \geq 4$ if w carries sites, $k \geq 6$ at an HT or binary leaf). A route is dead iff two or more of its ends die; with one dying end it is pinned and escapes after a time $\propto \sigma^{-(2k-6)}$; routes with $k \leq 3$ never trap. The state at the start of the plateau is σ times a function of the initialization direction (through the σ-free correlation flow of Section 4.3), so the criterion is computed from the initialization alone, and whether a run traps depends on its direction, not on σ. Under fan-in initialization (saddle start, partner with sites) an end dies with probability $I_{F_u/(F_u+F_w)}(n_u/2, n_w/2)$ (I the regularized incomplete beta function, F the fan-ins, n the numbers of physical directions normal to the learned pattern): 0.392 at an MPS end that is a tree leaf and 0.5 at the root end ($d = \chi = 2$). With balanced initialization ends that are tree leaves cannot die, and spare bond dimension protects an end only when it reaches that end's own route edge, which reconciles the MPS8 stalls at $\chi = 2R$ with the binary trees. empirical Without any gradient flow the criterion separates the 3,492 fan-in locality runs (89% of the predicted-dead runs stall, against 1.8% of the predicted-alive ones); in the pilot 102 of 109 predicted-dead runs die; on MPS4 at $\chi = 2$ all 150 predicted-dead runs stall or die, and none of the other 344. For the odeco teachers the trap is predicted only from the state after the second escape (20 of 20 MPS4 and 6 of 6 HT4 runs checked), not from the initialization. Known failures: $k = 4$ with spare inner channels, the marginal case $k = 6$ on HT4, $\sigma \geq 0.2$, and the multi-stage odeco traps. Details: Section D.1.8. known (the mechanism overlaps Furman et al., 2026, Thm G.1, on imbalance, and Kunin et al., 2025, Lemma B.1)

Odeco teachers: dead routes grow with K (Figure 11). empirical No run has a dead route at $K = 3$ (0 of 380). On MPS4 at $\chi = 3$ the separated teacher's outcome is fixed by the initialization direction: the same 4 of 10 seeds end with a dead route at all eight values of $\sigma \in [0.003, 0.1]$ (32 of 80 runs); with the close teacher 14 of 80 runs have a dead route and 8 are slow, not fixed by the direction. For the separated teacher the slices at both route ends decay at a rate $\propto \sigma^2$, constant per seed to a few percent, equal at the two ends and seed-dependent (0.26 to $1.06\,\sigma^2$): the signature of the mechanism (its magnitude is not tested); for the close teacher the per-seed rates change with σ, by factors 1.3 and 3. At $K = 7, 8$ with $\chi = 3$ at least 9 of 30 runs end with a dead route on binary8, 24/30 on HT4 and 14/30 on MPS8 (lower bounds: some slow runs end with slices still decaying). With $\chi = 6$ none does on binary8 (0 of 50), but 5 of 50 do on HT4 and 7 of 50 on MPS8 (34 more are slow), whose route ends sit on bonds capped at $d = 4$, one channel above the teacher rank. Re-gauged to $Q_e = 0$, the same initializations never end with a dead route where fan-in ones do (0 of 120 runs), but 69 of the 90 balanced runs at $K = 7, 8$ are slow. The routes die on the plateaus, not in the escapes (Section D.1.7). new

Pattern teachers: dead routes at $k \geq 4$. empirical A pattern-teacher run is stopped as stalled when, after a component has been learned, every gradient entry stays below 10^{-12} for 5,000 evaluations; its loss then sits at a partial fit. Under fan-in initialization at $\chi = 2$, 398 of 5,682 two-component runs stall, all with $k \geq 4$ (none of 5,112 with $k \leq 3$). The rate depends on the tree

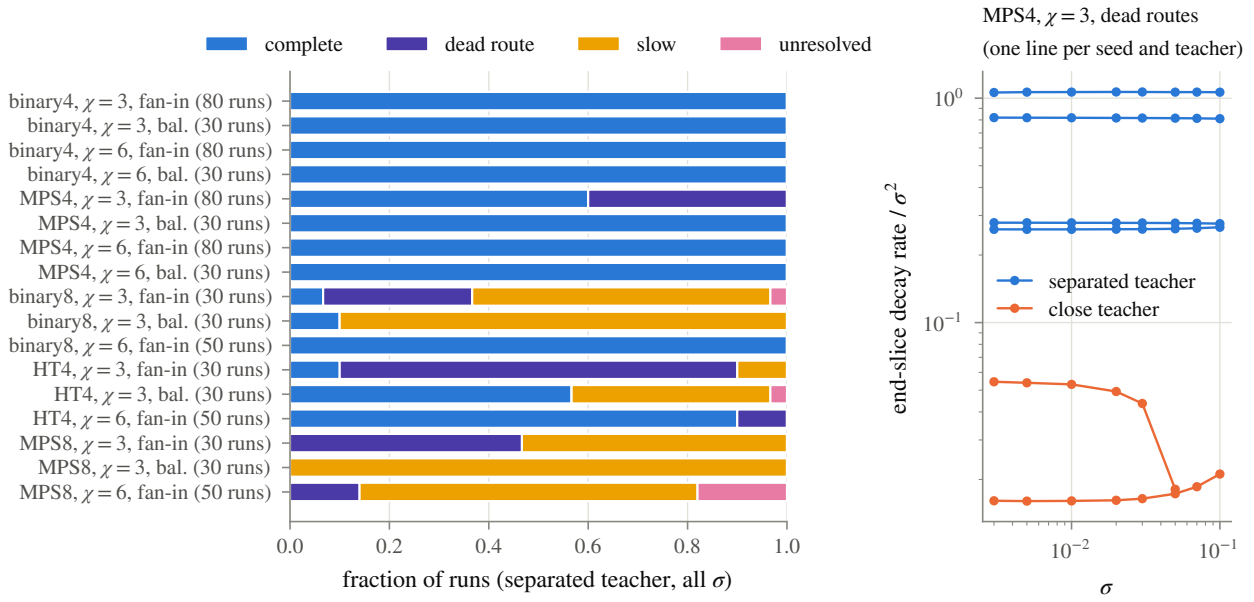


Figure 11: Odeco runs that do not complete. Left: fractions of run classes, separated teacher, all σ ($K = 7, 8$: $\sigma = 0.15$–0.3 at $\chi = 3$, 0.07–0.3 at $\chi = 6$), runs per row in the labels: *dead route* (a component never learned has a route-end slice below $10^{-8}\sigma$), *slow* (the budget ended with every such slice above it), *unresolved* (wall-clock limit); dead-route fractions at $K = 7, 8$ are lower bounds. Right: decay rate/σ^2 of the route-end slices of the component never learned (mean of the two ends), MPS4, $\chi = 3$, one line per seed and teacher with a dead route (blue: four seeds of the separated teacher, constant in σ; orange: two seeds of the close teacher, not constant), fitted from 1.5 times the last event to a slice of 10^{-9}, above the floating-point floor. Takeaway: no dead route at $K = 3$, fixed by the initialization at $K = 4$, frequent at $K = 7, 8$ except where no route end is capped (binary8 at $\chi = 6$) (source: Section B, Table 12).

more than on k (for $k = 5$: 7 of 624 binary and HT runs, 55 of 192 MPS runs), and stalls concentrate on MPS routes that end at the ends of the chain. The stalls checked are dead routes: rerun with a 60 times longer patience, all 37 flagged runs show no escape up to $t \approx 10^{10}$, with route-end amplitudes at roundoff. Balanced initialization cuts the stalls tenfold (169 to 16 of 1,008), and the checked balanced stalls are slow, not dead. Spare bond channels remove the stalls only on routes with an uncapped end; on routes with both ends capped they rise from 90 to 106 at $\chi = 4$. Counts and the effect on the exponents: Section D.1.7. new

The regime maps meet the same failures. empirical (confirming the prediction of Section A.8; mechanism conjecture, heuristic) With χ equal to the teacher rank R the last component needs a route through all K cores, which can die for $K \geq 5$ (already at $K = 4$ for some initializations, as above). In a stall test at $\chi = R = 3$ (Table 25) most runs on binary8 and every fan-in run on MPS8 do not learn the last component within the budget (9 of 18 fan-in MPS8 runs learn only the first), in the plain as in the centred model, so the initial function is not the cause; from fan-in initialization the route is dead (28 of 32 stalled plain runs), and balanced initialization removes the dead routes but not the stalls. In the plain-model $\sigma \times K$ sweeps 546 of the 795 stalled runs end with the R-th bond singular value at roundoff, a dead route (182 of the 795 stopped at the step limit). Spare channels remove the stalls only where no route end is capped: on binary8 at $d = 3$, $\chi = 6$ no run stalls at $\sigma \leq 0.63$ (30 runs), but on MPS8 every run stalls (50 of 50), its end bonds being capped at $d = 3$, the teacher rank; one spare channel per end is not enough either (MPS8 and HT4 at $d = 4$, above). With the centred model at $\chi = 4$ the stalls below the saddle-to-saddle

boundary sit at a partial fit, fixed by the initialization (Section 3.3). At practical budgets, stalls thus occupy part of the saddle-to-saddle side of every deep tree whose route ends are capped.

Long plateaus are not barren plateaus. derived From small initialization the gradient is $O(\sigma^{K-1})$ and the first plateau lasts $\propto \sigma^{-(K-2)}$: deep trees at small σ stall as if on a barren plateau. The cause differs: the gradient is small because the initialization is, not because it concentrates around zero over random initializations of fixed norm; with the fan-in normalization at $\sigma = 1$ the initial function has unit norm on every tree (Proposition 2.2). known (barren plateaus of tensor networks: global but not local losses of unitary-embedded MPS, Liu et al., 2022; absent for isometric MPS, TTNS and MERA, Miao and Barthel, 2024; Barthel and Miao, 2025)

What the mechanism explains, and what it does not. empirical The odeco runs show the signature of the mechanism where it was measured ($K = 4$, separated teacher), and balanced initialization leaves no dead route in either experiment. The first criterion of the reduced model (Proposition A.28) was only weakly associated with death in the pattern-teacher runs (30 of 55 runs with both ends unfavourable died, 7 of 67 with a favourable end); the criterion above replaces it, pending its verification. Open: the very late escapes of balanced MPS8 routes (delays of 10^5 and more), the mechanism of the rise of the stalls on capped routes with spare channels, and the steep exponents of some MPS routes with $k = 5$ (MPS8 $\{1, 5\}$ from the saddle, 3.67 ± 0.20, limited range), which go with shrinking route ends. Details: Section D.1.7.

5 Other regimes

This section gives the detail of the regimes other than saddle-to-saddle that Section 3 maps: lazy training and its failure below the cut dimension, the rich window between the two boundaries, and the limit of large bond dimension under three parameterizations. Runs that stall are treated with the failure modes of saddle-to-saddle training in Section 4.8. The section then asks how discrete steps, gradient noise and finite samples (Section 5.4) and the optimizers used in practice (Section 5.5) change the gradient-flow picture. Settings of the first three subsections: Sections B.5 and 3.1; supplementary results: Sections D.2 and D.3.

5.1 Lazy regime

At large σ the centred model trains like its linearization at θ_0: the function moves along the range of the tangent kernel Θ, which hardly changes. Lazy training is known in general known (Jacot et al., 2018; Chizat et al., 2019; Lee et al., 2019); what the tree adds is the kernel. At initialization $\mathbb{E}\Theta$ is a multiple of the identity at every bond dimension, but the rank of Θ is the dimension of the manifold of tensors with the given bond ranks, which reaches d^N only when every bond is full (Proposition A.46) proven new (the mean kernel on every tree and the full-bond condition) known (the rank as the dimension of the fixed-rank manifold: Holtz et al., 2012b; Uschmajew and Vandereycken, 2013).

What sets the lazy boundary. derived empirical On binary4 the kernel change has a zero-mean first-order part, linear in W_0, and a coherent second-order part (Proposition A.47). Near the boundary the measured δ_Θ is 0.96 times the law $\sqrt{(c_1\xi)^2 + (c_2\xi^2)^2}$ of (58) (median; 0.69–1.82), and at large χ the second-order part sets the boundary: the first-order estimate alone matches only up to $\chi = 12$ (at $\chi = 36$ it gives 0.59 against the measured 1.09). Hence $\sigma_{\text{lazy}} \propto \chi^{-1/2}$ rather than χ^{-1} (Section 3.2).

Lazy only with full bonds. derived empirical known (that a linearized model fits only the projection of the target on its tangent space: Chizat et al., 2019, Thm 2.5; Ghorbani et al., 2019, Thm 2) With a bond below its cut dimension the linearized flow fits only the projection of W^* on the range of Θ and stops at $\frac{1}{2}\|P_\perp W^*\|_F^2$, on binary4 a fraction $(1-\chi/d^2)^2$ of $L(0)$ on average (0.44, 0.31, 0.11 at $\chi = 3$, 4, 6; Section A.8). In addition the centred target $W^* + W_0$ is unrealizable at every σ, and gradient flow ends on its best approximation (Figure 12(a)): all 159 stalled centred runs at $\chi = 3$, 4, 6 end on the Eckart–Young minimum of $W^* + W_0$ across the cut 01|23 (to 0.04%). This floor rises like σ^{2K} at small σ (at $\chi = 3$ as (60) predicts, lower at $\chi = 4$, 6, where the spare bond rank absorbs part of W_0) and saturates at the lazy residual at large σ (Section D.2). Once it exceeds the convergence threshold (for every seed from $\sigma = 1.0$, 1.0, 1.6 at $\chi = 3$, 4, 6) the runs are stalled on a floor by the rule of Table 4, though their dynamics are kernel-like at large σ. A lazy regime therefore needs every bond at its cut dimension (binary4: $\chi \geq 9$), which grows like $d^{N/2}$ for the middle bonds.

Anisotropic, ill-conditioned kernels. derived empirical new (for TTNs; the mean kernels of MPS and TT models: Guo and Draper, 2021; Wesel and Batselier, 2025) A single initialization has an isotropic kernel only if every bond is far above its cut dimension; at a capped full bond the cores next to it are square Gaussian matrices and $\lambda_{\min}(\Theta)/\bar{\lambda} \sim 10^{-4}$ (Section A.8). The lazy fit then converges in a kernel time of order $(\bar{\lambda}/\lambda_{\min})\ln(1/\mathrm{tol})$. On the binary tree with $N = 8$, $d = 2$, the square full bonds ($\chi = 16$) are not lazy within the step limit: every run from $\sigma = 2.3$ stops there at the same seed-dependent loss, $L/L(0) = 0.05$–0.44, for every σ. With bonds over-parameterized twice most runs from $\sigma \approx 1.45$ on are lazy, though they too reach the step limit (Section D.2). At practical χ the lazy regime is thus kernel regression with a random, tree-dependent and anisotropic kernel, not with its mean.

Lazy by the kernel is not small motion. derived Under the standard parameterization the root of binary4 carries a share $\chi^2/(2d^2 + \chi^2)$ of the kernel, and its own kernel block does not depend on it: a lazy run can be a trained readout over fixed random features. In one checked run the root moved by more than 100% while $\delta_\Theta < 0.1$ empirical (Section A.8). Per-core changes should be reported next to δ_Θ (Section 5.3).

5.2 Rich, non-incremental regime

Between the two boundaries the runs converge and move their kernel (δ_Θ of the centred binary4 rich runs: 0.10 to 10^7, median 55), but their events overlap: some $\rho_i \leq 3$.

What sets the crossover. derived (aligned initialization) empirical (random) The plateaus between events grow like $\sigma^{-(K-2)}$ as $\sigma \to 0$ while the event widths stay $O(1)$, so the separation ratios fall through 3 as σ grows; for aligned initialization Proposition A.48 gives the boundary exactly, with the threshold exponent of Gissin et al. (2020) known. From random initialization the crossover is a seed property: on binary4 the per-seed boundaries at one χ span a factor 1.9–16 for $\chi \leq 12$ and 1.4–2.7 for $\chi \geq 16$ (six seeds). If both boundaries have finite limits in the function-space scale $\alpha = \sigma^K$, the rich window of Section 3.2 shrinks to a band of width $O(1/K)$ in $\ln\sigma$ conjecture. During an escape, components not yet learned rise transiently, in 116 of the 9,165 runs above $\frac{1}{2}$; events are therefore dated by their final rise (Section D.2).

Lazy, then rich. derived ($N = 4$) empirical known (analogues: mixed lazy and active dynamics, Tu et al., 2024; lazy-then-rich transitions, Lyu et al., 2024) The two-copy model is realizable at a deficient bond

when 2χ reaches the teacher's bond rank. At large σ it first fits the tangent part lazily and sits at the lazy residual, then escapes by rotating its bond subspaces, after a gradient-flow time $\propto \sigma^2$, a kernel time $\propto \sigma^{2K}$ (Section A.8). On binary4 at $\chi = 3$ (six seeds) the runs converge (rich) for $\sigma = 1.0$–4.0, end at $L/L(0) = 0.005$–0.05 at $\sigma = 6.3$, at 0.13–0.44 at $\sigma = 10$, and at the lazy residual from $\sigma = 25$ on (within 3.4% at $\sigma = 16$): the escape leaves any fixed budget as σ grows. A light root with a deficient bond behaved the same way, lazy and then rich, in the one seed checked (Section A.8) empirical.

Crowded first escapes. empirical In the two-copy model at small σ the two copies start identical, so their near-origin escapes start together and the first two events crowd: on binary4 at $\chi = 18$–36, $\min_i \rho_i$ at $\sigma = 0.010$ is only 2.7–14 (centred: 12–29) and grows far more slowly than $\sigma^{-(K-2)}$, so the two-copy sweep is no quantitative check of the boundary (Section D.2). On binary4 Adam's runs are rich, non-incremental for another reason: the first escape is as wide as its own time (Section 5.5).

5.3 Large bond dimension: NTK versus μP

As χ grows at fixed σ, the scales of the cores and their learning-rate multipliers decide whether the features still move. We compare three parameterizations, written as initial scales and multipliers η_v (Section 2.4, variant (e)): the *standard* one (SP: fan-in initialization, $\eta_v = 1$); *NTK* ($\eta_v = 1/F_v$, so that every core contributes equally to the expected kernel); and *μP-TTN*, the spectral condition of Yang et al. (2024) applied with the injective norm of each core, since a core only ever acts on product vectors: fan-in initialization except at the root, which is scaled by the inverse square root of the sum of its child-leg dimensions, $(2\chi)^{-1/2}$ on binary4, and $\eta_v = \chi_v/F_v$ (Section A.8) derived new (the spectral condition is known; factor-wise rules for structured layers: Qiu et al., 2024; Potapczynski et al., 2024).

Predictions. derived (binary4, centred, isotropic limit $\chi \gg d^2$, fixed σ; fixed $\xi = \|W^*\|_F/\alpha$ under SP and NTK) In each parameterization the first- and second-order constants c_1, c_2 of the law (58) carry the same power of χ, so the kernel change scales as χ^{-3} (SP), χ^{-1} (NTK) and χ^0 (μP-TTN); the parameters move as in Table 26 (constants in Section A.8). Under SP the root, with fan-in χ^2, carries almost all of the kernel and the model becomes linear in the root over fixed random features: lazy at any fixed σ as χ grows, centred or not. Under NTK the cores move by $O(\chi^{-1/2})$ or less, also lazy in the limit, as for the NTK of MPS known (Guo and Draper, 2021). Under μP-TTN every core's update stays of order one.

Measurements. empirical Binary4, $d = 3$, teacher $(3, 2, 1)$, centred, $\sigma = 1.5$ ($\xi = 1.11$), $\chi = 16$–128 (four values, 0.9 decade), six seeds; tensor kernel (Figure 12(b), (c); fitted exponents with standard errors in Table 26), confirming the predicted exponents in sign and size. Under SP the kernel change falls like $\chi^{-3.02\pm0.06}$, while at $\chi = 128$ the root still moves by 9% and the leaf cores by $2 \cdot 10^{-4}$ (medians). Under μP-TTN the kernel change approaches the predicted level, 5.1, 5.7, 6.1, 6.2 at $\chi = 16$, 32, 64, 128 against 6.2–6.3: feature learning survives. Under NTK the fit over $\chi = 16$–128 (-0.79 ± 0.04, 5.5 standard errors from -1) is pulled by the pre-asymptotic $\chi = 16$–32, where δ_Θ is still 0.4–0.6; between $\chi = 64$ and 128 the local slope is -1.01, with levels 0.250 and 0.124 against 0.240 and 0.120 from the second-order law with the NTK constants. The SP kernel and root and the NTK parameters and leaf cores match within 1.1 standard errors (the NTK root within 2.9); the μP-TTN kernel, parameters and leaf cores drift slightly (+0.085, −0.064, −0.065; 3.5–3.6 standard errors from 0, set by $\chi = 16$); the SP leaf cores (6.5) and parameter change (3.3 standard errors) are steeper than predicted. Further from the lazy limit ($\sigma = 1$) the slopes are shallower, so

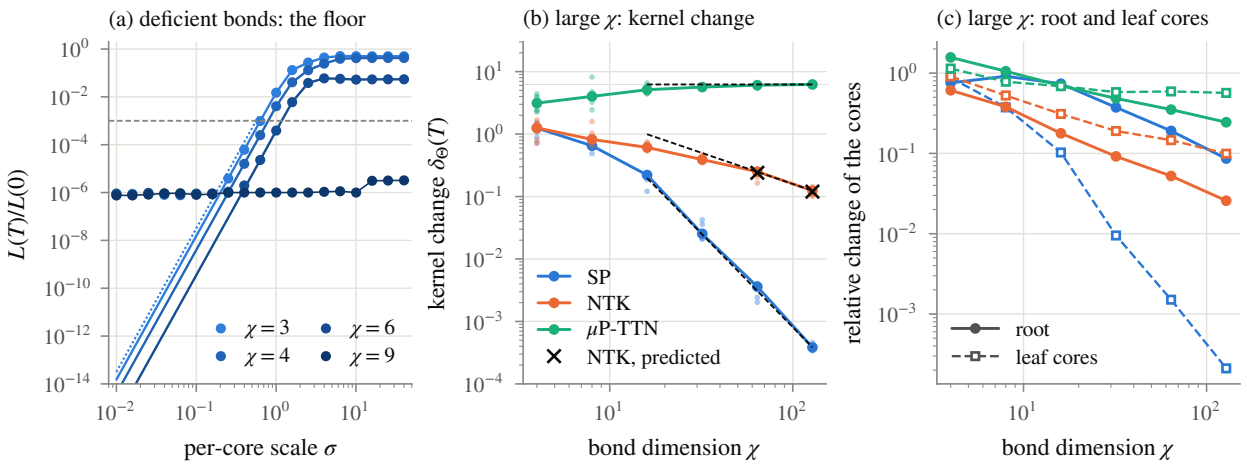


Figure 12: Lazy regime and large bond dimension; binary tree with $N = 4$, $d = 3$, teacher $(3, 2, 1)$, centred model, six seeds. (a) Final loss against σ (dots: medians) for $\chi = 3$, 4, 6 (deficient bonds) and $\chi = 9$ (full); solid: the Eckart–Young minimum of $W^* + W_0$ across the cut, median over seeds ($\chi = 9$: line through the medians); dotted ($\chi = 3$ only): the small-σ mean (60); dashed: the convergence threshold 10^{-3}. Runs stop at 10^{-6}; with deficient bonds they end on the minimum once it exceeds that, and are stalled above the threshold. (b) Kernel change (tensor kernel) against $\chi = 4$–128 at $\sigma = 1.5$ ($\chi = 4$, 8: deficient bonds; every run at $\chi = 4$ stalls): seeds (dots), medians (lines), predicted slopes -3, -1, 0 (dashed, through the $\chi = 128$ medians; the fits of Table 26 use $\chi = 16$–128), the predicted NTK levels (crosses) and the predicted μP-TTN level (grey band). (c) Relative change of the root (solid) and leaf cores (dashed), medians; colours as in (b). Takeaway: below the cut dimension the large-σ end is a floor, not a lazy fit; at large χ only μP-TTN keeps the kernel and the leaf cores moving. (source: Section B.5, Table 15)

the predicted exponents are limits as $\xi \to 0$; the plain model gives the same exponents within the errors (Section D.2). Scope: binary4 only; runs on the binary tree with $N = 8$ are incomplete and not interpreted. The μP-TTN limit itself is not derived; on binary4 it may be a two-population mean field with a pairwise coupling through the root conjecture.

5.4 Discrete steps, gradient noise and finite samples

Gradient flow is the limit of gradient descent (GD) as the step $\eta \to 0$ and of online SGD as $\eta/B \to 0$, and the population loss is the limit of the empirical loss as $n \to \infty$. How much of Section 4 survives finite values? GD keeps gradient flow's saddle sequence up to a sharp stability limit, with event times shifted by a computable $O(\eta)$; gradient noise does not shorten plateaus on average, but it lengthens later ones and can trap the dynamics at a saddle; and from a finite training set small initialization recovers the teacher with its bond ranks, while the lazy limit fails as dimension counting predicts. Setup (Section B.7): odeco teacher $(3, 2, 1)$, $d = \chi = 3$, fan-in initialization (five seeds) on the binary trees with $N = 4$ ($K = 3$) and $N = 8$ ($K = 7$) and the MPS with $N = 8$ ($K = 8$); GD and online SGD (fresh inputs at every step) run on the clock $t = \eta \times$(steps) and are compared with gradient flow from the same initial point. Details: Section D.3.

The stability limit. proven known (deep linear networks: Mulayoff and Michaeli, 2020; Ghosh et al., 2025) new (general trees) At the balanced global minimum the Hessian $J^\top J$ has largest eigenvalue $\lambda_{\text{bal}} = K s_1^{2-2/K}$, and every other aligned minimum is sharper (Proposition A.43; with $\chi_e = R$ every

other minimum, derived), so GD can converge to a global minimum only if

$$\eta < \eta_c = \frac{2}{K\, s_1^{2-2/K}} \tag{23}$$

(0.1541, 0.0435 and 0.0366 for the three trees). Along the saddle-to-saddle path the sharpness is $O(\sigma^{K-2})$ on the first plateau and $K \max_r s_r^{2-2/K}$ over the components learned so far derived (Section A.7): a step above η_c is felt only once component 1 is learned, and there is no catapult at the start (the condition $\eta\lambda_{\max}(H) > 2$ of Lewkowycz et al. (2020) fails near the origin). Measured empirical on binary $N = 4$ ($\sigma = 0.003$–0.3, five seeds, 18 steps from $0.01\eta_c$ to $3\eta_c$; Figure 13a): all 195 runs with $\eta \le 0.95\eta_c$ are GF-like and all 80 at 1.05–$1.3\eta_c$ oscillate. Below η_c, GD learned the components in gradient flow's order in all 765 runs, including 40 more seeds on binary $N = 4$ and both deep trees at $\sigma = 0.9$ (on the deep trees within a budget in which gradient flow learns one or two of the three components: on the MPS only the first component is compared in 36 of 45 runs).

Below the limit, GD is gradient flow on a shifted clock. derived new (for TTNs) known (the modified flow: Barrett and Dherin, 2021) To first order in η, GD follows the modified flow $\dot\theta = -(I + \frac{\eta}{2}H)\nabla L$, whose critical points are those of L: the saddles are gradient flow's, and each event moves by $\eta c_e + O(\eta^2)$, with c_e computed without free parameter from the variational equation along the gradient-flow path (Proposition A.44 and Section B.7). Measured empirical: over 526 comparisons at $\eta \le 0.1\eta_c$ the shift agrees with ηc_e within 0.3%, 1% and 5% in 66%, 90% and 97% of the cases, all 202 first events within 1%. Since c_1 grows like $\ln(1/\sigma)$ (asymptotically, conjecture in Section A.7; confirmed here) while $t_1 \propto \sigma^{-(K-2)}$, the relative shift of t_1 vanishes as $\sigma \to 0$: at $0.9\eta_c$ it falls from 0.11 at $\sigma = 0.3$ to 0.0036 at $\sigma = 0.003$ (Figure 13b). At small initialization every stable step thus reproduces gradient flow, the more closely the smaller σ.

Above the limit, and at larger initialization. derived empirical known (the scalar map and its two-cycle: Ghosh et al., 2025, Thm 1, MacDonald et al., 2025, Thm 5.3; edge of stability: Cohen et al., 2021; catapults: Lewkowycz et al., 2020) Just above η_c only the radial mode of component 1 is unstable, and GD on it is the balanced scalar map of Figure 13a (Section A.7): a two-cycle, then chaos, whose time-averaged excess loss the network matches to 0.9% while it learns the other components; from $1.5\eta_c$ on it diverges or is thrown back to the origin saddle (binary $N = 4$). At larger σ the minimum that gradient flow reaches is sharper and $2/\lambda_{\mathrm{GF}}$ replaces η_c, with $\lambda_{\mathrm{GF}} \propto \sigma^{3.965\pm0.014}$ on the lazy side (centred binary $N = 4$, $\chi = 9$), close to the scale $\sigma^{2(K-1)}$ of the tangent kernel; from $\sigma = 1$ on, steps up to 1.5–3 times $2/\lambda_{\mathrm{GF}}$ catapult to a flatter minimum and still reach gradient flow's loss (Figure 13c; Section D.3).

SGD: the first plateau keeps its law. derived new (for TTNs; framework: Li et al., 2017; Ziyin et al., 2024) The per-sample gradient carries the Jacobian $J = \partial W/\partial\theta$, which vanishes at the origin like the drift, so noise cannot drive the first escape and its exponent $K - 2$ is unchanged; on an aligned mode it only delays and spreads the escape (55), consistent with the experiments of Corlouer et al. (2026) on deep linear networks (their Fig. 2: order kept, SGD slower). Measured empirical with online SGD on binary $N = 4$ (1,780 runs, $B = 1$–256): on average no setting shortened the first plateau beyond its noise (11 of the 23 cell means of Figure 22a are negative, all within 2.5 standard errors of zero and none below -1.0%), while single runs scatter to both sides of gradient flow (earlier in 787 of the 1,600 runs from random initialization, by more than 5% in 141).

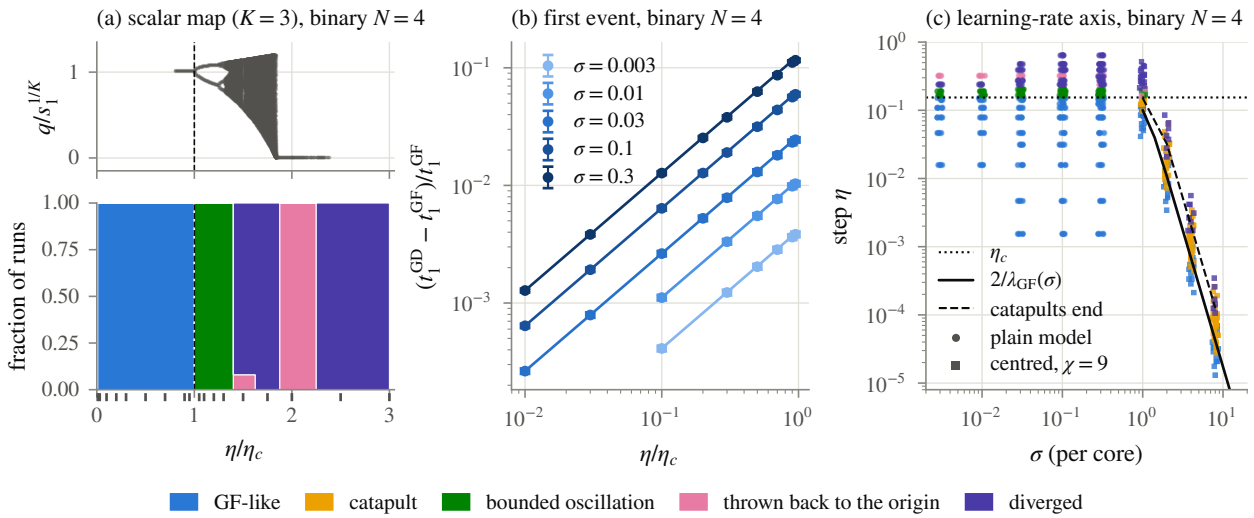


Figure 13: Gradient descent against gradient flow: binary tree, $N = 4$ ($K = 3$), $d = \chi = 3$, odeco teacher $(3, 2, 1)$, five seeds. (a) Top: attractor of the balanced scalar map $x \mapsto x + hx^{K-1}(1-x^K)$, $x = q/s_1^{1/K}$ with q the per-core amplitude of component 1, $h = \eta s_1^{2-2/K}$ (points at 0: returns to the origin). Bottom: outcome classes at the 18 steps marked below the axis, pooled over $\sigma = 0.003$–0.3 (15–25 runs per step). GF-like: monotone loss ending at gradient flow's loss (within 5% plus $10^{-3}L_0$); catapult: the same after a rise of the loss; thrown back: loss above $0.9L_0$ after the first event; bounded oscillation: a loss that keeps oscillating above gradient flow's level; diverged: above $10^6 L_0$. (b) Relative error of the first event time, mean ± s.e. over the five seeds; lines: $\eta c_1/t_1^{GF}$, no fit. (c) All runs in the (σ, η) plane, coloured by outcome class as in (a) (plain model with $\chi = 3$ and 9, $\sigma \le 0.3$; centred, $\chi = 9$, $\sigma \ge 1$): dotted, η_c; solid, $2/\lambda_{GF}$ (mean over seeds); dashed, the largest step at which a run still reaches gradient flow's loss. Takeaway: below η_c (at large σ, below $2/\lambda_{GF}$) GD is gradient flow with event times shifted by ηc_e; above it GD oscillates, diverges or, from $\sigma \ge 1$, catapults. (source: Section B.7, Table 17)

SGD: later plateaus are lengthened, and can trap. conjecture (heuristic law) known (mechanism: stochastic collapse, Chen et al., 2023) Once a component is learned, a route of $k \ge 3$ cores (Section 4.4) has no curvature, but its entries are coupled to the learned cores by zero-mean multiplicative noise of order one, which drives their log-amplitude down at a rate $c_{col}\eta/B$, with a collapse constant c_{col} that depends on teacher and tree. Proposition A.45 turns this into

$$\frac{T_{SGD}}{T_{GF}} = \frac{-\ln(1-x)}{x}, \qquad x = c_{col}\frac{\eta}{B}T_{GF}, \tag{24}$$

for a $k = 3$ plateau of gradient-flow length T_{GF}, and trapping for $x \ge 1$. Test empirical on binary $N = 4$ from the aligned route (component 1 learned, the route started at an amplitude that makes $T_{GF} = 50$–250 exact; 144 runs; Figure 22b): every escaped run was slower than gradient flow (114 of 114), none of the 30 runs at $(\eta/B)T_{GF} \gtrsim 0.025$ escaped (29 trapped, one diverged), and with one fitted $c_{col} = 32.6$ the condition $x = 1$ separates escaped from trapped settings in 23 of 24 cases; the law holds to about 10% for $T_{SGD}/T_{GF} \lesssim 1.5$, and within the errors only η/B matters. At the largest η/B the learned component also fell below $\frac{1}{2}$ in 19 of 24 runs, so the clean trap is the longest plateau ($T_{GF} = 250$) at half that η/B (6 of 6 runs, one dip). From random initialization the second plateau is lengthened as (24) predicts at $\sigma = 0.03$ but more than predicted at $\sigma = 0.1$ (Section D.3). Consequence: at fixed η/B a later plateau grows faster than $\sigma^{-(k-2)}$ and, by the law, becomes infinite at a finite σ conjecture; on the aligned route the ratio T_{SGD}/T_{GF} rises from 1.13 to 2.7 and then every run traps as T_{GF} grows empirical: SGD stalls where gradient flow is only

slow. By the same drift model a strict ($k = 2$) route still escapes if $|\lambda_-| > c_{\mathrm{col}}\eta/B$ conjecture: on the MPS with $N = 4$ and a pattern teacher $(3, 0.3)$ whose second component needs $k = 2$, SGD at $B = 1$ learned it in 5 of 5 runs, on average $8 \pm 5\%$ later than gradient flow empirical.

SGD: stability limit and forgetting. derived empirical new (the reduction to the pattern basis) known (the criteria: Wu et al., 2018; Chemnitz and Engel, 2025; exact mean square: Mulayoff and Michaeli, 2024, Thm 5) At the minimum the noise vanishes, and the exact mean-square limit (56) on binary $N = 4$ is $\eta^*_{\mathrm{ms}} = 0.00107$, 0.0161 and 0.112 for $B = 1$, 16, 256, against $\eta_c = 0.154$: per sample, small-batch SGD is about 150 times less stable than GD, because Gaussian product features have kurtosis 3^N and the coupling to the other patterns adds a factor 1.8. Well below this limit, noise on long plateaus erases learned components, which are then relearned: near its limit SGD breaks the saddle sequence rather than compressing it (Section D.3).

Finite samples: which interpolant? empirical new (for TTN regression) On the binary tree with $N = 8$, $d = 2$ ($W \in \mathbb{R}^{256}$) and an odeco teacher $(3, 2)$ with noise-free labels, GD on the empirical loss from small initialization recovers the teacher with its bond ranks from $n = 256$ samples at $\chi = 4$ (test error below 10^{-2} in 5 of 5 seeds), while the exact lazy limit, the minimum-norm solution of the model linearized at initialization, fails as dimension counting predicts derived: at $\chi = 4$ the teacher lies outside the tangent space and the error is worse than predicting zero at every n (Section D.3, Figure 23). A bias of small initialization towards low (hierarchical) rank is known for tensor factorizations known (Razin et al., 2021, 2022; tensor-train completion, empirically: Milanesi et al., 2021; matrices: Arora et al., 2019b).

When does gradient flow describe practical training? empirical derived (a summary of the results above; one teacher; GD on trees with $N \leq 8$, SGD on binary $N = 4$) GD follows gradient flow whenever $\eta < 2/\lambda_{\mathrm{GF}}(\sigma)$, with event times off by ηc_e and, from small initialization, the same saddle sequence; this limit is η_c at small initialization and falls like $\sigma^{-2(K-1)}$ on the lazy side (measured at $K = 3$: $\sigma^{-3.97}$), where steps up to 1.5–3 times larger still reach gradient flow's loss through a catapult. SGD needs in addition $\eta < \eta^*_{\mathrm{ms}}(B)$ (about $10^{-3}B$ here for $B \leq 16$), $c_{\mathrm{col}}(\eta/B)T \ll 1$ for every later plateau of length T, and η a small fraction of η^*_{ms} when learned components must survive long plateaus (no erasure in 72 runs of up to 400 time units at $\eta \leq 0.076\eta^*_{\mathrm{ms}}$; one in six runs of 1000 time units at $0.15\eta^*_{\mathrm{ms}}$). Because plateaus grow like $\sigma^{-(k-2)}$, at fixed η/B SGD eventually stalls or forgets where gradient flow is only slow.

5.5 Optimizers used in practice

Tensor-network models are trained with Adam (Kingma and Ba, 2015), with ALS and DMRG sweeps (White, 1992; Holtz et al., 2012a) (in machine learning often with a gradient step on the merged two-site core instead of an exact local solve, Stoudenmire and Schwab, 2016) or with Riemannian methods (Novikov et al., 2018; Steinlechner, 2016), which use the geometry of fixed-rank tensor manifolds (Holtz et al., 2012b; Uschmajew and Vandereycken, 2013). Which features of the gradient-flow picture of Section 4 survive them? Table 6 collects the answers, Figure 14 shows them from one initial point, and the evidence follows. Setup (Section B.8): every method starts from the same points, 42 on binary4 ($d = 3$, $\chi = 4$, odeco teacher $(3, 2, 1)$, seven $\sigma \in [0.002, 0.128]$, six seeds) and six seeds per σ (0.01–0.8) on the deep trees with $N = 8$, binary ($K = 7$) and MPS ($K = 8$), with the same teacher and with the pattern teacher E8 (patterns 0^8, $0^3 1^2 0^3$ and $010^4 10$, weights 1, 0.4, 0.7, whose weaker components need $k = 2$ and $k = 6$ cores on the MPS). GD and

heavy ball ($\beta = 0.9$) use effective steps $\eta/(1-\beta) \le 0.27\eta_c$, Adam $\epsilon = 10^{-8}$ unless stated; each method runs on its own clock. Details: Section D.3.

GD and heavy ball reproduce gradient flow. empirical new (for TTNs) GD with $\eta = 0.13\eta_c$ on binary4 has gradient flow's exponents to within 0.003 and its order from all 42 initial points; its first-escape delay grows by 0.0199 ± 0.0000 per e-fold of $1/\sigma$, the asymptotic slope $\eta(K-1)/2 = 0.020$ that follows from Proposition A.44 (asymptotically, conjecture in Section A.7; confirmed here). Heavy ball's delay grows $18.9 \approx (1+\beta)/(1-\beta) = 19$ times faster (conjecture: the momentum analogue). On the deep trees ($\sigma = 0.8$) GD and heavy ball escape after 1.001–1.006 and 1.01–1.11 times gradient flow's time.

Adam: same first component, but the first escape forgets σ. derived new (for TTNs; in aligned deep matrix factorization the sign-descent escape time is likewise bounded as the initialization vanishes: Sabanayagam and Lucey, 2026, Thm 2 and Prop. 6) With $\epsilon = 0$ Adam's step $-\eta\hat{m}_n/\sqrt{\hat{v}_n}$ is invariant under a common rescaling of past gradients known (Kingma and Ba, 2015), and near the origin ∇L is homogeneous of degree $K-1$; so the iterates from $c\theta_0$ with step $c\eta$ are c times those from θ_0 with step η, and Adam sees σ only through σ/η. Its first step is $-\eta \operatorname{sign} \nabla L(\theta_0)$, after which $|\theta| \sim \eta$ and the initial gradient weighs $\sim (\sigma/\eta)^{K-1}$ in the averages, so the trajectory has a limit as $\sigma/\eta \to 0$ and the number n_1 of steps to the first escape stays bounded as $\sigma \to 0$ (in the sign-flow limit $\eta \to 0$ at fixed β_1, β_2 known (Ma et al., 2022; Adam as sign descent: Balles and Hennig, 2018), ηn_1 tends to a constant). Measured empirical: first-escape exponents 0.009–0.013 on binary4 for four learning rates of Adam, and 0.012 ± 0.004 and 0.001 ± 0.019 for RMSProp; on binary $N = 8$, 0.049 ± 0.022 and 0.145 ± 0.033 ($\eta = 10^{-3}$, 10^{-2}; fitted over $\sigma \in [0.25, 0.8]$, 0.09 and 0.17 over $\sigma \le 0.4$: small but not zero), where gradient flow has 4.991 ± 0.002 ($\sigma \le 0.4$, prediction 5). The first component survives empirical: Adam and RMSProp pick gradient flow's first component in 94–100% of the resolved deep-tree runs (87–98% on binary4) and keep E8's locality order on the MPS (the weaker $k = 2$ component before the stronger $k = 6$ one, in 4–5 of 6 seeds, as gradient flow); the full order of gradient flow holds in 87–98% of the binary4 runs but only 67–83% of the deep-tree runs. On binary4 the first escape is as wide as its own time, so 79–100% of the runs fail $\rho_i > 3$ and are rich, non-incremental by Table 4; on binary $N = 8$ most are saddle-to-saddle.

Adam's ϵ brings the gradient-flow law back. derived new (for TTNs; concurrently, the same law in aligned deep matrix factorization: Sabanayagam and Lucey, 2026, Prop. 7) Where $|\partial_j L| \ll \epsilon$ Adam is gradient flow with $t = \eta n/\epsilon$: from a small σ it crawls along the gradient-flow path until the gradient reaches ϵ, then escapes as in the sign regime,

$$n_1 \simeq \frac{\epsilon}{\eta} T_{\mathrm{GF}}(\sigma) + n_{\mathrm{sign}}(\eta), \tag{25}$$

so the law $T_{\mathrm{GF}} \propto \sigma^{-(K-2)}$ returns, with prefactor ϵ/η, below the crossover $\sigma_\times$ where $\epsilon T_{\mathrm{GF}} = \eta n_{\mathrm{sign}}$. Test empirical on the odeco MPS with $N = 8$ (eight $\sigma \in [0.01, 0.3]$, six seeds, $\eta = 10^{-3}$, 10^{-2}; no fitted parameter): below the crossover measured over predicted n_1 is 0.9989–1.0003 at $\eta = 10^{-3}$ ($\sigma = 0.02$–0.03) and 1.0009–1.0062 at $\eta = 10^{-2}$ ($\sigma = 0.015$–0.02), and (25) predicts which runs escape in all 60 cases with $\sigma \le 0.05$; near $\sigma_\times \approx 0.05$–$0.1$ the deviations are systematic (Section D.3). There the crawl term is more than 98% of n_1, so the test checks that Adam is gradient flow on the clock $\eta n/\epsilon$ rather than the additive form. Only $K = 8$ was tested.

Table 6: What survives which optimizer. Binary $N = 4$ ($K = 3$): seven σ over 1.8 decades; binary $N = 8$ ($K = 7$) and MPS $N = 8$ ($K = 8$): two to eight σ over at most half a decade (1.5 decades for Adam's ϵ); six seeds each. "Gauge": the group (14). Exponent: of the first-escape time, $T \propto \sigma^{-\text{exponent}}$. Order: agreement with gradient flow from the same initial point, or counts over independent seeds for methods that give one trajectory per seed. Cost: median steps (sweeps, micro-steps) to $L < 10^{-3}L_0$, binary $N = 4$ at $\sigma = 0.008$, deep trees at $\sigma = 0.3$. S2S: saddle-to-saddle.

method	update equivariant under	plateaus	first escape vs. σ	learning order	cost
gradient flow	orthogonal maps of θ; conserves Q_e	saddle-to-saddle (S2S)	exponent 1.007 ($K = 3$), 4.99 ($K = 7$, $\sigma \leq 0.4$); predicted $K - 2$	reference	$3.6 \cdot 10^3$–$1.1 \cdot 10^4$ evaluations (to $10^{-6}L_0$)
GD, heavy ball (steps $\leq 0.27\eta_c$)	as gradient flow; Q_e drifts	as gradient flow	exponent 1.005, 0.970 ($K = 3$); delay $\propto \ln(1/\sigma)$	as gradient flow (100%)	T_{GF}/η for GD ($1.7 \cdot 10^4$, $K = 3$), $T_{\text{GF}}(1-\beta)/\eta$ for heavy ball
Adam, RMSProp	coordinate permutations, sign flips; gradient rescaling ($\epsilon = 0$)	sequential; rich ($K = 3$), S2S ($K = 7$, $\sigma \leq 0.4$)	σ-free, exponent 0.00–0.17; below $\sigma_\times$: (25)	first component as gradient flow: 94–100% (deep), 87–98% ($K = 3$); full order 87–98% ($K = 3$), 67–83% (deep)	$1.7 \cdot 10^2$–$2.1 \cdot 10^5$
re-gauged GD	gauge (cores reset to the root-centred canonical form)	none ($K = 3$); sequential ($K = 7$, 8)	σ-free, $\lvert\text{exponent}\rvert \leq 0.02$ ($K = 3$, 7)	largest first: 6/6 ($K = 3$, re-gauged every step), 2/6 and 5/6 ($K = 7$)	$2 \cdot 10^2$–$2 \cdot 10^4$; odeco MPS: budget
Riemannian GD ($\eta = 1$)	gauge (acts on W)	none	σ-free (exact)	one step ($K = 3$, 5 of 6 seeds); weakest last: 5/6 ($K = 7$)	2–7, each a dense Jacobian
Gram-precond. GD	gauge, up to damping (per-core Gauss–Newton blocks)	overshoot, then return	not σ-free: exponent 0.65 of the time to $10^{-3}L_0$ ($\eta = 0.01$); deep: 147–632 vs. 26–35 steps	mostly unresolved	$1.5 \cdot 10^2$–$4.3 \cdot 10^3$; odeco MPS: 9/24 converge
ALS, DMRG	gauge; invariant under per-core rescaling	none	σ-free (exact, proven)	none resolved (59/60 ties)	≤ 1 sweep (3–11 micro-steps)

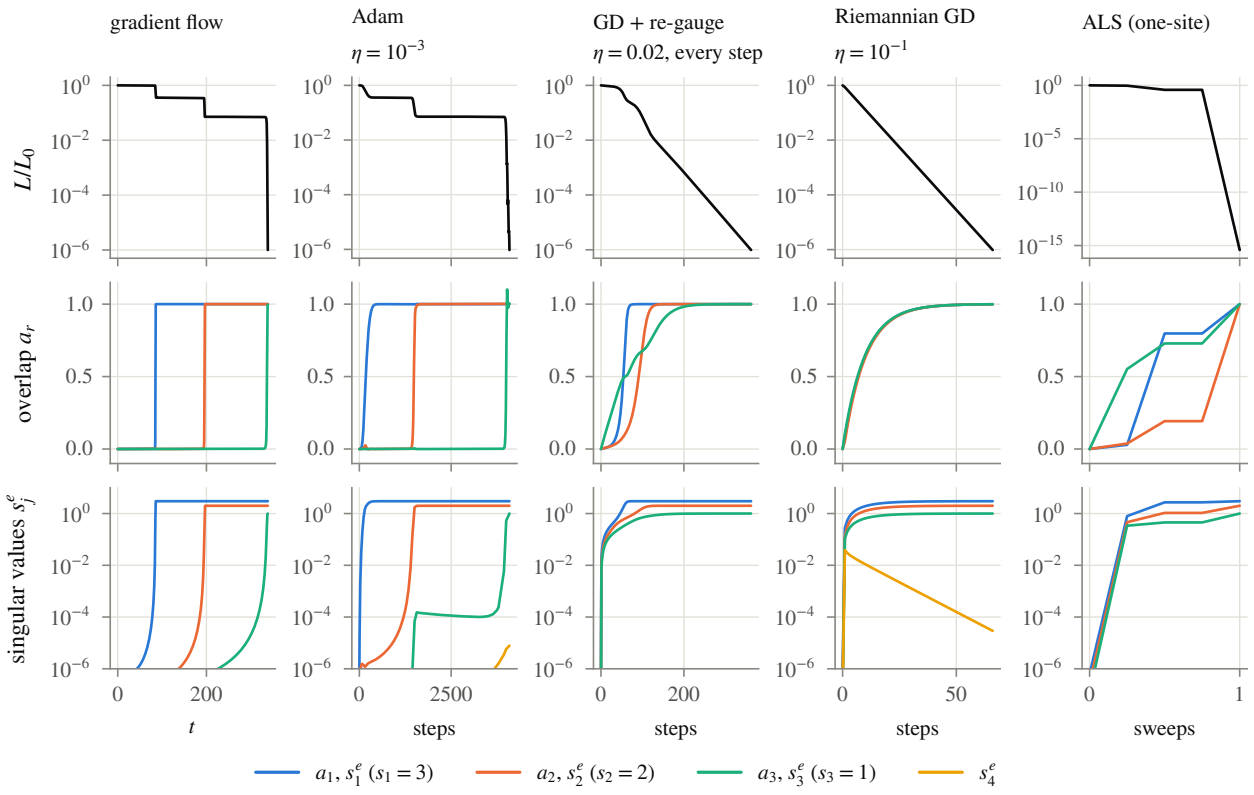


Figure 14: Five optimizers from one initial point: binary tree, $N = 4$ ($K = 3$), $d = 3$, $\chi = 4$, odeco teacher $(3, 2, 1)$, fan-in initialization, $\sigma = 0.008$, seed 0. Columns: gradient flow, Adam ($\eta = 10^{-3}$), GD re-gauged at every step ($\eta = 0.02$), Riemannian GD ($\eta = 0.1$), one-site ALS, each on its own clock. Rows: normalized loss, overlaps a_r, bond singular values of the cut 01|23 (colour: component r and singular-value index j). Gradient flow and Adam learn one component at a time in the same order, with plateaus at the partial-fit levels $L_1/L_0 = 0.36$ and $L_2/L_0 = 0.07$; the other three have none. This Adam run is saddle-to-saddle ($\rho_i = 7.1$, 45), which is atypical: most binary4 Adam runs are rich. The ALS run is the only one of the 60 ALS and DMRG runs (problem, method and seed combinations) in which no two components cross $a_r = \frac{1}{2}$ in the same micro-step. Takeaway: Adam keeps the sequential learning, the gauge-equivariant methods remove the plateaus. (source: Section B.8, Table 18)

No plateaus: sweeps and Riemannian GD. proven new (elementary) In the canonical form centred at core v the environment of v is an isometry, so the least-squares core is $\nabla_{G_v}\langle W, W^*\rangle$, a function of the other cores' frames, which are Q factors of QR decompositions and do not change when a core is multiplied by a positive number: every ALS and DMRG iterate is independent of σ. Riemannian GD, $W \leftarrow \mathrm{HSVD}_\chi(W - \eta P_{T(W)}(W - W^*))$ with $P_{T(W)}$ the projector onto the tangent space of the fixed-rank manifold, maps W_0 at $\eta = 1$ to $\mathrm{HSVD}_\chi(P_{T(W_0)}W^*)$, and that tangent space is the same at cW_0: it is exactly σ-free too. Measured empirical: ALS and DMRG fit the teacher within one sweep, with two components crossing $a_r = \frac{1}{2}$ in the same micro-step in 59 of 60 runs: no resolved order. Riemannian GD needs 2–7 steps; on binary $N = 4$ it fits all components within two steps (all three cross $\frac{1}{2}$ at step 1 in 5 of 6 seeds), but it stages them on binary $N = 8$, the weakest strictly last in 5 of 6 seeds for both teachers (3 and 2 of 6 on the MPS).

Re-gauged and Gram-preconditioned GD. derived empirical Re-gauging moves all of $\|W\|_F \sim \sigma^K$ into the root, whose environment becomes an isometry: the root fits the teacher's projection onto the current frames in a time $O(1)$ for any σ, and on the binary tree with $N = 8$, whose random frames keep only a small part of each component, learning is sequential again with σ-free event times. Gram preconditioning has no flat plateau but overshoots on its first step, and its time depends on σ (Table 6; details in Section D.3).

Interpretation. Two properties of an update decide what survives (Table 6, second column): how its step scales with the gradient, which near the origin is $O(\sigma^{K-1})$ (a step proportional to the gradient needs a time $\sigma^{-(K-2)}$, a step invariant under its rescaling does not, and an absolute scale such as Adam's ϵ brings the law back, derived), and the metric: ALS, DMRG and Riemannian GD see W_0 only through its frames, so they keep no σ (proven for ALS and DMRG, exact for Riemannian GD at $\eta = 1$), and because they move W at a rate of order one they keep no plateaus derived; re-gauged GD shares the equivariance and the σ-independence but not the rate (details in Section D.3).

6 Input covariance and teacher structure

Sections 4 and 5 use whitened inputs and orthogonally decomposable or pattern teachers. Here one complication is added at a time (covariance, correlations between sites, overlapping components, random TTN teachers, components that share a core, and finally real images), and we ask whether the saddle-to-saddle picture changes, and whether the baseline theory predicts the change after a substitution. An aligned covariance only reweights the components (Section 6.1). Each of the four ways we tried of making the components non-orthogonal in the metric of the loss (overlapping CP components, a misaligned covariance, random TTN teachers, correlated sites) changes the law of the escape from the rank-one saddle ($N = 4$, 8), and the tree decides how: strict and logarithmic on an MPS, of higher order on binary and HT trees, with escapes $\propto \varepsilon^{-1}$ from the saddle and, on binary4, $\propto \sigma^{-1}$ from random initialization; correlated sites were not run on HT. This is the central result (Sections 6.2 and 6.3). Rank-one plateaus sit at critical points of the bond-rank-constrained loss, blends rather than partial sums of components (on binary4 and MPS4; under correlated inputs only on binary4, under misalignment up to 30°; Sections 6.3 and 6.4); binary students learn bottom-up (Section 6.5); two smaller components that share a core can destabilize a learned larger one (Section 6.6); and on MNIST the same tree contrast appears with the data in place of the teacher (Section 6.7). Per-teacher tables and secondary checks are collected in Section D.4.

Setup (details in Section B.6). Gradient flow on the exact population loss, fan-in initialization, $\chi = 4$ unless stated. The *generic teachers* have $d = 2$: (a) CP teachers with $u_1^{(i)} = e_1$ and $u_2^{(i)} = \cos\phi\, e_1 + \sin\phi\, e_2$ at every site (overlap $\cos\phi$, weights $(1, 0.7)$); (b) an odeco teacher (weights $(1, 0.8)$ on e_1, e_2) under a misaligned covariance, the power-law spectrum of Section 2.3 rotated by an angle in the (e_1, e_2) plane; (c) random binary-tree and MPS teachers with bond ranks 2. Students are MPS, binary and HT trees with $N = 4$ and 8. Two protocols: random initialization at scale σ, or the *saddle start* of Section 4.4, the exact balanced bond-rank-one critical point of lowest loss, zero-padded, plus fan-in noise of scale ε. An escape is dated by an *onset event*, the first time the second singular value of some edge reaches 5% of the teacher's; that edge is the *realized edge*. Laws are fitted per seed; *asymptotic* fits use the three smallest σ or the five smallest ε; $\pm$ is the standard error over 4–10 seeds, statistical only, so agreements are cited at the 1–2% level.

6.1 Covariance aligned with the teacher: a reweighting

Prediction. Let the teacher be odeco with factors that are covariance eigenvectors, and let $\Lambda_r = \prod_i {u_r^{(i)}}^\top \Sigma_i u_r^{(i)}$ be the product of the covariance eigenvalues of component r. Near the origin the flow sees only the escape tensor $\Sigma W^* = \sum_r \Lambda_r s_r U_r$, so the *race* (which component escapes first, and when) is that of the whitened problem with weights $\Lambda_r s_r$; the *landscape* (critical points, plateau levels $\frac{1}{2}\sum_{r\notin S} \Lambda_r s_r^2$, saddle types) is that of $\Sigma^{1/2} W^*$, with weights $\Lambda_r^{1/2} s_r$ derived (Propositions A.39 and A.41). From an aligned initialization each component follows its whitened trajectory on the clock $\Lambda_r t$ proven known (Saxe-type modes: Saxe et al., 2014; depth two: Gidel et al., 2019; new: every tree) (Proposition A.40). When the two orderings disagree, a learned component j of binary4 is unstable toward r (a rotation through a strict saddle) iff $\Lambda_r s_r^2 > \Lambda_j s_j^2$; on MPS4 the rotation needs all four cores proven new (for TTNs; on binary4, a deep linear network, it follows from Achour et al., 2024, Thm 7, with Proposition A.39) (Proposition A.42).

Experiments. empirical On binary4 and MPS4 ($d = \chi = 4$, two teachers with the smaller component on the high-variance direction, 40 seeds; Section D.4) the aligned clocks hold to 10^{-9}; from random initialization the first component and first-escape time are those of the whitened race with weights Λs (95–100% of paired runs, times within 1.3%), so the covariance can reverse the order, while a control with the same landscape but the whitened flow differs in 10–53% of the seeds: the race is set by the flow metric. Exponents are unchanged; a learned component is replaced only on binary4 and only if the other has the larger Λs^2; plateau levels follow $\frac{1}{2}\sum_{r\notin S} \Lambda_r s_r^2$. The change is quantitative only and is predicted after two different substitutions new (for TTNs; in two-layer linear networks the split is implicit in Gidel et al., 2019, Thm 2, whose Corollary 2 states no ordering condition; related: escape along ΣW^* with a limited loss decrease, Rawal and DeWeese, 2026, Thm 4.2 and Prop. 4.5).

6.2 Generic teachers: the tree decides the saddle type

Prediction. derived new (Section A.5, "Generic teachers"; Proposition A.36 and Corollary A.37) After the first escape the flow sits near a balanced bond-rank-one critical point W_1, a local best rank-one approximation of W^* with value $\bar{s}$. A bond can gain rank through a two-core route if both of its cores have other legs of dimension above its rank. For odeco and pattern teachers the route's residual vanishes by orthogonality, which gives the $\sigma^{-(k-2)}$ laws of Section 4.4; for a generic teacher it does not. On an MPS every core carries a site, so W_1 is a strict saddle. Let M_e be the $d \times d$ matrix that W^* leaves on the two sites of the cores of edge e when the other sites are contracted with the factors of W_1, with singular values $s_j(M_e)$. Edge e contributes the Hessian

eigenvalues $-\bar{s}^{(N-2)/N} s_j(M_e)$, $j \geq 2$, each $\chi_e - 1$ times, in all $\sum_e (d-1)(\chi_e - 1)$ negative ones, and it escapes at the rate $\lambda_e = \bar{s}^{(N-2)/N} s_2(M_e)$, the analogue of λ_2 in (21): from the saddle start in a time $\lambda_e^{-1} \ln(1/\varepsilon) + O(1)$, and from random initialization with the slope $h_e = (1 - s_2(M_e)/\bar{s})/\lambda_e$ in $\ln(1/\sigma)$, where $s_2(M_e)/\bar{s}$ is the head start. On binary and HT trees the internal cores carry no site, so the cheapest route has $k = 3$: no negative eigenvalue and an escape $\propto \varepsilon^{-1}$ (or σ^{-1}). The landscape bound of Furman et al. (2026, Lemmas D.7–D.9) allows descent of order K at a rank-one point of an MPS; for generic teachers it is 2.

Experiments. Thirteen generic teachers (Table 27), each on the MPS, binary and HT student with the same N; five seeds; saddle start with ε from 10^{-8} (MPS) or 10^{-5} (binary, HT) to $5 \cdot 10^{-2}$, and random initialization for the six teachers with $N = 4$ on MPS4 ($\sigma = 3 \cdot 10^{-4}$–0.3) and binary4 (10^{-4}–0.1).

- *Saddle type.* derived and computed, new: all 13 MPS students are strict, with 5 (MPS4) and 17 (MPS8) negative eigenvalues, the count above; none of the 26 binary and HT students has an eigenvalue below -10^{-8}.
- *Escape from the saddle.* empirical new (Figure 15a,b) On MPS the log law wins in 65 of 65 seeds, and the slope equals $1/\lambda_e$ of the realized edge within 0.4% for all 13 teachers. On binary and HT students $t_{\text{esc}} \propto \varepsilon^{-1}$, with fitted exponents 0.9998–1.0005 in 23 of the 24 cases (1.0036 ± 0.0006 in the last, over a wider window), and the log law wins in no seed.
- *Random initialization, $N = 4$.* empirical new (Figure 15c,d) On MPS4 the log law holds, with slopes within 1.3% of h_e for the five teachers with a prediction; the head start matters (for one teacher 1.038 ± 0.005 against $h_e = 1.051$, and 2.78 without the head start). Binary4 gives σ^{-1} (fitted exponents 1.001–1.005). The curvature at the plateau does not depend on σ on MPS4 and falls $\propto \sigma$ on binary4, and the first escape keeps its $\sigma^{-(K-2)}$ law (per-teacher values in Section D.4).

One teacher (misaligned covariance, $N = 8$, 20°) first escapes to an intermediate saddle whose second singular values stay below the onset threshold; timed at that level, its MPS slope matches $1/\lambda_e$ to 0.24% (Section D.4). *Scope:* the escape from the rank-one saddle (the first rank increase), $N = 4$ and 8, $d = 2$; random initialization for $N = 4$ only. Later saddles on binary and HT students were not timed; there two-core routes can open on bottom edges at higher ranks, never for the root bipartition (Section A.5). Correlated inputs were never run on HT (Section 6.3).

6.3 Non-orthogonal components: a logarithmic second escape on MPS4

Three ways to make the components non-orthogonal in the metric of the loss are compared on MPS4 and binary4: overlapping CP components, a misaligned covariance and correlated sites. In each the rank-one plateau is a predicted critical point of the rank-constrained loss, a blend rather than a partial fit (except on MPS4 under correlated inputs, where it keeps the pure component, and beyond 30° of misalignment), and each turns the σ^{-2} second plateau of MPS4 into a logarithmic one (Table 7 and Figure 16).

Overlapping CP components. The rank-one critical points are those of $g(x) = \langle \bigotimes_i x_i, W^* \rangle$ on unit rank-one tensors proven. With two components of equal weight and overlap $\cos\phi$ at every site, an MPS has two rank-one maxima (one near each component) iff $\cos\phi < (N-2)/N$ and one blended maximum above proven (among symmetric points); unequal weights move the merger to a fold derived new (elementary) (Proposition A.35); on binary4 the rank-one point is the top singular

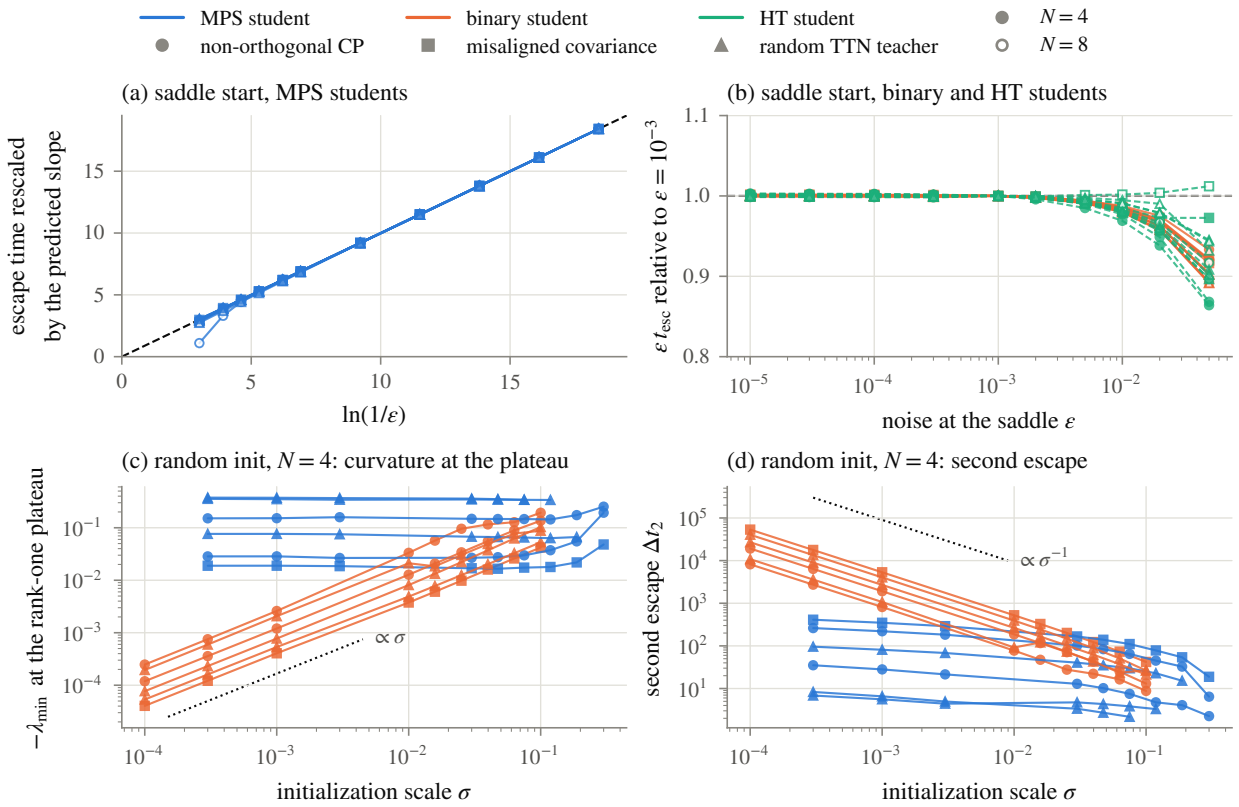


Figure 15: One generic teacher, three trees. (a) Saddle start, MPS students, 13 teachers: escape time rescaled by the predicted slope, λ_e times the time since the smallest ε, plus $\ln(1/\varepsilon_{\min})$ (mean over five seeds, λ_e of the realized edge), against $\ln(1/\varepsilon)$; dashed: the prediction. (b) Saddle start, binary (solid) and HT students (dashed), the same teachers except the misaligned one with $N = 8$ at 20°: $\varepsilon\, t_{\text{esc}}$ relative to its value at $\varepsilon = 10^{-3}$ (per seed, five seeds, then geometric mean); flat means $t_{\text{esc}} \propto \varepsilon^{-1}$. (c), (d) Random initialization, the six teachers with $N = 4$ on MPS4 and binary4 (runs on the major rank-one saddle with a resolved plateau, geometric means over one to five seeds, four or five at the three smallest σ): $-\lambda_{\min}$ of the gauge-projected Hessian at the plateau, and the second escape Δt_2. Takeaway: the same teachers give a strict saddle with a logarithmic escape at the predicted rate on the MPS, and a higher-order saddle on binary and HT trees (escape $\propto \varepsilon^{-1}$; from random initialization, binary4: $\lambda_{\min} \propto \sigma$, escape $\propto \sigma^{-1}$; random initialization was not run on HT) (source: Table 20).

pair of the 01|23 matricization (Remark A.23), a blend for every $\cos\phi > 0$. The plateau is never a component. empirical With $d = \chi = 4$ and overlaps up to 0.8, the measured rank-one minima are as predicted (two on MPS4 below the merger, one on binary4), and the first plateaus sit at these blends (Section D.4).

Misaligned covariance. Under a product covariance the critical points, critical values and saddle types are those of the whitened problem with teacher $\Sigma^{1/2}W^*$; escape rates are not, because the flow is preconditioned proven new (for TTNs) (Proposition A.39). With misaligned factors $\Sigma^{1/2}W^*$ is a non-orthogonal CP teacher, with factors $\propto \Sigma_i^{1/2} u_r^{(i)}$, while the escape is driven by ΣW^*, with factors $\propto \Sigma_i u_r^{(i)}$: the state must rotate after the escape, and misalignment alone can reverse the order derived new (Section A.6). empirical With $d = \chi = 4$ and weights $(1, 0.8)$, the first plateau is the predicted rank-one critical point of the Σ-loss for angles up to 30° (median relative level errors 10^{-6}–10^{-3}; at 45° most MPS4 runs have no resolved plateau). Binary4 runs first align with the

Table 7: Second escape on MPS4 from random initialization when the components are not orthogonal: slope of Δt_2 against $\ln(1/\sigma)$ over the three smallest σ (mean $\pm$ s.e. over seeds), for correlated sites its increment per halving of σ, against the prediction: h_e of Section 6.2 (CP overlap), the bound $1/|\lambda_{\min}|$ at the plateau (misalignment) and the two-stage law (correlation). "Log law": seeds in which it fits better than a power law. The orthogonal baselines (overlap 0, angle 0, $c = 0$) give $\Delta t_2 \propto \sigma^{-2.00}$. † Not yet asymptotic near the crossover overlap of order σ (Section A.5; conjecture). Settings as in Figure 16.

non-orthogonality	value	log law	measured	predicted
CP overlap $\cos\phi$	0.45	9 of 9	5.686 ± 0.016	5.682
	0.2	8 of 8	34.40 ± 0.20	34.14
	0.05	8 of 8	595 ± 11	$570^\dagger$
misalignment angle	20°	10 of 10	12.14 ± 0.03	bound 12.31
	5°	10 of 10	194.9 ± 1.2	bound 193.2, approached from above
correlation c (per halving)	0.4	4 of 4	2.133 ± 0.003	2.128
	0.3	4 of 4	4.616 ± 0.010	–
	0.2	4 of 4	14.67 ± 0.06	14.39, still falling
	0.1	4 of 4	113.9 ± 0.7	–

best rank-one approximation of ΣW^* and then turn to the saddle (75–100% of the runs from 10° on) known (in linear networks, binary4 being one: Atanasov et al., 2022); MPS4 runs do so only at 45°.

Correlated sites. In the one-factor model $x^{(i)} = (1-c)^{1/2} z_i + c^{1/2} z$ (independent standard Gaussians z and z_i, as in Section A.6) every site is whitened and two sites have covariance cI. The second-moment matrix is not a product, so no whitening substitution exists; the loss is computed with the dense matrix. The residual acquires *pairing* tensors such as $(x^{(0)} \cdot x^{(1)})\, x_r^{(2)} x_r^{(3)}$, of bond rank one on binary4 but d on MPS4 derived. On MPS4 the escape from the product saddle passes through binary4's first plateau point, which is strict on MPS4 but of higher order on binary4, so the second plateau grows by a fixed amount per halving of σ, set by the rates at the two saddles and the head start (the two-stage law of Section A.6) derived new; binary4 stays at σ^{-1}. empirical On binary4 ($d = \chi = 4$, weights $(1, 0.7)$, $\sigma = 0.05$) the first plateau falls from $0.33\, L(0)$ at $c = 0$ to $0.10\, L(0)$ at $c = 0.5$, always at the predicted critical value: a blend with pairing terms, no longer a partial fit (the MPS4 rank-one point keeps the pure component). The MPS4 plateaus separate only logarithmically: at $c = 0.4$ the separation ratio grows from 2.6 to 11 between $\sigma = 0.05$ and 0.0016, against 19 to 185 on binary4. HT students and $N > 4$ were not run with correlated inputs.

The second escape on MPS4. empirical new (overlap and misalignment; the correlated-site law and its first check are in Section A.6) With any of the three, the second escape of MPS4 from random initialization follows the log law in every seed, with the predicted slope where there is a prediction (at overlap 0.05 4% above it, not yet asymptotic; Table 7 and Figure 16), and the plateau curvature tends to its exact-saddle value as σ decreases; the orthogonal baselines give $\sigma^{-2.00}$. Under misalignment the slope is only bounded, by $1/|\lambda_{\min}|$ at the plateau, since the head start under the preconditioned flow was not derived. Binary4 keeps σ^{-1}: with correlated inputs (fitted exponents 1.005–1.011) and, in Section 6.2, for overlaps 0.2 and 0.45 and a 20° misalignment.

6.4 Random TTN teachers: plateaus are rank-constrained critical points

With bond ranks fixed, the critical points are the fixed points of one-site alternating least squares proven known (in substance: Eldén and Savas, 2011; Holtz et al., 2012a) (Proposition A.33); the hierarchical SVD is generically not one proven (Proposition A.34). The rank-one plateau should ap-

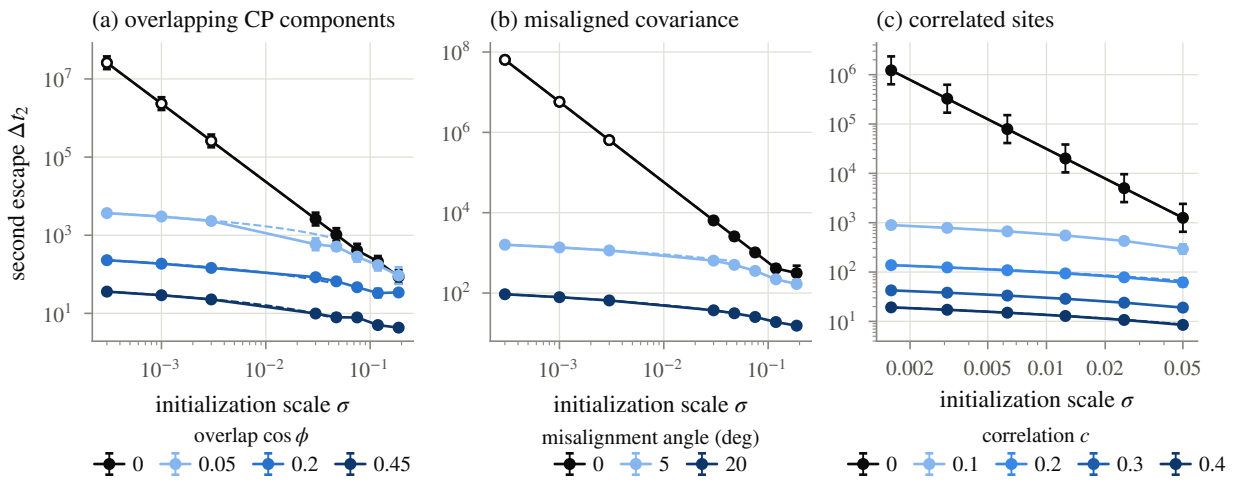


Figure 16: MPS4 from random initialization: the second escape Δt_2 (onset events; runs on the major rank-one saddle with a resolved plateau; geometric mean over seeds with its standard error) when the components of the teacher are made non-orthogonal in three ways. (a) CP overlap $\cos\phi$ ($d = 4$, weights $(1, 0.7)$, ten seeds, not all of which qualify at every σ); dashed: $h_e \ln(1/\sigma)$ (Corollary A.37), anchored at the smallest σ since the intercept is not predicted. (b) Misaligned covariance, angle in degrees ($d = 4$, ten seeds); dashed: slope $1/|\lambda_{\min}|$, an upper bound. (c) Correlated sites, correlation c ($d = 3$, weights $(1, 0.6)$, the four of five seeds on the major saddle, events on the singular values across 01|23); dashed: the two-stage law (Section A.6; not computed for $c = 0.1, 0.3$). Black: the orthogonal baseline, dashed $\propto \sigma^{-2}$; open markers: runs flagged by the float64-floor scan (dormant overlap 10^{-21}–10^{-12}); without them the baseline exponents are 2.008 ± 0.002 (overlap 0) and 2.000 ± 0.002 (angle 0). Takeaway: each of the three non-orthogonalities tested turns the σ^{-2} second plateau into a logarithmic one, with the predicted slope where there is a prediction (overlap 0.05: +4%, not yet asymptotic) (source: Table 20).

proach a local best rank-one approximation as a power of σ (conjecture: a heuristic crossing argument gives the exponent), and an MPS then gains rank bond by bond, first on the edge e^* with the largest $s_2(M_e)$, with slope h_{e^*} derived. Experiments empirical new (MPS4 student, $d = 3$, $\chi = 4$; five random MPS4 teachers with bond ranks $(2, 3, 2)$; eight σ from 10^{-3} to 0.2; six seeds): the rank-one plateau converges to the bond-rank-one critical point as a power of σ (exponents 0.07–0.15 above the heuristic; Section D.4); for three teachers that point is separated from every SVD-type truncation, while for the other two, and at the higher-rank plateaus, it lies within 0.0032 of one, so the plateaus do not discriminate. Ranks then grow one bond at a time, e.g. $(1, 1, 1) \to (2, 1, 1) \to (2, 2, 1) \to (2, 3, 1) \to (2, 3, 2)$; the first increase follows the log law in 26 of 26 seeds, with single-route slopes within 0.3–1.6% of h_{e^*}, always on e^* at $\sigma = 10^{-3}$ (later increases were not timed). In CP factorization small initialization follows a rank-one trajectory (Razin et al., 2021), and gradient descent on tensor-train completion shows a plateau followed by abrupt rank growth (Milanesi et al., 2021).

6.5 Teacher and student on different trees

A student bond must reach the rank of the teacher's matricization across it, bounded by products of bond dimensions over cuts of the teacher's tree proven known (standard min-cut bound; cf. Grasedyck and Hackbusch, 2011, Lemmas 7–8) (Section A.5): a binary8 student of an MPS8 teacher with bond ranks 2 and $d = 2$ needs rank 4 on the bonds above the bottom cores of sites 2, 3 and of sites 4, 5, whose cuts cross two teacher bonds. The routes give *bottom-up order* derived new: the root bipartition 0–3|4–7 (the teacher's middle bond) has a three-core route only after both halves have

bond rank above one below it, so it grows only after a bottom bond on each half, whatever its teacher singular value. Experiments empirical new (binary8 student, $\chi = 4$, twelve random MPS8 teachers with bond ranks 2, $\sigma = 0.3$, 0.4, 0.5 (two teachers at 0.4 only), eight seeds): in the 7 teachers whose root cut has a larger teacher singular value than both bottom bonds of a half, so that a weight order disagrees, bottom-up order holds in 163 of 168 runs (56 of 56 at $\sigma = 0.3$), including 117 of 120 for five teachers whose root cut is the strongest of all by 5.5–13%; the 5 exceptions grow the root after a bottom bond on one half only. Over all 12 teachers: 249 of 256. "Largest teacher singular value first" fails, and "a sibling pair first" is no rule (8–100% of the runs by teacher). Scope: $N = 8$, $\sigma = 0.3$–0.5; the reverse mismatch, an MPS8 student of binary8 teachers, was run only at a σ too large to test the asymptotic race (Section D.4).

6.6 Star teachers: forgetting a learned component

A learned component j without a two-core route to the rest is a strict saddle iff $\lambda_{\max}(\Gamma) > s_j$, Γ the weighted adjacency matrix of the piggyback graph, and then $\lambda_{\min} = -s_j^{(K-2)/K}(\lambda_{\max}(\Gamma) - s_j)$ proven new (for general trees; deep linear networks: Achour et al., 2024, Thm 7; related: active units can become dormant, Pesme and Flammarion, 2023, and Kunin et al., 2025, Lemma B.1) (Proposition A.22). The *star* has $q = 0000$ ($s_q = 2$) and $r = 1010$, $r' = 1001$ (weights $s_r = s_{r'}$, $d = 2$), which share the piggyback node: $\lambda_{\max}(\Gamma) = \sqrt{2}\, s_r$, so "$q$ learned" (θ_q) is strict iff $s_r > \sqrt{2}$, although both residual components are smaller and each needs three or more cores. Experiments ($\chi = 3$; MPS4, binary4, HT4; saddle start at θ_q with $\varepsilon = 10^{-6}$–10^{-2}, five seeds; random initialization at $s_r = 1.2$, 1.6, ten seeds):

- *Hessian* proven and computed: the formula holds to 10^{-15} on MPS4, binary4, HT4 and a binary8 analogue; there is no negative eigenvalue for $s_r \le 1.40$.
- *Threshold* empirical new (dynamics): started at θ_q with $\varepsilon = 10^{-6}$, no seed leaves it at $s_r = 1.4138$ ($\lambda_{\max}(\Gamma)/s_q = 0.9997$) and all do at 1.4146 (1.0003; three seeds each) on MPS4 and binary4; on HT4 the step lies between 1.40 and 1.42 for $\varepsilon \le 10^{-3}$. At $\varepsilon = 10^{-2}$ it moves to 1.43–1.44 on MPS4 and binary4, where the polynomial route that adds r beats the slow strict escape.
- *One or two strict stages* derived (from the computed Hessians) and empirical new: where one strict stage is timed ($s_r \ge 1.48$; every s_r on binary4) the slope of the escape time against $\ln(1/\varepsilon)$ is $1/\lambda_q$ within 0.21%, with $\lambda_q = -\lambda_{\min}$ at θ_q. Near the threshold on MPS4 the flow stops at a blended rank-one point that is itself strict, and the two times add (within 0.2%). On HT4 that point has no negative eigenvalue and is left in a time $\propto \varepsilon^{-1}$: a star forgets q in logarithmic time on MPS4 and binary4 but polynomially on HT4, the tree contrast of Section 6.2 again.
- *Random initialization* empirical new ($s_r = 1.6$): the flow never sits at θ_q (at most 0.13 time units at its loss level, one seed 1.7). The initialization direction sets the class, but not the outcome: q learned last; a_q rising to 0.78–1.00 and then, in most runs, decaying (on MPS4 one of these seeds keeps q for $\sigma \ge 0.025$, and three more do at $\sigma = 0.1$); or a blended start. Where q decays, the decay deepens steeply as σ decreases, not as a power law, and q is relearned last. Sub-critical stars ($s_r = 1.2$) keep q ($\min a_q \ge 0.991$). Per-class counts and the near-threshold values are in Section D.4.

6.7 Real data

Do the teacher–student results survive real inputs, with a non-product input law and no teacher? The empirical loss is $L = \frac{1}{2}\langle W, CW\rangle - \langle Y, W\rangle + \text{const}$, with the second-moment operator C of

the features, which is not a product, and the label tensor $Y = \mathbb{E}[y\Phi]$ (averages over the training set). Two predictions need no product law derived. Near the origin $L \approx L(0) - \langle Y, W\rangle$, so the first escape is that of Section 4.3 with $W^* \to T$ (Proposition A.41). At a bond-rank-one critical point the routes that can open depend only on bond ranks and site dimensions (Section A.5, "Generic teachers"): on a binary tree, whose internal cores carry no pixel, the cheapest routes join two sibling bottom cores ($k = 3$: a higher-order saddle, a plateau $\propto \sigma^{-1}$); on an MPS any adjacent pair of cores can open one ($k = 2$: strict, $\propto \ln(1/\sigma)$). These are the generic-teacher predictions of Section 6.2, with Y for W^* in the first escape; the route counting at the rank-one point does not depend on the target. The data decide which of the cheapest routes goes first, and the assignment of pixels to sites decides which pixel interactions lie on short routes.

Setup (details in Section B.9). MNIST 3 ($y = +1$) against 5 ($y = -1$): 4000 training and 1600 test images, cropped to 24×24 and block-averaged to 4×4 pixels ($N = 16$), with the features $\phi(x) = (\cos\frac{\pi x}{2}, \sin\frac{\pi x}{2})$ of Stoudenmire and Schwab (2016); square loss ($L(0) \approx \frac{1}{2}$), fan-in initialization, $\chi = 4$. Gradient flow runs through the first plateau to the time t_a at which the loss has fallen by 1%, then on the clock $\tau = t - t_a$ up to $\tau = 10^4$. Trees: binary16 ($K = 15$; two pixels per bottom core, none on the internal cores) and MPS16 ($K = 16$). Pixel orders: *quadtree* (Morton order; bottom cores carry horizontally adjacent pixel pairs, level-2 subtrees 2×2 blocks), *snake* (boustrophedon, for the MPS) and random permutations (*scrambled*). Main sweep: binary quadtree, snake MPS and three scrambled binary orders, seven σ from 0.05 to 0.5 (four for one scrambled order), seeds 0–4, plus whitened site features for three of the orders. Events are loss drops (peaks of $-\mathrm{d}L/\mathrm{d}t$; gaps on the τ clock) and rank events (a ratio s_j^e/s_1^e of data-weighted singular values crossing 0.05). Runs of one (order, seed) at different σ mostly reach the same rank-one critical point, so independent counts are per (order, seed) or per *unit* (order, seed, critical point); per-state counts (one state per run, repeating a critical point over σ) are given only next to them. $\pm$ is the standard error over seeds (over series for the free gap exponent below).

What survives.

- *First plateau: an identity, not evidence.* $t_a\sigma^{K-2}$ is constant per seed to four digits or better (exponents 13.000 ± 0.000 on binary16, 14.000 ± 0.000 on MPS16). As $\sigma \to 0$ it tends to a per-seed constant for any input law: with $\theta = \sigma\tilde{\theta}$ and $s = \sigma^{K-2}t$, $\tilde{\theta}$ follows the flow of $\langle Y, W\rangle$ from Z up to terms of relative order σ^K proven known (the exponent: Jacot et al., 2022; Kumar and Haupt, 2025a). It checks the integrator and the clock, not the theory.

- *The first binary route is a sibling pair* empirical new. After the escape, 96 of the 124 binary states lie within 0.05 (relative) of a rank-one product, the others mostly at $\sigma \geq 0.3$. In all 20 (order, seed) pairs the first two edges to gain a singular value belong to two sibling bottom cores, the predicted $k = 3$ route, and in all 96 rank-one states they cross the threshold together; 11 independent runs (nine orders, $\sigma = 0.1$) agree.

- *Route couplings pick the route* empirical new. At the rank-one state the *route coupling* μ_{uw} of two bottom cores is the norm of the residual's correlation between their unused feature directions, weighted by the other cores (62): the coupling of Remark A.27, not the size of the dormant route entries. It sets the rate of the route through them up to factors common to all routes with the same number $k_{uw} \in \{3, 5, 7\}$ of cores (Definition A.17). The sibling pair with the largest μ_{uw} rises first in 91 of the 96 rank-one states, in about 80–85% of the critical-point units (chance $\frac{1}{4}$; Section D.4) and in 11 of 11 independent runs. Unlike in the teacher runs, the couplings themselves decide these ties: the largest sibling coupling exceeds

the second by a median factor 60 (quadtree) and 5 (scrambled orders). Degree comes before coupling: in 95 of the 96 states (25 of the 26 units) the largest of the 28 couplings lies on a route with $k = 5$ or 7, yet a sibling pair rises first (Remark A.27).

- *The $k = 3$ gap is $A/\sigma + t_0$, not logarithmic* empirical new (Figure 17). Over the 11 series (one per order and seed) whose first route, one sibling pair, rises inside the second loss drop at every σ (at least four values over at least half a decade), the gap between the first two drops fits $A/\sigma + t_0$, the $k = 3$ law (21) with its $O(1)$ correction (Proposition A.25), with relative rms residual at most 0.060 and $t_0 < 0$ in all 11; the log law of a two-core route leaves 0.11–0.30. A free exponent (1.22 ± 0.03, range 1.06–1.36) mostly measures the offset and depends on the σ range. All 11 series use raw features; the verdicts also hold over all 124 binary runs (Section D.4).
- *Saddle types* empirical new, from the σ-scaling of $\lambda_{\min}(H_\perp)$ at plateau midpoints (Section 2.8). Binary16 (quadtree) at the rank-one plateau ($L \approx 0.27$): $|\lambda_{\min}| \propto \sigma^{0.87}$ to $\sigma^{0.89}$ on the three seeds with five or six σ (and $\sigma^{1.15}$ on a seed with three): higher order, though below the exponent 1 of $k = 3$ (Proposition A.26) at these σ. MPS16 at $L \approx 0.23$ (five seeds): $\lambda_{\min} = -0.054$ to -0.079, exponent 0.047 ± 0.016, a strict saddle; it comes after the first route has risen, with two or three eigenvalues below $-10^{-3}\lambda_{\max}$ at $\sigma \le 0.1$: the saddle of later two-core routes.
- *Logarithmic later routes on the MPS* empirical new. The first rank increase is on one edge in all 25 runs where it is resolved (in 10 of 35 the threshold is met at $\tau = 0$), within two time units of the first drop's peak: it merges with the escape and has no measurable gap. Of the later edges, timed by their first rise after the first drop, edges 3, 4 and 6–9 prefer the log law (28 of 29 seed fits; relative rms 0.03–0.11 against 0.17–0.36), edge 4 with slope 14.7 ± 1.5 per unit of $\ln(1/\sigma)$, comparable to $1/|\lambda_{\min}| = 13$–19. Edges 1 and 2 prefer a power law and edge 0 is mixed, which rank-one counting ($k = 2$ on every MPS edge) does not explain.
- *Plateaus creep* empirical: between drops the loss falls by a median 0.76 of the next drop (56 intervals in 20 binary runs, $\sigma = 0.1$). With drops dated by rate peaks the separation ratios on binary trees grow as $\sigma \to 0$ (quadtree: median smallest ratio 2.8 at $\sigma = 0.5$, 24 at 0.05; all quadtree runs saddle-to-saddle at $\sigma \le 0.07$), not on the MPS.

Pixel order: a pre-registered test. Among the binary orders of the main sweep the quadtree ends lowest at small σ: at $\sigma \le 0.14$ a scrambled order ends below the quadtree of the same seed in 1 of 50 runs (at $\sigma = 0.1$, $L(10^4) = 0.098 \pm 0.001$ against 0.19–0.23 for the three scrambled orders), while at $\sigma = 0.5$ scrambled orders end below it in 14 of 15 runs (the snake MPS ends lower still at every seed for $\sigma \le 0.3$). The quadtree puts most of $\sum \mu_{uw}^2$ on routes with $k \le 5$ (0.89 ± 0.01 at $\sigma = 0.05$; scrambled orders 0.17 ± 0.07) empirical. In independent runs, though, the equally local transposed quadtree (vertical pairs) ended above six random orders at seed 0. Before running any new order we registered a test of this share; the registration and the predictor values were committed before the runs (Section D.4), and the test is reported here as registered.

- *Design.* Hypothesis: across pixel orders of binary16 at $\sigma = 0.1$, the share $\mathcal{S}$ of $\sum_{uw} \mu_{uw}^2$ on routes with $k_{uw} \le 5$, evaluated at the best rank-one fit of the labels (no training), predicts (O1) the training loss at $\tau = 10^4$ and (O2) $\log_{10}$ of the time to $L \le 0.25$: Spearman correlation $r_S < 0$ with a 95% bootstrap interval that excludes 0, for each outcome, over all orders and within the new random orders. Orders: 20 new random ones and 7 structured ones (quadtree; quadtreeT, the Morton order of the transposed image; row-major; column-major; snake; a

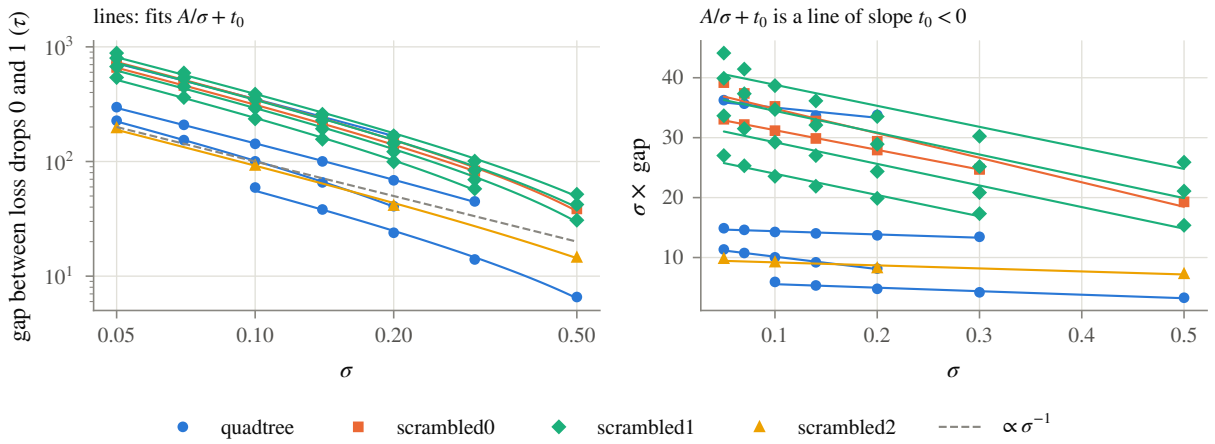


Figure 17: First route after the escape on binary16 (MNIST 3 against 5, $\chi = 4$): gap between the first two loss drops (τ clock) against σ, for the 11 series (one per order and seed; colour and marker: order) whose first route, one sibling pair, rises inside the second drop at every σ. Left: log–log axes; lines: $A/\sigma + t_0$ fitted with relative residuals; dashed: $\propto \sigma^{-1}$. Right: σ times the gap, in which $A/\sigma + t_0$ is a line of slope t_0. Takeaway: every series follows the $k = 3$ law with a negative $O(1)$ offset; a logarithmic law fails (source: Table 20).

Hilbert curve; mortonrev, bit-reversed Morton keys, which put every sibling pair two rows apart). Seeds 0–2: 81 runs, 74 new (the outcomes of quadtree, quadtreeT and row-major were seen before registration); outcomes averaged over seeds; O2 censored at $\tau = 10^4$ (4 of 81 seeds).

- *Result* empirical (pre-registered) new (Figure 18). O1: $r_S = -0.60$ $[-0.82, -0.22]$ over the 27 orders (permutation p-value 0.001) and -0.55 $[-0.85, -0.09]$ within the 20 random ones (p-value 0.015): *supported*. O2: -0.30 $[-0.59, +0.06]$ and -0.26 $[-0.63, +0.23]$: *not supported*. The registered secondary analyses agree (O1 without the three orders seen before registration: -0.51 $[-0.80, -0.08]$; with the three scrambled orders of the main sweep added: -0.60). Orders with the same pixel pairs share $\mathcal{S}$ (24 distinct values over 27 orders), so the within-random result is the cleaner test. Also registered: the rank-one search on the data does not always find the rank-one state that gradient flow reaches (quadtree: a fit at loss 0.310, while four of five runs plateau at 0.27), so $\mathcal{S}$ is a noisy proxy.
- *Deviations.* None in the design or the analysis; the compute budget of 8 processes was exceeded for a few minutes, and two procedural points (the timing of one commit and an interim summary) are listed in Section D.4.

The pixel order thus sets how low a binary TTN ends at fixed training time, and a route-coupling share computed from the data before training predicts the ranking of the orders by that loss (Spearman correlation -0.60; -0.55 within random orders), but not by their early speed. Locality in the image is neither necessary nor sufficient: the three orders with horizontal pixel pairs (quadtree, snake, row-major; $\mathcal{S} = 0.989$) rank 1, 2 and 4 of 27 by O1, but the equally local orders with vertical or mixed pairs (quadtreeT, column-major, Hilbert) rank 13–21; the transposed quadtree ends at twice the loss of the quadtree (0.189 against 0.097; the quadtree's outcome, the lowest, and two of the three transposed-quadtree seeds were seen before registration); and among random orders one with $\mathcal{S} = 0.994$ ranks third (4 of its 8 bottom pairs happen to be adjacent in the image, against 1.6 expected by chance) and one with $\mathcal{S} = 0.990$ and no adjacent pair ranks sixth. Tree tensor networks for images are often built to follow the pixel geometry (e.g. Cheng et al., 2019). Our

Table 8: Input covariance and teacher structure: what changes, and what predicts it. "Escape": the escape from the rank-one saddle.

complication	what changed (empirical)	predicted by
aligned covariance	quantitative only: order and times follow $\Lambda_r s_r$, plateaus and swaps $\Lambda_r s_r^2$; exponents unchanged	whitened problems with Λs (race) and $\Lambda^{1/2} s$ (landscape)
generic teachers	escape strict and logarithmic on MPS ($\leq 0.4\%$ from the saddle; $\leq 1.3\%$ from random init, MPS4, $d = 2$); $\propto \varepsilon^{-1}$ on binary and HT, $\propto \sigma^{-1}$ on binary4 (random initialization not run on HT)	two-core routes and their rates
misaligned covariance	plateaus at critical points of the Σ-loss; early alignment (binary4); MPS4 escape logarithmic	$\Sigma^{1/2} W^*$; slope only bounded
cross-site correlations	binary4: first saddle with pairing terms; MPS4: logarithmic two-stage escape	two-stage law
overlapping CP components	blended plateaus; MPS4 rank-one maxima merge at overlap $\frac{1}{2}$ (equal weights; fold near 0.36 for $(1, 0.7)$); logarithmic escape	rank-one landscape; h_e within 1% (overlap ≥ 0.2)
random TTN teachers	plateaus at rank-constrained critical points (3 of 5 teachers separated from SVD truncations); ranks grow bond by bond	critical points; h_{e^*} within 1.6%
teacher on another tree	bottom-up order on binary8 (163 of 168 discriminating runs)	routes
star teachers	started at the larger learned component ($\varepsilon \leq 10^{-3}$) the flow leaves it iff $\lambda_{\max}(\Gamma) > s_q$; from random initialization it never sits there; log time on MPS4, polynomial on HT4	piggyback graph; rates within 0.21%
real data (MNIST)	as for a generic teacher: binary first route a sibling pair, gap $A/\sigma + t_0$; MPS later routes logarithmic (6 of 9 edges); plateaus creep; the pixel order sets the final loss at small σ (locality in the image neither necessary nor sufficient)	route counting with $Y = \mathbb{E}[y\Phi]$ for W^*; route couplings (first route; ranking of orders, pre-registered)

explanation is a conjecture: an order helps by placing label-relevant pixel interactions on routes that escape within the horizon ($k = 5$, plateaus $\propto \sigma^{-3}$) rather than on routes with $k = 7$ ($\propto \sigma^{-5}$). *Exploratory, not registered:* the same share at the state each run actually reaches ranks O1 more strongly ($r_S = -0.83$ $[-0.91, -0.66]$; -0.85 $[-0.95, -0.60]$ within random orders) and O2 no better empirical.

Scope. The scaling claims rest on one task at 4×4 pixels, gradient flow and 4000 fixed images; 8×8 images (12 runs) show the same first routes but allow no fits, and whitened site features do not make the dynamics cleaner. The horizon $\tau = 10^4$ cuts the escapes along routes with $k \geq 5$ at $\sigma < 0.1$, so final losses are losses at fixed training time. Events are read from loss rates and spectral thresholds, so late events are only partly identified by route. Results on Fashion-MNIST, at $\chi = 8$, on the clock τ against the absolute clock, and at large σ are in Section D.4.

6.8 Summary

Only an aligned covariance leaves the qualitative picture unchanged (Table 8). When a complication tested breaks the orthogonality of the teacher's components in the metric of the loss (overlap, misalignment, correlated sites, random TTN teachers), the student's tree decides whether the escape from the rank-one saddle is logarithmic (MPS) or polynomial (binary; HT from the saddle start, never with correlated inputs). Stars, whose components are orthogonal, change which saddle is strict. On MNIST the same tree contrast appears with the label tensor Y in place of W^*, while the effect of the pixel order is only ranked, by a data-only route-coupling share (pre-registered). The baseline theory predicts each change after a substitution, except the slope under a misaligned

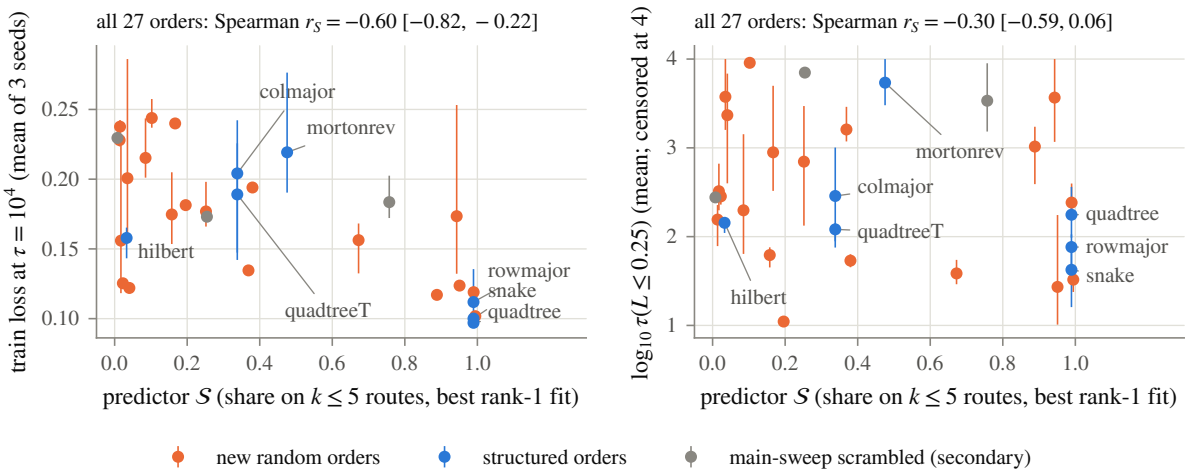


Figure 18: Pre-registered order test: 27 pixel orders of binary16 (orange: 20 new random orders; blue: 7 structured orders, labelled), $\sigma = 0.1$, seeds 0–2 (points: mean; bars: range over seeds); grey: the three scrambled orders of the main sweep (secondary analysis only). Horizontal axis: the predictor $\mathcal{S}$, computed from the data before training. Left: training loss at $\tau = 10^4$ (O1). Right: $\log_{10}$ of the time to $L \le 0.25$ (O2), censored at 4. Titles: Spearman correlation r_S over the 27 orders, 95% bootstrap intervals. Takeaway: $\mathcal{S}$ predicts the ranking of the orders by their final loss ($r_S = -0.60$), not by their early speed; orders with the same pixel pairs share $\mathcal{S}$, and local orders with vertical or mixed pairs rank only 13–21 of 27 (source: Table 20).

covariance from random initialization, which is only bounded, and, on MNIST, the creep of the plateaus and the power-law rises of MPS edges 0–2, which it does not explain.

7 Summary by parameter and open questions

7.1 How each parameter acts

Table 9 collects, for each parameter, what it does to the saddle sequence (which saddles are visited and in which order), to the plateau durations, and to the regime boundaries of Section 3, with the sections that give the evidence and its scope.

Table 9: How each parameter acts on gradient flow from small initialization, and on the regime boundaries; sections under each parameter, tags as there. σ: initialization scale (α: the function-space scale, the product of the core scales); K: number of cores; χ: bond dimension; d: local dimension; R: number of teacher components; k^*: cores of the cheapest route; s_1: largest teacher weight; s_j, s_r: weights of a learned and a residual component; $\mathcal{T}(Z)$: first-escape time computed from the initialization direction Z; σ_c, σ_{lazy}: saddle-to-saddle and lazy boundaries; Λ_r: input variance along component r (aligned covariance Σ); η: step size; B: batch size; λ: scale of the root core relative to the others; δ_Θ: relative change of the tangent kernel; binary4: binary tree with $N = 4$; full bonds: every bond at its cut dimension; two-core pairs: two components whose differing sites are carried by two cores.

parameter (sections)	saddle sequence	plateau durations	regime boundaries
initialization scale σ (4.3–4.7, 3.2)	At leading order none: the first escape direction, hence the first component and, for pairs differing on three or more cores, the order, depend on the initialization direction only derived empirical; two-core pairs become greedy as $\sigma \to 0$, and close weights separate only at small σ empirical	First plateau $\sigma^{-(K-2)}\mathcal{T}(Z)$ proven; later ones none, $\ln(1/\sigma)$ or $\sigma^{-(k^*-2)}$ for $k^* = 1$, 2, ≥ 3 derived empirical, with prefactors set by the route amplitudes when each plateau starts	The main knob: saddle-to-saddle, rich, then lazy with full bonds or a floor below them; binary4: $\sigma_c \approx 0.3/\sqrt{\chi}$ empirical, $\sigma_{\text{lazy}} \approx 6.5/\sqrt{\chi}$ at $\chi \geq 32$ derived empirical
cores K, tree shape (4.3–4.8, 3.3, 6.2, 6.5)	The tree sets k^* of every route, hence the next component (*locality beats weight*) and the saddle type; the number of cores carrying differing sites sets race versus greedy order; generic teachers: strict escapes on MPS, higher-order on binary and HT trees; binary8 students grow bottom-up derived empirical	First plateau: exponent $K - 2$, whatever the depth proven; median prefactor from 0.42 (HT3, $K = 5$) to 814 (binary16, $K = 15$), growing with K within each family empirical; later plateaus: exponent $k^* - 2$ (one teacher: logarithmic on MPS8, σ^{-3} on binary8, about σ^{-5} on HT8) empirical	σ_c rises with K (0.17 to 0.75 for $K = 3$–7); the rich window narrows; stalls and dead routes grow with K; with the initial function kept and spare channels σ_c does not tend to 1 (lower bounds up to $K = 15$) empirical; the centred limit is open conjecture
bond dimension χ (4.3, 4.7, 4.8, 3.2, 5.3)	Plateau levels and route degrees do not change with extra channels, as zero-padded critical points stay critical (Lemma A.2) proven; at $\chi = R$ the last component needs a route through all cores, which can die empirical; larger uncapped χ makes the order slowly greedier empirical	First-escape prefactor falls slowly with χ (log-log slopes -0.26 to -0.76) empirical; route exponents unchanged at $\chi = 4$; spare channels remove stalls only on routes with an uncapped end empirical	Both boundaries $\propto \chi^{-1/2}$ on binary4 empirical; lazy only with full bonds derived empirical; large χ: standard and NTK parameterizations lazy, the μP-type one not derived empirical
local dimension d (4.3, 4.7, 4.8, 3.3)	On an MPS it enters the order only through the bond caps (3) derived; at equal bonds no effect on the order ($\lvert\Delta P\rvert \leq 0.021$) empirical; a larger χ gives no spare channel to route ends on bonds capped at d (MPS chain ends, HT leaves), where the remaining stalls sit empirical	Binary4 with full bonds ($\chi = d^2$): median first-escape prefactor $\approx \pi/2$ for $d = 2, 3, 4$, independent of d derived empirical, while the norm-based time falls by a factor 1.9; MPS4: 1.39–1.50 empirical	Sets the cut dimensions, hence where a lazy regime exists derived; whether $\sigma_c(K)$ tends to $\sqrt{d}$ is open conjecture

continued on the next page

Table 9: How each parameter acts (continued).

parameter (sections)	saddle sequence	plateau durations	regime boundaries
teacher weights and structure (4.1, 4.4, 4.7, 6.2–6.6)	Weights decide the order only through the race (prefactor over weight) or, for two-core pairs, as $\sigma \to 0$ derived empirical; the sites where components differ decide the routes proven; a star (smaller components sharing a core) whose piggyback coupling exceeds s_j makes the learned saddle strict proven, and, started there, the flow forgets the larger learned component empirical (from random initialization it never sits there); non-orthogonal teachers give blended rank-one plateaus and ranks that grow bond by bond empirical	First plateau $\propto 1/s_1$ at fixed weight ratios proven; $k = 2$: rate $s_r s_j^{(K-2)/K}$ and head start $1 - s_r/s_j$; $k \geq 3$: coupling $s_r s_j^{(K-k)/K}$ derived; generic teachers: slopes set by the second singular value of the teacher restricted to the two cores of a bond derived empirical; close weights overlap unless σ is small empirical	Aligned initialization: the boundary depends on the weight ratios (Proposition A.48) proven; lazy boundary through $\|W^*\|_F/\alpha$; centred floor $\propto \sigma^{2K}/\|W^*\|_F^2$ derived
input covariance (6.1, 6.3, 6.3)	Aligned: race with weights $\Lambda_r s_r$, landscape with $\Lambda_r^{1/2} s_r$, which can reverse the order and, on binary4, swap a learned component derived empirical; otherwise plateaus at critical points of the Σ-weighted loss proven empirical	Aligned: exact clocks $\Lambda_r t$, exponents unchanged proven empirical; non-orthogonal in the Σ-metric: the MPS escape from the rank-one saddle becomes logarithmic, binary4 stays at σ^{-1} empirical	Not mapped
step size and optimizer (5.4, 5.5)	GD below η_c keeps gradient flow's sequence and order derived empirical; above it component 1 oscillates, then divergence or a return to the origin empirical; SGD near its mean-square limit erases learned components empirical; Adam keeps gradient flow's first component and mostly its order; ALS, DMRG and Riemannian GD have no plateaus derived empirical	GD: each event shifted by a computable $O(\eta)$ derived empirical; SGD: first-plateau law unchanged derived, later plateaus lengthened on average, or trapped empirical; Adam: first escape independent of σ until ϵ restores $\sigma^{-(K-2)}$ derived empirical; ALS, DMRG: iterates independent of σ proven	$\eta_c = 2/(K s_1^{2-2/K})$ at small σ, from the sharpness of the balanced minimum proven (no stable global minimum above it: derived for $\chi = R$); on the lazy side the limit set by the minimum gradient flow reaches falls like $\sigma^{-3.97}$ (binary4), catapults beyond it empirical; at fixed η/B SGD traps where gradient flow is only slow empirical (law: conjecture, Proposition A.45)
balanced or fan-in initialization (4.3, 4.4, 4.8, 3.4)	Balanced re-gauging keeps the first component in 88 of 94 seeds empirical; the escape direction is balanced either way proven; balanced initialization removes the dead routes in the runs checked, but not all stalls empirical	First-escape prefactor changes per seed by a factor 0.88–1.92, exponents unchanged empirical; the $k = 6, 7$ exponents are measured with balanced initialization only; some balanced routes escape 10^5 times later than usual empirical	Root at scale $\lambda\sigma$, fixed α: a light root is lazy at up to 100 times smaller α, with $\delta_\Theta \propto \lambda^2$ derived empirical; a heavy root acts as balance does empirical; the fan-in default is root-light

Landscape and clock. Two kinds of effect run through the table. Some parameters change the landscape: which saddles exist, at which levels, and of which type. These are the tree, the teacher and the covariance and, through the bond caps, χ and d. Others change only the clock, the time spent near each saddle: above all σ, which at leading order changes neither the direction of the

first escape nor the order of pairs that differ on three or more cores, and the step size below its stability limit. The balance of the initialization leaves the landscape and the exponents unchanged, shifts the prefactors and occasionally the first component, and decides whether a route survives its plateau. Optimizers can change both, and two properties of an update decide what survives (Table 6). The first is how its step scales with the gradient, which near the origin is $O(\sigma^{K-1})$ while the parameters must move by their own size: a step proportional to the gradient needs a time $\sigma^{-(K-2)}$, a step invariant under its rescaling (Adam) does not, and an absolute scale in the preconditioner, such as Adam's ϵ, brings the law back derived; Adam still follows the sign of the gradient, which one component dominates, so learning stays sequential (conjecture). The second is the metric. Under gradient flow W moves through $JJ^\top$ ($J = \partial W/\partial\theta$), which vanishes like $\sigma^{2(K-1)}$ at the origin, whereas Riemannian gradient descent (in continuous time the Gauss–Newton flow $\dot\theta = J^+(W^* - W)$) and the exact block minimizations of ALS and DMRG move W at a rate of order one. These methods are equivariant under the whole gauge group (14), including the rescaling of a core against its neighbour, and see W_0 only through its frames, so they keep neither plateaus derived nor σ proven; plateaus at singular points of a parametrization, and the better behaviour of the natural gradient there, are known for multilayer perceptrons known (Amari et al., 2006). Equivariance alone is not enough: Gram preconditioning is gauge-equivariant up to damping derived but moves all cores at once from their current values. Balance and the flattest minimum (Proposition A.43) rest on the conserved Q_e, the charges of the gauge symmetry under Euclidean gradient flow (Proposition 2.5); their measured relative drift is 10^{-11}, 10^{-5} and $3 \cdot 10^{-4}$ under gradient flow, GD and heavy ball (momentum flows conserve fewer quantities, with time- and velocity-dependent laws, Marcotte et al., 2024), while Adam, equivariant only under coordinate permutations and sign flips, breaks them at order one empirical. Where an entry of the table rests on one tree, one teacher or a single σ, the sections cited give its scope.

7.2 Open questions

The questions are ranked by their importance to the main question, the saddle-to-saddle dynamics of TTNs, and then by how directly they can be settled. For each we say what is known and, in one sentence, what would settle it.

1. **The whole saddle sequence.** Is the sequence of saddles fixed by an algorithm on the teacher and the tree, as conjectured for deep linear networks (Jacot et al., 2022) and proven for diagonal networks (Pesme and Flammarion, 2023)? The route rule (Remark A.27) is derived at leading order for each step and predicts the plateau levels measured with three and four components and, where they are long enough to measure, their laws, with one exception (Section 4.4); as an algorithm it is a conjecture: the saddle-order theorem covers one learned component only; with several, the fixed-rank block (couplings through learned superpositions) is not worked out; ties between routes are decided by random route amplitudes; and cheaper routes can insert intermediate plateaus by learning part of a component. What exists is bookkeeping: an implementation of the known rules that hops from saddle to saddle (Section 4.4) agrees with the measured sequences, but its prediction differs from the weight order without rotations in only 107 runs, and on the informative runs it scores 97.0% against that baseline's 95.8% empirical. What is missing is a predictive version. *What would settle it:* a rule that runs the per-initialization race (Remark A.7) at every saddle and includes the lower orders of Theorem A.31, compared run by run with gradient flow, together with a version of Theorem A.30 at saddles of higher bond rank (its proof needs the learned part to be a product across a core).

2. **Saddle orders below the route's, and long routes.** Where the order is $2\ell < k^*$ (Theo-

rem A.31), the escape should take a time $\propto \sigma^{-(2\ell-2)}$ instead of $\sigma^{-(k^*-2)}$ derived, and the learned component is then forgotten and relearned empirical; on one teacher only, the escape law was checked from the saddle and from small initialization, the forgetting from small initialization. Also open: the borderline case in which the largest eigenvalue of the generalized piggyback operator equals s_j ($k^* \geq 4$); Theorem A.31 without its index condition, or for signed and non-pattern residuals; the length of the plateau after the forgetting; and an exponent near 4 in half of the control seeds of the same check, where the route predicts 3. Long routes are only partly measured: $k = 6, 7$ over one decade, with balanced initialization only, on binary8 and HT4; MPS8 with $k = 6$ gives 4.51 ± 0.19 against 4; $k = 8$ was not measured; and at $k = 4$ a second negative Hessian eigenvalue of the route's order appears on the plateau, so for $k \geq 5$ such directions might change how $\lambda_{\min}$ scales with σ (Proposition A.26). *What would settle it:* pre-registered per-seed fits of the escape time from the saddle plus a perturbation of size ε (predicted $\propto \varepsilon^{-(\text{order}-2)}$) on every case with order below k^* and on controls, and balanced runs over more than a decade for $k \geq 6$, with plateau Hessians.

3. **Prefactors from random initialization.** The exponents are predicted, the prefactors are not, except that of the first escape, which is exact at leading order but has no closed form on trees with internal path amplitudes (binary8, HT, MPS with $N \geq 6$; the path-amplitude mode of Proposition A.13 is good to a factor 1.2–1.6 there). Later prefactors are set by the route amplitudes at the start of each plateau, which vary between seeds by factors above 20 at $k \geq 4$; for generic teachers, offsets of up to about 50 time units would decide the order of near-tied escapes at accessible σ conjecture (heuristic, Proposition A.38). *What would settle it:* a law for how route amplitudes are carried through each escape, tested against the per-seed prefactors already measured (Sections 4.3 and 4.4).

4. **Why routes die, and how often.** The reduced model of Proposition A.28 explains the decay of route ends at a rate $\propto \sigma^2$ and the role of balance, but its criterion, evaluated at the start of the plateau, is neither sufficient nor necessary (seven dead runs had a live end); balanced initialization removes dead routes yet delays some escapes by 10^5 or more; HT routes stall far less often than the heuristic predicts; spare channels raise the stalls on routes whose ends are capped, for no known reason; and whether the odeco runs at $K = 7, 8$ that end with live route ends face long plateaus or slowly dying routes is a conjecture (from fan-in initialization most stalls of the regime map are dead routes; Section 4.8). One route (MPS8, $D = \{1, 5\}$, from the saddle) is steeper than predicted (3.67 ± 0.20 against 3), with no mechanism. *What would settle it:* following the criterion of Proposition A.28 through the whole plateau, a reduced model with capped end bonds, and the dead fraction against σ at $k \geq 5$ for fan-in and balanced initialization (predicted to tend to the probability that all route ends are unfavourable, conjecture).

5. **The learning order in closed form.** The race of independent components is exact only on MPS; where cores without sites couple the components it still names the first component in 95–100% of the runs on binary8 and HT4 but in only 537 of 640 on binary4, and it has no closed form there. For pairs whose differing sites sit on two cores, the prefactor of P(smaller first) is not derived, the exponent $n(1-w)/w$ (weight ratio w) with $n = \chi_e$ for adjacent cores is a conjecture supported only at the smallest σ, and whether gradient flow has the leading-order exponent is open. That P(larger first) $\to \frac{1}{2}$ as $K \to \infty$, and that $P \to 1$ needs both d and χ to grow, are untested conjectures (Section 4.7, Proposition A.14). *What would settle it:* a leading-order emulation of the coupled race on binary and HT trees, and σ scans down to 10^{-6} with 1,000 seeds for adjacent pairs.

6. **Regime boundaries at large K and large χ.** With the initial function kept and spare channels, the saddle-to-saddle boundary does not tend to 1, the limit of aligned initialization: at $K = 8$–15 ($d = 2, 3$) all 97 per-seed lower bounds exceed both candidate laws, $\sigma_c^K = \text{const}$ and $\sigma_c^K/\sqrt{d^N} = \text{const}$, and 76 of 96 seeds stay saddle-to-saddle up to where the initial function overlaps the teacher, about $1.15\sqrt{d}$ empirical (this setup only). The proposed mechanism, the model unlearning its initial function on the plateau, is consistent with the data but not confirmed (conjecture). With the initial function removed and full bonds the two laws are not resolved, so whether $\sigma_c(K)$ tends to $\sqrt{d}$, a fixed amplitude per component, remains a conjecture (Section 3.3). Its $\chi^{-1/2}$ scaling is measured on one tree over less than a decade and is not derived, and that the rich window shrinks to a band of width $O(1/K)$ in $\ln\sigma$ is a conjecture (Sections 3.2 and 5.2). Under the μP-type parameterization the kernel keeps moving as χ grows on binary4, but the limit is not derived (a two-population mean field coupled through the root is a conjecture), the runs on binary8 are incomplete, and whether saddle-to-saddle dynamics survives $\chi \to \infty$ there is unknown (Section 5.3). *What would settle it:* the large-K limit of the centred, realizable problem, with budgets long enough at $K \geq 8$, a causal test of the unlearning mechanism, and the limiting μP dynamics on binary4 compared with the runs at $\chi = 16$–128.

7. **Adam and gradient noise on later plateaus.** Adam keeps gradient flow's first component, and mostly its order, and makes the first escape independent of σ, but no law is established for its later plateaus (per-seed local slopes from -0.4 to 5.5 on binary8, over half a decade), its ϵ law was tested only at $K = 8$, and that it stays sequential because one component dominates the sign of the gradient is a conjecture, as is the $(1+\beta)/(1-\beta)$ delay of heavy ball (Section 5.5). For SGD the law $T_{\text{SGD}}/T_{\text{GF}} = -\ln(1-x)/x$ is heuristic, holds to about 10% only for $T_{\text{SGD}}/T_{\text{GF}} \lesssim 1.5$, and its one coefficient is fitted (32.6, against 26 from a commuting-pair estimate); the rate at which noise erases learned components near the stability limit, and whether noise can revive a dead route, are open (Section 5.4). *What would settle it:* a sign-flow analysis of the route equations with and without ϵ, and the top Lyapunov exponent of the linearized route block under noise, tested on SGD runs started from stalled states.

8. **Covariance, teachers, real data and samples beyond the tested range.** Under a misaligned covariance the slope of the escape from random initialization is only bounded (Section 6.3); correlated inputs were run neither on HT trees nor on MPS with $N > 4$, where a theory check on MPS6 found a slow second stage that is not a single strict saddle; later saddles of generic teachers on binary and HT students were not timed (Section 6.2); and that the local maxima of g have bond rank one is a conjecture (Section A.5). On MNIST, the share of the data's route couplings on short routes correlates with the final loss across pixel orders in a preregistered test (Spearman correlation $r_S = -0.60$ over 27 orders), but not with the early speed; that it acts through which label-relevant interactions lie on routes of five rather than seven cores is a conjecture, and on the MPS tree shape and pixel order are confounded (Section 6.7). Small initialization recovers the teacher with its bond ranks from about 200–250 samples, 5–9 times the dimension of the teacher manifold, an unexplained threshold resting on one teacher per setting (Section 5.4). *What would settle it:* σ sweeps on HT4 and MPS6 under correlated inputs, timing every later route on binary16 in several pixel orders, and sample-size sweeps over teachers with manifolds of different dimension.

9. **Exact theory and scope.** The theorems of Kumar and Haupt (2025b) on the dynamics after an escape need non-degenerate critical points, which the gauge orbits exclude for $\chi \geq 2$; a quotient (Morse–Bott) version would make the escape path rigorous. Whether the per-bond conservation laws are all the conservation laws (the algorithm of Marcotte et al., 2023, run on

small trees) and the explicit metric $(JJ^\top)^{-1}$ of balanced TTNs are open. Beyond the model: input laws that are not products, classification and several outputs, pointwise nonlinearities inside the network, and what all of this implies for the choice of initialization and of tree in practice. *What would settle it:* the quotient version of those theorems, first for chains, and the route rule tested with a label leg on one core, as in MPS classifiers (Stoudenmire and Schwab, 2016).

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A Derivations

This appendix derives the theoretical statements of the main text not proved in Section 2.

Conventions. Notation is that of Section 2. Unless stated otherwise inputs are whitened ($\Sigma = I$), gradient flow has unit multipliers ($\eta_v = 1$), so that $L = \frac{1}{2}\|W - W^*\|_F^2$, and the teacher is a CP teacher (4). For two components j, r of a pattern teacher, $D_{jr} = \{i : \pi_j(i) \neq \pi_r(i)\}$ is the set of sites where they differ, $N(D_{jr})$ the set of (*terminal*) cores carrying a site of D_{jr}, $m_{jr} = |N(D_{jr})|$, and $\mathrm{St}(N(D_{jr}))$ the smallest subtree of $\mathcal{T}$ containing $N(D_{jr})$ (its Steiner subtree; compare Proposition 2.1(b)). At a learned saddle the learned component is j, with per-core amplitude $q_j = s_j^{1/K}$ (Proposition 2.4), and r is a component still to be learned. A ratio of two teacher weights, smaller over larger, is written ϱ. $\sigma_k(M)$ is the k-th singular value of a matrix M; σ is the initialization scale. "LO" means leading order in the small quantities of each statement; $\hat{\theta} = \theta/\|\theta\|$. A statement that combines a proven identity with a derived consequence carries both tags. Arguments the theory notes call *heuristic* (assumptions not controlled) are tagged conjecture, or empirical with the words "heuristic mechanism" where experiments support them. "Checked" refers to the numerical checks of the theory notes (`notes/theory_*.md`), which give all values; only the key ones are quoted here.

A.1 Conservation laws, gauge and invariant subspaces

Besides Proposition 2.5, three facts are used below.

Lemma A.1 (Equal norm shift). proven known (deep linear networks: Arora et al., 2018; Du et al., 2018) *Let $L(\theta) = \mathcal{L}(W(\theta))$ for a differentiable $\mathcal{L}$. Under gradient flow (13) (no mask), $\eta_v^{-1}\frac{\mathrm{d}}{\mathrm{d}t}\|G_v\|_F^2 = -2\langle\nabla_W\mathcal{L}, W\rangle$ for every core v, the same for all cores. For the population loss (6) the right-hand side is $2(\langle W, W^*\rangle_\Sigma - \|W\|_\Sigma^2)$; for the leading-order flow $\dot\theta = \nabla g$ of Section A.2 it is $2g$.*

Proof. W is linear in G_v, so $\langle G_v, \nabla_{G_v}L\rangle = \langle\nabla_W\mathcal{L}, W\rangle$, and $\frac{\mathrm{d}}{\mathrm{d}t}\|G_v\|_F^2 = -2\eta_v\langle G_v, \nabla_{G_v}L\rangle$. □

The lemma gives the common shift $y(t)$ in $\|G_v(t)\|_F^2 = \|G_v(0)\|_F^2 + y(t)$. Along the leading-order flow $\dot g = \|\nabla g\|^2 \geq 0$, so $y = 2\int_0^t g$ is convex: *all cores shrink, by the same amount, and only while* $g < 0$, i.e. only if $\langle W_0, W^*\rangle < 0$. The identity holds for any block of variables in which g is multilinear, e.g. the reduced cores of Proposition A.11 (checked).

Lemma A.2 (Unused bond channels are invariant). proven new (elementary) *Let $e = e_v$ with $p = p(v)$, and let $\mathcal{B}$ be a set of channels of e such that every entry of G_v and of G_p whose e-index lies in $\mathcal{B}$ vanishes. Then, for every loss of the form $\mathcal{L}(W(\theta))$ (any input law, population or empirical), the gradient vanishes on these entries, so gradient flow, gradient descent and SGD keep them at zero. In particular, a critical point of the problem with bond dimensions $r_e \leq \chi_e$, padded with zeros, is a critical point of the problem with bond dimensions χ_e.*

Proof. The derivative of W with respect to an entry of G_v with e-index $b \in \mathcal{B}$ contracts the other cores with G_p at e-index b, a vanishing slice; likewise for G_p. □

Lemma A.3 (Conservation laws under discrete steps). proven known (two-layer factorization: Du et al., 2018, App. B.1; the $O(\eta^2)$ change in deep linear networks: Arora et al., 2019a, App. D.2.3; scalar form: Hofmann, 2026, Prop. 14) *With the gradients $g_v = \nabla_{G_{v,(e)}}L$, $g_p = \nabla_{G_{p,(e)}}L$ matricized as in (15), one step $\theta \mapsto \theta - \eta\nabla L$ (unit multipliers) changes Q_e by exactly $Q_e^+ - Q_e = \eta^2(g_v g_v^\top - g_p g_p^\top)$; likewise for SGD. For one aligned component (Proposition 2.4), $Q = q_{v,1}^2 - q_{p,1}^2$ obeys $Q^+ = Q(1-\eta^2\Delta)$, $\Delta = (s_1 - \Pi_1)^2\Pi_1^2/(q_{v,1}^2 q_{p,1}^2)$: a step contracts the imbalance iff $0 < \eta^2\Delta < 2$.*

Proof. With $X = G_{v,(e)}$, $Y = G_{p,(e)}$, the proof of Proposition 2.5 gives $g_v X^\top = Y g_p^\top =: C$ for every loss that depends on θ through W; so the terms linear in η cancel in $(X - \eta g_v)(X - \eta g_v)^\top - (Y - \eta g_p)(Y - \eta g_p)^\top$. For the aligned component $g_v = -(s_1 - \Pi_1)\Pi_1/q_{v,1}$, and likewise for p. □

The change is negligible on plateaus and large during oscillations at large steps (Section A.7). At initialization with a common scale, $\mathbb{E}\,Q_e(0) = \sigma^2(1 - \chi_p/\chi_v)I$ (Section 2.6; χ_p is the parent's own bond dimension, $\chi_{\text{root}} = 1$): edges into the root are child-heavy ($\mathbb{E}\,Q_e \succ 0$), and HT leaf edges with $\chi_v = d = 2$ below $\chi_p = 3$ are parent-heavy ($\mathbb{E}\,Q_e = -\frac{1}{2}\sigma^2 I$).

A.2 Escape from the origin

Setting. Whitened inputs, a CP teacher, and gradient flow from the fan-in initialization $\theta_0 = \sigma Z$ (Equation (9); Z is the initialization at $\sigma = 1$); Section A.6 extends this to product covariances by $W^* \to \Sigma W^*$. Write

$$L(\theta) = \tfrac{1}{2}\|W^*\|_F^2 - g(\theta) + \tfrac{1}{2}\|W(\theta)\|_F^2, \qquad g(\theta) := \langle W(\theta), W^*\rangle. \tag{26}$$

g is homogeneous of degree K and linear in each core, while $\|W\|_F^2$ has degree $2K$. The *leading-order (LO) flow* $\dot\theta = \nabla g(\theta)$ drops a term of relative size $\|W\|_F/\|W^*\|_F = O(\sigma^K)$ on the first plateau. Let t^* be its blow-up time and $\nabla_{\mathbb{S}}g(\hat\theta) = \nabla g(\hat\theta) - Kg(\hat\theta)\hat\theta$ the gradient of g on the unit sphere.

Proposition A.4 (Spherical–radial decomposition). proven known (directional convergence: Kumar and Haupt, 2025a) *Along the LO flow, with the time change* $\mathrm{d}\tau = \|\theta\|^{K-2}\mathrm{d}t$,

$$\frac{\mathrm{d}\hat{\theta}}{\mathrm{d}\tau} = \nabla_{\mathbb{S}} g(\hat{\theta}), \qquad \frac{\mathrm{d}\ln\|\theta\|}{\mathrm{d}\tau} = K\, g(\hat{\theta}). \tag{27}$$

The direction follows the spherical gradient flow of g*, independent of* $\|\theta\|$ *and* σ*, and* $Q_e(\hat{\theta}(\tau)) = Q_e(\theta_0)/\|\theta(\tau)\|^2$.

Proof. By Euler's identity and the homogeneity of ∇g (degree $K-1$), $\frac{\mathrm{d}}{\mathrm{d}t}\|\theta\| = K\|\theta\|^{K-1} g(\hat{\theta})$ and $\frac{\mathrm{d}}{\mathrm{d}t}\hat{\theta} = \|\theta\|^{K-2}\nabla_{\mathbb{S}} g(\hat{\theta})$; Q_e is quadratic in θ. □

As $\sigma \to 0$ the escape needs $\|\theta\|$ to grow by a factor of order $1/\sigma$ (τ of order $\ln(1/\sigma)$), so the escape direction is $\lim_{\tau\to\infty}\hat{\theta}(\tau)$, a KKT point of g on the sphere (Kumar and Haupt, 2025a, Thm 1, which applies verbatim to TTNs with $K \geq 3$). The conservation laws bind only while $\|\theta\| \approx \|\theta_0\|$ ($\tau = O(1)$), the phase that decides the escape time and the KKT point reached; they do not change the balanced form of the limit (Proposition A.9).

Corollary A.5 (Blow-up time). proven known (the identity is the activation law of alternating gradient flows, Kunin et al., 2025, Eqs. 2–3, written in spherical time with homogeneity degree K and utility g; the exponent $K-2$ in analogues: Jacot et al., 2022; Gissin et al., 2020) new (its per-initialization form for TTNs) *With* $G(\tau) = \int_0^\tau g(\hat{\theta})\,\mathrm{d}\tau'$,

$$t^* = \|\theta_0\|^{-(K-2)} \int_0^\infty e^{-(K-2)KG(\tau)}\,\mathrm{d}\tau. \tag{28}$$

For $\theta_0 = \sigma Z$, $t^* = \sigma^{-(K-2)}\,\hat{t}_1(Z)/s_1$, *where* $\hat{t}_1(Z) := s_1\, t^*|_{\sigma=1}$ *depends on the initialization direction and on* W^*/s_1 *but not on* σ*; the exponent* $K-2$ *counts every core.*

Proof. By (27), $\|\theta(\tau)\| = \|\theta_0\| e^{KG(\tau)}$ and $\mathrm{d}t = \|\theta\|^{-(K-2)}\mathrm{d}\tau$. The LO field is homogeneous of degree $K-1$, so $\theta_\sigma(t) = \sigma\,\theta_1(\sigma^{K-2}t)$; replacing W^* by cW^* divides all times by c. □

There is no separate alignment time: the whole integral is alignment ($g(\hat{\theta})$ starts near 0).

Proposition A.6 (From the blow-up to the first event). derived new (for TTNs) *Assume that at the end of the LO phase the direction* $\hat{\theta}$ *has converged to the balanced KKT point of the first component learned, of weight* s*. Then, as* $\sigma \to 0$,

$$t_1 = t^* + \delta_K\, s^{-(2-2/K)} + o(1), \qquad \delta_K = -\frac{2^{(K-2)/K}}{K-2} + \int_0^{2^{-1/K}} \frac{b\,\mathrm{d}b}{1-b^K}, \tag{29}$$

with $\delta_3 = -0.850$, $\delta_4 = -0.266$, $\delta_5 = -0.047$, $\delta_6 = 0.072$, $\delta_7 = 0.148$ *and* $\delta_8 = 0.201$. *Hence* $t_1\sigma^{K-2}s_1/\hat{t}_1(Z) = 1 + O(\sigma^{K-2})$ *for every initialization.*

Derivation. Once $\|\theta\| = O(1)$ the remaining time does not depend on σ. Along a balanced KKT point the flow is the balanced mode $\dot{q} = q^{K-1}(s - q^K)$ of Proposition 2.4, whose half-time is $q_0^{-(K-2)}/((K-2)s) + \delta_K s^{-(2-2/K)} + O(q_0^2)$ (write $b = qs^{-1/K}$ and $1/(b^{K-1}(1-b^K)) = b^{-(K-1)} + b/(1-b^K)$); the first term is the LO blow-up time. □

Checked per initialization (within 0.8% of gradient flow for $K = 3, 4$, $\sigma \leq 0.03$); for $K = 6, 7$ at $\sigma = 0.15$–0.5 the direction has not converged (heuristically), so a 1% test needs $\sigma^{K-2} \leq 10^{-3}$.

Remark A.7 (Per-initialization prediction). proven at LO. Integrate in τ the state (u, ℓ, T) from $u(0) = \theta_0/\|\theta_0\|$, $\ell(0) = T(0) = 0$: $\mathrm{d}u/\mathrm{d}\tau = \nabla g(u) - Kg(u)u$ (renormalizing u), $\mathrm{d}\ell/\mathrm{d}\tau = Kg(u)$, $\mathrm{d}T/\mathrm{d}\tau = e^{-(K-2)\ell}$, up to $\ell = 40/(K-2) + 5$, and add the tail $e^{-(K-2)\ell}/((K-2)Kg(u))$. Then $t^* = \|\theta_0\|^{-(K-2)}T$ for the actual teacher (any number of components), and the dominant component of the final u is the first one learned as $\sigma \to 0$: a parameter-free prediction of t_1 (tested in Section 4.3).

Proposition A.8 (Maximum of g on the sphere). proven new (for TTNs; deep linear networks, where the injective norm is the spectral norm: Jacot et al., 2022, App. C.1.1) *Group the sites by the core that carries them (cores without sites form no group), and let* $\|W^*\|_{\mathrm{inj}} = \max\{\langle W^*, \bigotimes_v Y_v\rangle : \|Y_v\|_F = 1\}$ *be the injective norm of* W^* *seen as a tensor with one leg per group. Then*

$$g(\theta) \le \|W^*\|_{\mathrm{inj}} \prod_v \|G_v\|_F \le K^{-K/2}\|W^*\|_{\mathrm{inj}}\|\theta\|^K, \tag{30}$$

with equality on the unit sphere at the balanced bond-rank-one configurations $G_v = K^{-1/2} Y_v \otimes \varepsilon_1 \otimes \cdots \otimes \varepsilon_1$ *built from a maximizing* (Y_v). *For odeco teachers* $\|W^*\|_{\mathrm{inj}} = s_1$ *for every grouping; in general the grouping matters (for* $W^* = (I/\sqrt{d}) \otimes (I/\sqrt{d})$ *on the site pairs* $(0,1)$, $(2,3)$ *the unit-sphere maximum is* $3^{-3/2}$ *on binary* $N = 4$ *and* $4^{-2}/d$ *on the MPS).*

Proof. First inequality, by induction on the number of cores: for a leaf ℓ with parent p and the SVD $G_\ell = \sum_k \mu_k Y_k \otimes z_k$ across (physical legs | bond leg), $g = \sum_k \mu_k \langle W'_k, W^*[Y_k]\rangle$, with W'_k the network without ℓ and G_p contracted with z_k on the ℓ-leg, and $W^*[Y_k]$ the contraction of W^* with Y_k on the sites of ℓ. Since $\|W^*[Y_k]\|_{\mathrm{inj}} \le \|W^*\|_{\mathrm{inj}}$, induction bounds the sum by $\|W^*\|_{\mathrm{inj}} \prod_{v\neq\ell,p}\|G_v\|_F \sum_k \mu_k \|G_p \times_\ell z_k\|_F$, and Cauchy–Schwarz with $\sum_k \|G_p \times_\ell z_k\|_F^2 \le \|G_p\|_F^2$ closes the step. The second inequality is AM–GM. □

Proposition A.9 (KKT points are balanced). proven new (TTN form; linear chains: Kumar and Haupt, 2025a, Lemma 5; it also follows from the proof of Furman et al., 2026, Prop. F.10) *Every KKT point* θ^* *of* g *on the unit sphere with* $g^* := g(\theta^*) > 0$ *has* $Q_e = 0$ *on every bond and* $\|G_v\|_F^2 = 1/K$ *for every core.*

Proof. At a KKT point $\nabla g(\theta^*) = \lambda\theta^*$, and Euler's identity gives $\lambda = Kg^* > 0$. g is invariant under the gauge transformations (14), so $\langle \nabla g, \delta\theta\rangle = 0$ for every gauge direction $\delta\theta = (MG_{v,(e)}, -M^\top G_{p,(e)})$, i.e. $\lambda \operatorname{tr}\left((G_{v,(e)}G_{v,(e)}^\top - G_{p,(e)}G_{p,(e)}^\top)M\right) = 0$ for all M; hence $Q_e = 0$. Since g is linear in each core, $\langle G_v, \nabla_{G_v} g\rangle = g^*$, i.e. $\lambda\|G_v\|_F^2 = g^*$. □

Hence (checked to 10^{-15}) the near-origin escape direction is balanced, so of minimal norm in its fibre (Furman et al., 2026, Prop. F.10), and the $O(\sigma^2)$ initial imbalance does not affect it at LO.

Proposition A.10 (Second-order structure at a component). derived new (for general trees; for binary $N = 4$ the unstable direction toward the top component is Jacot et al., 2022, Lemma 15; strict and non-strict critical points of the loss: Achour et al., 2024) *For a pattern teacher with distinct weights let* θ_j *be the balanced bond-rank-one configuration of component* j *on the unit sphere (per-core amplitude* $K^{-1/2}$, *the pattern of* j *on the physical legs, channel 1 on every bond), so that* $g(\theta_j) = s_j K^{-K/2}$. *(a) If some component* r *has* $m_{jr} = 1$, θ_j *is not a critical point of* g *on the sphere. (b) Otherwise, assume that no two components* r *with* $m_{jr} = 2$ *share a terminal core. Each such* r, *with terminal cores* v, w, *contributes* n_{jr} *eigenvalues* $K^{1-K/2}(s_r - s_j)$ *to the Riemannian Hessian of* g *at* θ_j, *where* $n_{jr} = \chi_e$ *if* v *and* w *share the bond* e *and* $n_{jr} = 1$ *otherwise; all other eigenvalues are negative, except the* $\sum_e(\chi_e - 1)$ *gauge zero modes. So* θ_j *is a strict local maximum modulo gauge iff no* $m_{jr} = 1$ *and every* r *with* $m_{jr} = 2$ *has* $s_r < s_j$; *otherwise it has* $\sum_{r:\, m_{jr}=2,\, s_r > s_j} n_{jr}$ *unstable directions. Ties are decided at higher order; partners sharing a terminal core: Proposition A.22.*

Derivation. The Riemannian Hessian is $\nabla^2 g - s_j K^{1-K/2} I$. As g is linear in each core, $\nabla^2 g$ has only cross terms between cores $v \neq w$, which contract W^* with the pattern of j outside $S_v \cup S_w$: only components with $D_{jr} \subseteq S_v \cup S_w$ contribute. If $m_{jr} = 1$, $\nabla g(\theta_j)$ has a component along the pattern of r on the terminal core. Component j's own cross terms vanish on the gauge orbit and give $-2s_j K^{1-K/2}$ off it; uncoupled directions give $-s_j K^{1-K/2}$. For $m_{jr} = 2$, rotating G_v, G_w toward the pattern of r (channel c on the bond joining them if adjacent, channel 1 elsewhere) gives the cross term $s_r K^{1-K/2}$ for every c if v, w are adjacent and for $c = 1$ only otherwise; each surviving channel gives the block $K^{1-K/2}\left(\begin{smallmatrix} -s_j & s_r \\ s_r & -s_j \end{smallmatrix}\right)$, with eigenvalues $K^{1-K/2}(\pm s_r - s_j)$. □

Checked on 19 pattern-teacher cases and 7 odeco settings. For odeco teachers m_{jr} is the number of cores carrying sites (N on MPS and HT trees, $3, 4$ on binary $N = 6, 8$, and 2 on binary $N = 4$, whose bottom cores are not adjacent, $n = 1$): every odeco component is a local maximum on MPS, HT and binary $N \geq 6$ trees (*tensor-like*: one basin per component), only the largest on binary $N = 4$ (*matrix-like*); pattern teachers whose components differ on exactly two cores are matrix-like on every tree. Spherical flows from 200 random initializations (odeco teachers, five trees) ended at component configurations empirical; other local maxima for non-odeco teachers: Section A.5.

Proposition A.11 (Only the teacher slices move)**.** proven new (for TTNs) *For an odeco teacher* $\nabla_{G_v} g = \sum_r s_r (\bigotimes_{i \in S_v} u_r^{(i)}) \otimes E_v^{(r)}$, $E_v^{(r)}$ *the contraction of all other cores with* $u_r^{(i)}$ *on their sites. Under the LO flow the part of* G_v *orthogonal on its physical legs to every* $\bigotimes_{i \in S_v} u_r^{(i)}$ *is therefore frozen; the moving variables are the reduced cores* $H_v^{(r)}$ *(G_v contracted with the* $u_r^{(i)}$ *on its physical legs; a core without sites is shared by all components), and* $g = \sum_r s_r \mathcal{W}_{\mathrm{red}}(H^{(r)})$, $\mathcal{W}_{\mathrm{red}}$ *the scalar contraction of the reduced tree. Under (9) the* $H_v^{(r)}$ *have i.i.d.* $\mathcal{N}(0, \sigma^2/F_v)$ *entries, independent across* r *(a fan-in scalar TTN with per-core scale* $\sigma d^{-|S_v|/2}$*); on an MPS the components have disjoint moving variables and evolve independently at LO.*

Proof. The physical legs of $\nabla_{G_v} g$ lie in the teacher span, so g depends on G_v only through the $H_v^{(r)}$, which are i.i.d. Gaussian as orthonormal contractions of i.i.d. Gaussians. □

The same holds for pattern teachers (one slice per distinct pattern on each core). Most of $\|\theta_0\|$ is frozen: $\mathbb{E}\|H_v^{(r)}\|_F^2 = \sigma^2 \chi_v d^{-|S_v|}$, against $\mathbb{E}\|G_v\|_F^2 = \sigma^2 \chi_v$.

Corollary A.12 (The Kumar–Haupt time is a lower bound)**.** proven *(bounds)* empirical *(sizes)* new (the lower bound and its sizes; t_{KH} itself is the escape time of Kumar and Haupt, 2025b, text before Thm 11) *Let the teacher be odeco and let* r_{m} *be the norm of the moving variables at initialization. Then, for every initialization,*

$$t^* \;\geq\; \frac{K^{K/2-1}}{(K-2)s_1}\, r_{\mathrm{m}}^{-(K-2)} \;\geq\; \frac{K^{K/2-1}}{(K-2)s_1}\, \|\theta_0\|^{-(K-2)} =: t_{\mathrm{KH}}, \tag{31}$$

with equality iff the initial direction already maximizes g; $t_{\mathrm{KH}} = 1/(K(K-2)g_*\|\theta_0\|^{K-2})$, $g_* = s_1 K^{-K/2}$, *is, up to* $O(1)$, *the escape time of Kumar and Haupt (2025b, text before Thm 11) from an initialization along a maximizer. Under fan-in initialization the median of* t^*/t_{KH} *ranges from 2.5–5.2 (binary,* $N = 4$, $d = 2$–4*) to 182–12378 (MPS,* $N = 8$*), growing with* d, χ *and* K.

Proof. By Proposition A.8, $g \leq g_*$ on the unit sphere, so $\ln\|\theta(\tau)\| \leq \ln\|\theta_0\| + K g_* \tau$ and (28) gives $t^* \geq t_{\mathrm{KH}}$. The moving variables follow the LO flow of the same g restricted to them, so the same argument with their norm $r_{\mathrm{m}} \leq \|\theta_0\|$ gives the first inequality. □

Proposition A.13 (Escape along an imbalanced product path). derived known (the trichotomy: Rawal and DeWeese, 2026, Thm 4.2; new: the TTN reduction and the constants) *If the escape follows a rank-one path whose per-core amplitudes $x_v > 0$ obey $\dot{x}_v = s\prod_{w\neq v} x_w$, then the $x_v^2 - x_w^2$ are conserved and, with $x_v^2 = c_v^2 + y$,*

$$\hat{t} = \frac{1}{2s}\int_0^\infty \prod_{v=1}^{K} (y + c_v^2)^{-1/2} \mathrm{d}y. \tag{32}$$

Let m internal cores start at amplitude γ and the other $K-m$ (the ends) at $c_e \gg \gamma$. As $\gamma \to 0$: for $m = 0$ and equal amplitudes c, $\hat{t} = c^{-(K-2)}/((K-2)s)$ (integrating the mode equation of Saxe et al., 2014, eq. 15); for $m = 1$ and ends at c, $\hat{t} \to \frac{1}{2}\mathrm{B}(\frac{1}{2}, \frac{K-2}{2})c^{-(K-2)}/s$ (B is Euler's beta function), a finite limit; for $m = 2$, $\hat{t} = \ln(1/\gamma)/(s\prod_e c_e) + O(1)$; for $m \geq 3$, $\hat{t} \simeq \gamma^{-(m-2)}/((m-2)s\prod_e c_e)$.

The integral follows from Lemma A.1 ($\dot{y} = 2s\prod_w (y + c_w^2)^{1/2}$). Binary $N = 4$ and MPS $N = 3$ have $m = 1$, MPS $N = 4$ $m = 2$ (only qualitatively: its internal amplitude is not small), binary $N = 8$ and HT $N = 4$ $m = 3$, the MPS $m = N - 2$. The escape time is set by the internal path amplitudes, while $\|\theta_0\|$ is dominated by frozen slices and by internal cores whose norm is spread over $\sim \chi^2$ directions: hence t_{KH} fails most at large K. That the end directions stay frozen is empirical (median cosine 0.92–1.00; a minority rotate strongly). Given it, on binary $N = 4$ (reduced network $g = s_1 a^\top C b$, $a, b \in \mathbb{R}^{\chi_b}$, $\chi_b = \min(\chi, d^2)$) the median is $t_1 \approx \pi d/(2s_1\sigma\sqrt{\chi_b})$ for $d \leq \chi_b \leq d^2$, with $\mathrm{sd}\ln\hat{t}_1 \approx 1/(2\sqrt{\chi_b})$ derived new (0.80–1.00 of measured medians empirical).

Learning order is decided in the spherical phase by the basin of the initial direction.

Proposition A.14 (Race on trees whose cores all carry sites). proven *(at LO)* new *Let the teacher be odeco and the tree an MPS. By Proposition A.11 the components evolve independently at LO, and component r blows up at $\sigma^{-(K-2)}\hat{t}_{1,r}/s_r$, where the $\hat{t}_{1,r}$ are i.i.d. copies of the prefactor of Corollary A.5 for a single component of unit weight. The first component learned is* $\arg\min_r \hat{t}_{1,r}/s_r$, *and for two components with $\varrho = s_2/s_1 < 1$*

$$P(\textit{larger first}) = P\Big(\frac{\hat{t}_{1,2}}{\hat{t}_{1,1}} > \varrho\Big) = \frac{1}{2} + \frac{1}{2}P\Big(\Big|\ln\frac{\hat{t}_{1,2}}{\hat{t}_{1,1}}\Big| < \ln\frac{1}{\varrho}\Big), \tag{33}$$

which does not depend on σ. In the spherical flow the basin of component r is exactly the set of initial directions with $\hat{t}_{1,r}/s_r < \hat{t}_{1,r'}/s_{r'}$ for all $r' \neq r$.

Proof. The components are independent at LO and the $\hat{t}_{1,r}$ are exchangeable. □

The race agreed with gradient flow on 100 of 100 initializations (MPS, $N = 4$). empirical: $P(\text{larger first}) \approx F_{\mathcal{N}}(\ln(1/\varrho)/S)$ within ± 0.05 in all tensor-like settings computed ($F_{\mathcal{N}}$ the standard normal distribution function, S the standard deviation of $\ln(\hat{t}_{1,2}/\hat{t}_{1,1})$), and $\mathrm{sd}\ln\hat{t}_1 \approx c\sqrt{K-2}$ on MPS ($c = 0.27$–0.49). Hence conjecture: $P(\text{larger first}) \to \frac{1}{2}$ as $K \to \infty$ at fixed ϱ, d, χ, and $P \to 1$ needs $d \to \infty$ as well as large χ (compare the largest-first (deflation) order of over-parameterized orthogonal tensor decomposition, proven for a modified gradient flow and observed for plain gradient flow, Ge et al., 2021). When some cores carry no site (binary $N = 8$, HT) the race predicts the first component on 94–100% of initializations empirical.

Proposition A.15 (Matrix-like pairs). *Let the teacher have two components, the smaller j and the larger r, with $m_{jr} = 2$, and let $n = n_{jr}$ and $\varrho = s_j/s_r$. (a)* derived known (binary $N = 4$, the limit $P \to 0$: Jacot et al., 2022, Thm 2; their probability-one proof for depth three assumes a full-rank gradient at

the origin) θ_j *is a saddle of the spherical flow with* n *unstable directions while* θ_r *is a local maximum (Proposition A.10), so* $P(\textit{smaller first}) \to 0$ *as* $\sigma \to 0$. *(b)* conjecture *(heuristic)* new *As* $\sigma \to 0$,

$$P(\textit{smaller first}) \propto \sigma^{n(1-\varrho)/\varrho}. \tag{34}$$

Heuristic argument for (b). A trajectory that escapes near θ_j passes close to it, where the deviations along the n unstable directions grow in τ at rate $K^{1-K/2}(s_r - s_j)$ while $\mathrm{d}\ln\|\theta\|/\mathrm{d}\tau = K^{1-K/2}s_j$, i.e. like $(\|\theta\|/\|\theta_0\|)^{(1-\varrho)/\varrho}$. As $\|\theta\|/\|\theta_0\| \sim 1/\sigma$ at the escape, all n initial deviations must be below $\sim \sigma^{(1-\varrho)/\varrho}$, which for a smooth density has probability $\propto \sigma^{n(1-\varrho)/\varrho}$. □

Checks of (b): in an LO emulation (MPS, $N = 4$, $n = 1, \dots, 4$) the slopes $\mathrm{d}\ln P/\mathrm{d}\ln\sigma$ are ordered as n and are 0.67–1.03 of $n(1-\varrho)/\varrho$, steepening toward it as $\sigma \to 0$. For gradient flow (binary $N = 4$, $\varrho = 0.7$) the estimated P(smaller first) = 0.155, 0.064, 0.024 at $\sigma = 0.1$, 0.01, 0.001 (emulation probability times gradient flow's agreement rate) has local slopes 0.38, 0.43 against 0.43 empirical; whether gradient flow has the LO exponent is open. A wrong order is corrected through a strict saddle (Remark A.24); tests in Section 4.7.

A.3 Saddles after the first escape

Setting and assumptions. After the first escape, gradient flow passes near a *learned saddle* θ_S representing the partial fit $W_S = \sum_{j\in S} s_j U_j$. We assume: (A1) whitened inputs; (A2) a pattern teacher with distinct patterns (orthonormal U_r); (A3) θ_S is a balanced ($Q_e = 0$) critical point with bond ranks $r_e = \mathrm{rank}(W_S)^{\langle e\rangle}$, padded with zeros to χ_e (Lemma A.2 lifts criticality from r_e to χ_e; with one learned component θ_S is critical iff no residual has $k = 1$, Proposition A.21(c); small initialization gives $Q_e = O(\sigma^2)$); (A4) $\chi_e \geq r_e + 1$ on every bond whose rank must grow; (A5) leading order in the entries that vanish at θ_S (of size σ_{eff} on the plateau). In a basis of each bond with the learned channels first, entries whose bond indices are all learned are *old*, the others *new*; $c_e = \chi_e - r_e$. Let $\mathcal{R} = W^* - W_S$ and $L_S = \frac{1}{2}\|\mathcal{R}\|_F^2$. With one learned component j, balance forces equal core norms: each core is $q_j = s_j^{1/K}$ times the pattern of j on its sites and channel 1 on its bonds proven.

Expansion around the saddle. For a set $\mathcal{J}$ of cores and perturbations $\delta = (\delta_v)_v$, let $W_{\mathcal{J}}[\delta]$ be the tensor with the cores in $\mathcal{J}$ replaced by the δ_v and the others at θ_S. Multilinearity gives exactly $W(\theta_S + \delta) = \sum_{\mathcal{J}} W_{\mathcal{J}}[\delta]$, $W_\emptyset = W_S$, and

$$L(\theta_S + \delta) = L_S - \sum_{\mathcal{J}\neq\emptyset} \langle \mathcal{R},\, W_{\mathcal{J}}[\delta]\rangle + \tfrac{1}{2}\Big\| \sum_{\mathcal{J}\neq\emptyset} W_{\mathcal{J}}[\delta]\Big\|_F^2. \tag{35}$$

Lemma A.16 (Routes). proven new *(i) If* δ_v *has only new entries,* $W_{\{v\}}[\delta_v] = 0$*: new entries carry no Gauss–Newton cost at second order. (ii) For new entries,* $W_{\mathcal{J}}[\delta] \neq 0$ *requires every new index of every* δ_v *to sit on a bond inside* $\mathcal{J}$ *and to be matched there; so every bond leaving* $\mathcal{J}$ *carries a learned index, and every connected component of* $\mathcal{J}$ *has at least two cores. For a connected set* $\mathcal{C}$ *(a* route*),* $W_{\mathcal{C}}[\delta] = \mathcal{W}_{\mathcal{C}}(\delta) \otimes \bigotimes_{f\in\partial\mathcal{C}} T_f$*, where* $\mathcal{W}_{\mathcal{C}}(\delta)$ *is the small TTN formed by the* δ_v *(internal bond dimensions* c_e*),* $\partial\mathcal{C}$ *the set of bonds leaving* $\mathcal{C}$*, and* T_f *the contraction of the learned network beyond* f *(a map from the* r_f *learned indices of* f *to tensors on the sites* H_f *beyond* f*). Every proper subset* $\mathcal{C}' \subsetneq \mathcal{C}$ *has* $W_{\mathcal{C}'}[\delta] = 0$*:* a route pays no Gauss–Newton cost at the order of its gain.

Proof. A new index of δ_v meets, in the neighbouring core, a perturbation or the zero block of a learned core (Lemma A.2), and in a proper subset of a route some new index meets a zero block. □

Disconnected sets $\mathcal{J}$ also contribute, possibly below the route degree (MPS $N = 5$, $D = \{0, 4\}$: degree 4), but each connected component $\mathcal{J}_i$ pays the Gauss–Newton cost $\frac{1}{2}\|W_{\mathcal{J}_i}\|_F^2$ at the order of the gain; for one learned component and two blocks the quadratic form is positive semidefinite iff $s_j \geq s_r$, the condition of Proposition A.22 derived. In general (one learned component) whether disconnected sets descend before the route is decided by the generalized piggyback operator of Section A.4 proven.

Definition A.17 (Local residual)**.** For a route $\mathcal{C}$, $\mathcal{R}_\mathcal{C} := \langle \mathcal{R}, \bigotimes_{f \in \partial\mathcal{C}} T_f \rangle$ (contraction over all sites beyond $\partial\mathcal{C}$) is a tensor with one mode per core of $\mathcal{C}$, made of its physical legs and its learned boundary indices; then $\langle \mathcal{R}, W_\mathcal{C}[\delta] \rangle = \langle \mathcal{R}_\mathcal{C}, \mathcal{W}_\mathcal{C}(\delta) \rangle$.

Proposition A.18 (Escape degree)**.** derived known (mechanism: Kumar and Haupt, 2026, Thm 8 (a lower bound on the plateau, for active weights that start near a local minimum of the restricted loss with a Łojasiewicz inequality), and the accumulated utility of Kunin et al., 2025; new: the degree is set by the tree) *The new entries first lower L at order*

$$k^* = \min\{|\mathcal{C}| : \ \mathcal{C} \text{ a route}, \ \mathcal{R}_\mathcal{C} \neq 0\}. \tag{36}$$

At LO (old entries frozen at θ_S) the new entries of a minimizing route follow the gradient flow of a k^*-core TTN that fits the target $\mathcal{R}_\mathcal{C}$ from small initialization: the escape from the saddle is the first-escape problem of Section A.2 for the route, with $K \to k^*$ and $W^* \to \mathcal{R}_\mathcal{C}$. This needs the old entries to sit at a local minimum of the fixed-rank problem; it fails at the strict saddles of Proposition A.22, where the old entries move first. It also needs the generalized piggyback operator below the learned weight (Theorem A.30); otherwise, from $k^* = 5$ on, a smaller cluster of cores plus old rotations can escape first (Theorem A.31).

Proposition A.19 (Steiner identity)**.** proven *(identity and upper bound on the order; order $= k^*$ under the condition of Theorem A.30)* new (the route is the Steiner subtree of the residual's sites, which can be a proper subtree of a connected component of rank-deficient bonds; descent along a whole such component (model rank below target rank): Furman et al., 2026, Lemma D.8; with several such components the smallest gives the least suppressed escape: their Fig. 4) *Let $S = \{j\}$. For every route $\mathcal{C}$,*

$$\mathcal{R}_\mathcal{C} = q_j^{K-|\mathcal{C}|} \sum_{r:\, N(D_{jr}) \subseteq \mathcal{C}} s_r \, U_r|_\mathcal{C}, \tag{37}$$

where $U_r|_\mathcal{C}$ is U_r restricted to the sites of $\mathcal{C}$. Hence $k^* = \min_r |\mathrm{St}(N(D_{jr}))|$; the route of component r has degree $k = |\mathrm{St}(N(D_{jr}))|$ and coupling

$$c_r = s_r \, q_j^{K-k} = s_r \, s_j^{(K-k)/K}. \tag{38}$$

The bonds whose rank must grow to add r are exactly the bonds of $\mathrm{St}(N(D_{jr}))$, so k is one plus their number, and $k = 1$ iff D_{jr} lies on a single core. With route entries proportional to t, $L - L_S = -c_r t^k + O(t^{2k})$: the saddle has order at most k^*. No curve descends faster if the generalized piggyback operator stays below s_j (Theorem A.30; always for $k^* = 2$, for $k^* \in \{3, 4\}$ with a positive definite fixed-rank block, and for two-component teachers in the correct order); otherwise the order can be smaller, from $k^* = 5$ on (Theorem A.31 and Corollary A.32).

Proof. Beyond a boundary bond f the learned network is $T_f = q_j^{n_f} U_j|_{H_f}$ (n_f cores, sites H_f, $\sum_f n_f = K - |\mathcal{C}|$). As $\mathcal{R} = \sum_{r \neq j} s_r U_r$ and U_r contracted with T_f gives $q_j^{n_f} \prod_{i \in H_f} \langle u_r^{(i)}, u_j^{(i)} \rangle$ ($q_j^{n_f}$ if $\pi_r = \pi_j$ on H_f, else 0), U_r survives iff $N(D_{jr}) \subseteq \mathcal{C}$, and a connected $\mathcal{C}$ containing $N(D_{jr})$ contains its Steiner subtree. The rank statement is Proposition 2.1(b). For the descent curve let δ_v be t times the pattern of r on S_v, with a new channel on the bonds inside $\mathcal{C}$ and channel 1 on $\partial\mathcal{C}$: then $W_\mathcal{C}[\delta] = t^k q_j^{K-k} U_r$, every other $W_\mathcal{J}[\delta]$ vanishes (Lemma A.16), and $L - L_S = -c_r t^k + \frac{1}{2} q_j^{2(K-k)} t^{2k}$. □

Table 10: Route degree $k = |\mathrm{St}(N(D))|$ (Proposition A.19) for a two-component pattern teacher whose components differ on the sites D (numbered from 0), on the trees of Table 2; $m = |N(D)| = 2$ unless shown. With the larger component learned first, $k = 1$: no plateau; $k = 2$: strict saddle, plateau $\propto \ln(1/\sigma)$; $k \geq 3$: higher-order saddle, plateau $\propto \sigma^{-(k-2)}$.

D ($N = 4$)	binary $K = 3$	MPS $K = 4$	HT $K = 7$	D ($N = 8$)	binary $K = 7$	MPS $K = 8$	HT $K = 15$
$\{0,1\}$	1 ($m = 1$)	2	3	$\{0,1\}$	1 ($m = 1$)	2	3
$\{1,2\}$	3	2	5	$\{1,2\}$	3	2	5
$\{0,2\}$	3	3	5	$\{3,4\}$	5	2	7
$\{0,3\}$	3	4	5	$\{0,4\}$	5	5	7
$\{0,1,2,3\}$	3	4 ($m = 4$)	7 ($m = 4$)	$\{2,5\}$	5	4	7
				$\{1,6\}$	5	6	7
				$\{0,7\}$	5	8	7
				$\{0,\ldots,3\}$	3	4 ($m = 4$)	7 ($m = 4$)
				all	7 ($m = 4$)	8 ($m = 8$)	15 ($m = 8$)

Table 10 lists k for two-component teachers. For Tucker decompositions, Frandsen and Ge (2022, proof of Lemma 7) exhibit, at balanced points, descent of order 2, 3 and 4 when 2, 3 and 4 cores must move (no lower bounds).

Lemma A.20 (Several learned components). proven new *Let E_+ be the set of bonds e with* $\mathrm{rank}(W_S + U_r)^{\langle e \rangle} > \mathrm{rank}(W_S)^{\langle e \rangle}$. *(i) E_+ is connected. (ii) If $E_+ = \emptyset$, one core can produce U_r: $\nabla L(\theta_S) \neq 0$.*

Proof. (i) If the restriction of U_r to one side of a bond lies outside the learned column space, so does its restriction to any larger side. (ii) A walk from any core toward a branch on which U_r is new cannot return, so it ends at a core all of whose branches are old, which alone can produce U_r. □

With several learned components, adding r exactly needs $k_{\mathrm{ex}} = |E_+| + 1$ cores, and $k^* \leq k_{\mathrm{ex}}$ derived, strictly when the learned space of a bond holds a superposition of learned patterns that overlaps $U_r|_{H_f}$ without containing it: a cheaper route then first learns the projection, adding a plateau at the best fit of an intermediate rank profile (*partial learning*, Remark A.27). When $k^* = k_{\mathrm{ex}}$ the coupling is $c_r = s_r \prod_{f \in \partial \mathcal{C}} \| T_f^\top U_r|_{H_f} \|$ in the balanced gauge derived.

Proposition A.21 (Hessian at a balanced learned saddle). proven *(at θ_S)* new (the block structure, the rate and its multiplicity; that a two-core descent curve gives a strict saddle follows from Furman et al., 2026, Lemmas D.8–D.9, though they do not state it; deep linear networks: Achour et al., 2024) *(a) The Hessian at θ_S is block diagonal between old and new entries; the old (*fixed-rank*) block is the Hessian of the problem with bond ranks fixed at r_e (Gauss–Newton term plus the piggyback couplings of Proposition A.22). (b) The new (*rank-increasing*) block vanishes except on two-core routes: a route $\{u, v\}$ through the bond e contributes $\left(\begin{smallmatrix} 0 & -B \\ -B^\top & 0 \end{smallmatrix}\right)$, $B = \mathcal{R}_{\{u,v\}}$, for each of the c_e new channels, i.e. eigenvalues $\pm\sigma_k(B)$ with multiplicity c_e; with one learned component the negative one is $-\lambda$,*

$$\lambda = \sigma_{\max}(\mathcal{R}_{\{u,v\}}) = s_r \, s_j^{(K-2)/K}. \tag{39}$$

(c) Hence θ_S is a strict saddle iff the fixed-rank block has a negative eigenvalue or a route with $k^* = 2$ exists. If $k^* = 1$ it is not critical (one learned component: $\|\nabla L(\theta_S)\| = s_r q_j^{K-1}$); if $k^* \geq 3$ and the fixed-rank block is positive semidefinite, the Hessian has no negative eigenvalue: a higher-order saddle. (d) The gauge orbit through θ_S has dimension $\sum_e [\chi_e^2 - (\chi_e - r_e)^2]$, and the Hessian kernel has dimension $\sum_e r_e^2 + \dim \theta_{\mathrm{new}} - 2 \sum_{k^*=2 \text{ routes}} c_e$ ($\dim \theta_{\mathrm{new}}$: the number of new entries).

Proof sketch. An old–new cross term of $\langle \mathcal{R}, W_{\mathcal{J}} \rangle$ needs an old index to meet a new one, which gives zero; the Gauss–Newton cross term vanishes by Lemma A.16(i). On the new entries the only second-order terms are $-\langle \mathcal{R}, W_{\{u,v\}}[\delta] \rangle$ for adjacent pairs, $\mathcal{R}_{\{u,v\}} = q_j^{K-2} \sum_r s_r U_r|_{\{u,v\}}$ for one learned component (37). Gauge generators on new–new blocks fix θ_S. □

Checked in 21 (tree, D) cases (λ to six digits); λ is also the escape rate of a $k = 2$ route.

Proposition A.22 (Piggyback block and piggyback graph)**.** proven new (the graph criterion on general trees; deep linear networks: Achour et al., 2024, Thm 7; odeco components are the isolated local maximizers of $W^*(u,u,u)$ on the unit ball in the symmetric third-order case: Anandkumar et al., 2014, Thm. 4.2) *Let $S = \{j\}$. A* piggyback direction *keeps the learned bond index and rotates the physical vectors of one terminal core u toward a site pattern w on S_u. (i) For one residual r with $m_{jr} = 2$ and terminal cores u, v the rotation block has diagonal $q_j^{2(K-1)}$, off-diagonal $-s_r q_j^{K-2}$ and eigenvalues $q_j^{K-2}(s_j \mp s_r)$; for $m_{jr} \geq 3$ it is $q_j^{2(K-1)} I \succ 0$. (ii) In general it is*

$$H_{\mathrm{rot}} = q_j^{K-2}(s_j I - \Gamma), \tag{40}$$

where Γ is the weighted adjacency matrix of the piggyback graph*: its nodes are the pairs (u, w) of a terminal core and a rotated site pattern, with one edge between $(u, \pi_r|_u)$ and $(v, \pi_r|_v)$, of weight s_r, for every residual component r with $N(D_{jr}) = \{u, v\}$. Hence $\lambda_{\min}(H_{\mathrm{rot}}) = q_j^{K-2}(s_j - \lambda_{\max}(\Gamma))$, and θ_j is a strict saddle iff $\lambda_{\max}(\Gamma) > s_j$ or a route with $k^* = 2$ exists.*

Proof. Rotating core u by ε_u toward w creates a mixed tensor (pattern w on S_u, j elsewhere) of norm $q_j^{K-1}\varepsilon_u$; those of different nodes are orthonormal, so the Gauss–Newton part is diagonal ($q_j^{2(K-1)} = s_j q_j^{K-2}$). A residual r with terminals $\{u, v\}$ couples two rotations through $-\langle \mathcal{R}, W_{\{u,v\}} \rangle = -s_r q_j^{K-2} \varepsilon_u \varepsilon_v$; residuals with $m_{jr} \geq 3$ enter at order m_{jr}; the radial part is $\succeq 0$. □

With one learned component j and one residual r the saddle is thus not critical for $k = 1$, strict for $k = 2$ ($\lambda_{\min} = -s_r q_j^{K-2}$), of higher order for $k \geq 3$ if $m \geq 3$ or $s_j > s_r$, and strict via the piggyback block ($\lambda_{\min} = -(s_r - s_j) q_j^{K-2}$) for $k \geq 3$, $m = 2$, $s_j < s_r$ proven. A *star* (residuals $r_1, \ldots, r_n$ sharing one node) has $\lambda_{\max}(\Gamma) = (\sum_i s_{r_i}^2)^{1/2}$ and a triangle of equal weights s has $2s$, so θ_j can be strict although every residual is smaller than s_j and every route has $k^* \geq 3$ (checked to six digits). Dynamics empirical: for $j = 0000$ (weight 2) and $r = 1010$, $r' = 1001$ (weight 1.6 each) on MPS $N = 4$ ($\lambda_{\max}(\Gamma) = 2.263$), gradient flow passes near j in 4 of 5 seeds and in all 5 forgets it, learns r, r' and relearns j; with weights 1.2 ($\lambda_{\max}(\Gamma) = 1.70$) it plateaus at j.

Remark A.23 (Binary tree with $N = 4$). proven known (Saxe et al., 2014; Jacot et al., 2022; Achour et al., 2024) With the bottom cores as matrices $\tilde{A}, \tilde{B} \in \mathbb{R}^{d^2 \times \chi}$ (sites $(0,1)$ and $(2,3)$ as rows), $W^{\langle e \rangle} = \tilde{A} G_{\mathrm{root}} \tilde{B}^\top$ on both bonds, and L is exactly the loss of a depth-3 linear network fitting the $d^2 \times d^2$ matrix $(W^*)^{\langle e \rangle}$. Its critical points are SVD truncations onto subsets of singular pairs (Achour et al., 2024, Prop. 1); from small initialization the flow leaves the origin along the top singular pair (Jacot et al., 2022, Thm 2), and that the later saddles are the successive truncations is their Conjecture 3. Critical points that select non-top singular values are strict, balanced ones that select the top values are not (Achour et al., 2024, Thm 7, for full-rank targets with distinct singular values; the computation above covers ours).

Remark A.24 (After a wrong first escape). derived new Two-component teachers, smaller component j learned first; ε and b are the piggyback and route amplitudes when the plateau starts. (1) $k \geq 3$, $m = 2$ (matrix-like pair), if the rotation beats the direct add route (always as $\sigma \to 0$): the learned

bond index rotates from j to r after $\Delta t \approx \ln(1/\varepsilon)/((s_r - s_j)s_j^{(K-2)/K})$, the loss plateaus at $\frac{1}{2}s_j^2$ with $a_j \approx 0$, and j is relearned through its route after a time $\propto \sigma_{\text{eff}}^{-(k-2)}$, σ_{eff} set by the largest new-index seed. (2) $k = 2$: r is added directly, at the larger rate $s_r s_j^{(K-2)/K}$, and j is kept. (3) $m \geq 3$ (tensor-like pair): no rotation; r is added through its route after $\Delta t \approx b^{-(k-2)}/((k-2)s_r s_j^{(K-k)/K})$, and a_j stays at 1. This needs $N(D_{jr})$ to have no split into two sets of degree below $k/2$; with one (from $k = 5$ on, e.g. HT with $N = 4$ and $D =$ all sites) the order is $2\ell^* < k$ (Corollary A.32), and at small enough σ a cluster escape is expected to make j forgotten and relearned (derived, Section A.4). So a correction through a strict saddle exists only for matrix-like pairs, where the wrong order becomes rare as $\sigma \to 0$ (Proposition A.15). Checked on binary and MPS $N = 4$, before the random streams of teacher and initialization were separated.

Escape times. Let the route of component r have $k = |\mathcal{C}|$ cores and amplitudes b_v (its new entries: the pattern of r on the physical legs, a new channel on internal bonds, channel 1 on $\partial\mathcal{C}$). With the old entries frozen, one learned component and $A := q_j^{K-k}$, (35) and Lemma A.16 give

$$\dot{b}_v = A \prod_{u \in \mathcal{C}\setminus v} b_u \Big(s_r - A \prod_{u \in \mathcal{C}} b_u\Big), \qquad a_r = \frac{A}{s_r} \prod_{u \in \mathcal{C}} b_u, \tag{41}$$

a depth-k mode with target s_r/A that conserves the differences $b_u^2 - b_v^2$.

Proposition A.25 (Escape times with head start)**.** derived known (dormant amplitudes carried across phases: Kunin et al., 2025, Eqs. 2–3 and Sec. 2.2, and, in diagonal linear networks, Pesme and Flammarion, 2023, Algorithm 1; the $k = 2$ head start (weaker directions are $O(\sigma^{1-s_r/s_j})$ once the stronger one is learned, two-layer linear networks): Zhang et al., 2026, Thm 4; the trichotomy in k: Rawal and DeWeese, 2026, Thm 4.2; $k = K$: the decoupled modes of Saxe et al., 2014) new (only the closed forms for TTN routes, in particular the effective amplitude for $3 \leq k < K$) *Let the initialization be aligned: every core of the learned component j and of the route starts at amplitude σ, and the hanging cores are aligned with the shared pattern. Let $\Delta t_2 = t(a_r = \frac{1}{2}) - t(a_j = \frac{1}{2})$. As $\sigma \to 0$,*

$$k = 2: \quad \Delta t_2 = \frac{(1 - s_r/s_j)\ln(1/\sigma) + O(1)}{s_r\, s_j^{(K-2)/K}}, \tag{42}$$

$$k \geq 3: \quad \Delta t_2 = \frac{\sigma^{-(k-2)}\,(1 - s_r/s_j)}{(k-2)\, s_r\, s_j^{(K-k)/K}}(1 + o(1)), \tag{43}$$

and $\Delta t_2 = O(1)$ (no plateau) for $k = 1$. For $k \geq 3$ the route enters the plateau with the effective amplitude $\sigma_{\text{eff}} = \sigma(1 - s_r/s_j)^{-1/(k-2)}$; for $k = K$ it never sees j, and r escapes at $\sigma^{-(K-2)}/((K - 2)s_r)$.

Derivation. The route exists from the start, with the then-small learned cores in its hanging part, so it grows during the rise of j with the coupling $c(t) = s_r q_j(t)^{K-k}$, where $\dot{q}_j \simeq s_j q_j^{K-1}$ while q_j is small. For $k = 2$, $\ln b$ grows by $\int c\,\mathrm{d}t = \int s_r q_j^{-1}\mathrm{d}q_j/s_j = (s_r/s_j)\ln(1/\sigma) + O(1)$, so the route enters the plateau at $\ln b_{\text{eff}} = -(1 - s_r/s_j)\ln(1/\sigma) + O(1)$ and needs a further $\ln(1/b_{\text{eff}})/\lambda$, with λ from (39). For $k \geq 3$ the balanced route obeys $\frac{\mathrm{d}}{\mathrm{d}t}b^{-(k-2)} = -(k-2)c(t)$, so at the end of the rise of j $b_{\text{eff}}^{-(k-2)} = \sigma^{-(k-2)} - (k-2)\int c\,\mathrm{d}t = \sigma^{-(k-2)}(1 - s_r/s_j) + O(1)$, and the plateau lasts $b_{\text{eff}}^{-(k-2)}/((k-2)s_r s_j^{(K-k)/K})$; as $q_j^{K-k}\mathrm{d}t = q_j^{1-k}\mathrm{d}q_j/s_j$, this head start accrues while $q_j \approx \sigma$. □

The factor $1 - s_r/s_j$ is the head start (without it the $k = 2$ slope would be $1/\lambda$; Section 4.4). Checked to 10^{-4} with aligned initialization; with fan-in initialization the $k = 2$ slope is 1.066 against

1.065, per-seed exponents -0.97 to -1.02 ($k=3$) and -1.99 to -2.00 ($k=4$). An unbalanced route with positive product has $\Delta t = \int_0^\infty \mathrm{d}\beta/(2c\prod_{v\in\mathcal{C}}(b_v^2+\beta)^{1/2})$ derived (as in Proposition A.13); a negative product lengthens the plateau. With random initialization the prefactors are set by the route amplitudes at each plateau start (seed-to-seed factors above 20 for $k\geq 4$) empirical.

Proposition A.26 (Signatures of the route degree)**.** derived known (the exponent $2-2/k$ of a balanced product of k factors: Arora et al., 2019b; Razin et al., 2022; new: its use to read k off TTN plateaus) *(i) Per seed (fixed initialization direction, σ varied), $\Delta t_2 \propto \sigma^{-(k-2)}$ for $k\geq 3$. (ii) On a balanced route, (41) gives $\dot{a}_r = kA^{2/k}s_r^{2-2/k}a_r^{2-2/k}(1-a_r)$, so the local growth exponent is*

$$\gamma(a_r) = \frac{\mathrm{d}\ln \dot{a}_r}{\mathrm{d}\ln a_r} = 2 - \frac{2}{k} - \frac{a_r}{1-a_r}, \tag{44}$$

that is $1, \frac{4}{3}, \frac{3}{2}, \frac{8}{5}$ for $k=2,\dots,5$ as $a_r\to 0$. The route is balanced only once $b^2 \gg |b_u^2 - b_v^2| = O(\sigma^2)$, so k is read in the window $a_r \in [10^{-4}, 10^{-2}]$ (a smaller γ at smaller a_r indicates an unbalanced route). (iii) At a plateau point (route amplitudes small but nonzero) each active route contributes one negative Hessian direction modulo gauge, with

$$\lambda_{\min} \simeq -s_r s_j^{(K-2)/K} \quad (k=2), \qquad \lambda_{\min} \simeq -(k-1)\, s_r A\, b^{k-2} = O(\sigma_{\mathrm{eff}}^{k-2}) \quad (k\geq 3), \tag{45}$$

where $b = (s_r a_r/A)^{1/k}$ (for $k\geq 3$ on a balanced route). The count of one negative eigenvalue per route holds for $k\leq 3$ only: at $k=4$ a second negative eigenvalue of the same order appears empirical (heuristic mechanism: new-index pairs that connect to learned structure add indefinite terms of order σ^2, the same order as the route eigenvalue).

In (iii) the route block of the Hessian is $-s_r A b^{k-2}(\mathbf{1}\mathbf{1}^\top - I)$, whose $k-1$ positive directions are gauge. Checked on MPS $N=4$ (γ within 0.01; $|\lambda_{\min}|/\lambda_{\max} \propto \sigma^0, \sigma^1, \sigma^2$ for $k=2,3,4$; (45) within a factor 1.4 for $k=3$, 3.5 for $k=4$), so the saddle type is read from the σ-scaling of $\lambda_{\min}$ or from γ.

Remark A.27 (The whole sequence). derived at LO for each step; conjecture (heuristic) as an algorithm. new (1) The first escape ends near a critical point of the bond-rank-one problem; it is a single component j only if θ_j is a local maximum of the spherical problem (Proposition A.10, with the piggyback graph when several partners share a core); a star with $\lambda_{\max}(\Gamma) > s_j$ sends the flow past j toward a superposition. (2) At a later saddle whose fixed-rank block has a negative eigenvalue (wrong order, stars) the next event is a rotation of the learned structure or a $k=2$ route, whichever is more unstable, in time $\propto \ln(1/\sigma)$ (Proposition A.18 does not apply); otherwise it is the route of smallest k^* (logarithmic beats power law, smaller powers beat larger ones), ties being decided by the coupling and head start $c\, b_{\mathrm{eff}}^{k-2}$, dominated for $k\geq 3$ by the random route amplitudes, provided the generalized piggyback operator stays below the learned weight; if a level $\ell^* < k^*/2$ exceeds it, the order $2\ell^*$ competes with k^* (Corollary A.32; a leading-order race). (3) The escape raises the bond rank by one on the bonds inside the route, and the next saddle is the best approximation of W^* with the new rank profile (for pattern teachers without superpositions, the partial sum of the learned components), whose loss is the plateau level. Since k^* depends on the tree, *locality beats weight* (teacher B4, Section 4.4). The number of cores that must change at once plays the role of the leap of Abbe et al. (2023) (the number of new support coordinates that SGD must align with at once, which sets a runtime $\tilde{\Theta}(d^{\max(L,2)})$ in the input dimension d for a leap L in their conjecture; Furman et al., 2026, Remark D.10, draw the same analogy for the order of their descent curves); specific to the TTN is that it is set by the tree, and that the plateau scales with the initialization, $\sigma^{-(k-2)}$, not with the dimension.

Proposition A.28 (Conservation-law trap). derived *(mechanism, in a reduced model)* new (the route-end condition; in the landscape, a point that is not of minimum norm, with dead leaf columns and live core entries, can be a spurious local minimum: Furman et al., 2026, Thm G.1) *Let a route end at a leaf core, on the bond e (child v, parent p); let ν be the new channel used by the route, $Q_{\nu\nu} = \nu^\top Q_e \nu$, x the leaf's new column, and split the parent's slice along ν into its route entry y_{route} and the* visible *entries y_{vis}, which connect through the other side to learned structure. Conservation gives $\|x\|^2 - \|y_{\text{vis}}\|^2 - \|y_{\text{route}}\|^2 = Q_{\nu\nu}$. On the plateau the Gauss–Newton term makes x and y_{vis} decay together, at a rate of order $|Q_{\nu\nu}| = O(\sigma^2)$, so the leaf column decays to zero iff the end is* unfavourable, *$Q_{\nu\nu} + \|y_{\text{route}}\|^2 < 0$. The route's drive on x is proportional to the product of the other route amplitudes, so while another end is favourable it pins x at a positive value and the route escapes, slowly. In the reduced model a permanent stall ($x \to 0$ down to roundoff) therefore requires every end of the route to be unfavourable.*

The dead state is linearly stable if $\gamma_1\gamma_2 > \kappa^2$, with decay rates $\gamma_i \sim |Q_i| \sim \sigma^2$ and mutual drive $\kappa \sim \sigma^{k-2}$ (conjecture, heuristic): stable as $\sigma \to 0$ for $k \geq 5$, never for $k = 3$. On HT leaf bonds ($\chi_v = d = 2$ below a parent bond of dimension 3) fan-in initialization gives $Q_{\nu\nu} = \sigma^2(X_2/2 - X_6/4)$ (X_n independent chi-squared, n degrees of freedom; ν independent of the initialization), so $\mathbb{E}Q_{\nu\nu} = -\sigma^2/2$; then $P(Q_{\nu\nu} > 0) \approx 0.30$ and about half of the two-ended routes have both ends unfavourable (conjecture, heuristic). Data empirical (HT $N = 4$, teacher B4, $\sigma = 0.2$, nine fan-in initializations): all 6 runs with all ends unfavourable stalled with dead leaves, the 2 with one favourable end had the other end pinned; balanced initialization ($Q_e = 0$, same W_0) kept the routes alive (3 of 3, before the random streams were separated). conjecture: balanced initialization always avoids the trap (a property of unbalanced initialization, not of the landscape; in another model, four-layer matrix factorization of a target with equal singular values, exactly balanced initialization fails to converge with probability at least $\frac{1}{2}$, Luo et al., 2025, Thm 2), and for $k \geq 5$ the stalled fraction tends to P(all ends unfavourable) as $\sigma \to 0$. The locality experiments contradict parts of this picture (dead runs with a live route end, no stall in 144 fan-in HT4 runs with $k = 5$; Section 4.8).

A.4 Saddle order: lower bound and sharpness

Proposition A.19 gives a descent curve of order k^* at a learned saddle, an upper bound on the saddle order (as Lemma D.8 of Furman et al., 2026 does for a whole connected component of rank-deficient bonds). Here we ask when no curve descends faster. Lower bounds on saddle orders are known in special cases only: in deep linear networks the strict/non-strict classification (Achour et al., 2024, Thm 7) (order 2 versus more than 2), and for one parity saddle of a binary tree (Furman et al., 2026, Prop. 5.3).

Setting. One learned component j: θ_S represents $W_S = s_j U_j$ (balanced or not), $\mathcal{R} = W^* - W_S$ is any residual with $\langle U_j, \mathcal{R}\rangle = 0$ (a pattern teacher only where stated), inputs are whitened. The *saddle order* of θ_S is the largest p with $L(\theta) - L_S \geq -C\|\theta - \theta_S\|^p$ near θ_S; measuring the distance to the local gauge orbit instead gives the same p, because the gauge acts linearly and preserves L. Let P_v project onto the physical vector of core v in U_j (the pattern of j on S_v), $P_v^\perp = I - P_v$, and for $B \subseteq \mathcal{V}$ let $\Pi_B = \bigotimes_{v\in B} P_v^\perp \otimes \bigotimes_{v\notin B} P_v$. These 2^K *sector* projectors are orthogonal and sum to the identity, and the sector $\mathcal{H}_B = \text{range}(\Pi_B)$ satisfies $\mathcal{H}_{B_1} \otimes \mathcal{H}_{B_2} \cong \mathcal{H}_{B_1 \sqcup B_2}$ for disjoint sets. θ_S is critical iff $\Pi_{\{v\}}\mathcal{R} = 0$ for every core v. The *degree* of a set is $\deg B = |\text{St}(B)|$, the number of cores of its Steiner subtree, and $k^* = \min\{\deg B : \Pi_B\mathcal{R} \neq 0\}$ (equal to (36) for a pattern teacher, where $\Pi_B\mathcal{R} = \sum_{r:N(D_{jr})=B} s_r U_r$). Let $\mathcal{L}_\ell = \{B \neq \emptyset : \deg B \leq \ell\}$.

Definition A.29 (Generalized piggyback operator)**.** $\Gamma^{(\ell)}$ is the symmetric operator on $\bigoplus_{B\in\mathcal{L}_\ell}\mathcal{H}_B$ whose block (B_1,B_2) is the bilinear form $(x,y)\mapsto\langle\Pi_{B_1\sqcup B_2}\mathcal{R},\,x\otimes y\rangle$ if $B_1\cap B_2=\emptyset$ and zero otherwise; $\lambda_\ell=\lambda_{\max}(\Gamma^{(\ell)})$, nondecreasing in ℓ. $\Gamma^{(1)}$ is the piggyback matrix Γ of Proposition A.22. For a pattern teacher with all $s_r>0$, $\Gamma^{(\ell)}$ is, in the pattern basis, the weighted adjacency matrix of the *generalized piggyback graph*: its nodes are pairs (B,w) of a set $B\in\mathcal{L}_\ell$ and a pattern w on its sites that differs from j on every core of B, and every residual r adds an edge of weight s_r between $(B_1,\pi_r|_{B_1})$ and $(B_2,\pi_r|_{B_2})$ for every split $N(D_{jr})=B_1\sqcup B_2$ with $B_1,B_2\in\mathcal{L}_\ell$. A node with $|B|\geq 2$ is reached by new entries on $\mathrm{St}(B)$, a single core by an old rotation (Proposition A.22).

Theorem A.30 (Lower bound)**.** proven new (lower bounds were known for deep linear networks, Achour et al., 2024, Thm 7 (order 2 or more), and for one parity saddle, Furman et al., 2026, Prop. 5.3) *Let θ_S be critical. (a) If $\lambda_{\lceil k^*/2\rceil-1}<s_j$, there are $C,r_0>0$ with $L(\theta)-L_S\geq -C\|\theta-\theta_S\|^{k^*}$ for $\|\theta-\theta_S\|<r_0$; with the route of Proposition A.19 ($\chi_e\geq 2$ on the bonds of one minimal Steiner subtree) the saddle order is exactly k^*. (b) (Level version.) If $2\leq\ell\leq\lceil k^*/2\rceil$ and $\lambda_{\ell-1}<s_j$, then $L(\theta)-L_S\geq -C\|\theta-\theta_S\|^{\min(k^*,2\ell)}$ near θ_S.*

Theorem A.31 (Sharpness)**.** proven new *Let the teacher be a pattern teacher with all $s_r>0$, let $\ell<k^*/2$, and let X be a set of nodes of the generalized piggyback graph on $\mathcal{L}_\ell$ whose adjacency matrix has spectral radius $\rho_X>s_j$. Assume the index condition: on every bond e, χ_e-1 is at least the number of nodes $(B,w)\in X$ with $|B|\geq 2$ and e a bond of $\mathrm{St}(B)$. Then there is a curve with $\|\theta(t)-\theta_S\|\leq t$ and $L(\theta(t))-L_S\leq -ct^{2\ell}$: the saddle order is at most $2\ell<k^*$.*

Proof sketch of Theorem A.30. (Full proof: `notes/theory_saddle_order.md`, sections 3.1-3.5.) Let $\omega=\langle U_j,\,W\rangle$ ($=s_j$ at θ_S) and $\mu_B=\Pi_B W/\omega\in\mathcal{H}_B$. Sector orthogonality gives the exact identity

$$L-L_S=\tfrac{1}{2}(\omega-s_j)^2+\sum_{B\neq\emptyset}\Big[\tfrac{\omega^2}{2}\|\mu_B\|^2-\omega\langle\Pi_B\mathcal{R},\,\mu_B\rangle\Big]. \tag{46}$$

Families $(f_B)_{B\subseteq\mathcal{V}}$, $f_B\in\mathcal{H}_B$, form a commutative algebra under $(f*g)_B=\sum_{B_1\sqcup B_2=B}f_{B_1}\otimes g_{B_2}$; the cumulants $\kappa=\log\mu$ are given by the set-partition moment-cumulant formulas. *Locality of cumulants:* for $|B|\geq 2$, $\|\kappa_B\|\leq C_0\prod_{v\in\mathrm{St}(B)}\|\nu_v\|$, ν_v the new entries of core v. Indeed, if $\nu_v=0$ the contraction factorizes at v, μ is a $*$-product of factors supported on $\{v\}$ and on the branches at v, its logarithm is a sum of such factors, and none contains B; analyticity turns this vanishing into the product bound. Now fix ℓ, call a set *low* if its degree is at most $\ell-1$ and *high* otherwise, and let $\varepsilon=\|\theta-\theta_S\|$. High cumulants are $O(\varepsilon^\ell)$, low sets carry no residual ($\ell-1<k^*$), and the low part of the Gauss–Newton sum is at least $(1-C\varepsilon)\|\kappa_{\mathrm{low}}\|^2$. In the gain, the terms with two low cumulants sum exactly to $\frac{1}{2}\langle\kappa_{\mathrm{low}},\,\Gamma^{(\ell-1)}\kappa_{\mathrm{low}}\rangle$; every other term contains a single cumulant on a set of degree at least k^* ($O(\varepsilon^{k^*})$), or two high cumulants ($O(\varepsilon^{2\ell})$, after Young's inequality for one low and one high), or is ε times the quadratic form. So $L-L_S\geq\frac{\omega}{2}(s_j-\lambda_{\ell-1}-O(\varepsilon))\|\kappa_{\mathrm{low}}\|^2-O(\varepsilon^{\min(k^*,2\ell)})$, which gives (b); (a) is $\ell=\lceil k^*/2\rceil$. Balance, pattern structure, bond dimensions and the gauge are not used. □

Proof sketch of Theorem A.31. (Full proof: section 4 of the note.) Take a Perron vector $x\geq 0$, $\|x\|=1$, of the adjacency matrix on X. For $0<\eta\leq 1$, rotate each single core v of a node $(\{v\},w)$ by $\eta x_{v,w}$ toward w (old entries), and give each node $n=(B,w)$ with $|B|\geq 2$ a *polymer*: new entries of size $q_j(\eta x_n)^{1/|\mathrm{St}(B)|}$ on the cores of $\mathrm{St}(B)$, with the pattern w on B, the pattern of j elsewhere, and a spare index on the bonds of $\mathrm{St}(B)$ that no other node of X uses there (the index condition). A nonzero contraction term then uses whole polymers with disjoint Steiner subtrees, so $\omega=s_j$ exactly, the Gauss–Newton term at order η^2 is $\frac{s_j^2}{2}\|x\|^2$, and the gain at order η^2 comes

exactly from the edges of the graph, $\frac{s_j}{2}x^\top A_X x = \frac{s_j}{2}\rho_X$. Hence $L - L_S = -\frac{s_j}{2}(\rho_X - s_j)\eta^2 + O(\eta^3)$ while $\|\theta - \theta_S\| = O(\eta^{1/\ell})$. □

Corollary A.32 (When the saddle order is k^*)**.** proven new *With one learned component j and the route of Proposition A.19: (i) $k^* = 2$: order 2. (ii) $k^* \in \{3, 4\}$: then $\Gamma^{(\lceil k^*/2\rceil - 1)} = \Gamma$, so the order is k^* if $\lambda_{\max}(\Gamma) < s_j$, i.e. if the fixed-rank block is positive definite modulo gauge, and 2 (strict saddle) if $\lambda_{\max}(\Gamma) > s_j$; at equality it is 3 for $k^* = 3$ (positive semidefinite Hessian) and open for $k^* = 4$. (iii) Two-component teachers (one residual r): every node of the graph has at most one edge, so $\lambda_\ell \in \{0, s_r\}$. In the correct order ($s_r < s_j$) the order is k^* on every tree. In the wrong order it is k^* if no split $N(D_{jr}) = B_1 \sqcup B_2$ has both degrees below $k^*/2$, and otherwise exactly $2\ell^* < k^*$, ℓ^* the smallest $\max(\deg B_1, \deg B_2)$ over such splits (lower bound by Theorem A.30(b) at level ℓ^*, upper bound by Theorem A.31). (iv) In general, for a pattern teacher with the index condition, the order is $\min(k^*, 2\ell^*)$, ℓ^* the first level below $k^*/2$ with $\lambda_{\ell^*} > s_j$, provided $\lambda_{\ell^*-1} < s_j$.*

So the order can drop below k^* only for $k^* \geq 5$ (given a positive definite fixed-rank block; the boundary case $k^* = 4$, $\lambda_{\max}(\Gamma) = s_j$, is open), and only through a residual with at least three terminal cores, two of which lie in a set of degree below $k^*/2$. Two examples on the HT tree with $N = 4$ ($K = 7$) and wrong order: three-site D have $k^* = 6$ and no low split (two leaves already have degree $3 = k^*/2$), so the order is 6; $D = \{0, 1, 2, 3\}$ has the split into the two leaf pairs (degrees 3, 3), so the order is 6, not $k^* = 7$. The binary tree with $N = 8$ and $D =$ all sites is the same case.

Counterexample with a positive semidefinite Hessian. proven (order) empirical (numerics) MPS with $N = 6$, $d = \chi = 2$, $j = 000000$ ($s_j = 1$), residuals $r_1 = 110001$ and $r_2 = 110010$ with weight 0.8 each. The route of r_2 has $k^* = 5$ (that of r_1 has 6); the gradient vanishes and the Hessian is positive semidefinite (eigenvalues 0, 1, 6), since no residual has two terminal cores and $\Gamma^{(1)} = 0$. At level 2 the node $(\{0, 1\}, 11)$ is joined to $(\{5\}, 1)$ through r_1 and to $(\{4\}, 1)$ through r_2, a star with $\lambda_2 = 0.8\sqrt{2} = 1.131 > 1$: by Theorem A.30(b) and Theorem A.31 the order is exactly $4 < k^*$. The descent combines new entries of size t on the bond between cores 0 and 1, which point at the *non-residual* pattern 110000, with old rotations of cores 4 and 5 of size t^2. Measured minimum over the sphere of radius t: $-0.0350\,t^4$ (slope 4.000), against $-(2 \cdot 0.8^2 - 1)/8 = -0.0350$; with weights 0.6 ($\lambda_2 = 0.85$) the order is 5 ($-0.0107\,t^5$, the route of r_2). Single wrong-order residuals give counterexamples to "order $= k$" for two-component teachers: $r = 110001$ with weight 1.5 on the same MPS ($k^* = 6$, order 4) and $r = 1111$ with weight 1.5 on the HT tree with $N = 4$ ($k^* = 7$, order 6, $-t^6/432$). The orders predicted by Theorems A.30 and A.31 were measured on 32 instances ($k^* = 2$ to 7, MPS, binary and HT trees, pattern and random residuals), 12 of them with independent code, with slopes within 0.003 of the prediction (one slowly converging case). Dynamics derived: where Theorem A.31 applies at level ℓ, the old rotations follow the cluster adiabatically and the cluster obeys $\dot{b} = c\,b^{2\ell-1}$, so the escape takes $\propto \sigma^{-(2\ell-2)}$ instead of $\sigma^{-(k^*-2)}$ once it beats the route. On the counterexample, gradient flow from θ_S plus a perturbation of size σ escapes in $\propto \sigma^{-2}$ (local exponents 1.98 to 2.04 at the smallest σ, 5 seeds) along 110000, then forgets j and relearns it later; from small initialization only some seeds visit θ_S, and in those the σ^{-2} law sets in at a seed-dependent σ empirical.

A.5 Teachers that are not orthogonally decomposable

For a general teacher the saddles are no longer partial sums of components. $\mathcal{M}_{\mathbf{r}}$ is the manifold of tensors with bond ranks exactly $\mathbf{r} = (r_e)_e$ (Holtz et al., 2012b; Uschmajew and Vandereycken, 2013). For $W \in \mathcal{M}_{\mathbf{r}}$ and $e = e_v$, $p = p(v)$, U_e and V_e are the column and row spaces of $W^{\langle e \rangle}$, the environment spaces of e are $E_e^- = (\bigotimes_{c \in \mathrm{ch}(v)} U_c) \otimes \mathbb{R}^{d_v}$ and $E_e^+ = (\bigotimes_{c \in \mathrm{ch}(p) \setminus \{v\}} U_c) \otimes \mathbb{R}^{d_p} \otimes V_p$

($d_v = \prod_{i \in S_v} d_i$; no V_p at the root), and $\widetilde{W}_e^* := P_{E_e^-} (W^*)^{\langle e \rangle} P_{E_e^+}$ is the *environment-projected teacher*.

Proposition A.33 (Critical points of the rank-constrained problem). proven known (in substance, for the Tucker format: the first-order optimality conditions of best multilinear-rank approximation, Eldén and Savas, 2011 (HOOI, De Lathauwer et al., 2000, iterates their dominant-subspace version), and the fixed points of one-site ALS, Holtz et al., 2012a; tangent spaces: Uschmajew and Vandereycken, 2013) *$W \in \mathcal{M}_{\mathbf{r}}$ is a critical point of L restricted to $\mathcal{M}_{\mathbf{r}}$ iff, for every bond e,*

$$W^{\langle e \rangle} = P_{U_e} \widetilde{W}_e^* = \widetilde{W}_e^* P_{V_e}, \tag{47}$$

i.e. iff U_e is spanned by r_e left singular vectors of $\widetilde{W}_e^*$ (not necessarily the top ones) and $W^{\langle e \rangle}$ is the corresponding SVD truncation, i.e. iff W is a fixed point of one-site alternating least squares. Padded with zeros it is critical for every $\chi_e \geq r_e$ (Lemma A.2).

Proof sketch. At full bond ranks the differential with respect to G_v spans $\mathcal{V}_v = E_e^- \otimes V_e$ ($e = e_v$), so W is critical iff $P_{\mathcal{V}_v} W^* = W$ for every v, i.e. $\widetilde{W}_e^* P_{V_e} = W^{\langle e \rangle}$ at v and $P_{U_e} \widetilde{W}_e^* = W^{\langle e \rangle}$ at p. If $P_U A = A P_V = X$ has rank $\dim U$, $P_U A A^\top = X X^\top$ is symmetric: U is a left singular space of A. □

Proposition A.34 (When is the hierarchical SVD critical?). proven *(a, b)* derived *(c)* new (the ε^4 gap; the HSVD is Grasedyck, 2010) *Let W_H be the HSVD truncation of W^* at ranks $\mathbf{r}$. (a) W_H is critical iff its frames satisfy (47), which generic teachers violate (parts discarded at other bonds leak into each bond's environment). (b) If the best rank-r_e approximation of W^* across every bond e is one and the same tensor W (consistent truncations, e.g. an odeco teacher truncated to the same components on every bond, or binary $N = 4$), then $W_H = W$ is critical and the global minimizer over bond ranks at most $\mathbf{r}$. (c) If $W^* = W + \varepsilon E$ with W consistent and E generic, the nearby critical point W_c has $\|W_H - W_c\|_F = O(\varepsilon^2)$ and $L(W_H) - L(W_c) = \frac{1}{2}\|W_H - W_c\|_F^2 (1 + O(\varepsilon)) = O(\varepsilon^4)$.*

For (b), $\|W^* - X\|_F \geq \|W^* - W\|_F$ for every X with one bond rank at most r_e; for (c), both sets of frames are first-order rotations of those of W with the same driving terms, $W^* - W_c$ is normal to $\mathcal{M}_{\mathbf{r}}$ at W_c and $W_H - W_c$ tangent up to curvature (checked for $\varepsilon \leq 0.2$).

The rank-one plateau. derived new The origin plateau ends along a KKT direction of g, and the global maximum of g has bond rank one (Proposition A.8). That local maxima do too, and that the flow reaches one, are conjectures (true at every rank-one plateau computed); then the next plateau is a critical point of the bond-rank-one problem, not an HSVD truncation. If every core carries at most one site (MPS, HT) it is a local best rank-one approximation $\bar{s} \bigotimes_i w_i$, $\bar{s} = \langle W^*, \bigotimes_i w_i \rangle$, at level $\frac{1}{2}(\|W^*\|_F^2 - \bar{s}^2)$; on binary $N = 4$ it is the top singular pair of the matricization. In the cases computed it lies 6–12% below the HSVD truncation empirical, which is not general.

Proposition A.35 (Two non-orthogonal components). proven *(stationarity; threshold among symmetric critical points)* derived *(fold)* new (elementary) *Let $W^* = s_1 U_1 + s_2 U_2$ with N sites, $s_1 \geq s_2$ and overlaps $\langle u_1^{(i)}, u_2^{(i)} \rangle = \cos \phi_i \in [0, 1)$, at most one of them zero. Every rank-one critical point has w_i in the plane of $u_1^{(i)}, u_2^{(i)}$, at angle ψ_i from $u_1^{(i)}$, with value $s_1 \prod_i \cos \psi_i + s_2 \prod_i \cos(\phi_i - \psi_i)$. For equal weights and overlaps $\cos \phi$, the symmetric point $\psi_i = \phi/2$ has Hessian eigenvalues $-2c^{N-2}[c^2 - (N-1)(1-c^2)]$ along $(1, \ldots, 1)$ and $-2c^{N-2}$ on its complement ($c = \cos \frac{\phi}{2}$, unit weight), so there are two local maxima, one near each component, iff $\cos \phi < (N-2)/N$, and one merged maximum otherwise (asymmetric maxima excluded numerically). For $\varrho = s_2/s_1 < 1$ the maximum near U_2 disappears at a fold, $\cos \phi_{\text{fold}} \approx \frac{N-2}{N-1} (\frac{\varrho}{N-1})^{1/(N-2)}$ for small ϱ (1–13% below the exact values). The rank-one plateau is never one of the components.*

Proposition A.36 (Rates at the rank-one saddle of an MPS). derived *(rates)* proven *(bound)* new *Let the student be an MPS ($K = N$, core k carries site k) at the balanced, zero-padded rank-one critical point $W_1 = \bar{s} \bigotimes_i w_i$, a local maximum of the rank-one problem, and let $M_e = W^* \times_{i \notin \{k,k+1\}} w_i$ be the $d \times d$ matrix of the bond e between cores k and $k+1$. Then (w_k, w_{k+1}) is the top singular pair of M_e (singular value $\bar{s}$), the route block of e is $\bar{s}^{(N-2)/N}(M_e - \bar{s}\, w_k w_{k+1}^\top)$, with Hessian eigenvalues $-\bar{s}^{(N-2)/N}\sigma_m(M_e)$, $m = 2, \dots, d$ (each $\chi_e - 1$ times), and the escape rate of bond e is*

$$\gamma_e = \bar{s}^{(N-2)/N}\, \sigma_2(M_e), \qquad \text{with} \quad \sigma_m(M_e) \le \sigma_m((W^*)^{\langle e \rangle}) \tag{48}$$

by interlacing (M_e compresses $(W^*)^{\langle e \rangle}$ by isometries).

For two components with $d = 2$, $\sigma_2(M_e) = \det M_e / \bar{s}$ with $\det M_e = s_1 s_2 \sin\phi_k \sin\phi_{k+1} a_e b_e$, $a_e = \prod_{i \neq k,k+1} \cos\psi_i$, $b_e = \prod_{i \neq k,k+1} \cos(\phi_i - \psi_i)$: *the bond that grows first is the one whose two sites differ most while the other sites agree most*, not the one with the largest teacher Schmidt value (checked to four digits).

Corollary A.37 (Second escape). derived known (the degree-two case of the activation rule of alternating gradient flows, with carried-over amplitudes: Kunin et al., 2025, Eq. 3) new (the TTN rates: Proposition A.25 with $s_j \to \bar{s}$ and $s_r \to \sigma_2(M_e)$) *Assume $\sigma \to 0$, balanced cores and route projections $\propto \sigma$ at initialization, and let $\kappa_e = \sigma_2(M_e)/\bar{s}$. During the first escape the route of bond e grows by the factor $\sigma^{-\kappa_e}$, so the first bond to gain rank is $e^* = \arg\min_e h_e = \arg\max_e \sigma_2(M_e)$, and*

$$t_{e^*} - t_1 = h_{e^*} \ln(1/\sigma) + O(1), \qquad h_e = \frac{1 - \kappa_e}{\gamma_e}. \tag{49}$$

Here t_e is when σ_2 across e first exceeds $f \in [0.03, 0.1]$ times the teacher's (slopes checked to 1%).

Proposition A.38 (Later escapes on an MPS: an accumulated-utility race). derived *(as $\sigma \to 0$, under (A-carry))* known (the rule: alternating gradient flows, Kunin et al., 2025; new: the TTN routes and their rates) *Give every pending two-core route ρ the log-amplitude $\ell_\rho = \ln x_\rho / \ln(1/\sigma) \in [-1, 0]$ and use the clock $\vartheta = (t - t_1)/\ln(1/\sigma)$. Then $\ell_\rho(0) = -1 + \kappa_\rho$; while the flow sits at the saddle S_n, $\mathrm{d}\ell_\rho/\mathrm{d}\vartheta = \gamma_\rho(S_n)$, the rate of the route at the exact critical point S_n; ρ escapes when $\ell_\rho = 0$, and the flow moves in time $O(1)$ to S_{n+1}, with the rank of its bond raised by one. (A-carry): at each transition a pending route's amplitude projects onto its new unstable direction with a σ-independent factor $p \neq 0$.*

Event times are $t_e - t_1 = \vartheta_e \ln(1/\sigma) + C_e + o(1)$ with initialization-dependent offsets C_e (up to about 50 time units), so near-tied later escapes come in a seed-dependent order at accessible σ (conjecture, heuristic). Unlike greedy low-rank learning (Li et al., 2021), the race keeps memory across saddles. Checked (two components, overlaps $(0.2, 0.5, 0.7, 0.3)$): predicted bond order 0, 2, 1 with slopes 6.823, 12.178, 14.467, confirmed to 0.5% at $\sigma \le 10^{-6}$ in 7 of 8 seeds.

Generic teachers: dependence on the tree. derived new At any critical point the local residual of an adjacent pair $\{u, v\}$ through the bond e annihilates the learned subspaces on both sides, so it lives on complements of dimensions $\mathsf{d}_u - r_e$, $\mathsf{d}_v - r_e$ (d_u: the dimension of the legs of u other than e), where a generic teacher makes it nonzero: *a route with $k^* = 2$ (strict saddle, logarithmic escape) exists whenever some adjacent pair has $\mathsf{d}_u, \mathsf{d}_v > r_e$* (and $\chi_e > r_e$). (i) On an MPS every core has a physical leg, so a generic teacher gives one polynomial plateau (at the origin) and then only logarithmic ones, with ranks growing bond by bond (consistent with the single long plateau

followed by an abrupt, nearly simultaneous rise of the leading unfolding singular values seen with gradient descent on tensor-train completion, Milanesi et al., 2021, Fig. 3(b)). (ii) On binary and HT trees the internal cores carry no site and, at the rank-one saddle, have rank-one outer legs, so the cheapest routes join two sibling bottom cores through their parent ($k = 3$): escapes from the rank-one saddle are polynomial even for generic teachers, and the root bipartition grows only after both halves (bottom-up order); two-core routes can open later on bottom edges, never for the root bipartition. (iii) Odeco and pattern teachers suppress two-core routes by orthogonality, so power-law later plateaus need components orthogonal on the relevant site blocks; for nearly orthogonal ones (overlaps below a crossover $c_\times = O(\sigma)$ up to logarithms) they persist at accessible σ (conjecture, heuristic). Checked on one random CP teacher: strict saddles with logarithmic gaps on MPS $N = 4$, $\lambda_{\min} \propto \sigma$ and gaps $\propto \sigma^{-1}$ on the binary tree.

Teacher on another tree. proven known (standard (min-cut bound); cf. Grasedyck and Hackbusch, 2011, Lemmas 7–8) A student bond e must reach $\operatorname{rank}(W^*)^{\langle e \rangle}$, which for a TTN teacher is at most the minimum, over cuts of the teacher's tree that separate the two sides of e, of the product of the cut dimensions; a binary student with $N = 8$ of an MPS teacher with bond dimension r, for example, needs rank $\min(d^2, r^2)$ on the bonds of sites $\{2, 3\}$, $\{4, 5\}$, i.e. about r^2 separate saddles. A binary student with $N = 4$ is the deep linear network of Remark A.23 for any teacher: plateaus at $\frac{1}{2}\sum_{i>m} \sigma_i((W^*)^{\langle (01\mid 23) \rangle})^2$, escapes of degree three ($\propto \sigma^{-1}$), as many plateaus as the rank of the matricization. One teacher thus has different plateau levels and exponents on an MPS and on a binary tree derived new (tests: Sections 6.3 to 6.5).

A.6 Input covariance

Let $\Sigma = \Sigma_1 \otimes \cdots \otimes \Sigma_N$ (cross-site correlations: end of this section), and let Ξ multiply the physical leg of each site i of every core by $\Sigma_i^{1/2}$; Ξ is linear, symmetric and invertible, and $W(\Xi\theta) = \Sigma^{1/2} W(\theta)$.

Proposition A.39 (A product covariance is a preconditioned whitened problem)**.** proven new (for TTNs) *(a) $L(\theta) = \tilde{L}(\Xi\theta)$ with $\tilde{L}(\tilde{\theta}) = \frac{1}{2}\|W(\tilde{\theta}) - \Sigma^{1/2} W^*\|_F^2$. (b) In the variables $\tilde{\theta} = \Xi\theta$ gradient flow reads $\dot{\tilde{\theta}} = -\Xi^2 \nabla \tilde{L}(\tilde{\theta})$ from the anisotropic initialization $\Xi\theta_0$: a whitened problem with teacher $\Sigma^{1/2} W^*$ and the preconditioner Σ_i on every physical leg. (c) $\nabla L(\theta) = \Xi \nabla \tilde{L}(\Xi\theta)$ and $\nabla^2 L(\theta) = \Xi\, \nabla^2 \tilde{L}(\Xi\theta)\, \Xi$ at every θ. Critical points, critical values, Morse indices and saddle types are therefore those of the whitened problem with teacher $\Sigma^{1/2} W^*$ (Sylvester's law of inertia), but escape rates are eigenvalues of $\Xi^2 \nabla^2 \tilde{L}$, not of $\nabla^2 \tilde{L}$. (d) The gauge group acts on bond legs only, so Proposition 2.5 holds unchanged in θ: small initialization is balanced in θ, not in $\tilde{\theta}$. (e) Plain whitened gradient flow is recovered iff $\Xi^2 \propto I$, i.e. iff every $\Sigma_i = \kappa_i I$ and $\prod_{i \in S_v} \kappa_i$ is the same for every core (equal to 1 on binary and HT trees, whose internal cores have factor 1); for $\Sigma_i = \kappa_i I$ time is rescaled by $\prod_i \kappa_i$.*

Proof. (a) is $W(\Xi\theta) = \Sigma^{1/2} W(\theta)$; (b) and (c) are the chain rule through the symmetric map Ξ (checked to $2 \cdot 10^{-8}$; without the preconditioner the flow deviates by 4–100%). □

Proposition A.40 (Aligned covariance: exact clocks)**.** proven known (Saxe-type modes: Saxe et al., 2014; depth two: Gidel et al., 2019; new: every tree) *Let the teacher be odeco with factors that are eigenvectors of the site covariances, $\Sigma_i u_r^{(i)} = \lambda_{r,i} u_r^{(i)}$, and let $\Lambda_r = \prod_i \lambda_{r,i}$. On the aligned family of Proposition 2.4: (a) $L = \frac{1}{2}\sum_r \Lambda_r (\Pi_r - s_r)^2$; (b) the family is invariant and $\dot{q}_{v,r} = \Lambda_r (s_r - \Pi_r) \prod_{u \neq v} q_{u,r}$, so each component evolves as in the whitened problem with its own clock, $\Pi_r(t; \Sigma) = \Pi_r(\Lambda_r t; I)$, the same for every core; (c) from a balanced start q_0, $t_{1/2,r} = q_0^{-(K-2)}/((K-2)\Lambda_r s_r) +$*

$\delta_K/(\Lambda_r s_r^{2-2/K}) + O(q_0^2)$ *(Proposition A.6); (d) with the set S learned the loss is* $\frac{1}{2}\sum_{r\notin S}\Lambda_r s_r^2$*, and the events of j and then r are separated by* $(1-\frac{\Lambda_r s_r}{\Lambda_j s_j})/((K-2)\Lambda_r s_r q_0^{K-2})$.

Proof. $\langle U_r, \Sigma U_{r'}\rangle = \delta_{rr'}\Lambda_r$, so $\Sigma(W-W^*) = \sum_r \Lambda_r(\Pi_r - s_r)U_r$; argue as for Proposition 2.4. □

Three orderings coexist derived new: by escape speed $\Lambda_r s_r$, by loss drop $\frac{1}{2}\Lambda_r s_r^2$, and the whitened one by s_r. With $s_1 > s_2$ and aligned initialization, component 2 escapes first iff $\Lambda_2/\Lambda_1 > s_1/s_2$ (LO in q_0); in the *mismatch window* $s_1/s_2 < \Lambda_2/\Lambda_1 < (s_1/s_2)^2$ the faster component is the smaller loss drop, so the first plateau lies above the reduced-rank-regression optimum: the visited saddles are not the reduced-rank solutions of Gidel et al. (2019), whose Corollary 2 states no ordering condition. With power-law spectra $\lambda_k \propto k^{-\beta}$ and component r on eigenvector k_r at every site, $\Lambda_2/\Lambda_1 = (k_1/k_2)^{\beta N}$, so the pairing $k_r = r$ reinforces the whitened order.

Proposition A.41 (Random initialization: what replaces the teacher)**.** derived new (TTN form of the covariance-weighted first escape of Jacot et al., 2022; Atanasov et al., 2022; Rawal and DeWeese, 2026) *Near the origin $L = L_0 - \langle W, \Sigma W^*\rangle + \frac{1}{2}\|W\|_\Sigma^2$, and the fan-in initialization law of θ does not depend on Σ. Hence: (a) the escape from the origin (direction, time and learning order of Section A.2) is that of the whitened problem with the* escape tensor $\Sigma W^* = \mathbb{E}[f^*(x)\Phi(x)]$ *in place of W^*; for aligned covariance $s_r \to \Lambda_r s_r$, and for a single (or dominant) aligned component* $\mathrm{d}\ln t_1/\mathrm{d}\ln\Lambda = -1$ *exactly, for every K, tree and split of Λ over the sites; (b) the landscape (critical points, plateau levels, saddle types) is that of $\Sigma^{1/2}W^*$ (Proposition A.39); for aligned covariance $s_r \to \Lambda_r^{1/2} s_r$; (c) the problems (Σ, W^*) and $(I, \Sigma W^*)$ have the same escape at LO; for an aligned component their half-times, each relative to its own target, differ by the $O(1)$ offset $-\delta_K s_r^{-(2-2/K)}(\Lambda_r^{-(2-2/K)} - \Lambda_r^{-1})$.*

Without the preconditioner the exponent would be wrong ($-1/2$ with isotropic initialization). Checked: exponent -1.000 (binary, MPS, HT), offset -0.2218 against -0.2219. The order stays random (Proposition A.14), at rate ratio $\Lambda_2 s_2/(\Lambda_1 s_1)$.

Proposition A.42 (Piggyback block under aligned covariance)**.** proven *(at the saddle)* new *Let j be learned (balanced, $q_j = s_j^{1/K}$) and let r differ from j on $D = D_{jr}$, with two terminal cores c_1, c_2. With $\mu_x = \prod_{i\in D\cap S_{c_1}}\lambda_{x,i}$, $\nu_x = \prod_{i\in D\cap S_{c_2}}\lambda_{x,i}$ for $x \in \{j, r\}$ and $\lambda_{\mathrm{sh}} = \prod_{i\notin D}\lambda_{j,i}$, the Hessian on the two rotations is*

$$H_{\mathrm{pb}} = \frac{\lambda_{\mathrm{sh}}}{s_j^{2/K}}\begin{pmatrix} s_j^2\mu_r\nu_j & -s_r s_j\mu_r\nu_r \\ -s_r s_j\mu_r\nu_r & s_j^2\mu_j\nu_r \end{pmatrix}. \tag{50}$$

It has a negative eigenvalue iff $\Lambda_r s_r^2 > \Lambda_j s_j^2$, for any split of the eigenvalues over the cores. If c_1, c_2 are adjacent ($k = 2$), its rate is below the add-route rate $b = \Lambda_r s_r s_j^{(K-2)/K}$, so with comparable seeds r is added directly: a rotation needs $m = 2$ and $k \geq 3$.

Proof. At the learned amplitude, up to constants, $L = \frac{1}{2}s_j^2\lambda_{\mathrm{sh}}(\mu_j\cos^2\varphi + \mu_r\sin^2\varphi)(\nu_j\cos^2\psi + \nu_r\sin^2\psi) - s_j(\Lambda_j s_j\cos\varphi\cos\psi + \Lambda_r s_r\sin\varphi\sin\psi)$ in the terminal-core rotations φ, ψ, so $\det H_{\mathrm{pb}} \propto \Lambda_j s_j^2 - \Lambda_r s_r^2$. For adjacent terminals the off-diagonal entry is $-b$ ($\lambda_{\mathrm{sh}}\mu_r\nu_r = \Lambda_r$), and a symmetric 2×2 matrix with positive diagonal h_1, h_2 and off-diagonal $-b$ has $\lambda_{\min} > -b$, since $(\frac{h_1+h_2}{2} + b)^2 - (\frac{h_1-h_2}{2})^2 - b^2 = h_1h_2 + b(h_1+h_2) > 0$. □

In the mismatch window, if the faster component wins the first escape derived: on binary $N = 4$ ($m = 2$, $k = 3$) the learned term rotates to the other component at bond rank one, the loss plateaus at $\frac{1}{2}\Lambda_2 s_2^2$, and the faster component is relearned later; on MPS $N = 4$ and every tree with $m \geq 3$ there is no rotation, and the loss plateaus at the higher level $\frac{1}{2}\Lambda_1 s_1^2$ (rotation rates checked).

Misaligned covariance. proven (identities and inequality) derived (order, to leading order in the cross overlaps) new For every CP teacher, exactly, $\Sigma W^* = \sum_r s_r \Lambda_r^{\mathrm{esc}} \widetilde{U}_r$ and $\Sigma^{1/2} W^* = \sum_r s_r (\Lambda_r^{\mathrm{loss}})^{1/2} \widehat{U}_r$, with $\widetilde{U}_r = \bigotimes_i \Sigma_i u_r^{(i)} / \|\Sigma_i u_r^{(i)}\|$, $\widehat{U}_r = \bigotimes_i \Sigma_i^{1/2} u_r^{(i)} / \|\Sigma_i^{1/2} u_r^{(i)}\|$, $\Lambda_r^{\mathrm{esc}} = \prod_i \|\Sigma_i u_r^{(i)}\|$ and $\Lambda_r^{\mathrm{loss}} = \prod_i \langle u_r^{(i)}, \Sigma_i u_r^{(i)} \rangle$; by Cauchy–Schwarz $\Lambda_r^{\mathrm{esc}} \geq \Lambda_r^{\mathrm{loss}}$, with equality iff every $u_r^{(i)}$ is an eigenvector of Σ_i. Components escape in the order of $s_r \Lambda_r^{\mathrm{esc}}$ along $\widetilde{U}_r$ and then realign to U_r without an extra plateau (the TTN form of the early alignment of Atanasov et al., 2022), while the loss drops are $\frac{1}{2} s_r^2 \Lambda_r^{\mathrm{loss}}$: misalignment alone can reverse the order. It also opens routes of degree below k^* derived, with couplings of order $\epsilon^{n_{\mathrm{out}}}$ at per-site misalignment ϵ (n_{out} sites outside the route).

Cross-site correlations. In the one-factor model $x^{(i)} = \sqrt{1-c}\, z_i + \sqrt{c}\, z_0$, with independent $z_0, \ldots, z_N \sim \mathcal{N}(0, I_d)$, every marginal is standard and two sites have covariance cI. (i) proven new (the model is the standard one-factor Gaussian model; new: its tree contraction) Given z_0 the sites are independent with second moments $M_i(z_0) = (1-c)I + c z_0 z_0^\top$, so $L = \mathbb{E}_{z_0}[\frac{1}{2}\|W - W^*\|^2_{\otimes_i M_i(z_0)}]$, a polynomial of degree $2N$ in z_0, is exact by Gauss–Hermite quadrature with $(N+1)^d$ product-metric tree contractions; a site-aligned component has the weight $\Lambda_{\mathrm{eff}}(c, N) = \sum_k \binom{N}{k} (1-c)^{N-k} c^k (2k-1)!!$. (ii) derived The first plateau survives for every $c < 1$ (the origin is a degree-K critical point for any input law; rank-constrained critical points remain critical, Lemma A.2). (iii) proven (rates at the saddle) new On an MPS with even N and a site-aligned odeco teacher, the correlations open strict two-core routes at the product saddle on every bond, with rate $\lambda_2 \approx s_1^{(N-2)/N} s_2 c^{N-2} ((N-3)!!)^2$ at small c; whitened power-law plateaus therefore become logarithmic once c exceeds a crossover $c^* = O(\sigma)$, up to logarithms (heuristic mechanism: empirical for $N = 4$; conjecture for $N \geq 6$, checked on one example with $N = 6$). (iv) derived On MPS $N = 4$ this escape passes through the first plateau point of the binary tree (with junk pairing components; strict on the MPS), and with spectral events the second plateau grows by $\ln 2\, [(1 - \kappa_c)/\lambda_2 + 1/\lambda'']$ per halving of σ, with λ'' the rate at that intermediate saddle and $\kappa_c \in (0, 1)$ the head start of the first escape (2.128 predicted, 2.13 measured, at $c = 0.4$).

A.7 Discrete steps and gradient noise

Odeco teacher with $s_1 > s_2 > \ldots$, whitened Gaussian inputs, $q_r = s_r^{1/K}$; gradient descent (GD) and online SGD as in Section 2.5, recorded at $t = \eta \times$ (number of steps).

Proposition A.43 (Hessian at aligned global minima: the pattern rule)**.** proven known (binary $N = 4$ is a depth-3 linear network: spectrum for square layers, Ghosh et al., 2025, Lemma 1; top eigenvalue at the flattest minimum, Mulayoff and Michaeli, 2020, Thm 1; new: general trees) *At a global minimum $H = J^\top J$, $J = \partial W / \partial \theta$. At an aligned minimum (Proposition 2.4 with $\Pi_r = s_r$), an elementary perturbation of core w changes W by a gain times a unit rank-one* pattern*: the chosen vectors at the sites of w, the teacher vector ℓ_c at every site below a child c whose bond index is ℓ_c, and ℓ_p at every site outside the subtree of w. The gain is $\prod_{u \neq w} q_{u,\ell(u)}$, $\ell(u)$ the label of the branch of w that contains u (zero if a label exceeds R). Distinct patterns are orthonormal, so*

$$\mathrm{spec}(H) \setminus \{0\} = \Big\{ \lambda_P = \sum_{w \in \mathrm{prod}(P)} \mathrm{gain}_w(P)^2 \;:\; P \text{ a pattern} \Big\}, \tag{51}$$

where $\mathrm{prod}(P)$ is the set of cores that can produce P. The radial mode of component r (pattern U_r, produced by every core) has $\lambda = s_r^2 \sum_v q_{v,r}^{-2} \geq K s_r^{2-2/K}$, with equality iff the component is balanced. At the balanced minimum $\lambda_P \leq |\mathrm{prod}(P)|\, q_1^{2(K-1)}$, hence (nondegenerate if $s_2 < s_1$)

$$\lambda_{\max}(H) = K\, s_1^{2-2/K}. \tag{52}$$

Consequences derived new (for TTNs; deep linear networks: Mulayoff and Michaeli, 2020, Thm 1): GD is linearly stable at the balanced minimum iff $\eta < \eta_c := 2/(K s_1^{2-2/K})$ (tested in Section 5.4). Imbalanced aligned minima are at least as sharp (AM–GM; proven; deep linear networks: Mulayoff and Michaeli, 2020), and with $\chi_e = R$ so is every gauge-equivalent minimum (derived; random GL gauges were at least 1.43 times sharper): for $\eta > \eta_c$ no global minimum is linearly stable (derived for $\chi_e = R$; for binary $N = 4$ with $\chi \geq d^2$ it follows from Mulayoff and Michaeli, 2020, Thm 1; conjecture otherwise). The minimum reached by gradient flow, and its sharpness λ_{GF}, are fixed by $Q_e(0)$ (a strongly imbalanced weaker component can carry the top mode); for $2/\lambda_{\mathrm{GF}} < \eta < \eta_c$ GD oscillates, the imbalance contracts (Lemma A.3), and GD converges to a flatter, more balanced minimum, with sharpness just below $2/\eta$ on binary $N = 4$ (empirical; known (scalar networks: Kreisler et al., 2023, Thms 3.2–3.3; two-factor factorization, limit sharpness at most $2/\eta$: Wang et al., 2022, Thm 3.2; beyond the edge, deep matrix factorization balances while oscillating: Ghosh et al., 2025, Prop. 2)). Hessians checked to $6 \cdot 10^{-15}$ on five trees.

Sharpness along the saddle-to-saddle path. derived new On the first plateau the Hessian is indefinite and $O(\sigma^{K-2})$. During the event of a balanced component of amplitude a its radial eigenvalue $(2K-1)a^{2K-2} - (K-1)s_r a^{K-2}$ rises to $K s_r^{2-2/K}$ without overshoot, and the $K-1$ transverse ones $(s_r - a^K)a^{K-2}$ stay below $K s_r^{2-2/K}$. At a later plateau that has learned the set S the radial mode of the largest learned component is an exact eigenvector: the sharpness is $K \max_{r \in S} s_r^{2-2/K}$, up to a relative excess $O(\sigma^4)$ from the $O(\sigma^2)$ imbalance. So at a fixed step η a plateau is linearly unstable iff it has learned a component with $s_r > s_c(\eta) = (2/(\eta K))^{K/(2K-2)}$, and GD first becomes unstable at that component's event.

Edge of stability. derived known (edge of stability: Cohen et al., 2021; the scalar map and its two-cycle, given implicitly: Ghosh et al., 2025, Thm 1; the amplitude below follows from the normal form of MacDonald et al., 2025, Thm 5.3 and Prop. B.3; $K = 2$: Hofmann, 2026) new (only the TTN form of the reduction and the closed-form time-averaged loss (53)) For $\eta_c < \eta < 2/\lambda_2(H)$ only the radial mode of component 1 is unstable, and GD restricted to it (exactly for aligned initialization) is the balanced scalar map $x \mapsto x + hx^{K-1}(1-x^K)$, $x = a/s_1^{1/K}$ (a the per-core amplitude), $h = \eta s_1^{2-2/K}$, which doubles its period at $hK = 2$, i.e. at $\eta = \eta_c$. To third order around $x = 1$ the period-two orbit $1 \pm \delta_{\mathrm{p2}}$ has $\delta_{\mathrm{p2}}^2 \simeq 3(hK-2)/(4(K-1)(5K-4))$, so for $\eta_c < \eta < \min(2/\lambda_2, 1.245\eta_c)$ the time-averaged loss is

$$L_{\mathrm{EoS}} \simeq \frac{3K^2 s_1^2}{4(K-1)(5K-4)} \Big(\frac{\eta}{\eta_c} - 1 \Big). \tag{53}$$

Later events still occur and lower the loss. Checked (binary $N = 4$, $\eta = 1.05\eta_c$): late loss $1.45 \cdot 10^{-2}$ against $1.452 \cdot 10^{-2}$ (map) and $1.53 \cdot 10^{-2}$ (53); divergence at 1.5–$1.65\eta_c$.

Proposition A.44 (Shift of event times under gradient descent)**.** derived new (for TTNs; the modified flow is known: Barrett and Dherin, 2021) *To order η, GD follows the modified flow $\dot\theta = -(I + \frac{\eta}{2}H)\nabla L$. Its component along the gradient-flow path reparametrizes time, and since $\frac{\mathrm{d}}{\mathrm{d}t}\ln\|\nabla L\| = -\nabla L^\top H \nabla L / \|\nabla L\|^2$ along gradient flow,*

$$t_{\mathrm{GD}} - t_{\mathrm{GF}} = \frac{\eta}{2} \ln \frac{\|\nabla L(t_{\mathrm{event}})\|}{\|\nabla L(0)\|} + (\textit{transverse part}) + O(\eta^2). \tag{54}$$

For aligned initialization the path is exact and the event of component r moves by $\frac{\eta}{2}\ln(f(a_1)/f(a_0))$, $f(a) = a^{K-1}(s_r - a^K)$, a_0, a_1 the per-core amplitudes at initialization and at the event: plateau exponents and prefactors are those of gradient flow, and no saddle is skipped or reordered while the gradient-flow gaps exceed the differences of the shifts $+O(\eta^2)$.

For generic initialization $\|\nabla L(0)\| \propto \sigma^{K-1}$, so $\Delta t_1/\eta \approx \frac{K-1}{2}\ln(1/\sigma) + \text{const}$ asymptotically (conjecture, heuristic). At a strict saddle the growth per step is $1+\eta|\lambda_-|$ instead of $e^{\eta|\lambda_-|}$, so $t_{\rm GD}/t_{\rm GF} = \eta|\lambda_-|/\ln(1+\eta|\lambda_-|) \in [1, 1.82]$ for $\eta|\lambda_-| < 2$. Checked: aligned shifts within 1.3%, generic slopes 0.69–0.70 per halving of σ (predicted 0.69).

SGD on the first plateau. derived new (the TTN delay law; framework: stochastic differential equations for SGD, Li et al., 2017, 2019, and Type-II saddles, Ziyin et al., 2024; consistent with the experiments of Corlouer et al., 2026 (Fig. 2: order kept, SGD slower)) For online SGD with batch B the per-sample gradient is $\langle W - W^*, \Phi(x)\rangle J^\top \Phi(x)$; near the origin the gradient covariance is $O(s_1^2 3^N \|\theta\|^{2K-2})/B$ and the drift $O(s_1\|\theta\|^{K-1})$, so in the clock $\mathrm{d}\tau = \|\theta\|^{K-2}\mathrm{d}t$ the noise vanishes: noise cannot drive the first escape. For one aligned balanced component, $\mathrm{d}a = s_1 a^{K-1}\mathrm{d}t + \sqrt{\eta/B}\, s_1 \kappa_{\rm n}^{1/2} a^{K-1}\mathrm{d}\mathcal{B}_t$ ($\mathcal{B}_t$ a standard Brownian motion, $\kappa_{\rm n} = 3^N - 1 + \sum_{r\geq 2} s_r^2/s_1^2$), whose hitting time is inverse Gaussian at leading order:

$$\mathbb{E}t_1^{\rm SGD} - t_1^{\rm GF} \simeq \frac{\eta}{2}(K-1)\Big(1 + \frac{\kappa_{\rm n}}{B}\Big)\ln\frac{a_{1/2}}{a_0} > 0, \qquad \mathrm{sd}(t_1^{\rm SGD}) \simeq \Big(\kappa_{\rm n}\frac{\eta}{B}t_1^{\rm GF}\Big)^{1/2}, \tag{55}$$

with $a_{1/2} = (s_1/2)^{1/K}$. Both corrections vanish relative to $t_1^{\rm GF} \propto a_0^{-(K-2)}$: the exponent is unchanged. Checked: within 10–15% for aligned, order of magnitude for generic initialization.

SGD at the minimum. derived (exact mean-square threshold) proven (the inequality) new (the reduction to the pattern basis; general criteria are known: Ziyin et al., 2024; exact mean square, Mulayoff and Michaeli, 2024, Thm 5, of which (56) is the evaluation in the pattern basis; one-dimensional, Wu et al., 2018; top Lyapunov exponent, Chemnitz and Engel, 2025) In the eigenbasis of (51) linearized SGD at the minimum acts on one coordinate per pattern, with the products $Z_P(x) = \prod_i (t_{P,i}\cdot x^{(i)})$ of its site vectors; the second-moment operator maps diagonal matrices to diagonal ones ($\mathbb{E}[Z_R^2 Z_P Z_Q] = 0$ for $P \neq Q$) and preserves the positive semidefinite cone, which gives the exact mean-square threshold

$$\eta_{\rm ms}^*(B) = \frac{2}{\lambda_{\max}(G_B)}, \qquad G_B = \Big(1 - \frac{1}{B}\Big)\Lambda_H + \frac{1}{B}\Lambda_H^{1/2}\mathcal{K}\Lambda_H^{1/2}, \qquad \mathcal{K}_{PQ} = \prod_i m(t_{P,i}, t_{Q,i}), \tag{56}$$

with $\Lambda_H = \mathrm{diag}(\lambda_P)$, $m = 3$ if the two site vectors coincide and 1 if they are orthogonal (Gaussian sites). Jensen's inequality gives $\eta_{\rm as} \geq \eta_{\rm ms}^*$ for the almost-sure divergence threshold; empirical: the ratios are about 2 ($N = 4$, $B \leq 16$) and 8–21 (binary $N = 8$), and SGD near the minimum diverges at $\eta_{\rm as}$.

Proposition A.45 (SGD lengthens later plateaus). conjecture *(heuristic law)* known (mechanism: stochastic collapse, Chen et al., 2023) *At a later saddle with a route of degree $k \geq 3$ the population Hessian vanishes on the route and the per-sample gradients vanish at zero route amplitude. Zero-mean multiplicative noise coupling the route entries to mixed learned entries gives the log-amplitude of the coherent route a drift $-\gamma_c$, $\gamma_c = c_\gamma \eta/B$; added to the route equation, $\dot b = s_r b^{k-1} - \gamma_c b$. For $k = 3$, with $T_{\rm GF} = 1/(s_r b_0)$,*

$$\frac{T_{\rm SGD}}{T_{\rm GF}} = \frac{-\ln(1-x)}{x}, \qquad x = \gamma_c T_{\rm GF}, \tag{57}$$

and the route is trapped (decays) for $x \geq 1$; for general k escape needs roughly $(k-2)\gamma_c T_{\rm GF} < 1$. As b_0 scales with the initialization, at fixed η/B SGD stalls on a later degenerate plateau once σ is small enough; strict saddles escape if $|\lambda_-| > \gamma_c$.

Solving $\dot{w} = -s_r + \gamma_c w$ for $w = 1/b$ gives (57). Checked on binary $N = 4$ only (weights $(1, 0.6)$, Gaussian sites; 21 runs): delays 1.02–2 times T_{GF}, trapping for $x \geq 1$, never faster than gradient flow, $c_\gamma \approx 1.35$ empirical (1.08 from a commuting-pair estimate; about 33 for weights $(3, 2, 1)$, $d = \chi = 3$: c_γ depends on teacher and tree). Near the stability limit SGD breaks the saddle sequence instead empirical: with weights $(1, 0.6)$ no learned component was erased in 13 runs at $\eta \leq 0.25\eta^*_{\mathrm{ms}}$, but with $(3, 2, 1)$ one was erased (mostly relearned) in 20 of 36 longer runs at 0.25–$0.30\,\eta^*_{\mathrm{ms}}$ (Section 5.4); at 0.7–$1.2\,\eta^*_{\mathrm{ms}}$ components were repeatedly erased and relearned, or runs diverged.

A.8 Regime boundaries

Here $d_i = d$, and $\Theta = \sum_v \eta_v J_v J_v^\top$ on $\mathbb{R}^{d^N}$, $J_v = \partial W / \partial G_v$, is the tensor-space tangent kernel that drives the function-space dynamics of (13) (for whitened inputs the probe-set kernel of Table 3 is $\Phi(x_p)^\top \Theta \Phi(x_{p'})$). Recall $W^{\langle e_v \rangle} = T_v C_v^\top$ (Section 2.1); the initialization is (9) with scales σ_v, $\alpha = \prod_v \sigma_v$. A bond e is *full* if χ_e reaches its cut dimension $\min(d^{n_e}, d^{N-n_e})$ (n_e sites below e).

Proposition A.46 (Tangent kernel at initialization). proven new (on every tree; for MPS and TT the bond-dimension-independent mean kernel is implicit in Guo and Draper, 2021; Wesel and Batselier, 2025; the rank in (c) is the manifold dimension of Uschmajew and Vandereycken, 2013) *(a)* $J_v J_v^\top = \bigotimes_{c \in \mathrm{ch}(v)} T_c T_c^\top \otimes I_{d_v} \otimes C_v C_v^\top$ *(legs in site order). (b) For all bond dimensions,* $\mathbb{E}\,\Theta = \frac{\alpha^2}{d^N} \sum_v \frac{\eta_v F_v}{\sigma_v^2} I$ *and* $\mathbb{E}[W_0 \otimes W_0] = \frac{\alpha^2}{d^N} I$*; with* $\sigma_v = \sigma$ *and* $\eta_v = 1$ *the kernel scale is* $\bar{\lambda} := \mathbb{E}\,\mathrm{tr}\,\Theta / d^N = \sigma^{2(K-1)} \sum_v F_v / d^N$ *(core* v *contributes the term* F_v*). (c) If the bond dimensions are consistent (each* χ_e *at most the product of the other leg dimensions of both of its cores) and the initialization is generic,* $\mathrm{rank}\,\Theta = \sum_v F_v \chi_v - \sum_e \chi_e^2$ *(capped at* d^N*), the dimension of the manifold* $\mathcal{M}_\chi$ *of tensors with bond ranks exactly* χ_e *(dense in the set with ranks at most* χ_e*; for inconsistent bond dimensions, first reduce each* χ_e *to its effective value);* $\mathrm{rank}\,\Theta = d^N$ *iff every bond is full.*

Proof sketch. (a) W is linear in G_v. (b) The cores are independent, T_c and C_v involve disjoint cores, and the initialization law is invariant under orthogonal maps of each physical leg (irreducible); the traces follow by induction from the leaves (Proposition 2.2). (c) $\mathrm{rank}\,\Theta$ is the dimension of the tangent space of $\mathcal{M}_\chi$ at W_0, parameters minus gauge (Uschmajew and Vandereycken, 2013, Thms 2 and 4). □

Consequences derived: with a bond that is not full, the linearized (lazy) dynamics fits only the projection of the target on the range of Θ and stops at $\frac{1}{2}\|P_\perp W^*\|_F^2$, with $\mathbb{E}\|P_\perp W^*\|_F^2 / \|W^*\|_F^2 = (1 - \chi/d^2)^2$ for binary $N = 4$ (compare Chizat et al., 2019, Thm 2.5 and the remark after it, and Ghorbani et al., 2019, Thm 2). $\Theta \approx \bar{\lambda} I$ needs every χ_e far above its cut dimension; at capped full bonds $\lambda_{\min}(\Theta)/\bar{\lambda} \sim 10^{-4}$ empirical, so lazy runs need uncapped $\chi \geq 2d^2$ (Section 5.1); at practical χ the lazy regime follows a random, tree-dependent kernel.

Proposition A.47 (Lazy boundary of the centred model, $N = 4$). derived new (the TTN constants; the criterion is a threshold on the relative kernel change, as in Geiger et al., 2020, refining the a-priori relative scale of Chizat et al., 2019, Sec. 1.2; a coherent second-order kernel change is known in wide networks: Bordelon and Pehlevan, 2022) *Binary tree with* $N = 4$*, standard parameterization, centred model; let* $\xi = \|W^*\|_F / \alpha$*, assume* $\chi \gg d^2$ *and keep the leading order in* d^2/χ*. Then*

$$\delta_\Theta^2 \approx (c_1 \xi)^2 + 2\zeta c_1 c_2 \xi^3 + (c_2 \xi^2)^2, \qquad c_1 = \frac{4d^3\sqrt{d^2+1}}{\chi^3}, \quad c_2 = \frac{2d^4\sqrt{2d^2\nu + 2}}{\chi^3}, \tag{58}$$

with $\nu = \sum_r s_r^4/(\sum_r s_r^2)^2$ *and a seed-dependent correlation* $\zeta \in [-1,1]$*: a zero-mean first-order term, linear in* W_0*, and a coherent second-order term* $\propto (W^*)^{\langle e\rangle}((W^*)^{\langle e\rangle})^\top$ *(the bottom cores drift toward the singular directions of the target). The boundary* $\delta_\Theta = \delta_*$ *lies in* $[\xi_c^+, \xi_c^-]$*,* $\xi_c^\pm = (\mp c_1 + \sqrt{c_1^2 + 4c_2\delta_*})/(2c_2)$*, and for large* χ *(*$c_1^2 \ll c_2\delta_*$*)*

$$\sigma_{\text{lazy}} = \Big(\frac{\|W^*\|_F^2 c_2}{\delta_*}\Big)^{1/6} \propto \chi^{-1/2} \tag{59}$$

($6.50\,\chi^{-1/2}$ *for* $d = 3$*, weights* $(3,2,1)$*,* $\delta_* = 0.1$*), not* χ^{-1} *as the first-order term alone would give. In the two-copy model the first-order term cancels,* $\delta_\Theta \approx \frac{1}{4}c_2\xi^2$*, and* $\sigma_{\text{lazy}}^{(2)} = 2^{-1/3}\sigma_{\text{lazy}}$*.*

Checked empirical: predicted $\xi_c \in [2.17, 2.81]$, $[6.67, 7.31]$ at $\chi = 32$, 64 against 2.25–2.82, 6.73–7.53 in gradient flow. The boundary moves with the threshold as $\sigma_{\text{lazy}} \propto \delta_*^{-1/(Ks)}$, $s = \mathrm{d}\ln\delta_\Theta/\mathrm{d}\ln\xi \in [1,2]$ (1.5–1.8 measured); the root can move by more than 100% while $\delta_\Theta < 0.1$.

Deficient bonds. derived new If some bond is not full, $W^* + W_0$ generically has more bond rank than the student can carry: the centred model is unrealizable at every σ. At small σ (first order in W_0 around a smooth point W^* of $\mathcal{M}_\chi$, i.e. $\chi_e = \operatorname{rank}(W^*)^{\langle e\rangle}$, with σ^K far below the smallest teacher Schmidt value) the best reachable point is $W^* + P_T W_0 + O(\|W_0\|_F^2)$ (P_T, $P_\perp$: tangent and normal projectors of $\mathcal{M}_\chi$ at W^*), so the loss floors at $\frac{1}{2}\|P_\perp W_0\|_F^2$, with mean, by Proposition A.46(b),

$$\frac{\mathbb{E}L_{\text{end}}}{L(0)} \approx \Big(1 - \frac{\dim\mathcal{M}_\chi}{d^N}\Big)\frac{\sigma^{2K}}{\|W^*\|_F^2}. \tag{60}$$

A typical run meets the convergence rule $L < 10^{-3}L(0)$ only below $\sigma_{\text{floor}} = [10^{-3}\|W^*\|_F^2/(1 - \dim\mathcal{M}_\chi/d^N)]^{1/(2K)}$ (0.56 for binary $N = 4$, 0.74 for $N = 8$; $d = \chi = 3$, weights $(3,2,1)$); runs above it are stalled (unrealizable). At large σ the floor tends to $\|P_\perp W^*\|_F^2/\|W^*\|_F^2$ ($P_\perp$ at W_0). In the two-copy model (realizable iff $2\chi_e \geq \operatorname{rank}(W^*)^{\langle e\rangle}$) a deficient bond gives, at large σ, a lazy plateau at that level and then a saddle-free escape by rotation of the bond subspaces: adiabatic elimination gives $t_{\text{esc}} = 2\vartheta\sigma^2/\|W^*\|_F^2$, $\vartheta = 0.37$–0.70, a kernel time $\propto \sigma^{2K}$ (derived for $N = 4$; conjecture, heuristic, for general K; compare Tu et al., 2024; Lyu et al., 2024).

Proposition A.48 (Saddle-to-saddle boundary for aligned initialization)**.** proven *(given the exact aligned reduction)* known (mode equation: Saxe et al., 2014; threshold exponent $K/(K-2)$ and a product threshold that stays finite as $K \to \infty$, so that $a_{0c} \to 1$: Gissin et al., 2020, Thm 2, with the $K \to \infty$ mode solution in their Thm 1; new: the constants for the criterion of Section 2.8, including their $K \to \infty$ limits) *Let every component of an odeco teacher start at product amplitude* $u_0 = q_0^K$*, let* $I_K(v_0, v_1) = \frac{1}{K}\int_{v_0}^{v_1} v^{-2+2/K}(1-v)^{-1}\mathrm{d}v$ *and* $\omega_K = I_K(0.1, 0.9)$*. Then* $t_r = s_r^{-(2-2/K)} I_K(u_0/s_r, \frac{1}{2})$ *and* $w_r = s_r^{-(2-2/K)}\omega_K$*, so, with* $\varrho_i = s_{i+1}/s_i$*,*

$$\rho_i = \frac{I_K(u_0/s_{i+1}, \frac{1}{2}) - \varrho_i^{2-2/K} I_K(u_0/s_i, \frac{1}{2})}{\omega_K} \simeq \Big(\frac{u_0}{s_{i+1}}\Big)^{-(1-2/K)} \frac{1-\varrho_i}{(K-2)\,\omega_K} \quad (u_0 \to 0), \tag{61}$$

and the boundary $\rho_i = 3$ *lies at* $u_{0c} \simeq s_{i+1}[(1-\varrho_i)/(3(K-2)\omega_K)]^{K/(K-2)}$*. As* $K \to \infty$*,* $KI_K(v_0, v_1) \to \frac{1}{v_0} - \frac{1}{v_1} + \ln\frac{v_1}{v_0} + \ln\frac{1-v_0}{1-v_1}$ *and* $K\omega_K \to \frac{80}{9} + 2\ln 9 = 13.283$*: stays finite (*$1.306 \cdot 10^{-2}$ *for weights* $(3,2,1)$*), and* $a_{0c} = u_{0c}^{1/K} = 1 + \ln u_{0c}/K + O(K^{-2}) \to 1$*.*

Values: $\omega_3 = 2.009$, $\omega_5 = 1.612$, $\omega_7 = 1.318$; for weights $(3,2,1)$, $a_{0c} = 0.066$, 0.324, 0.480 at $K = 3, 5, 7$. Aligned gradient flow reproduces (61) to four decimals at $K = 3$ and 7.

Random initialization. derived (t_1) conjecture (later gaps; large K) empirical (boundary) By Corollary A.5 and Proposition A.6, $t_1 = \sigma^{-(K-2)}\hat{t}_1(Z)/s_1 + O(1)$. For teachers whose escape routes span the whole tree (odeco), later gaps scale the same way if the dormant amplitudes entering each plateau stay $\propto \sigma$ (exact for aligned initialization), so $\rho_i \approx R_i\sigma^{-(K-2)}$ with seed-dependent R_i; in general a later gap scales as $\sigma^{-(k-2)}$, k the route degree (Section A.3). The boundary is a seed distribution: median per-seed $\sigma_c = 0.27$–0.31 for binary $N = 4$, $d = \chi = 3$, weights $(3, 2, 1)$, and 0.71 for $N = 8$, $d = 2$, $\chi = 16$, weights $(3, 2)$; as aligned initialization with per-core amplitude $\kappa_K\sigma$, $\kappa_3 \approx 0.22$ ($d = 3$) and $\kappa_7 \approx 0.69$ ($d = 2$). conjecture: $\kappa_K \to 1$, so $\sigma_c(K) \to a_{0c}(K) \to 1$; the data (all at $d = 2$ for large K) do not distinguish this from $\kappa_K \to d^{-1/2}$, i.e. $\sigma_c \to \sqrt{d}$ (Section 3.3). For matrix-like pairs the transient overlap of a component not yet learned peaks at a value $\propto \sigma^{2(1-\varrho)}$ (conjecture, heuristic), hence events are timed by their final rise. With χ equal to the teacher rank R and $K \geq 5$, the last component needs a route through all K cores, which the child-heavy root bonds of fan-in initialization (Section A.1) can kill before it escapes (conjecture, heuristic). empirical: it was not learned in 2 of 4 runs at $K = 5$ and 2 of 2 at $K = 7$ ($d = 3$); balanced initialization removed these stalls at $K = 5$ (leaving long plateaus) but not at $K = 7$; spare channels ($\chi \geq 2R$) avoided stalls in the runs made (one at $d = 3$; 26 at $d = 2$).

Large bond dimension. derived new (the spectral condition is Yang et al., 2024; factor-wise rules for structured layers: Qiu et al., 2024; Potapczynski et al., 2024) The input of a core is always a product of vectors, so its signals are controlled by its injective norm, not by the spectral norm of its matricization. The spectral condition with the injective norm keeps the fan-in initialization of every non-root core, multiplies the root by $(\sum_\ell m_\ell)^{-1/2}$ (m_ℓ: its child-leg dimensions) and sets $\eta_v \propto \chi_v/F_v$ ("μP-TTN"). At fixed ξ ($N = 4$) this gives $\delta_\Theta \propto \chi^0$, against χ^{-3} (standard) and χ^{-1} (NTK parameterization; compare Guo and Draper, 2021); the factor-wise rule makes the initial kernel of binary $N = 8$ vanish like χ^{-2} and slows the initial fit about linearly in χ empirical.

Imbalance. derived (aligned mode) new (compare two-layer linear networks: Kunin et al., 2024) With the root at scale $\lambda_{\text{root}}\sigma$ and the other cores at σ (fixed α), the root bonds have $\mathbb{E}Q_e = \sigma^2(1 - \lambda_{\text{root}}^2/\chi_e)I$, balanced at $\lambda_{\text{root}} = \sqrt{\chi_e}$ (fan-in initialization is root-light). A small root ($\lambda_{\text{root}} \ll 1$) makes training lazy when $s \ll \alpha/\lambda_{\text{root}}$ (s a teacher weight), with $\delta_\Theta \propto \lambda_{\text{root}}^2$, but only with full bonds (with a deficient bond: lazy, then rich). A large root freezes, leaving a degree-$(K-1)$ model, which does not make training lazy for $K \geq 3$ (unlike the two-layer case). Tests: Sections 3.2 to 3.4 and 5.3.

B Experimental details

This appendix gives the settings of every experiment: the problems (trees, dimensions, teachers), the initializations, the dynamics and their tolerances, the sweeps, seeds and budgets, how runs are classified, and which runs were excluded. Section B.1 lists what all experiments share. Each later subsection names its experiment folder, whose README lists every command in order and whose `configs/` holds the YAML configurations, the notes with its findings, and the reviews whose errata are applied in the main text. Run counts are of finished runs. Raw outputs are not in the repository; Section C says how to regenerate them.

B.1 Common settings

Model and loss. Every experiment trains the model of Section 2.1 on the exact population loss (6), computed by tree contractions (Section 2.3), except where stated: the finite-sample sweeps

(empirical loss), the correlated-input runs (dense second-moment matrix) and the real-data runs (empirical loss on a fixed training set). Inputs are whitened Gaussians ($\Sigma_i = I$) unless stated, so the loss is half the squared Frobenius distance to the teacher, and only the second moments of the features enter.

Precision and processes. All computations are in double precision: with $\sigma = 10^{-2}$ and $K = 7$ the initial tensor has norm about 10^{-14} and the gradients are of order 10^{-12}. A dormant component whose escape signal falls below about 10^{-15} relative is seeded by roundoff, not by the initialization; the experiments that can reach this floor scan for it and flag such runs (Section 2.5). Every process runs PyTorch with a single thread (`torch.set_num_threads(1)`): the tensors are small, and many threads slow their linear algebra down by large factors on a loaded machine. Parallelism comes from running many single-threaded processes.

Integrators and tolerances. The reference integrator is the adaptive Dormand–Prince 5(4) pair of `ttn.dynamics.integrate_gf` with relative and absolute tolerances 10^{-8} and 10^{-12} (Section 2.5). Past the first escape the flow is stiff: learned directions have Hessian eigenvalues of order one while a plateau lasts $\propto \sigma^{-(k-2)}$, so an explicit method needs a number of steps that grows with the plateau. Most experiments therefore integrate with SciPy's LSODA and the exact Hessian (by automatic differentiation) as its Jacobian; short runs and references use Dormand–Prince or DOP853; Radau is a fallback. Table 11 gives each experiment's choice. Event times come from root finding on the integrator's dense output, from re-integrating the event window on a fine linear grid, or from dense merged records (log-spaced and linear output times plus the fine rows of Section 2.5); each subsection says which. Accuracy is checked by the conservation drift (16), below 10^{-6} (exceptions are reported per experiment), by the exact aligned solutions (Section 4.2), by reruns at tighter tolerances, and by independent implementations in the reviews.

Seeds and random streams. Teacher and initialization draw from independent random streams: the teacher's generator is `ttn.teachers.teacher_generator`(s), which mixes the teacher seed s through a seed sequence, so that a teacher and an initialization with equal seeds share no Gaussian draws. Before this separation (library commit 57d5262), equal seeds made a random TTN teacher proportional to the student's initialization and tied an odeco teacher's first site basis to the first core; experiment runs from that time support no claim and were discarded or rerun (each subsection lists its exclusions), and two numerical checks of the theory that predate the separation are marked as such in Remark A.24 and Proposition A.28. Initialization seeds are consecutive integers, the same at every σ of a sweep, so that a seed fixes the initialization direction while σ varies.

Initializations. *Fan-in*: (9) with one scale σ for all cores, or per-core scales where stated. *Balanced*: the same tensor W_0 in the gauge with $Q_e = 0$ on every bond. For each bond, with $M = G_{v,(e)}G_{v,(e)}^\top$ and $N = G_{p,(e)}G_{p,(e)}^\top$, the symmetric positive definite P with $PMP = N$ gives the gauge transformation $P^{1/2}$ that makes the two Gram matrices equal; since balancing one bond changes its neighbours, sweeps over the bonds repeat until $\max_e \|Q_e\|_{\max} < 10^{-13} \max_v \|G_v\|_F^2$ (`gauge_balance` in `experiments/s2s_core/s2s.py`; the locality, regime-map and optimizer experiments have their own implementations). A gauge transformation leaves W unchanged, so a balanced run and its fan-in twin start from the same function and differ only in the conserved quantities. *Aligned*: cores diagonal in the teacher's basis (Proposition 2.4; `ttn.network.aligned_init`). *Saddle starts*: an exact balanced saddle, aligned for pattern teachers or found by L-BFGS on the rank-constrained problem for generic teachers, plus a fan-in perturbation of scale σ or ε.

Centred and two-copy models. The centred model trains on the target $W^* + W_0$ (`ttn.teachers.ShiftedTeacher`), so that $f_\theta - f_{\theta_0}$ is trained on f^* (Section 2.4). Written directly, its loss is a difference of terms of size $\|W_0\|_F^2$, with a cancellation error of about $10^{-16}\|W_0\|_F^2$ that is harmless at small α but ruins large ones. The library therefore writes $W(\theta) - W_0$ as one TTN on doubled bonds, in half-sums and half-differences of the two networks, so that nothing cancels (`ttn.dynamics.Problem`); the regime map uses the direct form only for $\alpha < 10^{-2}$ (Section B.5). The two-copy model $W(\theta^+) - W(\theta^-)$ is one TTN with doubled bonds and block-diagonal cores (`ttn.network.antisymmetric_double`), trained with the gradient mask of Remark 2.3 and contracted in the same half-sum and half-difference form (`ttn.network.sum_difference_cores`).

Recording, events and Hessians. Diagnostics are recorded on a logarithmic time grid (200 times by default; other grids where a subsection says so) plus the fine rows of Section 2.5, and merged for events (`ttn.dynamics.merged_rows`). Events and widths follow Section 2.8; which of its crossing rules an experiment uses is given in Table 11. Hessians are computed by automatic differentiation at plateau midpoints (minimum parameter speed between two events), raw and with the gauge directions projected out (`ttn.diagnostics.project_out_gauge`, relative cutoff 10^{-3}), and the saddle type is read from how $\lambda_{\min}$ scales with σ. The relative kernel change uses the probe kernel (in the regime map, 128 Gaussian inputs from generator seed 12345; on real data, the first 128 test images) and, where the dense Jacobian fits in memory ($d^N \dim\theta \le 2\cdot 10^7$), the tensor-space kernel $J\,\mathrm{diag}(\boldsymbol{\eta})J^\top$.

Compute. Each remote experiment ran on one rented CPU machine with 32 to 384 virtual cores, as many single-threaded processes in parallel; some fix rounds ran on the same machines (machines, hours and costs in `notes/spend.md`). Pilots, reviews, the fix rounds of the regime-map and real-data experiments, and the pre-registered order test of Section 6.7 ran on a local workstation. No GPU was used.

B.2 Saddle-to-saddle experiments

Settings of Sections 4.1 to 4.3 and of the odeco runs of Section 4.8 (experiment s2s-core; the locality experiment is in Section B.3). Experiment folder `experiments/s2s_core/` (README with every command in order; YAML configurations in `configs/`); findings in `notes/s2s_core.md`; reviews in `notes/reviews/s2s_core_round1.md` and `notes/reviews/s2s_core_round2_verification.md`, whose errata E1–E21 are applied in Section 4. Numbers that are not single entries of a results table are computed by `report_numbers.py` into `results/report_numbers.csv`. The order experiment of Section 4.7 is described in Section B.4.

Problems. Whitened Gaussian inputs, exact population loss, double precision. Odeco teachers (`ttn.teachers.odeco_teacher`) with $R = 3$, $d = 4$ at every site and Haar-random site bases from teacher seed 1000, a stream independent of the initialization seeds; weights $(3, 2, 1)$ (separated) and $(1, 0.85, 0.7)$ (close). Trees binary4, MPS4, binary8, HT4 and MPS8 ($K = 3, 4, 7, 7, 8$); nominal bond dimension $\chi = 3$ or 6, capped by (3) (at $\chi = 6$: MPS4 bonds $(4, 6, 4)$, MPS8 end bonds and HT4 leaf bonds 4, all other bonds 6). Initializations: fan-in (9) with seeds 0–9; *balanced*, the same W_0 re-gauged to $Q_e = 0$ on every bond (`s2s.gauge_balance`); *aligned*, the same per-core amplitude on every core and component (`ttn.network.aligned_init`). The leading-order sweeps also use MPS, binary and HT trees with up to 16 sites (Table 12).

Table 11: Integrators, event rules and seeds by experiment (details in the subsections named). LSODA: SciPy's LSODA with the exact Hessian as Jacobian; DP5(4): Dormand–Prince 5(4); tolerances relative, absolute. Event rules for overlaps (Section 2.8): *final rise*, the last upward crossing of $\frac{1}{2}$ before a_r first reaches 0.9; *persistent*, the last upward crossing after which a_r stays above $\frac{1}{2}$; *first*, the first crossing.

experiment	integrator (tolerances)	events	seeds: teacher; initialization
s2s-core (B.2)	LSODA (10^{-8}, $10^{-11}\sigma$; 10^{-6}, $10^{-9}\sigma$ for the longest plateaus); aligned runs DP5(4) and LSODA (10^{-10}, 10^{-14}); leading-order flow DOP853	final rise, refined on linear grids	1000; 0–9
locality (B.3)	LSODA (10^{-9}, 10^{-14}), Radau fallback; pilot: own code, LSODA (10^{-10}, $10^{-13}\sigma$)	persistent (pilot: first), root finding	1000; 0–7
order (B.4)	first events DP5(4) (10^{-9}, 10^{-14}); full runs LSODA (10^{-10}, 10^{-15}; three or four components 10^{-9}, 10^{-14})	first, root finding	none (standard basis); 0–399 or 0–999
regime map (B.5)	DP5(4) (10^{-8}, 10^{-12}), LSODA for long plateaus	final rise, re-integrated on a 200-point grid	$1000 + s$; $s = 0$–5
covariance, teacher (B.6)	DOP853 (10^{-9}, 10^{-14}; single-σ runs); LSODA (10^{-9}, 10^{-14})	onset of bond spectra, $f = 0.05$; aligned covariance: first persistent	independent stream; 5–40 seeds
steps, noise, samples (B.7)	references DOP853 (10^{-10}, 10^{-14}); LSODA (10^{-8}, 10^{-12}) at $\chi = 9$; GD and SGD by the experiment's own loops		101; 0–4 (5–44 for the order check)
optimizers (B.8)	gradient flow by LSODA (10^{-9}, 10^{-14}), Radau fallback; other methods by their own iterations	persistent, root finding	100; 0–5
real data (B.9)	DP5(4) (10^{-8}, 10^{-14}) to the first escape, then LSODA with the Gauss–Newton matrix	loss-rate peaks and rank events	data seed 0; 0–4

Dynamics. Gradient flow by SciPy's LSODA (`ttn.dynamics.integrate_gf`) with the exact Hessian as Jacobian, instead of the default integrator of Section 2.5: past the first escape the flow is stiff, and an explicit method needs a number of steps $\propto \sigma^{-(K-2)}$ per plateau (Radau, tried on one $K = 7$ run, did not finish in 25 minutes). Tolerances: relative 10^{-8} and absolute $10^{-11}\sigma$ (sweeps `main`, `first`, `balanced`), or 10^{-6} and $10^{-9}\sigma$ (`large`, `k78`, whose plateaus last up to 10^7); on binary8 at $\chi = 6$, $\sigma = 0.07$, the looser tolerances move the event times by $7 \cdot 10^{-7}$ relative. Aligned runs: Dormand–Prince 5(4) and LSODA with tolerances 10^{-10} and 10^{-14}. Events are last upward crossings of $a_r = \frac{1}{2}$ and widths final rises from 0.1 to 0.9 (Section 2.8), both refined by re-integrating the event window on linear grids until the rise holds at least 40 samples (`s2s.zoom_rise`); no duration is read off the logarithmic output grid. The plateau midpoint is the time of minimum parameter speed between two events; the parameters there are saved, and the Hessian, raw and gauge-projected (`ttn.diagnostics.project_out_gauge`), was recomputed from them after the library's projection fix (`recompute_hessians.py`). Bond spectra are recorded on the output grid and, in all but the first 332 complete runs of `main`, in the event windows. The conservation drift (per integration chunk, as in (16)) has median $9 \cdot 10^{-10}$ and exceeds 10^{-6} in 11 of 1,576 runs (at most $3.3 \cdot 10^{-6}$; ten MPS8 runs and one binary8 run).

Leading-order flow. $T(Z)$ and r_1 (Remark A.7) come from `s2s.correlation_flow_blowup`: the flow $\dot{\theta} = \nabla g$ from the initialization at $\sigma = 1$, by DOP853 with relative tolerance 10^{-10}, integrated in segments until the second-largest component amplitude $|\langle W, U_r \rangle|$ is below 10^{-6} of the largest and no longer growing relative to it, with a cap at amplitude 10^{30}; the remaining time of a balanced mode from the stop is added. The 40 binary4 close-teacher jobs and 2 separated binary4 jobs at $\chi = 6$ stopped at the cap with the direction still converging; T is converged regardless (the

time left beyond amplitude A is $\propto A^{-(K-2)/K}$), and an independent integration of the spherical flow agrees to $1.5 \cdot 10^{-11}$ (binary4) and $1.7 \cdot 10^{-10}$ (MPS4) relative on all 160 binary4 and MPS4 jobs (verification review). Round-1 values of T, from a stop at amplitude 10^6, were biased by up to $2.2 \cdot 10^{-3}$ relative (binary4, close teacher) and are superseded.

Sweeps and run counts. Table 12.

Table 12: Sweeps of the s2s-core experiment, one configuration file each in `experiments/s2s_core/configs/` except the last two rows. Seeds 0–9 unless stated; "sep." and "close": the two teachers.

sweep	content	runs
`aligned`	five trees, sep., $\chi = 3$, balanced aligned initialization; Dormand–Prince and LSODA	10
`main`	five trees, $\chi = 3, 6$, both teachers, fan-in; $K = 3, 4$: 8 values of σ in $[0.003, 0.1]$; $K = 7, 8$: 6 values in $[0.07, 0.4]$ at $\chi = 3$, plus 50 runs at $\sigma = 0.03$, 0.05 ($\chi = 3$) and 76 runs at $\chi = 6$ ($\sigma = 0.03$–0.1) made before these were dropped or moved to `large`, all kept	1,126
`large`	binary8, HT4, MPS8, $\chi = 6$, sep., fan-in, $\sigma \in \{0.07, 0.1, 0.15, 0.2, 0.3\}$, wall-clock limit 2400 s	150
`k78`	binary8, HT4, MPS8, $\chi = 3$, sep., fan-in and balanced, $\sigma \in \{0.15, 0.2, 0.3\}$, no amplitude stop	180
`balanced`	binary4, MPS4, $\chi = 3, 6$, sep., balanced, $\sigma \in \{0.003, 0.01, 0.03\}$	120
`first`, `first_k78`	first escape only, 8 values of σ: binary4, MPS4 ($\sigma = 0.002$–0.1; $\chi = 3, 6$, both teachers, fan-in and balanced); binary8, HT4, MPS8 ($\sigma = 0.01$–0.4; sep., fan-in, $\chi = 3, 6$, MPS8 at $\chi = 3$ only; binary8 also close and balanced at $\chi = 3$, 6–9 seeds); 231 (initialization, teacher) pairs	1,848
`lo_d`	leading-order flow only: binary4, MPS4, $d = 2, 3, 4$, $\chi = d^2$, one component of weight 1, 200 seeds	1,200
`lo_chi`	leading-order flow: the five trees, $\chi = 1$–16 (binary8 to 12) with one component of weight 1, and $\chi = 3, 6, 12$ with weights $(3, 2, 1)$; 100 seeds	5,400
`lo_K`	leading-order flow: MPS with $N = 3$–$8, 10, 12$, binary with $N = 4, 6, 8, 12, 16$, HT with $N = 3, 4, 6, 8$ ($K = 3$–15), $\chi = 3$, weights $(3, 2, 1)$, 50 seeds	850
continuation	stalled or unresolved runs continued from their final state (`continue_stalled.py`)	23
tolerance check	9 runs of `k78` rerun with tolerances 10^{-8}, $10^{-11}\sigma$ (`tolcheck.py`)	9

Budgets and run classes. Runs stop at $L \leq 10^{-9}L(0)$; at $t_{\max}$, 10^3 times the RMS-overlap escape time of the weakest component; at a wall-clock limit (300 s in `main`, 400 s in `balanced`, 2400 s in `large` and `k78`, so the number of unresolved runs depends on the machine) or a step limit per output chunk; and, except in the first ∼310 runs of `main`, when a component never learned has a route-end slice below $10^{-8}\sigma$ for a whole output chunk (a dead end, after which nothing is physics). `main` also stopped when every dormant amplitude $|\langle W, U_r\rangle|$ stayed below 10^{-15} for a chunk; at $K = 7, 8$ this was too strict (the amplitude is a product of K per-core amplitudes and can be 10^{-15} while each of them is 10^{-3}), so those runs are classed unresolved unless a slice is dead, and the cells were rerun without it in `k78`. Classes (`analyze_main.py`): *complete* (all three components learned); *trapped* (a component never learned has a route-end slice below $10^{-8}\sigma$ after the first escape); *stalled* (the time budget ended with every such slice above it); *unresolved* (wall-clock limit or the amplitude stop). The classes do not depend on the tolerance: the 9 runs of the tolerance check keep their class and order, and their event times change by at most $3.6 \cdot 10^{-6}$ relative. At $K = 7, 8$ the trapped fractions are lower bounds, since some stalled runs end with slices still decaying (`slice_trends.py`).

Fits. Exponents are fitted per seed, ln of the time against $\ln \sigma$ with at least four values of σ, on run sets classified by the saddle sequence (`analyze_main.fit_sets`): t_1 on the runs whose first component is the seed's most frequent one; $t_2 - t_1$ on the runs whose first two components are the seed's most frequent pair, with $\rho_1 > 3$; $t_3 - t_2$ on complete runs in the seed's most frequent order with $\rho_1, \rho_2 > 3$. The Hessian slopes use the same sets. Each cell comes from one sweep (`large` at $K = 7, 8$ with $\chi = 6$, `main` otherwise); sweeps with different tolerances are never pooled. The cut $\rho_i > 3$ matters only for binary4 with the close teacher, where it leaves some seeds with four or five values of σ (about 0.8 decade); without it, overlapping and wrong-order runs are mixed in and the second gap gives 0.92 ± 0.11 at $\chi = 3$ and 1.12 ± 0.05 at $\chi = 6$ (verification review, E9). First-escape exponents are also given with the offset $\delta_K s_{r_1}^{-(2-2/K)}$ of (19) subtracted from t_1.

Continued runs. `continue_stalled.py` integrates a stalled or unresolved run further from its saved final state, with the escape time resolved like every event. Three samples: the 8 stalled `k78` runs with the shortest predicted remaining plateau (3 seeds); a seeded random sample (`--seed 1`) of 8 of the 74 `k78` runs (73 stalled, 1 trapped) whose estimate is at most 100 times the time already integrated, one of them also in the first sample; and 8 MPS8 runs of `large`, spread over the ratio of estimate to elapsed time (0.2–420). The budget multiplier of the third sample was not recorded, and the README command selects a different set (verification review, E13); its results are reported as measured.

Excluded and superseded runs. The scipy pilots made before the library existed, part of them before the teacher and initialization streams were separated, support no claim. Superseded after review round 1: T from the amplitude-10^6 stop (kept as `T_v1` in the raw records), gap fits on pooled run sets, Hessians with the earlier gauge projection, and decay rates fitted into the floating-point floor (now fitted above 10^{-9} absolute). No run was dropped: runs that do not complete are reported in their class. The raw records (3,417 files, about 220 MB; local copy `data/s2s_core/`) are not in the repository; Section C says how to regenerate them.

B.3 Locality of later escapes

Settings of Sections 4.4 and 4.8. Experiment folder `experiments/locality/` (README; one YAML configuration per sweep in `configs/`, written by `make_configs.py`); findings, and the changes made after two review rounds, in `notes/locality.md`, sections 7–10; reviews in `notes/reviews/locality_round1.md` and `notes/reviews/locality_round2_verification.md`.

Problems. Whitened Gaussian inputs, exact population loss, double precision, the library model (`ttn.dynamics.Problem`). Pattern teachers (`ttn.teachers.pattern_teacher`) with $d = 2$ and random orthonormal site bases, teacher seed 1000, from a stream independent of the initialization seeds. Two-component teachers have patterns 0^N and the indicator of D, weights $(1, 0.5)$, also $(1, 0.25)$, $(1, 0.75)$ and $(2, 1)$, on 60 (tree, D) pairs. Multi-component teachers (patterns as site values, weights in brackets): TA 0000 (1), 1100 (0.6), 1110 (0.35); TB as TA plus 1111 (0.2); N4 0000 (1), 1100 (0.6), 1111 (0.4); B4 0000 (1), 0110 (0.3), 1001 (0.6); E8 0^8 (1), $0^3 1^2 0^3$ (0.4), $010^4 10$ (0.7); TC 0^8 (1), $1^2 0^6$ (0.6), $1^4 0^4$ (0.35), 1^8 (0.2); TD 0^8 (1), $0^3 1^2 0^3$ (0.6), $1^2 0^6$ (0.35). Bond dimension $\chi = 2$, except $\chi = 3$ for B4, N4 and E8, and $\chi \in \{4, 8\}$, $d = 4$ in the robustness sweep. Trees: MPS4, MPS6, MPS8 ($K = 4, 6, 8$), binary4, binary8 ($K = 3, 7$), HT4, HT8 ($K = 7, 15$), star3 ($K = 4$: a root without sites and three children carrying sites $\{0, 1\}$, $\{2, 3\}$, $\{4, 5\}$) and midroot6 ($K = 6$: the MPS6 chain rooted at the core of site 3).

Protocols and dynamics. *Small*: fan-in initialization (9). *Saddle*: the exact balanced saddle $s_1 U_1$ (`ttn.network.aligned_init`) plus a fan-in perturbation of per-core scale σ. *Balanced*: the fan-in W_0 re-gauged so that $Q_e = 0$ on every bond. Gradient flow by SciPy's LSODA (`solve_ivp`) with the exact Hessian as Jacobian, relative and absolute tolerances 10^{-9} and 10^{-14}, instead of the default integrator of Section 2.5; Radau as fallback (fewer than 1% of runs). Event times are located by root finding on the dense output; event r is the last upward crossing of $a_r = \frac{1}{2}$, and a run with an overlap that crosses $\frac{1}{2}$ more than once is a *rotation*. Budgets: $t_{\max} = 10^{14}$ and a wall-clock limit of 1500 s per run (900 s for the $k \geq 6$ rows of the main sweep, 2400 s in the $k \geq 6$ extension and the stall diagnosis). A run is *stalled* when, after an overlap has passed $\frac{1}{2}$, every gradient entry stays below 10^{-12} for more than 5,000 consecutive evaluations (300,000 in the stall diagnosis, which also keeps the trajectory). The Hessian is computed by automatic differentiation at the plateau midpoint (minimum gradient norm between consecutive events), raw and with the gauge directions projected out (`ttn.diagnostics.project_out_gauge`, relative cutoff 10^{-3}). An earlier cutoff, 10^{-10}, kept the generators of nearly unused bond channels, whose normalized direction overlaps the escape route, and biased the $k = 2$ values to -0.44 to -0.46 instead of -0.5000; all Hessians were recomputed (`hess_recompute.py`). The fixed projection is still biased on the short small-initialization $k = 2$ plateaus at $\sigma \geq 10^{-3}$ (-0.411 and -0.502 against raw -0.437 and -0.581), so raw values are quoted there; raw and projected σ-exponents of $\lambda_{\min}$ differ by up to 0.11.

Sweeps. Table 13; initialization seeds 0–7 (0–4 for the $k \geq 6$ rows of the main sweep).

Table 13: Sweeps of the locality experiment (Sections 4.4 and 4.8), one configuration file each in `experiments/locality/configs/`; tables and fits use only the runs of these configurations.

sweep	content	runs
`twocomp`	60 (tree, D) pairs, small and saddle (HT8 saddle only), weights $(1, 0.5)$; 7 values of σ in $[10^{-4}, 0.1]$ for $k \leq 3$ (lower end 10^{-4} to $8 \cdot 10^{-3}$), 6 values in $[0.01, 0.2]$, $[0.03, 0.3]$, $[0.08, 0.5]$, $[0.15, 0.6]$, $[0.2, 0.6]$ for $k = 4, 5, 6, 7, 8$	5,682
`twocomp_balanced`	the 24 cases with $k \geq 4$ (small), balanced initialization	1,008
`twocomp_chi4`	the same 24 cases, fan-in, $\chi = 4$	1,005
`robust`	MPS4, binary4, HT4 (6 cases with $k \leq 4$): $(\chi, d) \in \{(4,2), (2,4), (4,4), (8,4)\}$, weights $(1, 0.25)$, $(1, 0.75)$; MPS4 $\{1, 2\}$ also from the saddle	1,840
`k5_extension`	MPS6 $\{1, 3, 5\}$, MPS8 $\{1, 5\}$, saddle and balanced, $\sigma \in \{0.012, 0.019, 0.03\}$	96
`khigh_extension`	binary8 $\{0, 2, 4\}$, HT4 $\{0, 1, 2\}$, HT4 all sites, MPS8 $\{1, 6\}$; balanced, 8 values of σ over one decade	256
`lambda2_K`	MPS4, MPS6, MPS8 with $k = 2$, weights $(2, 1)$, saddle, 7 values of σ in $[10^{-4}, 0.1]$	168
`multicomp`	TA, TB, N4, B4 on MPS4, binary4, HT4 (HT4 also balanced); E8 and TC on MPS8 and binary8; TD on MPS8; small, 6 values of σ	1,008
`multicomp_wide`	B4, N4, TA, TB on MPS4 (7–9 values of σ down to 10^{-5}); TC, TD on MPS8 from the saddle (7 values in $[10^{-4}, 0.1]$)	384
`stall_diagnosis`	22 (case, σ) pairs with stalls and escaping controls, longer patience	170
total		11,617

Analysis. Run classes: greedy, other order, rotation, stalled, timeout, incomplete, error. Fits as in Section 4.4 (per seed, at least three σ; rotations excluded; the log law is fitted on the residuals of $\ln \Delta t_2$, like the power law). Several components: plateaus detected by `ttn.diagnostics.find_plateaus` on $L(t)$ (linear time, separation ratio above 3), matched to a predicted level within 5%; laws of levels too short to detect from the time spent within 2% of the level. Bond events: effective ranks just after the event ($a_2 = 0.9$) against the plateau midpoint, threshold 0.05 times the teacher's

singular value (0.03, 0.1 as sensitivity); rank profiles of several components in the middle of the time spent within 2% of each predicted level. Growth exponent: least squares of $\ln \dot{a}_2$ against $\ln a_2$ over $a_2 \in [10^{-4}, 10^{-2}]$.

Accuracy. The conservation drift $\max_e \|Q_e(T) - Q_e(0)\|_{\max}$ is at most 10^{-6} in every run (99.9% below $3.5 \cdot 10^{-8}$, median $1.2 \cdot 10^{-9}$), against Gram entries of order one after learning. Relative to the initial imbalance $|Q_e(0)| \sim \sigma^2$ it is at most $1.6 \cdot 10^{-4}$ in every run with $k \geq 4$, where the imbalance matters (Section 4.8), but up to 0.5 for $k \leq 3$ at $\sigma \leq 10^{-3}$, where σ^2 is below the absolute tolerance; those runs never stall, and their gaps agree with an independent integration to 10^{-7}–10^{-9}. LSODA (tolerances 10^{-9} and 10^{-10}), Radau and RK45 agree on a $k = 6$ gap to $3 \cdot 10^{-8}$. A reviewer's independent code (dense contraction, automatic differentiation, Radau at relative tolerance 10^{-10}) reproduced the gaps of 11 rows to 10^{-7}–10^{-9}, the gauge-projected $\lambda_{\min}$ to $7 \cdot 10^{-11}$ and the Steiner counts of all 60 (tree, D) pairs.

Excluded runs. One `lambda2_K` run (MPS8, seed 7, $\sigma = 10^{-4}$) was lost to the runner's start-up check, which compared two implementations of the velocity at the exact saddle without an absolute floor. Six slow runs did not finish within their time limit; they belong to rows reported as not measured ($k = 8$) or as spanning 0.6 decades (HT8, $k = 7$). Round-1 tables (rows on σ grids outside the committed configurations, rotation runs pooled, Hessians with the biased projection, a bond-rank statistic taken at the end of the run) are superseded. The raw run records and trajectories (every accepted integration step of every run, 8 GB) are not in the repository; Section C says how to regenerate them.

Local pilot. The pilot in `experiments/pilot_h2_locality/` (README; code `h2sim.py`, `run_sweep.py` and `analyze.py`) is an independent implementation of the same model, used as a cross-check in Section 4.4. $d = \chi = 2$; MPS4, binary4, MPS6, binary8; teacher seed 1000, weights $(1, 0.5)$; $D = \{0, 1\}$, $\{0, N-1\}$, all sites, $\{0, 2\}$, $\{0, 3\}$, $\{0, 4\}$; fan-in initialization with seeds 0–7 ($N = 4$; $\sigma \in [10^{-3}, 0.1]$ on MPS4, $[2 \cdot 10^{-4}, 0.05]$ on binary4) and 0–5 (MPS6, $\sigma \in [0.05, 0.3]$; binary8, $[0.07, 0.3]$): 784 runs, 19 of them rerun with a 100 times longer horizon, plus 340 runs from the saddle (seeds 0–3, five σ). Gradient flow by LSODA with the exact Hessian (relative tolerance 10^{-10}, absolute $10^{-13}\sigma$); events are first crossings of $a_r = \frac{1}{2}$, located by root finding; t_1 and t_2 agree with LSODA at 10^{-12}, Radau, BDF, DOP853 and RK45 to $1.8 \cdot 10^{-7}$ relative (47 comparisons). A run *collapses* when, at $t \geq 30\, t_1$, the largest second bond singular value is below 10^{-9} of its value after the first escape and still falling, or at roundoff; horizons $10^5 \sigma^{-(K-2)}$ ($N = 4$) and $10^4 \sigma^{-(K-2)}$. Fits per seed, mean $\pm$ s.d. over seeds; gap fits for $k \geq 3$ use runs with a distinct intermediate plateau. A first sweep, made before the teacher and initialization streams were separated, was discarded.

B.4 Order of learning

Settings of Section 4.7. Experiment folder `experiments/order/` (README with every command and the run order; one YAML configuration per sweep in `configs/`); findings in `notes/order.md`, whose sections 0 and 6–11 were revised after the review `notes/reviews/order_round1.md` (section 11 lists the changed claims); verification of the revision in `notes/reviews/order_round2_verification.md`, whose errata E1–E10 are applied in Section 4.7; running report `agent-runs/order/REPORT.md`.

Problems. Whitened Gaussian inputs, exact population loss, double precision. Teachers with weights $(1, \varrho)$, or $(1, \varrho, \ldots, \varrho^{R-1})$ with $R = 3, 4$: odeco, or pattern teachers whose first component has the first basis vector at every site and whose second differs from it on the sites D (MPS6: $D = \{0, 5\}$, $\{2, 3\}$, $\{0, 1, 2\}$, $\{0, 1, 2, 3\}$; MPS4: $\{0, 2\}$, $\{1, 2\}$). All teachers use the standard site basis: the fan-in initialization is invariant in law under an orthogonal rotation of any physical leg, so this changes no order statistic, and no teacher randomness enters. Trees binary4, MPS4, MPS6, binary8 and MPS8 ($K = 3, 4, 6, 7, 8$); $d \in \{2, 4\}$; nominal $\chi \in \{2, 4, 8\}$ capped by (3) (28 distinct settings), or uncapped $\chi = 1$–32. Fan-in initialization (9) with seeds $0, \ldots, n-1$, the same for every ϱ and σ; the d test uses the disjoint ranges from 200000 ($d = 2$) and 300000 ($d = 4$), the exchangeability check the range from 100000.

Dynamics. *First events* (`sweep.py`): gradient flow by SciPy's RK45 (Dormand–Prince 5(4)) with relative and absolute tolerances 10^{-9} and 10^{-14}, stopped by event location at the first crossing of $a_r = \frac{1}{2}$ by any component; 1,105 MPS8 runs of the main sweep that reached the float64 resolution of t were re-integrated with a restarted clock (`repair.py`; no winner changed). *Leading-order race* (`common.lo_blowup`): for each component r, the flow of $\langle W, U_r \rangle$ from the initialization at $\sigma = 1$ until $\|\theta\| = 10^4 \|\theta_0\|$, plus the radial remainder of the blow-up time; the predicted winner is $\arg\min_r \hat{t}_{1,r}/s_r$. *Full runs* (`h1b.py`): LSODA with the exact Hessian as Jacobian, tolerances 10^{-10} and 10^{-15}, dense output, event times by root finding on it and a fine record window around the events; a run stops below $10^{-10} L(0)$, at $t_{\max} = 10^9$ (10^{10} for the MPS6 and binary8 saddle starts, 10^{12} for the natural MPS4 runs), or when some component's slice of some core falls to 10^{-14} times the largest core norm (*stalled*). The threshold is justified empirically: in the 22 runs without any stall stop, every learned component kept a slice ratio of at least $6.7 \cdot 10^{-6}$ before its event, while 25 of the 29 unlearned ones fell below 10^{-14} (17 to an exact zero by underflow), so the rule had no false positive; the slices keep decaying rather than stopping at a roundoff floor (verification review). The Hessian is computed at the plateau midpoint (minimum parameter speed between the two events), raw and gauge-projected (`ttn.diagnostics`, relative cutoff 10^{-3}); at $\sigma = 10^{-2}$ the projection is biased for the adjacent MPS4 pair (-0.58 to -0.80 against the raw -0.71 to -0.95), so raw values are quoted. In the natural adjacent runs 57 of 150 wrong plateaus are near-ties without a resolved midpoint; no claim uses their Hessians. *Several components* (`multi.py`): LSODA with the exact Hessian, tolerances 10^{-9} and 10^{-14}, exact location of every first crossing; a run stops below $10^{-8} L(0)$ (converged), at $t = 10^9$ (unfinished) or by the slice criterion (stalled). The round-1 run `multi` (raw folder `multi2`) stopped instead when an overlap fell below 10^{-25}, which can also fire while a component is forgotten, and flagged runs with an overlap below 10^{-14} as seeded by roundoff, which wrongly excluded 18–19 binary8 runs per ϱ. The three-component rerun `multi3` reproduces it run for run (same status and event times) and changes only that flag; `multi` is used only for the four-component counts, which `multi_nostop` checks without any stall stop.

Sweeps. Table 14.

Analysis. P with Wilson 95% intervals; power laws $P = C\sigma^\gamma$ by binomial maximum likelihood over the whole measured range of σ, with standard errors from a bootstrap over seeds (the seeds are shared across σ), local slopes between consecutive values of σ and fits over $\sigma \leq 0.01$ as checks (`analyze_sigma.py`); escape times from the saddle fitted per seed against $\ln(1/\sigma)$ and as a power law, compared by their residuals (`analyze_h1b.py`); three components compared with the race model on the same runs, with $z = (P_{\mathrm{GF}} - P_{\mathrm{model}})/\mathrm{s.e.}$, and the excluded runs reported with their own race orders (`analyze_multi.py`). The verification review recomputed the power laws (and

Table 14: Sweeps of the order experiment (Section 4.7), one configuration file each in `experiments/order/configs/` (the `*` rows group several). GF: gradient-flow runs; LO: leading-order race only.

sweep	content	runs
`main`	binary4, MPS4, MPS6, binary8, MPS8; $d \in \{2,4\}$, $\chi \in \{2,4,8\}$ (28 settings); $\varrho \in \{0.5, 0.7, 0.8, 0.9, 0.95, 0.99\}$; $\sigma = 0.01$; seeds 0–399; LO race and initialization statistics per seed	67,200 GF
`b4_sigma`	binary4, $\chi \in \{2,4\}$, $\varrho \in \{0.5, 0.7, 0.8, 0.9, 0.95\}$, 7 values of $\sigma \in [10^{-4}, 0.1]$, seeds 0–999	70,000 GF
`m4_sigma`, `b8_sigma`	MPS4 (5 values of $\sigma \in [0.003, 0.3]$) and binary8 (4 values in $[0.01, 0.3]$), $\chi = 4$, $\varrho \in \{0.5, 0.8, 0.95\}$, seeds 0–399	10,800 GF
`pat_mps6_*`	MPS6 pattern pairs (D above), $\chi = 4$, $\varrho \in \{0.7, 0.8, 0.9\}$, $\sigma \in \{0.003, 0.01, 0.03, 0.1\}$, seeds 0–399	19,200 GF
`pat_mps4_sep`, `_adj`	MPS4, $D = \{0,2\}$ and $\{1,2\}$, $\chi = 4$, $\varrho \in \{0.8, 0.9\}$, $\sigma \in \{10^{-5}, \ldots, 10^{-1}\}$ (5 values), seeds 0–999	20,000 GF
`d_test_*`	binary8 (all bonds 4) and MPS4 (all bonds 2), $d = 2$ and 4, $\varrho \in \{0.5, 0.8\}$, $\sigma = 0.01$, 1,000 seeds each	8,000 GF
`chi_scan`, `chi_gf`	MPS4, MPS6, MPS8, binary8, uncapped $\chi \in \{1, 2, 4, 8, 16, 32\}$, 500 initializations; GF check on MPS4 and binary8 at $\chi = 16$, 32, $\varrho \in \{0.8, 0.9, 0.95\}$, $\sigma = 0.01$, 400 seeds	12,000 LO; 4,800 GF
`symmetry_check`	MPS4 and MPS8, $d = 4$, $\chi = 2$, seeds 100000–101499	3,000 LO
`h1b_saddle_*`	saddle starts, 6 values of $\sigma \in [10^{-7}, 10^{-2}]$, seeds 0–7: binary4 ($\varrho \in \{0.5, 0.7, 0.8, 0.9\}$), MPS4 $D = \{0,2\}$ and $\{1,2\}$ ($\varrho \in \{0.5, 0.8, 0.9\}$); MPS4 odeco ($\varrho \in \{0.5, 0.8\}$, 5 values of $\sigma \in [0.01, 0.1]$); MPS6 and binary8 odeco ($\varrho = 0.8$, 5 values in $[0.05, 0.2]$ and $[0.15, 0.35]$, seeds 0–5)	619 GF
`h1b_natural_*`	wrong-order seeds of `b4_sigma` ($\chi = 4$, $\varrho \in \{0.7, 0.8, 0.9\}$, $\sigma \in \{10^{-4}, 10^{-3}, 10^{-2}, 0.1\}$), of `pat_mps4_sep` and `_adj` ($\varrho \in \{0.8, 0.9\}$, $\sigma \in \{10^{-4}, 10^{-3}, 10^{-2}\}$) and of `main` (MPS4, $d = 2$, $\chi = 4$, $\varrho \in \{0.5, 0.8\}$); at most 40 (20 for `main`) per (ϱ, σ)	851 GF
`multi3`	three components, $\varrho \in \{0.8, 0.9\}$, $d = \chi = 4$; $\sigma = 0.05$ (binary4), 0.1 (MPS4), 0.2 (MPS6), 0.25 (binary8); seeds 0–199	1,597 GF
`multi` (`multi2`)	as `multi3`, with three and four components and the overlap stall criterion	3,190 GF
`multi_nostop`	four components, $\varrho = 0.8$, MPS6 and binary8, no stall stop, $t \leq 10^9$, seeds 0–11	22 GF

fits over the three smallest σ) and the three-component statistics from the raw runs with its own code, and added a paired comparison of each run's order with its own LO race (McNemar test), which Section 4.7 quotes. Figures 9 and 10 come from `plot_report.py`, from committed tables only (Section C).

Accuracy. Tightening the RK45 tolerance to 10^{-11} leaves the winner unchanged in 500 of 500 initializations (100 per tree, $\varrho = 0.8$, $\sigma = 0.01$) and moves t_1 by at most $1.0 \cdot 10^{-9}$ relative; the conservation drift at the event is below $3.5 \cdot 10^{-11}$ (`tol_check.py`). The LO escape direction is balanced: at $\|\theta\| = 10^4 \|\theta_0\|$ every core holds $1/K$ of $\|\theta\|^2$ to within $5.0 \cdot 10^{-5}$, and the per-bond imbalance relative to $\|\theta\|^2$ is below $3.3 \cdot 10^{-5}$ (five trees, five seeds each; `balance_check.py`). The review reproduced every number it recomputed, seed by seed, with the committed code and with independent code (numpy contractions, DOP853, its own LO blow-up times and exact Hessians): P on MPS4 at six values of ϱ (400 seeds), the binary4 wrong-order counts in every recomputed (ϱ, σ) cell, the race agreement, and the wrong-saddle eigenvalues.

Excluded runs. Missing from the tables: 3 of 1,600 `multi3` runs and 2 of 24 `multi_nostop` runs, still integrating after 1.5 h; 10 of 3,200 `multi` runs; and one binary8 saddle start ($\sigma = 0.3$, seed 0), stopped after 80 minutes, after which the table was rebuilt from the 29 stored trajectories (`rebuild_h1b_csv.py`). The raw outputs (JSON lines per sweep and `.npz` trajectories, 346 MB) are not in the repository; Section C says how to regenerate them.

B.5 Regime-map experiments

Settings of Section 3 and Sections 5 and 5.1 to 5.3. Experiment folder `experiments/regime_map/` (README with the run order; YAML configurations in `configs/`); findings in `notes/regime_map.md`, whose section 3 (fix round 1) overrides its sections 0–2; review in `notes/reviews/regime_map_round1.md`; theory in `notes/theory_regimes.md` and Section A.8. Every number of Section 3 and of Sections 5 and 5.1 to 5.3 is an entry of the committed tables in `results/` (per-seed boundaries), `results/fix1/` (fix round 1) and `results/report/` (written from the raw runs by `report_tables.py`), or an aggregate of them in `results/report/numbers.csv` (`report_numbers.py`).

Problems. Whitened Gaussian inputs, exact population loss, double precision, the library model (`ttn.dynamics.Problem`). Odeco teachers with weights $(3, 2, 1)$ at $d = 3$ or $(3, 2)$ at $d = 2$, Haar-random site bases from teacher seed $1000 + s$ for initialization seed s (an independent stream); fan-in initialization (9). Bond dimensions: binary4 uncapped ($\chi = 3$–128); the other trees capped by (3), except the variant of the binary tree with $N = 8$ with bonds 8 above two sites and 32 above four. Models: *centred*, target $W^* + W_0$ through the cancellation-free shifted teacher, or, for $\alpha < 10^{-2}$, through the direct loss $\frac{1}{2}\|W - W^* - W_0\|_F^2$ (cancellation error about $10^{-16}\alpha^2$; on two configurations both forms gave identical labels, event times to 10^{-4} and kernel changes to six digits); *plain*; *two-copy* (`antisymmetric_double` with the gradient mask of Remark 2.3). Imbalance: root scale $\lambda\sigma$ at fixed $\alpha = \lambda\sigma^3$. Large χ: the three parameterizations of Section 5.3 as initial scales and multipliers η_v.

Dynamics. Gradient flow (`ttn.dynamics.integrate_gf`) by Dormand–Prince 5(4) with tolerances 10^{-8} and 10^{-12}, or LSODA with the exact Hessian as Jacobian for long plateaus (the $\sigma \times K$ sweeps, the stall test, and $\sigma < 0.7$ in the sweeps of the binary tree with $N = 8$). A run stops when $L < 10^{-6} \cdot \frac{1}{2}\|W^*\|_F^2$, at the time budget $T = 50\, t_{\mathrm{esc}} + 10^4/\max(\bar{\lambda}, 1)$, with $t_{\mathrm{esc}} = (0.3\sigma)^{-(K-2)}/((K-2)s_{\min})$ the aligned escape time of the weakest component and $\bar{\lambda} = \operatorname{tr}\Theta(0)/d^N$, or at a step limit ($10^5$ steps by default; $3 \cdot 10^4$ or $2 \cdot 10^4$ in the $N = 8$ and $\sigma \times K$ sweeps, 10^4 for $K \geq 10$). Each final rise is re-integrated from the saved parameters on a 200-point linear grid (`run_regime.refine_events`); the event time is the last upward crossing of $a_r = \frac{1}{2}$ before a_r first reaches 0.9, the width the final rise from 0.1 to 0.9. Runs written before this rule are relabelled from their recorded log and fine rows (`analysis.relabel_last_crossing`). Deep in the saddle-to-saddle regime at $K \geq 7$ the widths are not resolved; separation ratios there are lower bounds, and no label depends on them.

Labels, kernels and boundaries. Labels of Table 4 with convergence measured against $\frac{1}{2}\|W^*\|_F^2$ (Remark 2.6), at the nine threshold pairs $\delta_* \in \{0.05, 0.1, 0.2\}$, $\rho \in \{2, 3, 5\}$; a run that ends on a floor below 10^{-3} counts as converged. $\delta_\Theta^{\mathrm{probe}}$ uses 128 Gaussian probe inputs (generator seed 12345) and the weights $\mathrm{diag}(\boldsymbol{\eta})$ (multipliers times mask); δ_Θ uses the tensor-space kernel $J\,\mathrm{diag}(\boldsymbol{\eta})J^\top$, computed in the run when $d^N \dim\theta \leq 2 \cdot 10^7$, and for earlier runs from their final parameters by `fix_round1.py` (`tensor`). Per-seed boundaries: `analysis.seed_rho_boundary` (log-log interpolation of the crossing of $\min_i \rho_i$), `seed_s2s_boundary` (label brackets; status "unresolved" when

the next run is stalled) and `seed_kernel_boundary` (crossing of δ_* among converged runs); slopes against χ are pooled least-squares fits with standard errors from the scatter, checked by per-seed fits. Stall types come from the R-th bond singular value of W at the end, minimized over edges (`analysis.stall_type`); most centred runs recorded only the spectra of $W - W_0$ and are not typed.

Sweeps and run counts. Table 15; six seeds (0–5) everywhere.

Table 15: Sweeps of the regime-map experiment, one configuration each in `experiments/regime_map/configs/` (`stall_test` has two); runs after de-duplication, from `results/report/inventory.csv`.

sweep	content	runs
`sigma_chi_b4`	binary4, $d = 3$; centred, plain, two-copy; $\chi = 3, 4, 6, 9, 12, 18, 24, 36$; 19 values of $\sigma \in [0.01, 40]$	2,736
`chi_b4`	binary4; centred, χ = 9–128 (9 values), 22 values of $\sigma \in [0.004, 4]$; two-copy, χ = 16–128, $\sigma \in [0.25, 4]$	1,428
`sigma_chi_b8`	binary8, $d = 3$, centred and plain, $\chi = 3, 6, 24$, $\sigma \in [0.3, 4]$	227
`sigma_chi_b8_d2`	binary8, $d = 2$, centred, $\chi = 2, 4, 8, 16$ and bonds 8/32, $\sigma \in [0.3, 10]$	375
`sigma_K_centred_d2`	$d = 2$, centred, $\chi = 4$ on binary4, 8, 16 and MPS4, 5, 6, 8, 10, 12; full bonds on MPS6, binary8, MPS8; σ from $\max(10^{-8/K}, 0.02)$ to 2	827
`sigma_K_d2`, `sigma_K_full_d2`	the same trees, plain, $\chi = 2$ and 4; full bonds on MPS6, binary8, MPS8	1,293, 220
`sigma_K`	$d = 3$, plain, χ = 3, 6, binary8 and 16, MPS8, 10, 12	302
`imbalance_b4`	binary4, centred; $\chi = 24$ and 3: 9 values of α times 10 of λ; $\chi = 16$, $\alpha = 3.375$, 9 values of λ	1,134
`large_chi_b4`	binary4, centred and plain, χ = 4–128, SP, NTK, μP-TTN, σ = 1, 1.5	432
`large_chi_b8`	binary8, $d = 2$, plain, χ = 4–32, $\sigma = 1$ (not interpreted)	71
`stall_test`	binary8 and MPS8, $d = 3$, $\chi = 3$, σ = 0.3, 0.45, 0.6; fan-in and balanced; plain and centred	120
total		9,165

Checks. 1,323 runs regenerated locally with the final driver from the committed configurations (the binary4 saddle-to-saddle side and lazy windows, large χ at $\sigma = 1.5$, $\sigma \times K$ runs above the brackets) give the labels of the remote runs in every case, with median relative differences of $\min_i \rho_i$ at most $2.5 \cdot 10^{-4}$, of $\delta_\Theta^{\text{probe}}$ at most $2 \cdot 10^{-4}$ and of δ_Θ at most $7 \cdot 10^{-7}$ per sweep (`results/fix1/validation_local_vs_remote.csv`). The fixed analysis applied with the first-crossing rule reproduces the per-seed tables committed before the fix round to $5 \cdot 10^{-15}$. The conservation drift (16) has a median of 10^{-12}–10^{-9} per sweep and a maximum of $9.5 \cdot 10^{-7}$. Of the 2,736 runs of `sigma_chi_b4`, 317 stopped at the step limit, all at $\sigma \geq 4$: 283 are labelled stalled and 34 had converged.

Excluded and superseded runs. Pilot runs made before the teacher and initialization streams were separated were discarded. A first map of binary4 with SciPy's RK45 on a logarithmic output grid (`b4_sigma_chi`) is superseded: its widths, read off that grid, were too coarse. Runs unfinished when the remote machine was stopped are missing from the tables (13 of 240 in `sigma_chi_b8`, 15 of 390 in `sigma_chi_b8_d2`, 85 of 912 in `sigma_K_centred_d2`, among them the full-bond binary8 and MPS8 runs at large σ and binary16 at σ = 0.76–0.92; the $\chi = 6$ variants of the stall test); every count above is of finished runs. Compute: one remote machine with 384 vCPUs for 6.2 hours (`notes/spend.md`). The raw records (one row and one trajectory per run, 1.2 GB; local copy `data/`

`regime_map/`, with the fix-round reruns and the tensor-kernel cache in `data/regime_map_local/`) are not in the repository; Section C says how to regenerate them.

B.6 Covariance and teacher experiments

Settings of Section 6. Experiment folder `experiments/data_teacher/` (README with the run order; YAML configurations in `configs/`; the star and correlated-input sweeps are defined in `r2_star.py` and `r2_corr.py`); findings in `notes/data_teacher.md`; reviews in `notes/reviews/` (`data_teacher_round1.md`, `data_teacher_round2.md` and `data_teacher_round3_verification.md`, whose errata are applied in Section 6); theory in Sections A.5 and A.6.

Problems. Exact population loss, double precision, fan-in initialization (9); teacher seeds from a stream independent of the initialization seeds. Site covariances `ttn.losses.power_law_cov`(d, β) (eigenvalues $\propto k^{-\beta}$, trace d), rotated by θ in the (e_1, e_2) plane where misaligned.

- *Aligned covariance* ($N = 4$, $d = \chi = 4$): odeco teachers on covariance eigenvectors, T1 $(1, 0.8)$ and T2 $(1, 0.6)$ with the smaller component on e_1, $\beta = 0.25$; one-component teachers on e_1 or e_4 with $\beta = 0$–1.5 ($\Lambda = 0.008$–33); three components $(1, 0.7, 0.5)$ for the exact-clock check. Variants from the same initialization: (Σ, s), (I, s), $(I, \Lambda s)$, $(I, \Lambda^{1/2} s)$, and $(I, \Lambda^{1/2} s)$ from the mapped initialization (physical legs multiplied by $\Sigma_i^{1/2}$).
- *Round 1, generic teachers* ($N = 4$, $d = \chi = 4$): misaligned covariance with an odeco teacher on e_1, e_2 (weights $(1, 0.8)$), $\beta \in \{0.5, 1\}$, $\theta \in \{0, 10, 20, 30, 45\}^\circ$, also as the absorbed white problem from the mapped initialization; non-orthogonal CP teachers $u_2^{(i)} = ce_1 + (1 - c^2)^{1/2} e_2$, weights $(1, s_2)$; correlated inputs (one-factor model, odeco teacher on e_1, e_2, weights $(1, 0.7)$, $\rho = 0$–0.5), with the dense second-moment matrix from Isserlis' theorem (`correlated.py`, checked against Monte Carlo and the closed form for $N = 2$).
- *Random TTN teachers*: MPS4 teachers with $d = 3$, bond ranks $(2, 3, 2)$, bond index k weighted by 0.5^k, $\|W^*\|_F = 1$, teacher seeds 101–105, MPS4 student with $\chi = 4$; cross-tree: MPS8 teachers ($d = 2$, bond ranks 2, index weights 0.6^k, $\|W^*\|_F = 1$) on a binary8 student ($\chi = 4$), teacher seeds 201–203, then 203, 204, 209, 212, 218, and 491, 548, 643, 694, 846 (the five of seeds 200–999 whose root cut exceeds every bottom bond by the largest margins); binary8 teachers (seeds 201–203) on an MPS8 student.
- *Round 2, generic teachers* ($d = 2$, $\chi = 4$, Table 27): CP $c = 0.2$, 0.45 ($N = 4$) and 0.45, 0.7 ($N = 8$), weights $(1, 0.7)$; misaligned $\theta = 20^\circ$ ($\beta = 1$ at $N = 4$, $\beta = 2$ at $N = 8$) and 45° ($\beta = 2$, $N = 8$); random binary-tree teachers (seeds 302, 303) and random MPS teachers (seed 301 at $N = 4$, 303 at $N = 8$), bond ranks 2, bond index k weighted by q^k ($q = 1$ and 1.5), $\|W^*\|_F = 1$. Correlated inputs: $d = 3$, site-aligned odeco teacher $(1, 0.6)$, $\rho = 0$–0.4.
- *Stars* ($d = 2$, $\chi = 3$): $q = 0000$ (weight 2), $r = 1010$, $r' = 1001$ (weight w); the binary8 analogue has r, r' differing from $q = 0^8$ on sites $\{0, 2\}$ and $\{0, 4\}$.

Dynamics.

- Round-1 runs at a single σ (and the aligned-covariance sweeps): SciPy DOP853 on the flat parameter vector, tolerances 10^{-9} and 10^{-14}, 2000 logarithmic output times; checked against the library integrator (event times and losses to $7 \cdot 10^{-7}$), conservation drift at most $1.1 \cdot 10^{-9}$.

- σ sweeps: `ttn.dynamics.integrate_gf` (LSODA, tolerances 10^{-9} and 10^{-14}) in two stages: until $L < 0.98L(0)$, then on 3000 times geometric in $t - t_A$, with the recorder's fine rows; stopped at $L < 10^{-6}$–$10^{-4}L(0)$ or at $t_{\max}$ (up to 10^{10}).
- Saddle start: the lowest-loss bond-rank-one critical point (L-BFGS from 12 starts on the rank-constrained problem, balanced, zero-padded), stored in `results/r2_saddle_cores.pt`, since gauge copies with losses one ulp apart shift per-run times on another platform (not the laws); fan-in noise of scale ε; LSODA in chunks of doubling length until $\max_e s_2^e \geq 0.2$ or $t = 10^7$ (10^{11} for the small-ε extension; $\max_e s_2^e \geq 2 \cdot 10^{-3}$ for the re-timed misaligned teacher). Hessians at the exact saddle, raw and gauge-projected; saddle type by the sign of $\lambda_{\min}$ (threshold -10^{-6}); per-edge rates γ_e from the Hessian block of the two cores of the edge.
- Correlated inputs: SciPy LSODA with the exact Hessian as Jacobian, on the dense loss, in the two stages above. Online SGD check: fresh Gaussian batches, $B = 256$ and 4096, $\eta = 0.01$ and 0.002.

Events and fits.

- Random initialization: t_1 is the first time $L \leq \frac{1}{2}(L(0) + L_1)$, L_1 the level of the rank-one critical point visited (matched within 5% at the point of minimum speed; runs with no such point are *unresolved* and left out); Δt_2 is the first onset event after t_1 minus t_1. Onset event: $s_2^e \geq f\, s_2^{e,*}$ on some edge, $f \in \{0.03, 0.05, 0.1\}$ (and, from the saddle, at least 10 times its start value); event times from merged log and fine rows (within 0.5% of fine-row times). The re-timed misaligned teacher uses $\max_e s_2^e \geq 1.8 \cdot 10^{-4}$, where $s_2^e(0) \leq 1.8 \cdot 10^{-5}$.
- Correlated inputs: events at half the final singular values across 01|23, final-rise widths; Δt_2 is the time between the events of s_2^e and s_1^e. Stars: θ_q is left when $a_q < 0.99$ (departure) or $a_q < 0.9$ before any $a_r > \frac{1}{2}$; the two rules agree for every run with $\varepsilon \leq 10^{-3}$.
- Fits per seed (`lawfit.py`): $t = A \ln(1/\sigma) + B$ and $t \propto \sigma^{-p}$, both scored in log space; asymptotic fits through the three smallest σ (random initialization) or the five smallest ε (saddle start). Errors are standard errors over seeds; per-seed fit errors are not propagated, and changing the window moves slopes by 1–3%. Increments per halving: per-seed regression of Δt_2 on $\log_2(1/\sigma)$ with the actual σ.
- Rank-one predictions: `constrained.py` (L-BFGS on the bond-rank-constrained loss from many starts) and `predict_rank1.py`; slopes h_e from `theory_log_slope.py` and the exact-saddle Hessians; heuristic exponents α from `p1_exponents.py`.

Sweeps and run counts. Table 16; about 9,100 runs in all.

Checks and exclusions. A float64-floor scan (`check_float_floor.py`) takes the smallest dormant signal between the first escape and the next learning event (overlap envelope below 10^{-12}, or $\max_e s_2^e$ below 10^{-14} for TTN teachers): it flags only the odeco baselines ($c = 0$, $\theta = 0$) at $\sigma \leq 3 \cdot 10^{-3}$, whose exponents without the flagged runs are in `results/report_float_floor_refit.csv`. Excluded: the round-1 single-σ plateau durations (withdrawn), the pilot versions of `e4b` and `e4c`, seed 2 of MPS4 under correlated inputs (minor saddle at every ρ), and 14 HT8 runs at $\varepsilon \leq 3 \cdot 10^{-4}$ (CP, $c = 0.45$) stopped beyond $t = 10^8$. An independent verification recomputed the Hessians at 12 stored saddles with its own code and reproduced the committed saddle-start escape times on Windows to $4 \cdot 10^{-9}$ (MPS4) and $4 \cdot 10^{-5}$ (binary4). Raw outputs (about 10 GB, local copy

Table 16: Sweeps of the experiment of Section 6: configuration files in `configs/` of the experiment, except where a script is named.

sweep	problem	σ or ε, seeds	runs
`e1_order`	aligned, T1, T2, four variants, binary4, MPS4	$\sigma = 0.02$–0.2 (3 per tree), 40	1920
`e1_scaling_sigma`, `_lambda`	aligned, t_1 against σ and Λ	6 σ; 13 teachers; 10	360 + 520
`e2_misaligned`, `e4a_nonorth`, `e3_correlated`	round 1, one σ per tree	0.05, 0.1; 20	800 + 560 + 240
`e4a_sigma`, `e2_sigma`	MPS4, $c \in \{0, 0.05, 0.2, 0.45\}$; $\theta \in \{0, 5, 20\}^\circ$	9 σ, $3 \cdot 10^{-4}$–0.3; 10	360 + 270
`e4b_ttn`	random MPS4 teachers	8 σ, 10^{-3}–0.2; 6	240
`e4c_cross`, `r2_cross`, `r2_cross_b`	cross-tree, $N = 8$	0.3–0.5; 8	48 + 120 + 120
`r2_random`	6 generic teachers, MPS4, binary4	9 σ per tree; 5	540
`r2_saddle` (+`_b`, `_ext`, `_m20`)	13 generic teachers, 6 trees, saddle start	$\varepsilon = 10^{-8}$–$5 \cdot 10^{-2}$; 5	1950 (+54 fine rows)
`r2_corr.py`	correlated, $d = 3$, MPS4, binary4	6 σ, 0.0016–0.05; 5	300
`r2_star.py`	stars: saddle start; random init	$\varepsilon = 10^{-6}$–10^{-2}, 15 w; 6 σ, 10	468 + 240
`e3_sgd_check.py`	online SGD, binary4, $\rho = 0.3$	$\sigma = 0.05$; 3	9

`data/data_teacher/`) are not in the repository; the summary tables in `results/` suffice to redraw every figure.

B.7 Discrete steps, gradient noise and finite samples

Settings of Section 5.4. Experiment folder `experiments/optim_samples/` (README with the script list; YAML configurations in `configs/`); findings in `notes/optim_samples.md`; the two reviews in `notes/reviews/` (`optim_samples_round1.md` and `optim_samples_round2_verification.md`, whose errata are applied in Section 5.4); theory in Section A.7.

Problems. Whitened Gaussian inputs, exact population loss (empirical loss in the finite-sample sweep), double precision. The GD and SGD loops of the experiment are its own (PyTorch), checked against `ttn.dynamics` (final losses agree to 10^{-14}, event times to $5 \cdot 10^{-4}$; `validate_library.py`). Teacher seed 101, independent of the initialization seeds.

- *Main problems*: binary trees with $N = 4$ ($K = 3$) and $N = 8$ ($K = 7$), MPS with $N = 8$ ($K = 8$); $d = 3$, $\chi = 3$; odeco teacher with weights $(3, 2, 1)$ and Haar-random site bases; $\eta_c = 0.1541$, 0.0435, 0.0366. Fan-in initialization (9); *imbalanced*: $\sigma = 0.1$ with the root core at scale 2, 4 or 6. Centred model (Section 2.4, variant (b)) with $\chi = 9$, the cut dimension of binary $N = 4$.
- *Aligned starts* (`ttn.network.aligned_init`): every per-core amplitude 0.05 (SGD sweep); component 1 at its balanced value $3^{1/3}$ and components 2 and 3 at b_0 (route test); the balanced minimum plus a Gaussian perturbation of size 10^{-5} per entry (SGD stability test).
- *Strict route*: MPS with $N = 4$, $d = \chi = 3$, pattern teacher with weights $(3, 0.3)$ and patterns 0000, 1100 (second component needs $k = 2$), $\sigma = 0.1$.
- *Finite samples*: binary $N = 8$, $d = 2$, odeco teacher $(3, 2)$ (bond rank 2; teacher-rank TTN

with 52 parameters, manifold dimension 28), students with $\chi = 4$ (208 parameters, tangent dimension 112, the measured rank of the Jacobian) and $\chi = 16$ (832 parameters, all bonds full); binary $N = 6$ ($K = 5$), $d = 3$, teacher $(3, 2)$ (manifold dimension 48), $\chi = 4$ (tangent dimension 120). n inputs with data seed 500+seed and noise-free labels; seed s fixes both the data and the initialization.

Dynamics.

- *Gradient-flow references*: SciPy DOP853 with relative and absolute tolerances 10^{-10} and 10^{-14}, instead of the default integrator of Section 2.5; every upward crossing of $a_r \in \{0.1, 0.5, 0.9\}$ is located by the integrator's event finder; conservation drift (16) at most $2 \cdot 10^{-12}$. In the $\chi = 9$ sweep: LSODA (`ttn.dynamics.integrate_gf`, tolerances 10^{-8} and 10^{-12}, 3000 logged times), stopped at $L = 10^{-8}$; λ_{GF} is the top eigenvalue of the full Hessian at the end. Route test and strict route: LSODA as well.
- *GD*: η a multiple of η_c, or in the $\chi = 9$ sweep of each run's own $2/\lambda_{\mathrm{GF}}$, for a gradient-flow time T: 2800, 900, 300, 100, 40 for $\sigma = 0.003$, 0.01, 0.03, 0.1, 0.3 on binary $N = 4$, 150 (binary $N = 8$), 60 (MPS), 100 (imbalanced), and as long as the reference took to converge ($\chi = 9$). Loss and overlaps at every step; sharpness (full Hessian), conservation drift and bond ranks at 41 checkpoints; diverged: loss non-finite or above $10^6 L_0$.
- *Event shift*: c_e from the variational equation $\dot{\delta} = -H(\delta + \frac{1}{2}\nabla L)$, $\delta(0) = 0$, integrated along the reference path, $c_e = -\nabla a_r \cdot \delta / \dot{a}_r$ at the event (`predict_shift.py`); the Richardson check extrapolates the measured shifts at $0.01\eta_c$ and $0.03\eta_c$ linearly to $\eta = 0$.
- *Online SGD*: fresh Gaussian inputs every step, no label noise (except in the label-noise runs of the strict route, $\zeta \in \{10, 20, 30\}$), an independent generator for every (initialization, noise seed, B, η). Overlaps every 4 steps; runs continue to $1.3 \max(t_1^{\mathrm{GF}}, t_2^{\mathrm{GF}}) + 2$, capped at 150, 60, 40 time units for $\sigma = 0.03$, 0.1, 0.3. Route test: `ttn.dynamics.run_sgd` for up to $4T_{\mathrm{GF}}$, with T_{GF} from the one-dimensional aligned integral (checked against LSODA to 10^{-8}). Stability test: 25 time units; converged if the final loss is below the initial one, diverged if non-finite or above 10^3. Mean-square limits $\eta^*_{\mathrm{ms}}(B)$ from the dense second-moment operator (`sgd_stability.py`), equal to (56).
- *Finite samples*: GD on the empirical loss with step $1/\lambda_{\max}$ (power iteration on the training Hessian every 200 steps), the relative parameter change per step capped at 1% (near the origin $1/\lambda_{\max}$ is huge), $6 \cdot 10^4$ steps ($3 \cdot 10^5$ in the long centred runs); no run reached the training tolerance 10^{-8} (final training loss $2 \cdot 10^{-7}$–$2 \cdot 10^{-4}$ of its initial value for small initialization from $n = 256$). Lazy limit: minimum-norm least squares with the Jacobian at initialization (relative cutoff 10^{-12}), exact.
- Auxiliary computations: Hessians at balanced and gauged minima on five trees ($\chi = 3$, 5; `check_sharpness.py`); the scalar map for $K = 3, 4, 7, 8$, h/h_c from 0.8 to 3.2 in steps of 0.005, $6 \cdot 10^4$ iterations from $x = 0.3$, attractor from the last 2000 (`map1d.py`); Hessians at the second-plateau midpoint of the strict route ($\sigma = 0.01$–0.1, five seeds; `check_k2_saddle.py`).

Sweeps and run counts. Table 17; 4,165 runs in total, plus 150 exact lazy limits.

Events and classification.

- Overlap events, in the GD, gradient-flow and SGD sweeps and the route test: the last upward crossing of $a_r = \frac{1}{2}$, with $a_r \geq \frac{1}{2}$ at the end of the record (`common.record_events`, or the

Table 17: Sweeps of the experiment of Section 5.4: configuration files in the folder `configs/` of the experiment, except the last four rows, which are defined in their scripts. Seeds 0–4 unless stated; GF: gradient-flow references.

sweep	problem, σ	steps, batches, sizes	runs
`gd_sweep`	binary $N = 4$, $\sigma = 0.003$–0.3; binary and MPS $N = 8$, $\sigma = 0.9$; imbalanced	η/η_c: 18 values in $[0.01, 3]$ (10 for $\sigma \le 0.01$; 10 in $[0.1, 1.05]$ imbalanced)	700 + 50 GF
`gd_order`	binary $N = 4$, $\sigma = 0.03$, 0.1, 0.3, seeds 5–44	$\eta/\eta_c = 0.1$, 0.3, 0.5, 0.9	480 + 120 GF
`gd_centred_chi9`	binary $N = 4$, $\chi = 9$: centred $\sigma = 1$–16 (GF), 1, 2, 4, 8 (GD); plain $\sigma = 0.1$, 0.3	$\eta\lambda_{\mathrm{GF}}/2$: 10 values in $[0.5, 4]$	300 + 75 GF
`sgd_sweep`	binary $N = 4$, $\sigma = 0.03$, 0.1, 0.3; aligned start	$B = 1$–256 at $\eta = 5 \cdot 10^{-4}$; $\eta = 10^{-4}$–$2 \cdot 10^{-3}$ at $B = 4$, 10^{-3}–$2 \cdot 10^{-2}$ at $B = 64$; 10 noise seeds per initialization (20 aligned)	1780
`p9_sgd.py`	binary $N = 4$, aligned route, $b_0 = 0.01$, 0.005, 0.003, 0.002	$\eta/B = 3.125 \cdot 10^{-5} \cdot 2^j$, $j = 0, \ldots, 3$, at $B = 64$ (and 16 for $b_0 = 0.01$, 0.003); 6 noise seeds	144
`sgd_limit_test.py`	binary $N = 4$, balanced minimum	$\eta/\eta^*_{\mathrm{ms}} \in [0.5, 3]$ (8 values), $B = 1$, 16	80
`collapse_test.py`, `check_k2_saddle.py`	MPS $N = 4$, strict route	SGD $B = 1$, $\eta = 5 \cdot 10^{-4}$, $\zeta \in \{0, 10, 20, 30\}$; GF at six $\sigma \in [0.01, 0.1]$	21 + 30
`samples_sweep`	binary $N = 8$, $d = 2$: small $\sigma = 0.3$, 0.5, centred $\sigma = 1$ ($\chi = 4$, 16), centred $\sigma = 1.5$, 2 ($\chi = 4$); binary $N = 6$, $d = 3$: small $\sigma = 0.5$	$n = 16$–512 (11 values; 5 for $\sigma \ge 1.5$, and 256, 512 with $3 \cdot 10^5$ steps; 8 values 32–384 for $N = 6$)	385

integrator's event finder for the DOP853 references); first crossings and the number of crossings are kept to count transients; widths are final rises (the first crossing of 0.9 minus the last crossing of 0.1 before it). The strict-route scripts (`collapse_test.py`, `check_k2_saddle.py`) use the first upward crossing of $\frac{1}{2}$ on the logged trajectory. No experiment of this section uses spectral events: bond spectra enter only through effective bond ranks (singular values above 0.1 times the smallest teacher weight, at checkpoints and at the end) and, in the route test, through the second Schmidt value as the route amplitude.

- GD outcome classes, checked in this order (`analyze_gd.py`): *diverged*; *GF-like*: final loss at most $1.05L_{\mathrm{GF}}(T) + 10^{-3}L_0$, with $L_{\mathrm{GF}}(T)$ the reference at the same time, and no step raising the loss by more than $10^{-3}L_0$; *catapult*: the same final loss after such a rise; *thrown back to the origin*: not at that level and the loss back above $0.9L_0$ after the first event; *bounded oscillation*: over the last 20% of the steps (at most 2000) the total variation of the loss exceeds ten times its net change and the tail reaches above gradient flow's level, or the end point is linearly unstable ($\eta\lambda_{\max}/2 > 1.001$); *other*: none of these (it did not occur in the runs shown). GF-like includes 39 of the 195 small-initialization runs that end, like their references, on the last plateau ($L/L_0 = 0.071$).
- SGD: the first escape is the first time any component reaches $\frac{1}{2}$, compared with gradient flow's; a run *forgot* if its first-learned component later falls below $\frac{1}{2}$. Runs are classified before

pooling; means ± s.e. are across initializations. Second plateaus use only initializations with a resolved gradient-flow plateau ($\rho_1 > 3$) and no near-tie (the eventually second component does not peak between 0.15 and 0.6 during the first event), and only runs that kept the first component and gradient flow's order. Spread exponents: per initialization, least squares of $\ln \mathrm{sd}(t_1)$ against $\ln(\eta/B)$. Collapse law: weighted least squares (weights $1/\text{s.e.}^2$) of $T_{\mathrm{SGD}}/T_{\mathrm{GF}}$ over the 19 conditions in which every run escaped, one c_γ per law; *trapped*: no persistent event by $4T_{\mathrm{GF}}$.

- Finite samples: test error $\|W - W^*\|_F^2/\|W^*\|_F^2$; *recovered*: test error below 10^{-2} (thresholds 10^{-3} and 10^{-1} give the same picture); effective bond rank: singular values above 0.2, a tenth of the smallest teacher weight, averaged over the edges; the lazy limit depends only on the initial direction and is counted once per (setting, n, seed).

Excluded runs. Superseded after review round 1 and not analysed: the first GD and SGD sweeps, whose events were first crossings and whose SGD runs stopped right after the second crossing; the first route test, whose $B = 16$ and 64 runs shared noise streams (kept as `p9_sgd_round1.csv`); the $\chi = 3$ centred-model sweep, whose gradient-flow references stalled because $W^* + W_0$ has bond rank up to 6 there; and a first strict-route test on binary $N = 4$, where the pattern lies inside one core ($k = 1$). On MPS $N = 8$, seed 3 never learns component 1 within $T = 60$ under gradient flow and is left out of the oscillation-loss comparison; in the strict-route Hessians two runs at $\sigma = 0.01$ have no resolved plateau midpoint. Raw outputs (about 420 MB, local copy `data/optim_samples/`) are not in the repository; the summary tables in `results/` suffice to redraw every figure (Section C).

B.8 Optimizers used in practice

Settings of Section 5.5. Experiment folder `experiments/optimizers/` (one YAML configuration per sweep in `configs/`); findings and claims withdrawn after review in `notes/optimizers.md`, sections 2, 3 and 5.

Problems. Whitened Gaussian inputs, exact population loss (empirical loss in the finite-sample sweep), double precision. Teacher seed 100; initialization seeds 0–5, from a stream independent of the teacher's.

- *binary4*: binary tree, $N = 4$ ($K = 3$), $d = 3$, $\chi = 4$; odeco teacher with weights $(3, 2, 1)$ and Haar-random site bases.
- *binary8*, *MPS8*: $N = 8$ ($K = 7$ and 8), the same odeco teacher, $d = 3$, $\chi = 4$.
- *E8* on binary8 and MPS8: pattern teacher with $d = 2$, $\chi = 3$ and patterns $\pi_0 = (1, \ldots, 1)$ (weight 1), π_A equal to 2 at sites 4 and 5 (weight 0.4) and π_B equal to 2 at sites 2 and 7 (weight 0.7), 1 elsewhere. The cores that carry the differing sites span $k = 2$ (π_A) and $k = 6$ (π_B) cores on the MPS and $k = 5$ for both on the binary tree.

Initializations. Fan-in (9) with per-core scale σ; *balanced*: the same W_0 in the gauge with $Q_e = 0$ on every edge. All deep-tree baselines use the balanced version, because fan-in gradient flow converges by $t = 2 \cdot 10^6$ in none of the 12 MPS8 odeco runs (the conservation-law trap of Proposition A.28, verified at the route ends in 11 of the 12 stalls), against 10 of 12 (binary8 odeco), 11 of 12 (binary8 E8) and 11 of 12 (MPS8 E8) fan-in runs. Methods whose iterates depend on the gauge were run from both versions; re-gauged GD, Riemannian GD, ALS and DMRG depend only on W_0 and were run from the balanced one.

Dynamics. All methods stop at $L < 10^{-6} L_0$ (finite samples: empirical loss below 10^{-8} times its initial value) or at their budget.

- *Gradient flow*: SciPy LSODA with relative and absolute tolerances 10^{-9} and 10^{-14} and the exact Hessian as Jacobian (Radau as fallback), instead of the default integrator of Section 2.5; the crossings $a_r \in \{0.1, 0.5, 0.9\}$ are located by root finding, so event times are exact. Budgets: $t_{\max} = 8/\sigma$ (binary4), $2 \cdot 10^6$ (deep trees), $4 \cdot 10^6$ (binary8 scan), 10^6 (finite samples).
- *GD and heavy ball* (PyTorch): $\eta = 0.02$ and 0.002 with $\beta = 0.9$ on binary4 ($0.13\,\eta_c$ for both, counting heavy ball's effective step $\eta/(1-\beta)$); $\eta = 0.01$ and 0.001 on the deep trees (0.23–$0.27\,\eta_c$), at $\sigma = 0.8$ only, because gradient-flow plateaus at $\sigma = 0.3$ last 10^4–10^6 time units. Budgets $8/\sigma$ (binary4) and $2 \cdot 10^4$ (deep) gradient-flow time units, at most $5 \cdot 10^5$ steps.
- *Adam* (PyTorch, bias-corrected): $\beta_1 = 0.9$, $\beta_2 = 0.999$ (also 0.99 and 0.9 at $\eta = 10^{-3}$ on binary4), $\epsilon = 10^{-8}$ (10^{-30} in the ϵ sweep), $\eta \in \{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}, 10^{-2}\}$ on binary4 and $\{10^{-3}, 10^{-2}\}$ on the deep trees. *RMSProp*: $\alpha = 0.99$, $\epsilon = 10^{-8}$, $\eta \in \{10^{-4}, 10^{-3}\}$. Budgets 10^5–$3 \cdot 10^5$ (binary4), $3 \cdot 10^5$ (deep), $5 \cdot 10^5$ (binary8 scan) and $2 \cdot 10^5$ (ϵ sweep) steps.
- *Re-gauged GD*: $\eta = 0.02$ (binary4) and 0.01 (deep); QR sweeps from the leaves to the root bring the cores to the root-centred canonical form every 1 or 50 steps; $5 \cdot 10^4$, 10^5 and $2 \cdot 10^5$ (binary8 scan) steps.
- *Riemannian GD*: $\eta \in \{0.1, 1\}$; tangent projection by least squares with the dense Jacobian of $W(\theta)$ (relative cutoff 10^{-12}), so only small d^N are feasible; leaves-to-root HSVD retraction; 2000 steps.
- *Gram-preconditioned GD*: $\eta \in \{0.01, 0.1, 0.5\}$; each Kronecker factor of a core's Hessian block is damped by 10^{-3} times its mean eigenvalue (scale-free) or, at $\eta = 0.1$, by an absolute floor 10^{-4} or 10^{-2}; no divergence guard, because the first step overshoots by up to $10^{34} L_0$ before the loss returns; $5 \cdot 10^4$ (binary4; $2 \cdot 10^5$ with a floor) and 10^5 (deep) steps.
- *ALS* (one site) and *DMRG* (two sites, then SVD truncation to at most χ values with relative cutoff 10^{-12}): exact least squares; the orthogonality centre moves by QR along a depth-first tour of the tree; at most 50 sweeps.

Sweeps and run counts. Table 18 lists the sweeps; six initialization seeds per setting, 2529 runs in total.

Events and classification.

- Event r is the last upward crossing of $a_r = \frac{1}{2}$ after which a_r stays at least $\frac{1}{2}$; transient crossings are not events (Section 2.8). Width: final rise from 0.1 to 0.9. Saddle-to-saddle: every separation ratio $\rho_i > 3$ (Table 4).
- Discrete methods record loss and overlaps at every step for the first 2000 steps, then every $\max(1, n_{\text{steps}}/(5 \cdot 10^4))$ steps. A first escape or gap shorter than two recording intervals is unresolved and excluded from fits (and counted in the result tables). Two components that cross $\frac{1}{2}$ in the same step or micro-step have no resolved order. For Gram-preconditioned GD, events are defined only after the loss has returned below L_0; a component already above $\frac{1}{2}$ at that point has no resolved event.
- Exponents: per seed, a least-squares fit of $\ln T$ against $\ln \sigma$ over the σ at which the seed follows its most common learning order (for plateau durations, not for times to reach a loss

Table 18: Sweeps of the optimizer experiment (Section 5.5), one configuration file per sweep in `experiments/optimizers/configs/`.

sweep	problem	σ	methods	runs
`scaling_b4`	binary4, fan-in	$0.002 \cdot 2^j$, $j = 0, \dots, 6$	19 settings: gradient flow, GD, heavy ball, Adam (6), RMSProp (2), re-gauged GD (2), Riemannian GD (2), Gram (2), ALS, DMRG	798
`precond_b4`	binary4, fan-in	as above	Gram: 3 step sizes with scale-free damping, 2 absolute floors	210
`deep`	binary8, MPS8; odeco, E8; balanced and fan-in	0.3, 0.8	gradient flow, GD and heavy ball ($\sigma = 0.8$), Adam (2), RMSProp, Gram; from the balanced W_0 only: re-gauged GD (2), Riemannian GD (2), ALS, DMRG	864
`later_b8`	binary8 odeco, balanced	0.25, 0.3, 0.35, 0.4, 0.5, 0.6, 0.8	gradient flow, Adam (2; every step recorded for the first 2000 steps), re-gauged GD	168
`eps_mps8`	MPS8 odeco, balanced	0.01, 0.015, 0.02, 0.03, 0.05, 0.1, 0.2, 0.3	Adam: $\eta \in \{10^{-3}, 10^{-2}\}$, $\epsilon \in \{10^{-8}, 10^{-30}\}$	192
`implicit_bias2_b4`	binary4, $n \in \{45, 60\}$, data seeds 200–203	0.01, 0.1	gradient flow, Adam (2); pairs share draw and initialization	288
`adam_probe`	binary4, binary8, MPS8 E8	single runs	Adam and RMSProp internal state (illustration only)	9

level); mean ± standard error over seeds. A logarithmic law fitted in the same space serves as comparison (gradient flow on binary4: root-mean-square residual of $\ln T$ 0.005 for the power law, 0.62 for the logarithmic law).

- Orders are compared with gradient flow only where both are resolved, for the same problem, σ and seed. Re-gauged GD, Riemannian GD, ALS and DMRG give one trajectory per seed at every σ, so their orders are counted over the six independent seeds (at the smallest σ).
- ϵ *test* (25): $T_{\mathrm{GF}}(\sigma)$ is the seed's exact gradient-flow first-escape time at $\sigma = 0.3$ (`deep` sweep, same balanced initial point up to scale) times $(0.3/\sigma)^6$, and n_{sign} the median over σ of the same seed's escape with $\epsilon = 10^{-30}$ at the same η. Runs without an escape within $2 \cdot 10^5$ steps enter only through the check that the prediction exceeds the budget.
- Convergence for cost and time-to-loss tables: $L < 10^{-3} L_0$. Runs that hit their budget are reported as budget-limited, never as stalls. RMSProp hovers above $10^{-6} L_0$ on the deep trees within $3 \cdot 10^5$ steps; its events are used, its convergence flags are not.

Excluded runs. Runs replaced after review round 1 are kept in `superseded/` subfolders of the raw data and ignored by the analysis: gradient-flow runs whose event times were interpolated on an output grid (errors up to 1.7%), Gram-preconditioned runs stopped by a $10^6 L_0$ loss guard after one step, binary8 Adam runs recorded only every 10 steps, and the first finite-sample sweep (`implicit_bias_b4`, 288 runs, one data draw per initialization). The pilots (`pilot_calib`, `pilot_trap`) support no claim. Raw trajectories (one file per run, 2.2 GB) are not in the repository; Section C says how to regenerate them.

B.9 Real data

Settings of Section 6.7. Experiment folder `experiments/real_data/` (README with every command in order; YAML configurations in `configs/`); findings in `notes/real_data.md`, whose section 8 (fix round) supersedes its sections 4–7 and whose section 9 holds the pre-registered order test; review in `notes/reviews/real_data_round1.md`.

Data and model. MNIST (torchvision mirror) and Fashion-MNIST, fetched and downsampled by `download.py`: images cropped to the central 24×24 pixels and block-averaged to 4×4 (6×6 blocks) or 8×8 (3×3 blocks), intensities in $[0, 1]$. Tasks: MNIST 3 ($y = +1$) against 5 ($y = -1$); Fashion-MNIST sneaker (7) against ankle boot (9), chosen before any run for a visible shape difference at 4×4. Balanced subsamples of 4000 training and 1600 test images, data seed 0. Site features $\phi(x) = (\cos\frac{\pi x}{2}, \sin\frac{\pi x}{2})$ (*raw*), or per site $A_i\phi$ with $A_i\mathbb{E}[\phi\phi^\top]A_i^\top = I$ (*white*; uncentred second moment, eigenvalues floored at 10^{-3} of the largest), which keeps the function class. Model $f(x) = \langle W, \bigotimes_i \phi(x_i)\rangle$ and loss $\frac{1}{2n}\sum_j (f(x_j) - y_j)^2$; in the regime sweep the centred model trains $f - f_0$. Trees from `data.make_tree`: binary16 ($K = 15$) and MPS16 ($K = 16$), binary64 ($K = 63$) and MPS64 ($K = 64$) for 8×8 images; $\chi = 4$ (8 in one sweep, where the bottom bonds are capped at 4 by their two-pixel cut). Pixel orders (`data.pixel_order`; site s carries pixel order[s], pixels numbered row by row): quadtree (Morton key, column bit lowest), snake (boustrophedon rows), scrambled$k$ (random permutation from `numpy.random.default_rng`$(k + 12345)$), and for the order test quadtreeT (Morton order of the transposed image), rowmajor, colmajor, hilbert (Hilbert curve) and mortonrev (site s carries the pixel whose Morton key is s with its bits reversed). Fan-in initialization (9) at per-core scale σ with initialization seeds 0–4, independent of the data seed. Double precision, one thread per process.

Dynamics (`run.py`). Phase 1: Dormand–Prince 5(4) with relative and absolute tolerances 10^{-8} and 10^{-14}, the clock re-anchored ($t \to 0$, offset kept) whenever an accepted step is shorter than $10^{-9}t$, so that escapes at $t \sim 10^{15}$–10^{20} are resolved; it stops at the first accepted step at which the loss has fallen by 1% of $L(0)$ (time t_a). Phase 2: LSODA with the same tolerances on the clock $\tau = t - t_a$, with the Gauss–Newton part of the Hessian (closed-form per-sample Jacobians) as Newton matrix, up to $\tau = 10^4$ (3000 for 8×8 images; $\min(3000, 10^4$ kernel units) in the regime sweep, a kernel unit being $128/\operatorname{tr}\Theta$ at t_a). Records: 30 logarithmic times per decade, 1000 linear times (600 for 8×8) and the library's fine rows, merged for every event time. Conservation drift (cumulative, relative to the largest squared core norm) at most $4.6 \cdot 10^{-8}$, median $1.8 \cdot 10^{-9}$ over the 234 main runs. Two cases rerun with tolerances 10^{-10} and 10^{-16} (`tol_check`) agree in t_a to 7 digits, in the final loss to 6, and in later drop times to 0.1%; the first drop ($\tau \approx$ 10–15) moves by 0.2–0.8 time units, below every gap reported.

Events and analyses.

- *Loss drops* (`analysis.py`): peaks of $-\mathrm{d}L/\mathrm{d}t$ that stand out by a factor 3 against the minima separating them from higher peaks and lower the loss by at least 1% of $L(0)$; time: the peak; width: its full width at half maximum. Widths by progress (from 10% to 90% of a drop) are not used, since the loss creeps between drops and such widths measure the creep. Separation ratios and the S2S signature as in Section 2.8, with these drops.
- *Data-weighted bond spectra* (`data.function_spectra`): across edge e the network computes $f(x) = \sum_a u_a(x)v_a(x)$, with messages u_a from below the edge and v_a from above it; the Schmidt values are the square roots of the eigenvalues of $G_u^{1/2}G_vG_u^{1/2}$, where G_u and G_v are

the Gram matrices of the two messages on the training set. They are gauge invariant and equal the Frobenius spectra under a product law with identity second moments (checked to $2 \cdot 10^{-12}$). Frobenius spectra are not used: after the escape $\|W\|_F$ grows to 10^6–10^7 along directions the data barely see.

- *Rank events*: s_k^e/s_1^e crossing 0.05 (log-interpolated); an edge belongs to a drop if it crosses inside the drop's half-maximum window widened by one width on each side. Crossings already met at $\tau = 0$ are flagged as unresolved.
- *First-route gaps* (`scaling.py`): gap = drop 1 − drop 0 on the τ clock; a route's drop is the one whose widened window contains its rank crossing, and points where it rose between drops, merged with drop 0, or where another route rose first are excluded with their reason. Clean series: one (order, seed) whose first route, one sibling pair, rises inside drop 1 at every σ, with at least four σ over at least half a decade (11 series); full sweep: 91 of the 124 binary points kept (16 between drops, 2 merged, 15 another route first), 10 seed series. Fits per seed: power law by least squares in $\ln g$, log law and $A/\sigma + B$ by weighted least squares with weights $1/g$; all compared by the relative rms residual.
- *MPS edges*: time of each edge's first rise after the first drop, per seed over the seven σ, both laws fitted with relative residuals (`scaling.py`, `edge_scaling`).
- *Route amplitudes* (`route_amplitudes.py`). A binary TTN of bond rank one computes $f = c\prod_v q_v$, $q_v = g_v \cdot \Phi_v$, over its bottom cores v, Φ_v the features of the two pixels of v. For a pair (u, w) of bottom cores, with the residual $r_j = f(x_j) - y_j$ and $h_{uw} = c\prod_{v\neq u,w} q_v$,

$$\mu_{uw} = \|P_u M_{uw} P_w\|_2, \qquad M_{uw} = \frac{1}{n}\sum_j r_j\, h_{uw}(x_j)\, \Phi_u(x_j)\Phi_w(x_j)^\top, \tag{62}$$

  with $P_u = I - g_u g_u^\top/\|g_u\|^2$; k_{uw} is the number of cores of the tree path from u to w. Evaluated for all 28 pairs at the plateau midpoint (minimum of $\|\dot\theta\|$ between the first two drops, reached by re-integrating from the end of phase 1). States farther than 0.05 (relative) from a rank-one product are dropped (28 of 124). Units (`analysis.route_units`): rank-one states of the same (order, seed) whose normalized patterns $\mu_{uw}^2/\sum\mu^2$ differ by less than 0.2 in total variation and whose plateau losses differ by less than 0.01; 96 states form 26 units in the 20 (order, seed) pairs.
- *Hessians* (`plateau_hessians.py`): exact Hessian (dense residual tensor) at the plateau midpoint (minimum $\|\dot\theta\|$ between the first two drops), raw and gauge-projected (Section 2.8); exponents per seed over the runs at the common plateau level ($L \in [0.265, 0.275]$ binary, $[0.225, 0.235]$ MPS; $[0.31, 0.345]$ on Fashion-MNIST, pooled).
- *Regime sweep* (`regime_analysis.py`): kernel on the first 128 test images; labels by Table 4 without the convergence rule, which no $\chi = 4$ network meets on 4000 labels (lazy if $\delta_\Theta(T) < 0.1$; S2S if at least two drops, all $\rho > 3$; rich otherwise); the linearized model at $\theta(t_a)$ in closed form, with its floor $L_{\text{lin}}(\infty)$.
- *Order test* (`order_predictor.py`, `order_test.py`; registered analysis): the best rank-one fit $f = \prod_v g_v \cdot \Phi_v$ of the labels from 32 starts (start i: unit g_v from `default_rng`$(1000 + i)$, 50 steps of higher-order power iteration on $\frac{1}{n}\sum_j y_j \bigotimes_v \Phi_v(x_j)$, rescaling to the least-squares amplitude, then alternating least squares until the relative loss change is below 10^{-12} or 2000 sweeps), the lowest-loss fit balanced, S from (62) at that state; outcomes O1 (mean over seeds of $L(10^4)$) and O2 (mean of $\log_{10}\tau(L \le 0.25)$ on the running minimum, censored at

10^4); Spearman ρ over orders, percentile bootstrap intervals (10,000 resamples of orders) and two-sided permutation p-values (10,000 permutations of S).

Sweeps and run counts. Table 19; 507 runs of the experiment's own configurations, plus the round-1 reviewers' runs with the same code (among them 11 binary runs in nine orders and three MPS chains in other orders, $\sigma = 0.1$, seeds 0 and 1; local copy `data/review_real_data/`), which enter only where the text says "review runs". The main, regime, Fashion-MNIST, $\chi = 8$ and 8×8 sweeps ran on one 64-core machine (about 60 processes, as the runs are bound by memory bandwidth); the fix-round reruns and the order test ran locally (the order test: 8 processes, 70 minutes).

Table 19: Sweeps of the experiment of Section 6.7: configuration files in the folder `configs/` of the experiment. Raw features and $\chi = 4$ unless stated; scrambled orders are random pixel orders on binary16.

configuration	problem	σ, seeds	runs
`main` (+`main_extra`)	MNIST 3/5, 4×4: quadtree, snake MPS, scrambled0, scrambled1, scrambled2 (raw); quadtree, MPS, scrambled0 (white)	7 values 0.05–0.5 (scrambled1: 4, and 3 more in `main_extra`; scrambled2: 4; white: 5); 0–4	220 + 14
`regime`	centred model, quadtree	0.5, 0.7, 1, 1.4, 2, 2.8, 4; 0–4	35
`fashion`	Fashion-MNIST 7/9: quadtree, snake MPS, scrambled0	0.05, 0.14, 0.5; 0–4	45
`chi8`, `chi8b`	$\chi = 8$: quadtree, snake MPS, scrambled0 (one sweep, two runners)	0.05, 0.14, 0.5; 0–4	45
`img8`	8×8 images ($N = 64$): quadtree, snake MPS, scrambled0; $\tau \leq 3000$	0.1, 0.3; 0, 1	12
`tol_check`	tolerances 10^{-10}, 10^{-16}: quadtree, MPS	0.1; 0	2
`fix1_local`	reruns: four binary orders to $\tau = 10^4$; MPS phase 1	0.1; 0–4	25
`fix1_mps`	reruns: snake MPS, $\tau \leq 1000$	as `main`; 0–4	35
`extra_orders`	order test: 27 orders on binary16 (7 runs reused)	0.1; 0–2	74 (+7)

Checks and exclusions. The fix-round reruns reproduce the committed main-sweep values: t_a to $7 \cdot 10^{-15}$ (relative) and the final loss to 5 digits for the quadtree at $\sigma = 0.1$, and the timing of the first MPS rise within 0.07 time units (median 0.01). The round-1 review reproduced the first-plateau times to about 10^{-15}, the gaps within 1.3% and the Hessians and route amplitudes within 4%, and an independent reimplementation agreed. Excluded: the first-route points of other routes (above) and the 28 route-amplitude states that are not near rank one. The round-0 claims superseded by the fix round are listed in `notes/real_data.md`, section 8.13. Raw outputs (about 0.8 GB, plus 0.2 GB of local reruns; local copies `data/real_data/` and `data/real_data_local/`) are not in the repository; the JSON summaries in `results/` suffice to redraw the figures of Section 6.7.

C Reproducibility

Every figure, and every table and number computed from runs, is regenerated by a script in the repository from saved outputs. The analysis scripts of each experiment reduce its raw runs to small tables in `experiments/<name>/results/`, which are committed; the report figures, tables and derived numbers are drawn from these tables (`plot_report.py`, `report_numbers.py` or

`make_figures.py` of the experiment), and only the figures that plot trajectories also read raw runs (Table 20). The sweeps that produced the raw runs are configured by YAML files (Section B), seeded and resumable, and each experiment's README lists every step in order. All figure and table commands, and the analysis steps that feed them, were rerun on the saved outputs and compared with the committed files; `notes/reviews/reproducibility_check.md` records the result, including the few entries that do not regenerate exactly.

C.1 Software and data

The code is a uv project: Python 3.13, dependencies in `pyproject.toml` with exact versions in `uv.lock` (PyTorch 2.14.1, NumPy 2.5.3, SciPy 1.18.1, pandas 3.0.6, Matplotlib 3.11.2); `uv sync` installs it and `uv run pytest` runs the tests of the shared library `ttn/` (trees, networks, teachers, exact losses, integrators, diagnostics, plotting). All numerics are in double precision. The check above used the library at commit `3a642ef` (code hash `6d786ce7...`, the SHA-256 of `ttn/*.py` computed by `ttn.runner.code_hash`); runs made with `ttn.runner` store the commit and this hash in their metadata. Each experiment has a folder `experiments/<name>/` with a README (question, setup, commands in order, variants, results), `configs/`, its code and `results/`. Raw outputs (per-run records and trajectories) are not in git: the sweeps wrote them to `/root/outputs/<name>/` on rented machines, from where they were copied to `data/<name>/` (gitignored) and verified byte for byte before the machines were deleted; local reruns are in `data/<name>_local/` and the reviewers' reruns in `data/review_<name>/`. Scripts written by the reviewers and verifiers, the source of some numbers in `notes/reviews/`, are kept for provenance in `notes/reviews/scripts/`.

C.2 Figures

Table 20 gives, for each figure, the command that draws it and the sweeps whose raw runs it rests on (their configurations are in Section B). Figure 2 is drawn in TikZ (`report/tikz/trees.tex`) when the report is built.

Table 20: Figures of the report: the command that draws each one (from the repository root) and the sweeps whose raw runs it reads. Figures are copied into `report/figures/` by `report/collect_figures.sh` from the list in `report/figures.txt`.

figure	command	raw data (sweeps; Section B)
3	`uv run python experiments/regime_map/plot_report.py` draws this figure (`report_map`) and Figure 12 from committed tables in `results/`, `results/fix1/` and `results/report/`; the last are written by `report_tables.py --raw data/regime_map --tensor-cache data/regime_map_local/ tensor_kernel --force` (README; `--force` overwrites the committed tables)	`sigma_chi_b4`, `chi_b4`, `sigma_K_centred_d2`, `imbalance_b4` (Table 15); local copy `data/regime_map/`

continued on the next page

Table 20: Figures of the report (continued).

figure	command	raw data (sweeps; Section B)
4	`uv run python experiments/s2s_core/plot_report.py --raw-main data/s2s_core/main --raw-large data/s2s_core/large --out experiments/s2s_core/results` draws this figure (`report_runs`) and Figure 20; scripts below are in `experiments/s2s_core/`, run with `uv run python` and `--out experiments/s2s_core/results`; full order in the experiment's README	`main`, `large` (one run each; Table 12); local copy `data/s2s_core/`
19	`plot_aligned.py --raw data/s2s_core/aligned` (`fig_a_aligned`)	`aligned`
5	`plot_scaling.py` (`fig_c_first_escape_chi3_fanin`) from `first_runs.csv` (`analyze_first.py`); Table 21 is `main_exponents.csv`, written by `recompute_hessians.py` and `analyze_main.py`	`first`, `first_k78`; table: `main`, `large`
20	`plot_report.py` as above (`report_prefactor`), from `lo_d.csv`, `lo_chi_table.csv`, `lo_K_table.csv` (`lo_sweep.py`, `plot_lo.py`)	`lo_d`, `lo_chi`, `lo_K`
6, 21	`bash experiments/locality/finalize.sh R` writes every figure and table of Sections 4.4 and 4.8 into directory `R` (`LOCALITY_EAGER=1` without a C++ compiler; its steps open the absolute trajectory paths stored in the run records, `/root/outputs/locality/traj/`, so on another machine that directory must link to the copy). Single steps, with scripts in `experiments/locality/` run by `uv run python`: these two by `make_figures.py $O/twocomp --config experiments/locality/configs/twocomp.yaml --hessians R/hessians.csv --out R/twocomp`	`twocomp` (Table 13); `$O` is `LOCALITY_OUT`, local copy `data/locality/`
8	`plot_trees.py $O/multicomp --teacher TA --n 4 --sigma 0.03 --seed 0 --levels R/levels_runs.csv --out R/trees_TA.png`	`multicomp`, `multicomp_wide` (levels by `levels.py`)
7	`plot_bonds.py $O/twocomp --cases mps8:25 binary8:34 ht8:12s --sigma 0.05 --out R/bond_spectra.png`	`twocomp` (saved trajectories)
9	`uv run python experiments/order/plot_report.py` draws this figure (`report_order_first`) and the next from committed tables: `main_summary.csv` (`analyze_main.py`), `sigma_table.csv`, `sigma_fits.csv` (`analyze_sigma.py`; also Table 23), `chi_scan.csv`, `chi_gf.csv` (`analyze_chi.py`)	`main`, `b4_sigma`, `m4_sigma`, `b8_sigma`, `pat_mps4_*`, `pat_mps6_m3`, `chi_scan`, `chi_gf` (Table 14); local copy `data/order/`

continued on the next page

Table 20: Figures of the report (continued).

figure	command	raw data (sweeps; Section B)
10	as above (`report_order_wrong`), from `h1b_saddle_{b4,mps4_sep,mps4_adj}.csv` (`h1b.py`) and `report_examples.csv` (`plot_report.py --raw data/order`)	`h1b_saddle_*`
11	`uv run python experiments/s2s_core/plot_dormant.py --out experiments/s2s_core/results` (`fig_g_dormant`) from `main_runs.csv` and `dormant_decay.csv` (`analyze_main.py`, `analyze_dormant.py`)	`main`, `large`, `k78`, `balanced` (Table 12)
12	as Figure 3 (`report_other`)	`sigma_chi_b4`, `large_chi_b4`
13	`uv run python experiments/optim_samples/plot_report.py --out-dir experiments/optim_samples/results` draws this figure and the next two from committed tables, here `gd_sweep.csv`, `map1d_points.csv` and `centred_chi9_gf.csv` (written from the raw runs by `analyze_gd.py`, `map1d.py`, `analyze_centred.py`; order of all steps in the experiment's README)	`gd_sweep`, `gd_centred_chi9` (Table 17); local copy `data/optim_samples/`
22	as above; tables `sgd_summary.csv`, `sgd_second_plateau.csv` (`analyze_sgd.py`), `p9_sgd.csv` (`p9_sgd.py`), `p9_conditions.csv` and `p9_fit.csv` (`plot_p9.py`)	`sgd_sweep`, `p9_sgd.py`
23	as above; table `samples_summary.csv` (`analyze_samples.py`)	`samples_sweep`
14	`OPTIMIZERS_OUT=data/optimizers uv run python -m experiments.optimizers.make_figures --results-dir experiments/optimizers/results b4` (`OPTIMIZERS_OUT`: the raw runs, whose trajectories this figure plots)	`scaling_b4` (Table 18)
24	as above, with target `eps`	`scaling_b4`, `eps_mps8`, and `deep` for the gradient-flow reference times
15	`uv run python experiments/data_teacher/plot_report.py` draws this figure and the next from committed tables, here `r2_saddle_runs_all.csv`, `r2_saddle_fits.csv` (`analyze_r2_saddle.py`) and `r2_random_v2_runs.csv` (`analyze_law_v2.py r2_random`); it also writes the tables `report_first_escape.csv`, `report_lambda_scaling.csv` and `report_float_floor_refit.csv` quoted in Section 6 (order of all steps in the experiment's README)	`r2_saddle` (with `_b`, `_ext`, `_m20`), `r2_random` (Table 16); local copy `data/data_teacher/`

continued on the next page

Table 20: Figures of the report (continued).

figure	command	raw data (sweeps; Section B)
16	as above; tables `e4a_sigma_v2_runs.csv`, `e2_sigma_v2_runs.csv`, `e2_sigma_v2_summary.csv` (`analyze_law_v2.py`), `r2_corr_runs.csv`, `r2_corr_fits.csv` (`analyze_r2_corr.py`), `float_floor_check.csv` (`check_float_floor.py`)	`e4a_sigma`, `e2_sigma`, `r2_corr.py`
17	`uv run python experiments/real_data/make_figures.py --json experiments/real_data/results experiments/real_data/results` draws this figure (`first_route_gaps`) and the next from committed JSON only, here `fix1_analysis.json` (`fix1_analysis.py`, from `events_table.json` and the run summaries of `summarize.py`; order of all steps in the experiment's README)	`main`, `fix1_local`, `fix1_mps` (Table 19), review runs; local copies `data/real_data/`, `data/real_data_local/`
18	as above; table `order_test.json` (`order_test.py`, with the predictor `extra_orders_predictor.json` written by `order_predictor.py` before the runs)	`extra_orders`; `main` for the secondary analysis

C.3 Building the report

The report is built by `report/build.sh` (pdflatex, BibTeX, then pdflatex until the cross-references are stable), after `report/collect_figures.sh` has copied the figures listed in `report/figures.txt` from `experiments/*/results/` into `report/figures/`. `report/README.md` describes both scripts.

D Supplementary results

Material moved here from the main text: detailed tables, secondary checks, and the provenance and corrections of each result after review.

D.1 Saddle-to-saddle: supplementary results

Details of Section 4, in the order of its subsections. Tags as in the main text; settings in Sections B.2 to B.4.

D.1.1 Phenomenology

Scope. empirical Complete runs (all three components learned): binary4 320 of 320 (fan-in, both teachers, $\chi = 3, 6$) and 60 of 60 (balanced); MPS4 160/160 at $\chi = 6$, 106/160 at $\chi = 3$ and 60/60 balanced; binary8 50/50 and HT4 45/50 at $\chi = 6$ ($\sigma = 0.07$–0.3). At $\chi = 3$ few runs with $K = 7$ complete (binary8 14 of 180, HT4 12 of 150), and no MPS8 run learns its third component (0 of 284; 132 learn two; Section 4.8). The events of the two runs of Figure 4 are at $3.6 \cdot 10^3$, $4.4 \cdot 10^4$, $5.6 \cdot 10^5$ (MPS4) and $1.3 \cdot 10^5$, $9.8 \cdot 10^5$, $6.8 \cdot 10^6$ (binary8), with widths below 2.

Table 21: Plateau durations and saddle type, odeco teachers, fan-in initialization. Gaps: per-seed exponent p of $t_{i+1} - t_i \propto \sigma^{-p}$, mean ± s.e. over seeds (run sets: Section B.2), for $\chi = 3$ and 6 or, in rows "both", the separated and close teachers; brackets: range of the per-seed values (binary4, MPS4: over their four cells). $\lambda_{\min}$: mean per-seed slope of $\ln(-\lambda_{\min}(H_\perp))$ against $\ln\sigma$ at the first and second plateau midpoints ($\chi = 3$; 6, or the two teachers). Predicted: $K - 2$. Ten seeds per fit except binary4 close (9, 9; 8, 7), MPS4 gap 2 at $\chi = 3$ (6, 7), HT4 gap 2 (9), MPS8 (7; 3). σ: 0.003–0.1 (binary4, MPS4), 0.07–0.3 at $\chi = 6$, 0.03–0.4 (binary8) and 0.1 or 0.15 to 0.4 (MPS8) at $\chi = 3$; †: limited range (less than one decade, or fewer than five seeds; binary8 at $\chi = 3$: 1.1 decades for 11 of 20 seeds; some binary4 close-teacher seeds keep four or five values of σ, about 0.8 decade). Source: `experiments/s2s_core/results/main_exponents.csv`.

tree	teacher	χ	$t_2 - t_1$	$t_3 - t_2$	$\lambda_{\min}$ slopes
binary4	sep.	3, 6	1.003 ± 0.001, 1.007 ± 0.001	1.004 ± 0.003, 1.011 ± 0.002	1.05, 1.02; 1.08, 1.04
($K = 3$)	close	3, 6	1.004 ± 0.010, 0.999 ± 0.008 [0.95, 1.06]	1.016 ± 0.011, 1.014 ± 0.006 [0.99, 1.09]	1.17, 1.12; 1.18, 1.17
MPS4	sep.	3, 6	2.000 ± 0.000, 2.001 ± 0.001	1.990 ± 0.005, 1.998 ± 0.001	2.01, 1.99; 2.00, 2.00
($K = 4$)	close	3, 6	2.001 ± 0.001, 2.001 ± 0.001 [2.00, 2.01]	2.004 ± 0.002, 2.001 ± 0.001 [1.97, 2.01]	2.01, 2.00; 2.02, 2.00
binary8	sep.	6†	5.020 ± 0.009 [4.98, 5.05]	4.969 ± 0.022 [4.88, 5.09]	5.10, 5.07
($K = 7$)	both	3	4.93 ± 0.05, 4.92 ± 0.04 [4.65, 5.13]	–	5.03; 5.03
HT4 ($K = 7$)	sep.	6†	5.10 ± 0.08 [4.83, 5.78]	4.75 ± 0.14 [3.93, 5.33]	5.13, 4.88
MPS8	sep.	6†	5.95 ± 0.05 [5.79, 6.20]	not reached	5.99
($K = 8$)	both	3†	6.0 ± 0.3, 6.2 ± 0.6 [5.14, 7.21]	not reached	5.88; 6.07

Plateau levels and spectra. empirical Largest relative deviation of the loss from (18) at plateaus between resolved events ($\rho_i > 3$): $8.5 \cdot 10^{-3}$ (binary4, separated teacher, $\sigma = 0.1$), $2.0 \cdot 10^{-2}$ (binary4, close teacher, at $\sigma = 0.003$ on a plateau with $\rho = 5.1$), $2.8 \cdot 10^{-3}$ (MPS4; $8.1 \cdot 10^{-3}$ balanced), and $2.1 \cdot 10^{-5}$ and $1.5 \cdot 10^{-4}$ at $K = 7, 8$ ($\chi = 6$, 3); at $\sigma = 0.003$ at most $7.4 \cdot 10^{-7}$, except on binary4 with the close teacher. In runs with the saddle-to-saddle signature the top $|S|$ singular values of every edge equal the learned weights to a relative $7.7 \cdot 10^{-3}$ (binary4, close), $9.0 \cdot 10^{-4}$ (binary4, separated; $1.7 \cdot 10^{-9}$ at $\sigma = 0.003$) and $3.5 \cdot 10^{-4}$ (other trees). The rank steps count the singular values above fs_r, r the component being learned, with $f = \frac{1}{2}$ or a nearby value that separates s_r from the learned weights; the 34 exceptions among 531 runs are 28 binary4 runs with the close teacher at $\sigma \geq 0.05$, whose events overlap, and 6 HT4 runs.

Separation of the events and the gap fits. empirical Every separation ratio exceeds 3 in all complete runs on MPS4 (326), binary8 (83) and HT4 (74), and in 218 of 220 on binary4 with the separated teacher; with the close teacher on binary4 in 5 of 20 runs at $\sigma = 0.1$, rising to 20 of 20 at $\sigma = 0.003$, as close weights make escapes overlap unless σ is small. Table 21 lists the per-seed fits; binary8 at $\chi = 3$, over up to 1.1 decades, gives 4.93 ± 0.05. At the plateau midpoints the raw Hessian gives the same slopes as $H_\perp$ to 0.004, and per seed the ratio of $|\lambda_{\min}|$ to the balanced-route value reaches 20 at $K = 7, 8$.

The strict wrong-order saddle on binary4. empirical With the close teacher on binary4 ($\chi = 3$), seed 8 learns in the order 1-3-2 at seven of eight values of σ (0.005–0.1), and seed 5 at two. At the plateau where 1 and 0.7 are learned and 0.85 is not, $\lambda_{\min}$ is -0.106 to -0.146 at the nine midpoints (raw: -0.113 to -0.149) and flat in σ (seed 8: slope 0.03), against -0.133; the next event follows within about one width ($\rho = 0.76$–1.25), and this gap does not grow as $\sigma \to 0$ (seed 8: $\propto \sigma^{0.08}$). At $\sigma = 0.003$ seed 8 learns greedily.

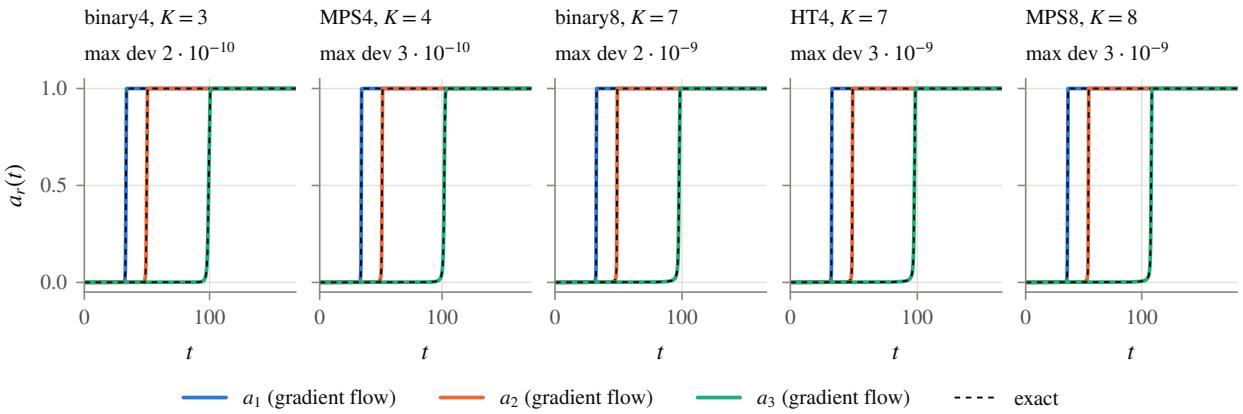


Figure 19: Aligned balanced initialization: overlaps from gradient flow on the full TTN (solid, Dormand–Prince, tolerances 10^{-10} and 10^{-14}) and the exact mode solutions (dashed); separated teacher $(3, 2, 1)$, $d = 4$, $\chi = 3$; one run per tree, with per-core amplitude $q_0 = 0.01$ (binary4), 0.07 (MPS4), 0.29 (binary8, HT4) and 0.34 (MPS8). Panel titles: largest deviation over time and components. Takeaway: with aligned initialization every tree learns the components in the order of their weights, at times $\propto 1/s_r$, as the decoupled modes of a deep linear network do (source: Section B, Table 12).

D.1.2 Aligned initialization: integrator test

empirical Separated teacher, $d = 4$, $\chi = 3$, balanced aligned initialization with per-core amplitude $q_0 = 0.01$ (binary4), 0.07 (MPS4), 0.29 (binary8, HT4) and 0.34 (MPS8), so that the events fall near $t = 33$, 50, 100 on every tree. Gradient flow on the full TTN (Dormand–Prince 5(4), tolerances 10^{-10} and 10^{-14}) follows the exact solutions (Figure 19): the largest deviation of any overlap is $2.0 \cdot 10^{-10}$ (binary4), $3.2 \cdot 10^{-10}$ (MPS4), $2.3 \cdot 10^{-9}$ (binary8), $2.6 \cdot 10^{-9}$ (HT4) and $2.7 \cdot 10^{-9}$ (MPS8); event times agree on the same output grid to $1.3 \cdot 10^{-11}$ relative; the conservation drift (16) stays below $3 \cdot 10^{-14}$. LSODA, the integrator of all other runs of Section 4, deviates by at most $7.1 \cdot 10^{-7}$. The test checks contractions, gradients, integrators and event detection on all five trees.

D.1.3 Escape from the origin

Test, seed by seed. empirical The 231 pairs of initialization and teacher were run at eight values of σ: 0.002–0.1 on binary4 and MPS4 ($\chi = 3, 6$, both teachers, fan-in and balanced) and 0.01–0.4 at $K = 7, 8$ (mostly separated teacher, fan-in; Table 12). At the smallest σ the prediction (19) matches every run to a relative $8.5 \cdot 10^{-6}$ (binary4, separated teacher), $2.2 \cdot 10^{-3}$ (binary4, close teacher, whose direction has not converged at $\sigma = 0.002$), $6 \cdot 10^{-5}$ (MPS4) and $4 \cdot 10^{-8}$ (binary8, HT4, MPS8 at $\sigma = 0.01$). With the offset removed the per-seed exponents are 0.9996–0.9997 (± 0.0001; binary4, separated), 0.990–0.996 (binary4, close: 2.6–3.8 s.e. below 1, as an unconverged offset makes them), 1.9999–2.0000 (MPS4), 4.996–4.999 (binary8, HT4) and 5.9995 ± 0.0002 (MPS8).

Dependence of the prefactor on d, χ and K (Figure 20). The leading-order flow gives $\mathcal{T} = \lim_{\sigma \to 0} \sigma^{K-2} t_1$ per initialization cheaply, so these were measured on many initializations ($d = 4$, one component of weight 1, unless stated). *Physical dimension* empirical new (the closed form): at full bond $\chi = d^2$ (200 initializations per point) the binary4 medians are 1.544 $[1.468, 1.594]$, 1.569 $[1.525, 1.615]$ and 1.563 $[1.525, 1.587]$ (95% bootstrap intervals) for $d = 2, 3, 4$, independent of d and equal to $\pi/2$, the full-bond value of the closed form of Proposition A.13 (derived given end directions that do not rotate), while t_{KH} falls by a factor 1.9; $\mathrm{sd} \ln \mathcal{T} = 0.218 \pm 0.010$, 0.157 ± 0.009,

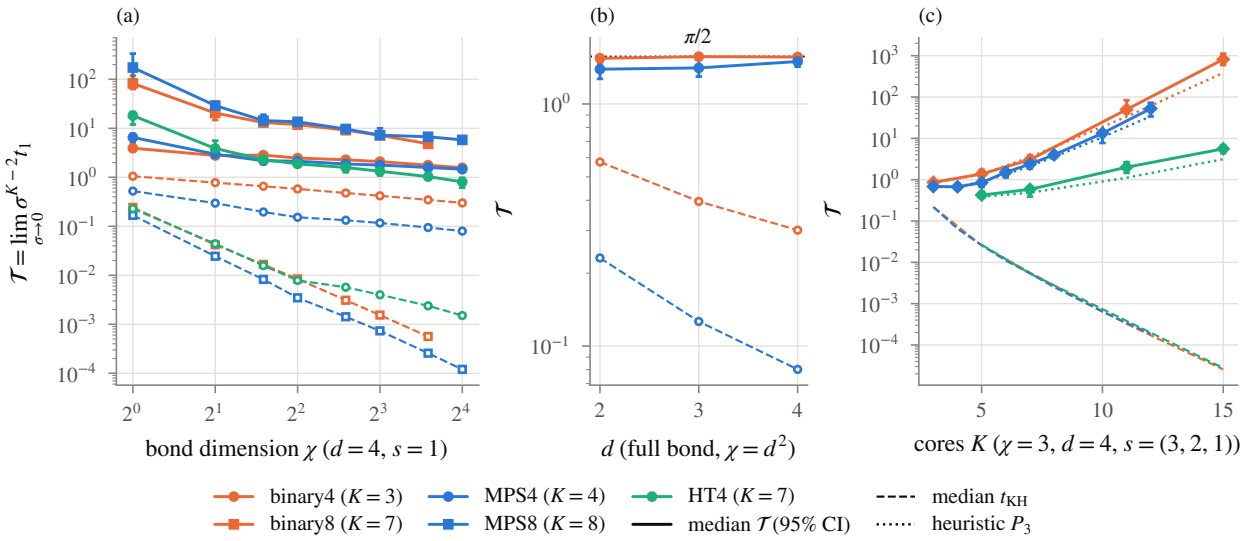


Figure 20: Prefactor $\mathcal{T} = \lim_{\sigma\to 0} \sigma^{K-2} t_1$ from the leading-order flow, medians over initializations with 95% bootstrap intervals. Colour: tree family (MPS blue, binary orange, HT green); marker in (a), (b): circles $N = 4$, squares $N = 8$. (a) Against χ: one component of weight 1, $d = 4$, 100 initializations per point; dashed (open markers): median Kumar–Haupt time t_{KH}. (b) Against d at full bond $\chi = d^2$, binary4 and MPS4, 200 initializations; dotted: $\pi/2$; dashed: t_{KH}. (c) Against K, 17 trees (diamonds; binary, MPS and HT with $N = 3$–16), $\chi = 3$, $d = 4$, weights $(3, 2, 1)$, 50 initializations; dotted: the RMS-overlap heuristic P_3; dashed: t_{KH}. Takeaway: weak χ-dependence, none on d at full bond, P_3 follows the growth with K, and t_{KH}, which uses only the norm of the initialization, is low by factors that grow with d, χ and K (source: Section B, Table 12).

0.135 ± 0.006 against the closed form's $1/(2\sqrt{\chi}) = 0.250$, 0.167, 0.125 (± 0.02). MPS4 gives 1.39 $[1.27, 1.58]$, 1.41 $[1.30, 1.52]$ and 1.50 $[1.43, 1.60]$ (another set of initializations: 1.30, 1.35, 1.46) while t_{KH} falls by 2.9. *Bond dimension* empirical new: the median falls slowly, with log-log slopes -0.335 ± 0.028 (binary4) and -0.264 ± 0.038 (MPS4) over $\chi = 4$–16 and -0.76 ± 0.08 (binary8, $\chi \le 12$), -0.69 ± 0.08 (HT4) and -0.72 ± 0.06 (MPS8) over $\chi = 2$–16, against -0.47 and -1.5 to -2.5 for t_{KH} (another set of initializations: -0.20 ± 0.05 for MPS4, -0.36 to -0.38 for binary4). *Number of cores* conjecture (heuristic) empirical (agreement) new: over 17 trees with $K = 3$–15 ($\chi = 3$, weights $(3, 2, 1)$, 50 initializations each) the median ranges from 0.42 (HT3, $K = 5$) to 814 (binary16, $K = 15$), growing with K within each family (binary4 0.86, MPS4 0.67; at $K = 7$, HT4 0.58 and binary8 3.04). The RMS-overlap heuristic $P_3 = d^{N(K-2)/(2K)}/((K-2)s_1)$, the escape time of a balanced aligned mode started at the root-mean-square overlap $\sigma^K d^{-N/2}$ of a fan-in W_0 with a unit rank-one tensor, tracks it within a factor 0.87–2.13 with no free parameter, while the median of $\mathcal{T}/t_{\mathrm{KH}}$ grows from 3.1 to $2.9 \cdot 10^7$. *Gauge balance* empirical new (minor): re-gauging the same W_0 to $Q_e = 0$ changes $\mathcal{T}$ per seed by a factor 0.88–1.92 (cell medians 0.99–1.18), keeps r_1 in 88 of 94 seeds and leaves the exponent unchanged.

D.1.4 Locality: two components

Head start. empirical MPS8 $\{3, 4\}$ from small initialization nearly ties in the fair comparison (residuals 0.004 and 0.005). On MPS4, $s_2 = 0.25$ gives slopes 3.023 ± 0.010 (small) and 4.04 ± 0.05 (saddle) against 3 and 4, and $s_2 = 0.75$ gives 0.367 ± 0.025 and 1.345 ± 0.016 against $1/3$ and $4/3$. With weights $(2, 1)$ from the saddle on MPS4, MPS6, MPS8 (8 seeds each), $\lambda_{\min} = -1.4142$, -1.5874, -1.6818 against $\lambda_2 = 2^{(K-2)/K} = 1.4142$, 1.5874, 1.6818, and the slopes are 0.712 ± 0.007,

Table 22: Plateau law after the first component, two-component pattern teachers ($d = \chi = 2$, weights $(1, 0.5)$): prediction (21) against measurement. p is the exponent of $\Delta t_2 \propto \sigma^{-p}$, fitted per seed; for $k = 2$ the slope of Δt_2 against $\ln(1/\sigma)$. Ranges are over rows (one row = tree, D, protocol); single rows give mean $\pm$ s.e. over seeds (median). ρ is the separation ratio (17). †: limited range (less than one decade of σ, or fewer than five fitted seeds; the reporting rule of Section 2.8). Sets D number the sites from 0 (source: Section B, Table 13).

k	rows	prediction	measured
1	binary4, binary8, star3 (4 cases)	no plateau	small: $\Delta t_2 \propto \sigma$, at most $8 \cdot 10^{-4}$, $\rho < 5 \cdot 10^{-4}$; saddle: $\Delta t_2 = 0.63$–0.65, independent of σ, $\rho = 0.34$–0.35
2	MPS, midroot6 (7 cases)	log, slope 1 (small), 2 (saddle)	0.980–1.009 (small), 2.002–2.071 (saddle); log law preferred in 14 of 14 rows
3	all nine trees (22 cases)	1	0.98–1.04 (42 rows)
4	MPS, star3, midroot6 (8 cases)	2	small 1.95–2.05 (MPS4 $\{0, 3\}$: 3 seeds†), balanced 1.94–2.02, $\chi = 4$: 1.99–2.03; saddle 1.48–2.39 (1.81–2.19 on never-stalling seeds)
5	binary8, HT4 (6 cases)	3	2.89–3.13, all protocols
5	HT8 $\{1, 2\}$	3	saddle 2.74 ± 0.37 (2.96); 3.10 ± 0.11 on never-stalling seeds
5	MPS6 $\{1, 3, 5\}$, MPS8 $\{1, 5\}$	3	$\sigma \in [0.03, 0.3]$: 2.86–3.76. $\sigma \in [0.012, 0.03]$†: MPS6 3.05 ± 0.06 (saddle), 3.15 ± 0.23 (balanced); MPS8 3.13 ± 0.25 (balanced, 4 seeds), $\mathbf{3.67 \pm 0.20}$ (saddle, 4 seeds). MPS8 small, never-stalling seeds: 2.97 ± 0.05. $\chi = 4$: 3.19 ± 0.15, 2.97 ± 0.04
6	binary8 $\{0, 2, 4\}$, HT4 $\{0, 1, 2\}$	4	balanced, one decade, 8 and 7 seeds: 4.007 ± 0.021, 4.061 ± 0.048 (fan-in†: 3.68 ± 0.20, 4.23 ± 0.10)
6	MPS8 $\{1, 6\}$	4	balanced, one decade: 4.51 ± 0.19
7	HT4, all sites	5	balanced, one decade, 8 seeds: 4.957 ± 0.061 (5.01)
6, 7	other rows (6 and 3 cases)	4, 5	fan-in and saddle†, 1–5 fitted seeds: 3.4–5.1 ($k = 6$), 5.3–6.3 ($k = 7$; HT8 $\{3, 4\}$: 5.74 ± 1.06, median 5.01)
8	MPS8 $\{0, 7\}$	6	not measured (≤ 2 fitted seeds per protocol†)

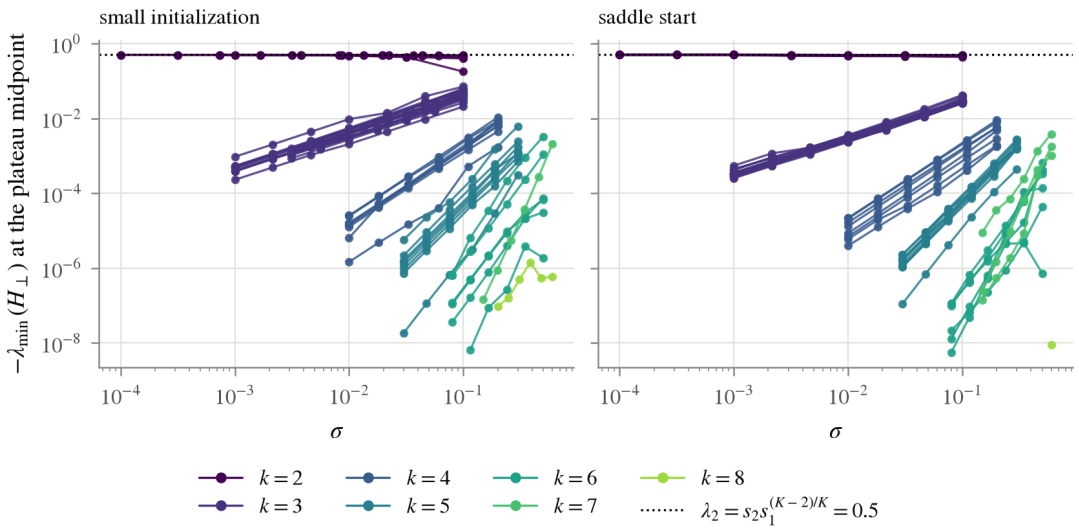


Figure 21: Saddle type from the Hessian at the plateau midpoint, same sweep as Figure 6. Plotted: $-\lambda_{\min}$ of the gauge-projected Hessian $H_\perp$ (median over seeds) against σ, one line per (tree, D), coloured by k; left small initialization, right saddle start; dotted: $\lambda_2 = s_2 s_1^{(K-2)/K} = 0.5$. For $k = 2$, $\lambda_{\min} = -\lambda_2$ at every σ: a strict saddle (the small-initialization curves drop at the largest σ, where the plateau is too short to reach the saddle). For $k \geq 3$, $\lambda_{\min} \to 0$ as $\sigma \to 0$, roughly as σ^{k-2}: higher-order saddles. Lines for $k \geq 6$ come from the rows of limited range of Figure 6. Takeaway: with the larger component learned first, negative curvature that survives $\sigma \to 0$ exists only for two-core routes (source: Section B, Table 13).

0.649 ± 0.016, 0.595 ± 0.003 against $1/\lambda_2 = 0.707$, 0.630, 0.595.

Saddle type. empirical Near rank-deficient states the gauge projection removes part of the escape direction (Section B.3), so raw values are quoted where the two differ; the fixed threshold of Section 2.8 would call $k = 3$ plateaus strict for $\sigma \gtrsim 0.01$–0.04. For $k = 2$ from the saddle, $\lambda_{\min} = -0.5000$, projected and raw (-0.2500 and -0.7500 for $s_2 = 0.25$, 0.75). From small initialization the raw value departs from $-\lambda_2$ where the plateau is too short to reach the saddle (-0.437 for MPS8 $\{3, 4\}$ at $\sigma = 0.008$; -0.581 against -0.75 for $s_2 = 0.75$). For $k \geq 3$ the exponent x of $\lambda_{\min} \propto \sigma^x$ ranges over rows from 0.81 to 1.06 for $k = 3$ (raw 0.92–1.06), 1.64–2.16 for $k = 4$, 2.82–3.76 for $k = 5$, 4.00 and 4.04 for $k = 6$ (balanced binary8 and HT4, one decade; 4.27 on MPS8; 3.19–5.23 in the rows of limited range), and 5.01 (balanced HT4) up to 6.5 (limited range) for $k = 7$; the median of $|x - p|$ is 0.02; raw and projected exponents differ by up to 0.11.

Bond spectra. empirical Outside the Steiner subtree s_2^e stays below $3 \cdot 10^{-3} s_2$ for $\sigma \leq 0.03$. The effective rank uses the threshold 0.05 times the teacher's value; with 0.03 and 0.1 the Steiner edges are the edges that gain rank in 99.6% and 99.9% of the events. The spread of the crossing times of the Steiner edges has a 95th percentile of 0.25 event widths and grows as $\sigma^{1.94}$.

Reading k from a single run. On a balanced route the new overlap grows as $\dot{a}_2 \propto a_2^{2-2/k}$ at small a_2, so the local growth exponent $\zeta = \mathrm{d} \ln \dot{a}_2 / \mathrm{d} \ln a_2$ gives $k = 2/(2-\zeta)$ (Equation (44)) derived known (the exponent $2 - 2/k$ of a balanced product of k factors: Arora et al., 2019b; Razin et al., 2022). Fitted over $a_2 \in [10^{-4}, 10^{-2}]$, the medians per row give $k = 2.00$, 3.0, 4.0, 5.1–5.3, 6.0, 7.0–7.7 for $k = 2$ (saddle start) to 7 empirical new (as a meter of k). Validity range: from the saddle, or from small initialization if the plateau is long enough for the window to lie after the escape transient.

The estimate fails for small-initialization $k = 2$ on MPS6 and MPS8 (2.4–3.3; plateau shorter than the transient), is biased high for $k \geq 5$ (HT8: 5.3 for $k = 5$, 7.7 for $k = 7$), and cannot separate $k = 7$ from 8, since $\mathrm{d}k/\mathrm{d}\zeta = k^2/2$ (per-run 5–95% ranges ± 1 at $k = 7$; MPS8 $\{0, 7\}$, $k = 8$: 7.9 from small initialization, 3 runs, 6.1 from the saddle, 2 runs).

Independent cross-check. empirical A pilot with separate code ($d = \chi = 2$, weights $(1, 0.5)$, MPS4, MPS6, binary4, binary8, fan-in initialization; mean $\pm$ s.d. of per-seed fits over 3–8 seeds) gives the first escape $t_1 \propto \sigma^{-(K-2)}$ to within 0.003 in the exponent for every seed; no plateau for $k = 1$; a logarithmic $k = 2$ plateau with slope 1.01 ± 0.01 on MPS4 ($\sigma = 10^{-3}$–0.1; predicted 1), rising to 2.3 ± 0.4 from the saddle (predicted 2); and $p = 1.00$–1.01 for $k = 3$, 2.00 (MPS4) and 2.05 ± 0.06 (MPS6) for $k = 4$, 3.00 ± 0.11 (binary8) for $k = 5$ and 4.97 ± 0.05 (binary8, limited range: under one decade) for $k = 7$. It also met the failure modes of Section 4.8: 95 of 332 runs with $k \geq 4$ (none with $k \leq 3$) never learn the second component, and on MPS6 the exponents for $k = 5$ and 6 are seed-dependent and up to two units steeper than $k - 2$ (3.75 ± 0.24 for $k = 5$, with 3 of 6 seeds escaping).

D.1.5 Several components: the saddle hopper

Source: `notes/saddle_hopper.md` (v2) and `experiments/saddle_hopper/` (README there); audit `notes/reviews/saddle_hopper_verification.md`, which finds the hopper reproducible but permissive. Section 9 of the note (the theorem check) was added after the audit and is not covered by it.

What it is. At each plateau the hopper computes the fixed-rank (old) block and the local residuals of the connected core sets, follows the most unstable direction, re-optimizes at the new bond-rank profile, and repeats; it adds the first escape of Section 4.3, the race of the $k = 2$ routes with the dormant amplitudes carried across saddles, the time laws and the failure flags. It is deterministic (a re-run is byte-identical) and reproduces the worked examples of the theory notes. Its predictions use neither σ nor the seed, and no rule is tuned per case; data-dependent elements are one rule for the 1-3-2 order of binary4 (made explicit after reading the odeco runs), the saddle start from the lowest rank-one attractor, and the checks of the causes of the mismatches, written after seeing them.

Scores empirical. Of the 11,432 scored locality runs, 96.9% follow the hopper's main branch (645 of them stalled runs admitted by its trap flag, set on every route with $k \geq 4$ under unbalanced initialization). On the informative runs (one main order, not stalled; 7,965 runs) it matches 97.0%, against 95.8% for the weight order without rotation; on the fan-in and balanced two-component runs the two predictions are identical (96.1% each); on the 107 multi-component runs whose main order is not the weight order (B4 on MPS4, E8 on MPS8) it matches 88.8%, against 0%. On the odeco runs it matches 96.6%, as does the weight order. Citable as a stall result: no stall at $k \leq 3$ (0 of 4,531 scored two-component runs with $k = 2$, 3). Where it commits to one order, that order is the weight order in all but the 107 runs, so as a whole-sequence validation it is bookkeeping of known theory.

Multi-saddle slopes empirical new. For the nested teacher TC on MPS8 from the saddle, the race across saddles predicts the log slopes 1.667, 5.356, 7.737, then a $k = 4$ plateau (exponent 2), then 80.26; measured: 1.718, 5.540, 7.581, 2.018, 78.92 (all five within 3.5%). From small initialization

TC misses at its second level (5.62 against 7.74), and TA and TB are 10.5% low at their first level (partial carry-over of the dormant amplitudes between saddles, not modelled).

The saddle-order theorem at the reachable saddles. proven (landscape: Theorems A.30 and A.31 applied by exact computation) Every first-escape candidate of the hopper gives a rank-one saddle, 587 in all, over 288 cases (tree, d, χ, teacher). At 147 of them $\lambda_{\max}(\Gamma) > s_q$ (q the learned component): strict saddles, wrong order or stars. At level $\ell = 2$ the generalized operator never exceeds s_q; at level $\ell = 3$ it does at 30 saddles, whose order is 6 instead of $k^* = 7$: locality HT4 ($\chi = 2$, 4) and HT8 with $D = \{0, 1, 2, 3\}$, odeco runs on HT4 and binary8 with the second or third component learned, and the binary8 runs of the order experiment (two components, $w = 0.5$, 0.8, 0.95; odeco with three and four components). All are reached after a wrong first escape; none is at a correct-order saddle or on an MPS. empirical The escape signature $x = (L_S - L)/(s_r^2 a)$, a the overlap of the escaping component r, is about 1 for a route and $1 - s_q/s_r$ for two symmetric clusters: in the odeco HT4 runs with the close teacher, seed 4 gives $x = 0.2998$, 0.2999, 0.289 at $\sigma = 0.07$, 0.15, 0.2 (prediction 0.300) and a gap exponent of 4.03, against σ^{-5} for the route; most other wrong-first seeds take the route (4.89–5.47). These escapes account for 19 of the 88 locality sequence mismatches, and for none of the steep MPS exponents and none of the stalls. 108 later escapes of the hopper leave saddles with two or more learned components, which the theorem does not cover.

D.1.6 Order of learning

The first component of the odeco runs. empirical The one run of the first-escape test whose first component is not r_1 (binary4, $\chi = 3$, close teacher, balanced, seed 6, $\sigma = 0.1$) learns the weakest component first. On binary4 the largest component is first in 639 of 640 first-escape runs and in all 380 full runs, which have 20 independent initialization directions. On the tensor-like trees the largest component is first in 400 of 640 runs on MPS4, 264/328 on binary8, 104/160 on HT4 and 64/80 on MPS8; for a given seed it is the same at every σ (10 of 10 seeds in every cell of the full runs). It is the LO flow's r_1 in all 231 runs at the smallest σ. The race names it in all 640 MPS4 and 80 MPS8 runs, in 328 of 328 binary8 and 152 of 160 HT4 runs, and in 537 of 640 binary4 runs.

Tensor-like pairs. empirical The race curves (22) lie at 0.66–0.89 ($w = 0.5$) and 0.55–0.66 ($w = 0.8$), with gradient flow 0.018 ± 0.019 below them over the 102 MPS cells, a fluctuation of the shared seeds. On binary8 P(smaller first) = 0.240, 0.4075, 0.4875 at $w = 0.5$, 0.8, 0.95 for every $\sigma \in [0.01, 0.1]$; at $\sigma = 0.3$ at most 1% change.

Number of cores, bond and physical dimension; the selection rule. empirical On MPS with $d = \chi = 2$, $\mathrm{sd} \ln \hat{t}_1 = 0.59$, 0.93, 1.20 for $K = 4$, 6, 8. With capped bonds the gain stops once the end bonds reach their cap d (MPS8, $d = 4$: P(larger first) = 0.60, 0.73, 0.76 at $\chi = 2$, 4, 8). With uncapped bonds $\mathrm{sd} \ln \hat{t}_1 \propto \chi^{-(0.48\text{–}0.54)}$ ($\chi \geq 4$), and its growth with K approaches $\sqrt{K-2}$. The test of d at equal bonds (binary8 and MPS4, independent seed sets) excludes differences above about 0.06 at 95%. With X_r the projection of the initialization onto the orbit of component r's balanced rank-one configuration, $\ln \hat{t}_{1,r} \approx \mathrm{const} - q \ln X_r$ with $q = (0.99\text{–}1.25)(K-2)$ (s.e. 1–4%).

Matrix-like pairs. empirical Table 23 lists the fitted exponents. Fits over the whole measured range lie below $(1-w)/w$ in 12 of 13 sets, by up to 24%: 2.5–3.3 s.e. at $w = 0.95$ and for MPS4 $D = \{0, 2\}$ at $w = 0.9$, otherwise within 1.8 s.e. For the adjacent pairs the fit over $\sigma = 10^{-5}$–10^{-3}

Table 23: Matrix-like pairs: exponent γ of $P(\text{smaller first}) = C\sigma^{\gamma}$, binomial maximum likelihood over the whole σ range, s.e. from a bootstrap over seeds (shared across σ); $d = 2$, $\chi = 4$ for the pattern pairs; weights $(1, w)$. Prediction (34): $n(1-w)/w$, $n = 1$ for non-adjacent cores, $n = \chi_e = 4$ (conjectured) for adjacent ones. –: not run, or (MPS6 $\{2,3\}$, $w = 0.7$) events at two values of σ only. Source: `experiments/order/results/sigma_fits.csv`.

pair	σ range, seeds	$w = 0.7$	0.8	0.9	0.95
binary4, $\chi = 2$	10^{-4}–0.1, 1,000	0.370 ± 0.034	0.237 ± 0.018	0.103 ± 0.008	0.042 ± 0.004
binary4, $\chi = 4$	10^{-4}–0.1, 1,000	0.42 ± 0.05	0.248 ± 0.018	0.102 ± 0.008	0.040 ± 0.004
MPS6, $D = \{0,5\}$	0.003–0.1, 400	0.46 ± 0.06	0.249 ± 0.029	0.101 ± 0.014	–
MPS4, $D = \{0,2\}$	10^{-5}–0.1, 1,000	–	0.240 ± 0.017	0.095 ± 0.006	–
prediction $(1-w)/w$ $(n = 1)$		0.429	0.250	0.111	0.053
MPS6, $D = \{2,3\}$ (adjacent)	0.003–0.1, 400	–	0.67 ± 0.09	0.22 ± 0.03	–
MPS4, $D = \{1,2\}$ (adjacent)	10^{-5}–0.1, 1,000	–	0.82 ± 0.06	0.31 ± 0.02	–
conjecture $4(1-w)/w$ $(n = \chi_e = 4)$		1.71	1.00	0.444	–

on MPS4 at $w = 0.9$ gives 0.40 ± 0.05; at $w = 0.8$ and on MPS6 (down to $\sigma = 0.003$) the exponents stay below $n(1-w)/w$. With two components binary4 is far greedier than a race of independent clocks (0.87 against 0.73 at $w = 0.8$).

After a wrong first escape. empirical Saddle starts: binary4 and MPS4 $D = \{0,2\}$ rotate (192 and 144 runs), MPS4 $D = \{1,2\}$ adds (144 runs, a_2 stays at 1), the odeco MPS4, MPS6 and binary8 keep the wrong order ($a_2 \geq 0.96$). At $w = 0.8$ the midpoint $\lambda_{\min}$ (raw) is -0.18559, -0.17890 and -0.89427 (three times) against -0.18566, -0.17888 and -0.89443, independent of σ; the escape slopes against $\ln(1/\sigma)$ are 5.466 ± 0.082, 5.5903 ± 0.0045, 1.1171 ± 0.0009 against 5.386, 5.590, 1.118 (mean $\pm$ s.e. over seeds; the binary4 excess comes from $\sigma = 10^{-2}$, and the local slope times $|\lambda_{\min}|$ tends to 1.0000 at $\sigma = 10^{-7}$), and a power law fits 11–88 times worse in rms residual (95–106 on the adding branch). On the kept branch the escape times scale as $t_1 \propto \sigma^{-1.98 \pm 0.03}$, $\sigma^{-4.26 \pm 0.13}$, $\sigma^{-5.3 \pm 0.3}$ on MPS4, MPS6, binary8 (MPS6 and binary8 limited range); one MPS6 seed of six stalls. Natural runs (up to 40 per (w, σ)): the scaled correction time $|\lambda_{\min}|(t_1 - t_2)$ has slope -0.02 ± 0.03 per unit of $\ln(1/\sigma)$, and a wrong plateau is resolved in at most 2.5% of the runs at $\sigma = 10^{-4}$–10^{-2}; the smaller component overshoots and falls to zero while the larger rises, then come a plateau at $\frac{1}{2}w^2$ and the relearning. At $\sigma = 0.1$, 17, 12 and 1 of 40 binary4 runs rotate at $w = 0.7$, 0.8, 0.9 (52 of 120 are in another class), with scaled times 0.85, 1.16, 0.38, and 5–7.5% show a wrong plateau. All 150 adjacent runs add, within 0.1–1 time units; on MPS4 odeco the wrong plateau is resolved and 39 of 40 runs add after it, while one stalls.

Three and four components. empirical One σ per tree (0.05 binary4, 0.1 MPS4, 0.2 MPS6, 0.25 binary8), keeping dormant components above the float64 floor; the last two lie outside the checked range of σ-independence. MPS4: 0.35 and 0.23 greedy against 0.36 and 0.25 for the model. Paired with each run's own LO race, 41 runs are greedy only in gradient flow and 43 only in the race; the own race names the first event in 97–100% of the runs. Runs that stalled or did not finish by $t = 10^9$ are excluded from both sides: at most 5 of 200 on MPS4 and binary8, about half on MPS6. With four components and $\chi = 4$, MPS6 stops before the fourth in 396 of 399 runs and binary8 in 389 of 393, and none of 22 runs without any stall stop learns all four by $t = 10^9$.

D.1.7 Failure modes

In the odeco runs the classes of Section B.2 called *trapped* and *stalled* are the dead routes and the slow routes of Section 4.8.

Odeco teachers. empirical With the close teacher on MPS4 at $\chi = 3$ the outcome is not fixed by the direction (one seed has a dead route at 6 of 8 values of σ). On HT4 at $\chi = 6$ the 5 dead routes of 50 runs come from one seed, at every σ. The classes are unchanged at a 100 times smaller absolute tolerance (9 of 9 runs). The optimizer experiment met the same failure on its odeco MPS8: fan-in gradient flow converged in none of 12 runs, with dead route ends in 11 (Section B.8; a fix-round result of that experiment, not re-verified independently). Balanced initialization gives no dead route where fan-in gives 12 of 30 and 47 of 90; 17 of those 47 fan-in runs complete when re-gauged, and 5 runs that complete from fan-in initialization are slow when re-gauged. The routes die on the plateaus, not in the escapes: across an escape a dormant amplitude $|\langle W, U_r\rangle|$ changes by a median factor 2.1–3.0 at $K = 3$ (it grows: the head start of Proposition A.25), 0.64–0.74 at $K = 4$ and 0.12–0.75 at $K = 7, 8$, and falls by more than 100 times in at most 1.3% ($K = 4$), 2% ($K = 7, 8$, $\chi = 6$) and 8.3% ($\chi = 3$; 11% for the later escapes on MPS8 with the separated teacher) of the escapes.

Slow routes: slow where tested. empirical (prediction derived) new A slow route with live ends should escape after about the plateau of a balanced route at its present amplitude, $b^{-(K-2)}/((K-2)s_r)$ with $b = (s_r|a_r|)^{1/K}$. Continued from their final state, 16 of 23 slow or unresolved runs escaped, at 0.24–2.19 times this estimate (all 8 with the shortest estimates, 6 of 7 further runs of a seeded random sample, 2 of 8 MPS8 runs at $\chi = 6$); 2 did not escape within three times it and 5 ran out of wall-clock time or failed. The estimate is not a bound (4 escapes came later). For the slow runs not continued (91 of 106 at $K = 7, 8$ with $\chi = 3$, 35 of 43 MPS8 runs at $\chi = 6$), "a long plateau, not a dead route" is a conjecture: their dormant slices are pinned at 0.01–0.04 σ on binary8 and balanced MPS8, but were still decaying in 6 of 13 balanced HT4 runs and 17 of 59 fan-in MPS8 runs.

Pattern teachers. empirical A stalled two-component run sits at the partial fit $L = s_2^2/2$. Stalls by k, both protocols pooled, fan-in initialization at $\chi = 2$: 130 of 768 runs ($k = 4$), 62/816 ($k = 5$), 133/360 (6), 29/118 (7), 44/60 (8). The dead-route check reran 22 (case, σ) pairs with a 60 times longer patience (170 runs); the 37 flagged runs end with route-end amplitudes of 10^{-17}–10^{-13}. The pilot, with separate code, found the same collapse to the rank-one saddle. Of 10 balanced stalls rerun with longer patience, 6 escaped very late (after $2 \cdot 10^9$–$5 \cdot 10^{10}$, about 10^5 times the usual plateau), 3 are pinned (route ends at $3 \cdot 10^{-10}$, $6 \cdot 10^{-9}$, $4 \cdot 10^{-8}$ at $t = 10^{11}$–$5 \cdot 10^{11}$) and one was in the middle of a late escape when its budget ended; all 16 balanced stalls are on MPS routes. Spare channels: on routes with an uncapped end the stalls fall from 79 at $\chi = 2$ to none at $\chi = 4$ (all 13 such cases escape); on routes with both ends capped they rise from 90 to 106, in 7 of the 8 such cases that stall at all (they fall only for MPS4 $\{0, 3\}$, from 30 to 6 of 48). midroot6 and MPS6, the same graph with different roots, share exponents but not stall rates ($D = \{0, 5\}$ at $\chi = 4$: 30 and 18 of 30): the root changes the fan-in statistics of Q_e. The odeco runs agree: $\chi = 6$ removes the dead routes on binary8, where every bond has twice the teacher rank, but not on HT4 and MPS8, whose route ends sit on bonds capped at $d = 4$, one channel above it.

Effect on the exponents. empirical Stalls depend on σ: in 20 of 33 fan-in rows with stalls a stalling seed escapes at other values of σ, so it enters the fits with fewer points. Restricting each

Table 24: Stalled runs among two-component runs with $k \geq 4$: small initialization, $d = 2$, the same 24 (tree, D) cases and σ grids in every column. Fan-in initialization at $\chi = 2$ and $\chi = 4$; balanced initialization ($Q_e = 0$, same W_0) at $\chi = 2$. Bottom rows: split by whether at least one end of the route sits on a bond below the cap (3) or both ends sit on bonds capped at d (MPS chain ends, HT leaves). Source: `experiments/locality/results/stall_capped_ends.csv` and `run_classes_by_sweep.csv`.

k	fan-in, $\chi = 2$	balanced, $\chi = 2$	fan-in, $\chi = 4$
4	67/384	2/384	18/384
5	20/384	5/384	0/384
6	58/180	5/180	59/180
7	8/30	0/30	12/30
8	16/30	4/30	17/27
all	169/1,008	16/1,008	106/1,005
route with an uncapped end (13 cases)	79	–	0
both route ends capped (11 cases)	90	–	106

row to seeds that never stall leaves the exponent unchanged in 27 of 46 rows with stalls and moves it by 0.1 or more in 9 fan-in rows (by up to 1.38; four of them keep only 1–3 seeds) and 3 other rows. This accounts for the off-target saddle-start $k = 4$ and HT8 rows of Table 22.

The reduced-model criterion. empirical Evaluated once, at the start of the plateau ($a_1 = 0.99$), in the 122 fan-in and saddle-start runs of the stall diagnosis, the criterion of the reduced model of Proposition A.28 (an end is unfavourable if $c = Q_{\nu\nu} + \|y_{\text{route}}\|^2 < 0$; dead if all ends are) is strongly associated with death (30 of 55 runs with both ends unfavourable died, 7 of 67 with a favourable end) but is neither sufficient (25 predicted-dead runs escaped) nor necessary (seven dead runs had a live end at the plateau start). It conflicts with the data in three places: (i) exactly balanced initialization leaves no dead route in either experiment (consistent), but on the pattern-teacher MPS8 routes the ends shrink to 10^{-10}–10^{-8} and escapes are delayed by 10^5 or more (the balanced odeco MPS8 runs are slow instead, with route slices near $0.03\,\sigma$); (ii) its heuristic frequencies (about half of the two-ended HT routes dead under fan-in initialization; fewer stalls on MPS with $\chi = d$) are contradicted: 0 of 144 small-initialization HT4 runs with $k = 5$ stall, against 30 of 48 on MPS4 $\{0, 3\}$ and 17 of 48 on MPS6 $\{0, 3\}$ (the odeco HT4 routes, with four leaf ends, end dead in at least 24 of 30 runs at $\chi = 3$); (iii) spare channels remove stalls only on routes with an uncapped end, here as in the regime maps, and the stalls rise on capped routes. The criterion of Section D.1.8 (pending verification) resolves (ii) and the drop in (iii).

Steep MPS routes. empirical (mechanism open) One row stays steep at small σ: MPS8 $\{1, 5\}$ ($k = 5$) from the saddle, $p = 3.67 \pm 0.20$ at $\sigma \in [0.012, 0.03]$ (median 3.83; limited range: 4 fitted seeds, 0.4 decade; 10 of 24 runs stalled), with a matching Hessian exponent (3.74 projected, 3.75 raw). Its fitted seeds escape at every σ, so it meets a refutation criterion of the theory (a per-seed exponent off by more than 0.2 at $\sigma \leq 0.05$ in runs without stall or sign correction), on thin evidence. In the steep MPS rows the route-end amplitudes at the plateau midpoint scale as σ^a with $a = 1.24$–1.91 and shrink during the plateau (median ratio of midpoint to first event 0.53–0.78, against 0.85–1.03 in on-target rows); but $a > 1$ also occurs in on-target rows (binary8 $\{0, 7\}$, balanced: $a = 1.34$, $p = 2.99$), so the steep exponents are associated with shrinking route ends, not shown to be caused by them. Neither the logarithmic correction expected if small ends caught up through the conservation laws (per seed it fits much worse than a free power law: median residual

Table 25: Stall test at χ equal to the teacher rank: $d = 3$, $\chi = 3$, odeco teacher $(3, 2, 1)$, $\sigma \in \{0.3, 0.45, 0.6\}$, six seeds; "balanced": the same W_0 with $Q_e = 0$ on every bond. Types of the stalled runs from the third bond singular value of W at the end, minimized over edges: dead route below 10^{-12}, decaying up to 10^{-4} (no stalled run is above); the centred runs did not record the spectra of W. Source: `experiments/regime_map/results/report/stall_test_summary.csv`.

model	tree (K)	initialization	converged	stalled: dead route	decaying	not typed
plain	binary8 (7)	fan-in	4/18	12	2	–
		balanced	6/18	0	12	–
	MPS8 (8)	fan-in	0/18	16	2	–
centred	binary8 (7)	fan-in	4/18	–	–	14
		balanced	8/18	–	–	10
	MPS8 (8)	fan-in	0/18	–	–	18
		balanced	3/12	–	–	9

of $\ln \Delta t_2$ 0.25–0.62 against 0.05–0.12) nor a law $\sigma^{-(2k-6)}$ (both ends pinned, $a = 2$; measured a is seed-dependent, between 1 and 2) describes the data.

Stall test of the regime maps. empirical The test of Table 25 ran budgets of $6 \cdot 10^4$–$1.6 \cdot 10^7$ time units. Fan-in initialization makes the root bonds child-heavy, $\mathbb{E}Q_e = \sigma^2(1 - 1/\chi)I \succ 0$, so with $\chi = R$ the route ends of the last component decay at a rate $\propto \sigma^2$ while its escape needs a time $\propto \sigma^{-(K-2)}$. From fan-in initialization the stalled route is dead: its third bond singular value ends below 10^{-12} in 28 of 32 stalled plain runs. With balanced initialization the route still decays and 12 of 18 runs do not escape: the third singular value is $1.4 \cdot 10^{-5}$ (median) at the end. In the plain-model $\sigma \times K$ sweeps the stalled runs are typed by the R-th bond singular value at the end (thresholds on the end value, not trends): below 10^{-12}, a dead route (546 of 795), between 10^{-12} and 10^{-4} (212) or above (37); 182 of the 795 stopped at the step limit, and some of the 37 at $\sigma \geq 0.77$ learned nothing. In the stall test 9 of 18 fan-in MPS8 runs learn only the first component. At $d = 2$ the MPS with full bonds still stall in up to 3 of 6 seeds. Some runs of the maps stop at the integrator's step limit: unfinished lazy fits at square bonds (Section 5.1) and some deep-tree runs, for example on MPS with $N = 8$ at $\sigma = 0.1$ (after $2 \cdot 10^4$ steps, at $t = 1.4 \cdot 10^9$ of a budget of $5.7 \cdot 10^9$).

D.1.8 Trap criterion from the initialization (pending verification)

Source: `notes/theory_trap.md` (v1) and `experiments/theory_checks/trap/` (README there), merged before its verification. Setting: a learned rank-one saddle of component q (every core $a\, p_v \otimes \bigotimes_f w_f$, $a = s_q^{1/K}$), the dormant component r with route C, $k = |C|$; an *end* u of C has one route neighbour, its *partner* w, on the route edge e.

Statements. (1) On the plateau the entries with all bond indices learned relax fast; the entries with one new bond index form two-core products whose Gauss–Newton cost is only their part normal to the learned manifold (lemma proven; reduced flow derived). (2) In this reduced flow $C_u = X_u^{\perp\top} X_u^{\perp} - X_w^{\perp\top} X_w^{\perp}$ is conserved, and the end's route entry survives iff C_u has a positive eigenvalue (proven within the reduced flow). (3) Identity: $C_u = \nu^\top Q_e^{(u)} \nu + \|Y_{\mathrm{nv}}\|^2 - \|X_{\mathrm{nv}}\|^2 + O(\sigma^3)$, the conserved bond imbalance (end minus partner) plus the partner's non-visible new entries minus the end's; the reduced model of Proposition A.28 kept only the route part of Y_{nv} (exact identity; checked to 3–4 digits). (4) A partner without sites gives no second-order cost: the end then decays

through a three-core cluster at a rate $\propto \sigma^4$; in general an end of visible degree j decays at a rate $\propto \sigma^{2(j-1)}$, so it can kill its route only if $k \geq 2j$: $k \geq 4$ for a partner with sites, $k \geq 6$ at HT and binary leaves, never for $k \leq 3$ (derived). (5) A route is dead iff two or more of its ends die; one dying end is pinned at $\propto \sigma^{k-3}$ and escapes after $\propto \sigma^{-(2k-6)}$; at $k = 4$ a σ-free competition, settled by a reduced route model, decides (derived; checked). (6) An end with more new directions on its route edge than its partner has normal physical rows generically never dies (spare channels protect only there; derived). (7) The plateau-start state is σ times a function of the initialization direction (checked on MPS4 $\{0, 3\}$: $\sigma = 0.02$ and 0.005 agree to 3 digits), so the outcome is a property of the direction (derived). (8) Balanced initialization: $C_u \geq 0$ at ends that are tree leaves, which cannot die; ends with extra legs (pattern teachers on MPS) still can (derived).

Per-run tests empirical. Every committed two-component run of the locality sweeps and of the pilot was rebuilt from its configuration and seed and predicted without gradient flow. Fan-in initialization, saddle start and small initialization (3,492 locality runs with $k \geq 4$, 587 stalled; the 191 wrong-order runs excluded), stalled over runs per predicted class: dead 408/457 (89%), pinned 141/934 (15%), alive 38/2,101 (1.8%); the reduced-model criterion of Proposition A.28 gives 417/1,143 (36%), 119/1,116 and 51/1,233 (4.1%). By k (dead, pinned, alive): $k = 4$: 203/232, 2/432, 5/796; $k = 5$: 73/79, 10/116, 9/1,089; $k = 6$: 86/98, 87/284, 18/150; $k = 7$: 25/27, 8/60, 0/42; $k = 8$: 21/21, 34/42, 6/24. Stall diagnosis (60 times the patience): predicted dead 30 dead, 6 escaped, 2 pinned; predicted pinned 5, 63, 0; predicted alive 2, 56, 0. Pilot: 102/109 predicted-dead runs die, 20/182 pinned and 2/146 alive. MPS4 at $\chi = 2$, fan-in (locality and pilot): all 150 predicted-dead runs stall or die, none of the other 344. Balanced initialization: 16 of the 50 predicted-dead runs stall, none of them dead (very late escapes instead); 4 of 990 predicted-alive runs stall. Per seed (cases with at least three values of σ): 447 of 462 predicted-alive seeds never stall, 45 of 65 predicted-dead seeds stall at every σ. Known failures (56 false alarms and 40 misses among 3,929 fan-in runs with the pilot): $k = 4$ with spare inner channels (MPS4 at $\chi = 4$: 20 of 46 predicted-dead runs escape); the marginal case $k = 6$ on HT4 $\{0, 1, 2\}$ (11 of 18 predicted-dead runs escape); large σ (24 predicted-alive runs on long MPS and midroot6 routes stall, all at $\sigma \geq 0.08$; two dead HT4 $\{1, 2\}$ saddle-start runs at $\sigma = 0.19$, 0.30; five dead runs predicted pinned, all at $\sigma \geq 0.19$); seeds near the boundary at $k = 4$–5.

Odeco teachers. empirical The dying component is the last one: the trap acts on the third plateau, after two escapes. The learning order is predicted from the initialization (the first component in 1,196 of 1,196 runs, the first two in order in 872 of 1,029), but the trap is not: from the initialization alone the criterion does not separate the classes. From the state after the second escape it does: MPS4 at $\chi = 3$, $\sigma = 0.03$, 10 of 10 seeds with the separated teacher and 10 of 10 with the close one; HT4 at $\chi = 3$, $\sigma = 0.2$, 6 of 6. A σ-free model of the second plateau would close the gap.

Probability under fan-in initialization. derived (saddle start, one new direction per route edge, partner with sites, independent Gaussian entries) With $n_u = d^{s_u} - 1$ normal physical rows of variance σ^2/F_u at the end and n_w of variance σ^2/F_w at the partner (F the fan-in, s the number of sites), $p_u = P(C_u < 0) = I_{F_u/(F_u+F_w)}(n_u/2, n_w/2)$: 0.392 at an MPS end that is a tree leaf ($d = \chi_e = 2$, any χ), 0.5 at an MPS root end at $\chi = 2$ (0.392 at $\chi = 4$); an internal MPS end has 0.5 at $\chi = 2$ and is protected at $\chi \geq 3$. For two ends ($k \geq 5$): $P(\text{dead}) = p_u p_{u'}$, $P(\text{pinned}) = p_u + p_{u'} - 2p_u p_{u'}$ (0.196 and 0.50 for an MPS chain with a leaf and a root end). A Monte Carlo with fresh seeds agrees with the closed form within two standard errors at every case with $k \geq 5$ empirical; the trap probability is 0.1–0.3 per run for routes whose two ends have partners with sites. The committed

seeds of MPS4 $\{0, 3\}$ (5 of 8 trapping) are a trap-prone draw (probability about 3%).

What it changes. The weak association of the reduced-model criterion (30 of 55 against 7 of 67) came from that criterion; with the new one the stall diagnosis gives 30 of 38 predicted-dead runs dead and 7 of 126 others. HT routes with $k = 5$ never stall (0 of 144) because HT leaves need $k \geq 6$; the drop of the stalls with spare channels is the protection of (6), while their rise on capped routes from small initialization is reproduced per run but its mechanism is not isolated. Steep MPS routes: a pinned end gives exponent $2k - 6$ and end amplitudes $\propto \sigma^{k-3}$, a partly decayed end values in between, qualitatively as measured ($k = 5$: 3.3–3.8; not tested per seed). Prior work: a conserved quantity that decides which side of a unit can vanish is Lemma B.1 of Kunin et al. (2025); Furman et al. (2026), Thm G.1, give a spurious minimum away from the minimum-norm set, and the trapped endpoints here are reached because the conserved imbalance, of order σ^2, is as large as the dormant entries.

D.1.9 Provenance and corrections after review

The numbers of Section 4 come from the committed result tables of the experiments `s2s_core`, `locality`, `order` and `regime_map` (Section C), checked against them in two verification reviews per experiment and a final review of the claims. Three numbers of Section 4.7 exist only in the independent recomputation of the order experiment (scripts in `notes/reviews/scripts/verify_order/`): the fits over the three smallest values of σ (`v_sigma.py`, `v_sigma2.py`, `v_sigma3.py`), the paired McNemar comparison of each run's order with its own LO race (`v_multi.py`), and the margin of the stall rule of the order runs, which had no false positive in 22 runs without any stall stop (`v_multi.py`). The same review reproduced $\mathcal{T}(Z)$ for the 160 binary4 and MPS4 first-escape jobs with an independent integration of the spherical flow, to $2 \cdot 10^{-10}$. The independent cross-check of the locality results is the pilot `experiments/pilot_h2_locality/` (its own code). Corrections after review: the norm law $t_1 \propto \|\theta_0\|^{-(K-2)}$, the first guess of the research plan, is only a lower bound (Section 4.3); the binary4 sequence, its σ^{-1} plateaus and its strict wrong-order saddle are known for deep linear networks and are reported as checks; the single-route consistency of the bond spectra is a check of the Steiner edges, not a prediction success rate; the law $\sigma^{-(2k-6)}$ for steep MPS routes was the pilot's conjecture; the claims withdrawn in the reviews are not restated.

D.2 Map of regimes and the lazy and rich regimes

Values behind Section 3 and Sections 5.1 to 5.3; settings in Section B.5. Boundaries are geometric means $\times/$ their geometric standard error over six seeds, or medians with quartiles or brackets.

Kernels and boundaries. The binary4 maps ($\sigma \times \chi$, χ sweep, imbalance) are labelled with the tensor kernel; the K sweeps, the maps of the binary tree with $N = 8$ and the large-χ tables with the probe kernel. Near the lazy boundary $\delta_\Theta^{\text{probe}}$ is 0.54–1.11 times δ_Θ (median 0.78; 43 runs at $\chi \geq 32$) empirical, so relabelling the binary4 maps from the probe to the tensor kernel changed 65 of their 5,298 labels (1.2%): 59 from lazy to non-lazy at the lazy boundary and 6 plain runs to lazy. It moved the per-seed lazy boundaries up by 1–9% in σ (centred: 4–9%; 14–15% at $\chi = 6$–9); the saddle-to-saddle boundaries do not depend on the kernel. Per seed, the saddle-to-saddle boundary is the crossing of $\min_i \rho_i = 3$, or a bracket from the last saddle-to-saddle run to the next run when that run is stalled; the lazy boundary is the crossing of 0.1 by the kernel change among converged runs.

Saddle-to-saddle boundary on binary4. empirical $0.237 \times /1.32$ at $\chi = 3$, $0.099 \times /1.51$ at $\chi = 4$, $0.137 \times /1.19$ at $\chi = 6$, $0.136 \times /1.11$ at $\chi = 9$, $0.075 \times /1.33$ at $\chi = 12$, $0.072 \times /1.07$ at $\chi = 18$, $0.067 \times /1.12$ at $\chi = 24$ and $0.043 \times /1.05$ at $\chi = 36$ ($\sigma \times \chi$ sweep), falling to 0.023 at $\chi = 128$ (χ sweep). At $\chi = 3$ it lies within the spread of 30 plain-model seeds (median 0.27–0.31, Section A.8). Per seed, $\min_i \rho_i$ falls like a power of σ with fitted exponents 1.015–1.048 in 96 of the 101 fits (all χ of both sweeps), against $K - 2 = 1$; the five exceptions (1.09–1.27) are seeds with transients above $\frac{1}{2}$. Deep in the saddle-to-saddle regime ($\chi = 36$, $\sigma = 0.01$) the widths of the final rises are 0.464–0.468, 0.798–0.801 and 2.010 against the aligned-mode values 0.464, 0.797 and 2.009 of Proposition A.48 for $s_r = 3, 2, 1$. Extrapolating $\rho \propto 1/\sigma$ from each seed's deepest run gives 0.86–1.23 times the measured boundary for 53 of 54 seeds: a consistency check of the σ^{-1} law, not a prediction of the boundary.

Lazy boundary on binary4. empirical With the tensor kernel (medians of six seeds) the lazy boundary is 1.13, 0.94, 0.82, 0.66 and 0.571 at $\chi = 32, 48, 64, 96, 128$, inside the predicted brackets $[1.10, 1.20]$, $[0.92, 0.96]$, $[0.80, 0.82]$, $[0.66, 0.67]$, and 0.07% below the lower edge of $[0.571, 0.577]$. At $\chi = 9, 12, 16, 24$, $\sigma_{\text{lazy}}\sqrt{\chi} = 8.96, 7.99, 7.43, 6.75$ (6.3%, 2.9%, 1.5% above the bracket at $\chi = 9, 12, 16$).

Slopes against χ. empirical Saddle-to-saddle boundary: -0.528 ± 0.053 over $\chi = 16$–128 (42 values, 0.9 decade; per-seed fits -0.528 ± 0.073) and -0.552 ± 0.049 over $\chi = 9$–128. Lazy boundary: -0.504 ± 0.011 over $\chi = 32$–128 and -0.553 ± 0.013 over $\chi = 16$–128, with $\chi = 16, 24$ not yet asymptotic. The slope $-\frac{1}{3}$ of $\sigma^3\chi/d^2$ lies 2.7 and 2.9 standard errors from the saddle-to-saddle slope in per-seed fits over $\chi \geq 16$ and $\chi \geq 32$ (3.7 in the pooled fit over $\chi \geq 16$, which treats seeds sharing a teacher across χ as independent). The rich window $\sigma_{\text{lazy}}/\sigma_c$ at $\chi \geq 32$ spans 19.5–24.2 with the tensor kernel and 18.4–22.8 with the probe kernel; the per-bond Gram scale $\chi\sigma^2/d^2$ is 0.010 at $\sigma\sqrt{\chi} = 0.30$ and 4.69 at $\sigma\sqrt{\chi} = 6.50$.

Number of cores. empirical Centred model, $d = 2$, $\chi = 4$: medians (quartiles) of the per-seed saddle-to-saddle boundaries are 0.17 (0.11–0.20) on binary4, 0.49 (0.32–0.58) on MPS4, 0.61 (0.48–0.80) on MPS5, 0.68 (0.68–0.74; four seeds) on MPS6 and 0.75 (0.61–0.93) on binary8. Read as aligned initialization with per-core amplitude $\kappa_K\sigma$, $\kappa_K = a_{0c}/\sigma_c$ grows from 0.39 ($K = 3$) to 0.65 ($K = 7$), as in the runs of Section A.8. At $K = 8, 10, 12, 15$ only 1, 1, 0, 0 of six seeds are resolved; the brackets have medians $[0.88, 1.16]$, $[0.80, 1.00]$, $[0.83, 1.03]$, $[0.63, 1.36]$. Classified by $\min_i \rho_i$ instead of the label, the run after the last saddle-to-saddle run is saddle-to-saddle-like ($\rho = 3.9$–642) for 13 of the 18 unresolved MPS seeds. At $K = 8$ the block then ends at a crossing in 5 of 6 seeds (median 1.01); at $K = 10$ and 12 such runs persist up to $\sigma \approx 1.0$–1.3 and the next runs learn nothing within the budget, so a crossing to rich is seen in 2 of 6 seeds at $K = 10$ (both at 0.90) and in none at $K = 12$. All these runs end on the floor ($L/L(0) = 0.001$–0.074), and 9 of the 13 learn the smaller component first: they do not measure the boundary of a realizable problem. On $K = 4$–7 both $\sigma_c^K = \text{const}$ and $\sigma_c^K/\sqrt{d^N} = \text{const}$ fit the medians ($\chi^2 = 0.79$ and 0.59, three degrees of freedom); adding the per-seed values whose bracket ends at a crossing ($K = 8$, two seeds at $K = 10$) leaves them indistinguishable ($\chi^2 = 7.5$ and 7.3, five degrees of freedom). At the smallest σ, 0, 1, 0, 3, 0, 5, 3 and 6 of six seeds stall on binary4, MPS4, MPS5, MPS6, binary8, MPS8, MPS10 and MPS12, and 5 of 5 on binary16, each at a partial fit with one component learned ($L/L(0) = 4/13$ or $9/13$).

Imbalance. empirical At $\chi = 16$, $\alpha = 3.375$ the per-seed local slopes of δ_Θ against λ are 1.97 ± 0.01, 1.85 ± 0.06, 1.55 ± 0.11 and 1.22 ± 0.09 between $\lambda = 0.01$, 0.03, 0.1, 0.3 and 1; the saturation level for $\lambda \geq 10$ is 0.59–0.61 in the medians and 0.51–0.75 over seeds.

Threshold sensitivity and the two-copy lazy boundary. empirical (predictions derived) Raising the separation threshold from 2 to 5 moves the saddle-to-saddle boundary of binary4 by a factor 2.14–2.34 ($\chi = 3$–36; means of six seeds, s.e. at most 0.08), against $(5/2)^{1/(K-2)} = 2.5$: the $O(1)$ part of ρ lowers the ratio at finite σ. Raising the kernel threshold from 0.05 to 0.2 moves the lazy boundary by 1.97 ± 0.08 at $\chi = 9$, down to 1.33 ± 0.01 at $\chi = 36$ and 1.30–1.35 at $\chi = 32$–128: with $\sigma_{\text{lazy}} \propto \delta_*^{-1/(Ks)}$, the local slope $s = \mathrm{d}\ln\delta_\Theta/\mathrm{d}\ln\xi$ grows from 1 (first order, ratio 1.59) towards 2 (second order, 1.26), and the slopes 1.5–1.8 at $\chi \geq 32$ predict 1.27–1.36. Only the boundaries move: across the nine threshold pairs at most 70 of the 912 centred binary4 labels change (7.7%), each to a neighbouring regime. The two-copy model has the same tangent space as the centred one, but its kernel changes only at second order, so its lazy boundary is $2^{-1/3} = 0.794$ times lower (Proposition A.47): measured 0.79 at $\chi = 32$, 64, 128 (tensor kernel). Its saddle-to-saddle side is not comparable (merged first events, Section 5.2).

Large K in realizable problems. empirical new Pre-registered sweep: binary trees with $N = 4$–16 ($K = 3$–15), odeco teacher $(3, 2)$ at $d = 2$ and 3, six seeds per cell; per seed, σ_c is the first crossing of $\rho = 3$ coming from small σ, on an adaptive grid refined until the bracket is a factor $2^{1/(K-2)}$ in σ. A run whose initial function already overlaps a teacher component by 0.1 is outside the domain of the event rule, and a seed that is saddle-to-saddle up to that scale gives only a lower bound. Plain model, $\chi = 4 = 2R$ (1,070 runs, 156 seeds): at $K \geq 8$, 85 of the 96 seeds are censored from above, 76 at the domain limit and 9 still saddle-to-saddle when the budget ended (at $d = 3$, 47 of 48); the other 11 cross at $\sigma_c = 0.86$–1.37 ($d = 2$, $K = 8$–12, including 4 of 6 seeds at $K = 12$; and 1.08 at $d = 3$, $K = 10$). Every censored lower bound at $K \geq 8$ (1.07–2.03; 1.18–2.03 at $K \geq 11$) exceeds 1 and both laws extrapolated from the resolved cells, so $\sigma_c \to 1$ is rejected for the plain model at $K \leq 15$. The bounds sit a median 3% below each seed's own domain limit, whose medians are 1.50 ($d = 2$) and 1.80 ($d = 3$) at $K \geq 11$, still rising: the domain limit is a product of K per-core factors and tends to about $1.15\sqrt{d}$ derived (checked by simulation), and only 38 of the 85 bounds exceed $\sqrt{d}$. The bounds rest on the sampled σ (a median of four runs inside the domain per seed); 14 of them, at $d = 3$, $K = 12$–15, rest on the rule for runs whose second component was not learned by the end of the budget. On the resolved cells ($d = 2$: $K = 3$–7 and 12; $d = 3$: $K = 3$–5) the slope of $\ln(\sigma_c^K/a_{0c}^K)$ against $\ln\sqrt{d^N}$ is 1.21 ± 0.22 at the separation threshold 3, favouring the fixed per-component amplitude by the pre-registered rule; at threshold 5 it favours it only with standard errors scaled by the fit's χ^2, which was not pre-registered (unresolved without); it is 0.72 ± 0.56 at threshold 2 and 0.61 ± 0.51 without the cells where the cores shrink (both unresolved): a censored selection that does not answer the large-K question.

Before the first escape the plain model's cores lose a median 0.1% of their squared norm where $x_W = \sigma^K\sqrt{d^N}/(s_2 d) < 0.1$, 10% where x_W is between 1 and 10, 34%, 52% and 72% in the next three decades and 92% where $x_W \geq 10^4$, and none in the centred model; that the cores shrink while $\|W\|_F^2 > \langle W, W^*\rangle$ follows from the exact identity $\frac{\mathrm{d}}{\mathrm{d}t}\|G_v\|_F^2 = -2(\|W\|_F^2 - \langle W, W^*\rangle)$ proven (standard). Near the top of the saddle-to-saddle block ρ flattens against σ (local exponents -3.7, -4.9, -6.4 at $K = 8$–10, $d = 2$, against -6, -7, -8). That this unlearning of W_0 lengthens the plateaus and lifts the plain boundary above both laws is consistent with these data but not tested causally conjecture; its registered signature also required the centred arm to follow the fixed per-component amplitude, which it did not (slope -0.07 ± 0.51, 2.1 standard errors from 1). Centred

model, full bonds (360 runs): medians 0.16, 0.39, 0.50, 0.65, 0.81, 0.83 at $K = 3$–8 ($d = 2$) and 0.15, 0.28 at $K = 3$, 4 ($d = 3$, every seed resolved); per-seed exponents of ρ against σ are -1.05, -2.13, -3.21, -4.93, -5.18, -6.93 at $K = 3$–8, against $-(K-2)$ known (the scaling; new as a check). The plain model agrees with the centred one up to $K = 4$ and lies above it from $K = 5$.

Transients. empirical During an escape, components not yet learned rise transiently (for matrix-like pairs the peak falls like $\sigma^{2(1-w)}$, w the weight ratio, conjecture, Section A.8). In 116 of the 9,165 runs of the regime maps such a transient crossed $\frac{1}{2}$ before the component was learned (14 more cross $\frac{1}{2}$ repeatedly only in a component never learned); with events dated at the first crossing, 15 runs would be rich instead of saddle-to-saddle. Events are therefore dated by the final rise (Section 2.8).

The floor below the cut dimension. empirical With χ equal to the teacher's bond rank ($\chi = 3$) the small-σ floor has mean $(1 - \dim \mathcal{M}_\chi / d^N)\, \sigma^{2K} / \|W^*\|_F^2$ (60), and the median of the six seeds is 0.48 times it; at $\chi = 4$, 6, where the spare bond rank absorbs part of W_0, the median floor is 0.18 and 0.044 times that expression, which applies only at $\chi = 3$ (centred runs at $\sigma \le 0.4$). At $\chi = 3$ the expression reaches the convergence threshold at $\sigma = 0.56$, against the measured onset $\sigma = 1.0$ from which every seed stalls. At large σ the floor saturates at the lazy residual: at $\sigma = 40$ the final loss of each of the 18 centred runs at $\chi = 3$, 4, 6 equals its residual computed at initialization to 0.13%, with a kernel change of at most $1.1 \cdot 10^{-3}$.

Lazy runs of the binary tree with $N = 8$. empirical With bonds over-parameterized twice ($d = 2$, bonds 8 above two sites, 32 above four), every run from $\sigma = 1.42$ on reaches the step limit, but the runs are lazy (below $10^{-3} L(0)$ by then) in 2 of 5 at $\sigma = 1.42$, 2 of 3 at 1.87, 4 of 5 or 6 at each σ from 2.47 to 7.6 and 2 of 5 at 10; the others end at $L/L(0) = 0.0014$–0.0054 with $\delta_\Theta^{\text{probe}} < 0.02$. The kernel boundary is 1.45 (probe kernel, median of four seeds).

Two-copy model at small σ. empirical On binary4 at $\chi = 18$, 24, 36 every two-copy run at $\sigma \le 0.04$ still has three events, but $\min_i \rho_i$ at $\sigma = 0.010$ is only 2.7–14 (centred: 12–29) and grows per seed like $\sigma^{-0.1}$ to $\sigma^{-1.0}$ between $\sigma = 0.04$ and 0.01 (centred: about σ^{-1}); the labels (5, 4 and 5 of six seeds saddle-to-saddle at $\sigma = 0.010$, 4, 3 and 2 at 0.016) do not follow the $\sigma^{-(K-2)}$ law.

Large bond dimension. empirical Fitted exponents of the measurements of Section 5.3: Table 26. At $\sigma = 1$ ($\xi = 3.74$ under SP, further from the lazy limit) the kernel slopes are shallower: SP -2.58 ± 0.06, NTK -0.62 ± 0.02, μP-TTN $+0.089 \pm 0.024$. The plain model gives the same exponents within the errors ($\sigma = 1.5$: -3.06 ± 0.07, -0.83 ± 0.04, $+0.095 \pm 0.024$): with full, over-parameterized bonds, centring does not change the scaling. The factor-wise rule of structured layers (Qiu et al., 2024; Potapczynski et al., 2024), applied to a TTN, makes the initial kernel of the binary tree with $N = 8$ vanish like χ^{-2} and slows the initial fit about linearly in χ (theory checks, two seeds, $\chi = 4$–16; Section A.8).

Provenance and corrections after review. Experiment folder `experiments/regime_map/`; findings in `notes/regime_map.md` (its section 3, the fix round, overrides sections 0–2; sections 4–5 hold round 2 and the tensor-kernel relabel); theory in `notes/theory_regimes.md` and Section A.8. Every number of Section 3, of Sections 5.1 to 5.3 and of this appendix is an entry of the committed tables in `results/`, `results/fix1/`, `results/report/` (aggregates in `results/report/numbers.csv`), `results/tensor_relabel/` and `results/round2/`; Table 26 is `results/fix1/large_chi_slopes_tensor.csv`, and its predicted parameter, root and leaf exponents are those of

Table 26: Exponents of χ at large bond dimension: binary4, $d = 3$, teacher $(3, 2, 1)$, centred model, $\sigma = 1.5$, $\chi = 16$–128, six seeds; least-squares slopes over the 24 runs with standard errors from the scatter; in brackets the predictions (kernel: Section A.8; parameters, root and leaf cores: the same scaling argument, see the provenance paragraph below). δ_Θ: tensor kernel; δ_θ: all parameters; root and leaf cores: relative Frobenius change (the spectral change of the μP-TTN root is predicted $O(1)$).

	δ_Θ	δ_θ	root	leaf cores
SP	-3.02 ± 0.06 (-3)	-1.63 ± 0.04 $(-\frac{3}{2})$	-1.02 ± 0.03 (-1)	-2.96 ± 0.07 $(-\frac{5}{2})$
NTK	-0.79 ± 0.04 (-1)	-0.52 ± 0.03 $(-\frac{1}{2})$	-0.93 ± 0.02 (-1)	-0.53 ± 0.03 $(-\frac{1}{2})$
μP-TTN	$+0.085 \pm 0.024$ (0)	-0.064 ± 0.018 (0)	-0.50 ± 0.01 $(-\frac{1}{2})$	-0.065 ± 0.018 (0)

`notes/theory_regimes.md`, section 4.4. Corrections after review (`notes/reviews/regime_map_round1.md`, `regime_map_round2_verification.md`, `regime_map_round3_verification.md` with its errata E1–E19, `final_review_claims_A.md`, `final_review_claims_B.md`): the lazy boundaries are compared with theory in the tensor kernel, and the binary4 maps are labelled with it; the $K \geq 8$ values of the first K sweep are brackets, not crossings; the extrapolation from each seed's deepest run is a consistency check, not a prediction; the stalled centred runs on the floor number 159; the large-K limit is stated only from the realizable second sweep; the small-σ floor formula is applied only at $\chi = 3$; the NTK exponent is quoted as a two-point local slope next to the full-range fit; and several large-χ exponents are reported with their distance from the prediction.

D.3 Steps, gradient noise, finite samples and optimizers

Details behind Sections 5.4 and 5.5; settings in Sections B.7 and B.8.

Gradient descent below and at the stability limit. empirical An independent code placed the edge of binary $N = 4$ between $0.99\eta_c$ and $1.01\eta_c$. Below η_c a larger step only enlarges transient excursions of near-tied components, which cross $\frac{1}{2}$ and fall back (at $\sigma = 0.1$ in 0.09 ± 0.04 of 45 initializations at $0.1\eta_c$ and 0.18 ± 0.06 at $0.9\eta_c$). Of the 526 event-shift comparisons at $\eta \leq 0.1\eta_c$, the 15 misses beyond 5% are later events, worst where $c_e \approx 0$ after a near-tied race and the $O(\eta^2)$ term dominates; Richardson extrapolation to $\eta \to 0$ recovers c_e to a median relative $1.2 \cdot 10^{-5}$ (57 events). At $0.9\eta_c$ the relative error of t_1 is 0.111, 0.056, 0.023, 0.0098 and 0.0036 for $\sigma = 0.3$, 0.1, 0.03, 0.01, 0.003 (five seeds).

Above the limit: edge of stability. derived empirical For $\eta_c < \eta < 2/\lambda_2(H)$ only the radial mode of component 1 is unstable, and GD on it is the balanced scalar map of Figure 13a (Section A.7): a two-cycle from η_c, period four from $1.245\eta_c$, chaos from $1.32\eta_c$, then returns to the origin interleaved with divergence ($K = 3$). On binary $N = 4$ component 1 oscillates while components 2 and 3 are still learned, and the time-averaged excess loss equals s_1^2 times the map's mean loss within 0.9% at 1.05–$1.3\eta_c$ (for example 0.4496 against 0.4496 at $1.2\eta_c$, 25 runs; (53) is its small-excess limit). The map's bounded chaotic window does not survive: the network diverges at 1.5–$1.75\eta_c$ (38 of 40 runs), is thrown back to the origin saddle and relearns its components again and again at $2\eta_c$ (25 of 25), and diverges from $2.5\eta_c$. The deep trees, started at $\sigma = 0.9$ where the gradient-flow states are 10–35% sharper than λ_{bal}, already oscillate or catapult at $0.9\eta_c$ in some runs (2 of 5 on binary $N = 8$, 3 of 5 on the MPS).

Larger initialization: catapults. empirical known (catapults: Lewkowycz et al., 2020; flatter, more balanced minima in scalar networks: Kreisler et al., 2023, Thms 3.2–3.3; limit sharpness at most $2/\eta$ in

two-factor factorization: Wang et al., 2022, Thm 3.2; balancing while oscillating beyond the edge: Ghosh et al., 2025, Prop. 2) Centred model on binary $N = 4$ at $\chi = 9$, the cut dimension, where all 45 gradient-flow references converge: $\lambda_{\mathrm{GF}} \propto \sigma^{3.965\pm0.014}$ over $\sigma = 1.41$–16 (five seeds), so $2/\lambda_{\mathrm{GF}}$ equals η_c for $\sigma \le 0.3$, $0.65\eta_c$ at $\sigma = 1$ and $2.7 \cdot 10^{-4}\eta_c$ at $\sigma = 8$ (Figure 13c). With steps of 0.5–4 times each run's own $2/\lambda_{\mathrm{GF}}$, all runs up to 0.95 times are GF-like at every σ, and monotone GD ends between 0.95 and 1.2 times. Beyond, small initialization ($\sigma = 0.1$, 0.3) oscillates and then diverges, since the balanced minimum is already the flattest; from $\sigma = 1$ on GD catapults: the loss spikes (by up to $2 \cdot 10^5 L_0$ at $\sigma = 8$) and then reaches gradient flow's final loss (86 of 90 runs below $10^{-6}L_0$) at a flatter minimum, of sharpness 0.58–1.00 times $2/\eta$ (per run), for steps up to 1.5 ($\sigma = 1$) and 3 ($\sigma = 2$–8) times $2/\lambda_{\mathrm{GF}}$; at 4 times all runs diverge. For $\sigma \ge 2$ the window thus ends between $6/\lambda_{\mathrm{GF}}$ and $8/\lambda_{\mathrm{GF}}$, 1.5–2 times the $4/\lambda_0$ of the two-layer model of Lewkowycz et al. (2020). Imbalanced small initializations ($\sigma = 0.1$ with the root core at scale 2, 4 or 6) reproduce the known selection: the 104 of 150 runs that converge end 0.05–3% below $\min(\lambda_{\mathrm{GF}}, 2/\eta)$.

SGD: the first plateau. derived empirical On an aligned mode (55) gives a mean delay of order $\eta(1 + 3^N/B)\ln(1/\sigma)$ and a standard deviation of about $(3^N t_1^{\mathrm{GF}}\eta/B)^{1/2}$, 3^N being the kurtosis of the product features (the exact factor is in Section A.7). With online SGD on binary $N = 4$ (1,780 runs; $B = 1$–256; five initializations with ten noise seeds each), at $B = 1$ the mean delays are $+0.24 \pm 0.10$, -0.01 ± 0.12 and $+0.13 \pm 0.07$ at $\sigma = 0.03$, 0.1, 0.3 (s.e. across initializations; escape times 24.2, 7.2, 2.3), against +0.17, +0.12, +0.08 predicted: consistent, but resolved only at $\sigma = 0.03$. The spread scales as a power 0.43–0.52 of η/B (five sweeps, s.e. 0.01–0.05; three s.e. below $\frac{1}{2}$ at $\sigma = 0.1$), against $\frac{1}{2}$. In the sweep over B at $\eta = 5 \cdot 10^{-4}$ (Figure 22a), 11 of the 23 cell means of the relative change of the first escape are negative, the largest $-1.0\% \pm 0.9\%$ ($\sigma = 0.1$, $B = 4$) and the most significant $-0.32\% \pm 0.13\%$ ($\sigma = 0.1$, $B = 128$); the positive ones reach +5% at $B = 1$. Run by run, the first escape comes earlier than gradient flow's in 787 of the 1,600 runs from random initialization (by more than 5% in 141), and the second plateau is shorter in 657 of the 1,529 runs that kept the first component; on the aligned route no escaped run was faster (114 of 114).

SGD: the later-plateau law. empirical The aligned-route test starts with component 1 learned and the route at per-core amplitude $b_0 = 0.002$–0.01, so that $T_{\mathrm{GF}} = 50$–250 is exact; $\eta/B = 3.1 \cdot 10^{-5}$–$2.5 \cdot 10^{-4}$, six noise seeds per point. One fitted $c_{\mathrm{col}} = 32.6$ (the commuting-pair estimate of `notes/theory_optimizer.md` gives about 26 for this teacher) yields $\chi^2 = 379$ for 18 degrees of freedom, against 535 for a pole $1/(1-x)$ and 929 for $1 + x/2$; but the effective c_{col} grows from 30 at small x to 45 near the trap. $B = 16$ and 64 agree within 1.8 combined s.e. at eight matched points, although $B = 64$ is higher at all three escaped points with $b_0 = 0.003$. At $\eta/B = 1.25 \cdot 10^{-4}$, $T_{\mathrm{SGD}}/T_{\mathrm{GF}} = 1.13$, 1.37 and 2.5–2.7 for $b_0 = 0.01$, 0.005, 0.003, and all runs are trapped at 0.002. From random initialization the second plateau is lengthened by $19.8 \pm 5.1\%$ at $\sigma = 0.03$, $B = 1$ (the 16 of 30 runs that kept component 1), against 17.7% from (24) with no further fit, but by $10.8 \pm 0.6\%$ against 4.4% at $\sigma = 0.1$.

SGD: stability and forgetting. empirical Started 10^{-5} from the minimum, SGD converged in 5 of 5 runs up to $1.5\eta^*_{\mathrm{ms}}$ and diverged in 9 of 10 at $2.5\eta^*_{\mathrm{ms}}$ ($B = 1$, 16), as the computed almost-sure threshold, about $1.9\eta^*_{\mathrm{ms}}$, predicts; η^*_{ms}/B is $1.07 \cdot 10^{-3}$ and $1.0 \cdot 10^{-3}$ at $B = 1$, 16 but $0.82 \cdot 10^{-3}$ and $0.44 \cdot 10^{-3}$ at $B = 64$, 256. In the random-initialization runs the first learned component fell back below $\frac{1}{2}$ in 20–66% of the runs at $0.47\eta^*_{\mathrm{ms}}$ (most in the longest runs), 4% at $0.38\eta^*_{\mathrm{ms}}$, 1 of 50 at $0.24\eta^*_{\mathrm{ms}}$ and none below; these are also the only runs whose order differs from gradient flow's

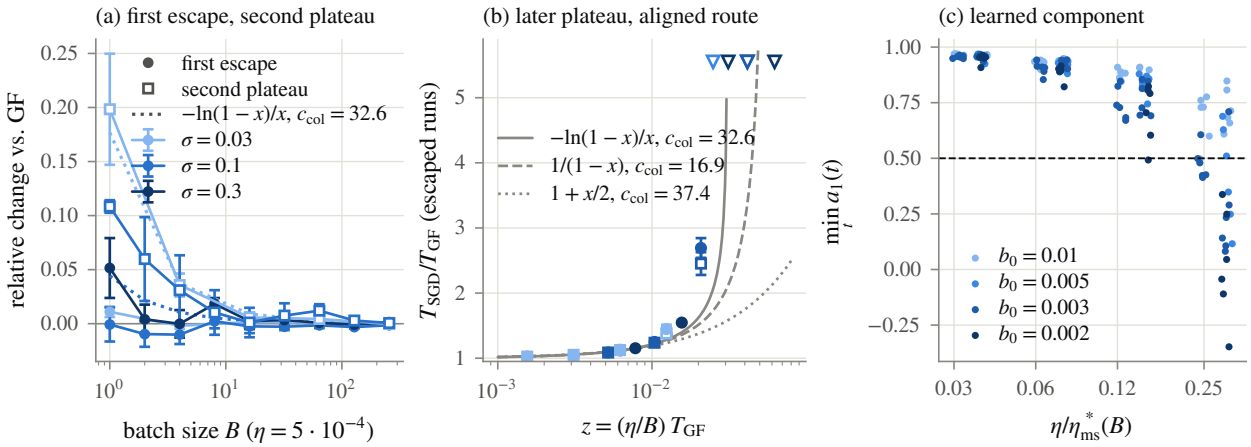


Figure 22: Online SGD, binary tree with $N = 4$ ($K = 3$), $d = \chi = 3$, odeco teacher $(3, 2, 1)$. (a) Change against gradient flow at $\eta = 5 \cdot 10^{-4}$ ($0.47\eta^*_{\text{ms}}$ at $B = 1$): first escape (filled; five initializations with ten noise seeds each) and second plateau (open; $\sigma = 0.03$, 0.1; runs that kept component 1), mean ± s.e. across initializations, $\sigma = 0.03$, 0.1, 0.3 from light to dark; dotted: (24) with c_{col} from (b). (b) Second plateau from the aligned route, mean ± s.e. over the escaped runs of six noise seeds; per-core route amplitude $b_0 = 0.01$, 0.005, 0.003, 0.002 from light to dark ($T_{\text{GF}} = 50$–250); circles $B = 64$, squares $B = 16$; triangles: all runs trapped; grey: three laws, one fitted collapse constant c_{col} each. (c) Minimum of a_1 during each run of (b) (colours as in (b)) against $\eta/\eta^*_{\text{ms}}(B)$, with $\eta^*_{\text{ms}}(B)$ the mean-square stability limit (56); dashed: $\frac{1}{2}$. Takeaway: SGD lengthens later plateaus as the collapse law predicts and traps at $x \geq 1$; near its stability limit it also erases the learned component. (source: Section B.7, Table 17)

(10) or is unresolved (19 at $B = 1$). On the aligned-route plateaus (Figure 22c), a_1 fell below $\frac{1}{2}$ in 0 of 72 runs at $\eta \leq 0.076\eta^*_{\text{ms}}$ (all escaped by $t = 405$), 1 of 36 at 0.12–$0.15\eta^*_{\text{ms}}$ (runs of up to 1000 time units) and 20 of 36 at 0.25–$0.30\eta^*_{\text{ms}}$, so the safe fraction $0.25\eta^*_{\text{ms}}$ seen with another teacher (Section A.7) does not carry over.

Finite samples. empirical derived Binary tree with $N = 8$, $d = 2$ ($W \in \mathbb{R}^{256}$), odeco teacher $(3, 2)$ of bond rank 2 (its rank-2 TTNs form a manifold of dimension 28), n Gaussian inputs with noise-free labels, five seeds; GD on the empirical loss (step $1/\lambda_{\max}$, $6 \cdot 10^4$ steps); students with $\chi = 4$, whose tangent space at a random point has dimension 112 (Proposition A.46), and $\chi = 16$, where all bonds are full; test error $\|W - W^*\|_F^2/\|W^*\|_F^2$ (Figure 23). Small initialization ($\sigma = 0.3$, 0.5) recovers the teacher with its bond ranks: test error below 10^{-2} in 5 of 5 seeds from $n = 256$ at $\chi = 4$ (about $2 \cdot 10^{-3}$ at 256 and $2 \cdot 10^{-4}$ at 384, limited by unfinished training) and from $n = 384$ at $\chi = 16$ ($\sigma = 0.5$ only), with mean effective bond rank 2.0–2.1. For $n \leq 96$ it fits the training data (to 1.3% of the initial loss) with a wrong tensor in 89 of 90 runs (mean test error 1.0–3.8). The exact lazy limit fails as dimension counting predicts derived: at $\chi = 4$ the teacher lies outside the tangent space, and the error is 1.7–32 at every n, worse than predicting zero; at $\chi = 16$ it is exact only from $n = d^N = 256$. The centred model at $\sigma = 1$, which is not lazy (its parameters move by 12–69%), reaches 0.01–0.04 at $n = 384$–512 with bond rank 2.4–2.9; at $\sigma = 1.5$ and 2 it had not converged after $3 \cdot 10^5$ steps and is not ranked. With $N = 6$, $d = 3$ ($W \in \mathbb{R}^{729}$, manifold dimension 48), small initialization recovers the teacher in 5 of 5 seeds at $n = 384$, while the $\chi = 4$ lazy limit is worse than predicting zero at every $n \leq 384$. The learned bond ranks are ordered small initialization < centred < lazy limit at every n, except at $\chi = 16$ for $n \geq 256$, where the lazy limit is exact. A linearized model can only fit the projection of the target on its tangent space known (Chizat et al., 2019, Thm 2.5; Ghorbani et al., 2019, Thm 2; lazy training of a tensor decomposition needs $\Omega(d^{N-1})$

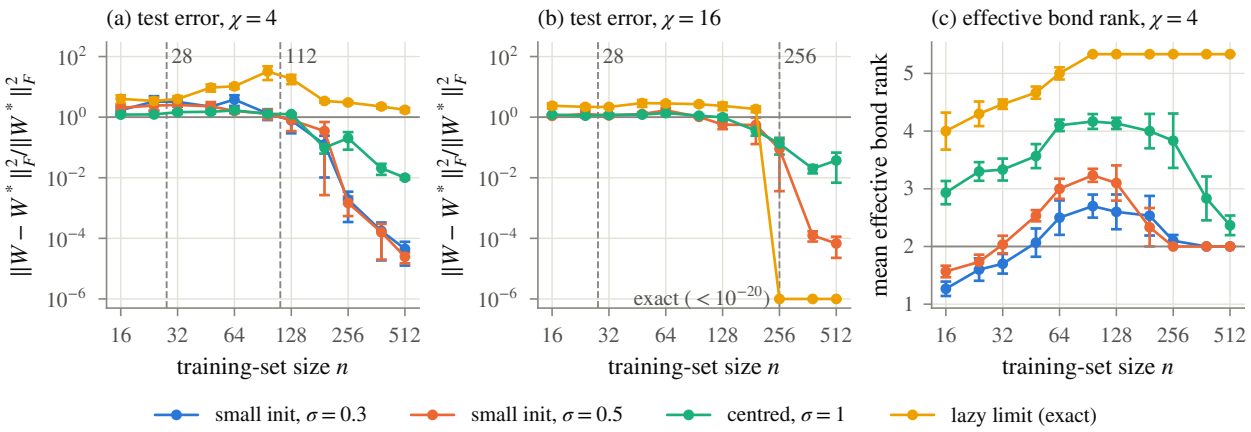


Figure 23: Finite training sets: binary tree, $N = 8$ ($K = 7$), $d = 2$, odeco teacher $(3, 2)$; mean $\pm$ s.e. over five seeds (data and initialization). (a), (b) Test error against n for $\chi = 4$ and 16 (lazy limit: kernel regression with the initial tangent kernel, values below 10^{-20} drawn at 10^{-6}); dashed: the teacher-manifold dimension 28, the tangent dimension 112 at $\chi = 4$, $d^N = 256$; grey: the zero predictor. (c) Effective bond rank (bond singular values above 0.2, a tenth of the smallest teacher weight, averaged over the six edges), $\chi = 4$; grey: the teacher's bond rank 2. Takeaway: small initialization recovers the teacher with its bond ranks from about 250 samples; the lazy limit cannot at $\chi = 4$. (source: Section B.7, Table 17)

components for a rank-one target: Wang et al., 2020, Thm 1). Scope: one teacher per setting, noise-free labels, GD with capped steps, and no monotone trend in σ within small initialization. The threshold of small initialization, 200–250 samples or 4–9 times the teacher-manifold dimension with $\chi = 4$ students, is unexplained.

Adam's first escape and its ϵ. empirical Figure 24 shows both regimes. On binary $N = 4$, (25) puts the crossover $\sigma_\times$ near 10^{-8} (derived, not tested), far below the $2 \cdot 10^{-4}$ at which the initial gradient falls under ϵ. On the MPS with $N = 8$ the prediction for each seed combines that seed's exact gradient-flow escape at $\sigma = 0.3$, scaled by $(0.3/\sigma)^6$, with its own escape at $\epsilon = 10^{-30}$. Every run that escaped within $2 \cdot 10^5$ steps is counted; at $\eta = 10^{-2}$, $\sigma = 0.03$ measured over predicted is 1.010–1.035, and near the crossover $\sigma_\times \approx 0.05$–$0.1$ the deviations run from -5% to $+17\%$.

Later plateaus and minima under Adam. empirical Later gaps depend weakly on σ on binary $N = 4$ (exponents 0.0–0.4). On binary $N = 8$ they grow as σ decreases, to a median $8 \cdot 10^4$ (up to $1.3 \cdot 10^5$) steps at $\sigma = 0.25$, but per-seed local slopes range from -0.4 to 5.5, so over half a decade no law is established (gradient flow: 4.72 and 4.85, prediction 5); with $\sigma \leq 0.4$, 83–100% of these runs are saddle-to-saddle. Adam and RMSProp break the conservation of Q_e and end at imbalanced minima (balance index $\|Q_e\|_F / (\|G_{v,(e)} G_{v,(e)}^\top\|_F + \|G_{p,(e)} G_{p,(e)}^\top\|_F) = 0.21$–$0.96$), 1.01–6.1 times sharper than the balanced value (52), the more so the larger the step. From balanced initializations (and on binary $N = 4$) gradient flow, GD and heavy ball end within 0.2% of it; from fan-in initialization on binary $N = 8$ they keep the initial imbalance (balance index up to 0.35) and end up to 1.07, 1.11 and 1.13 times sharper.

Re-gauged and Gram-preconditioned GD. derived empirical Re-gauging moves all of $\|W\|_F \sim \sigma^K$ into the root, whose environment becomes an isometry, so the root fits the teacher's projection onto the current frames in a time $O(1)$ for any σ; random frames keep only a fraction $\prod_e \chi / d^{n_e}$ of each component over the root's edges ($2.4 \cdot 10^{-3}$ on binary $N = 8$, 0.2 on $N = 4$), so on the

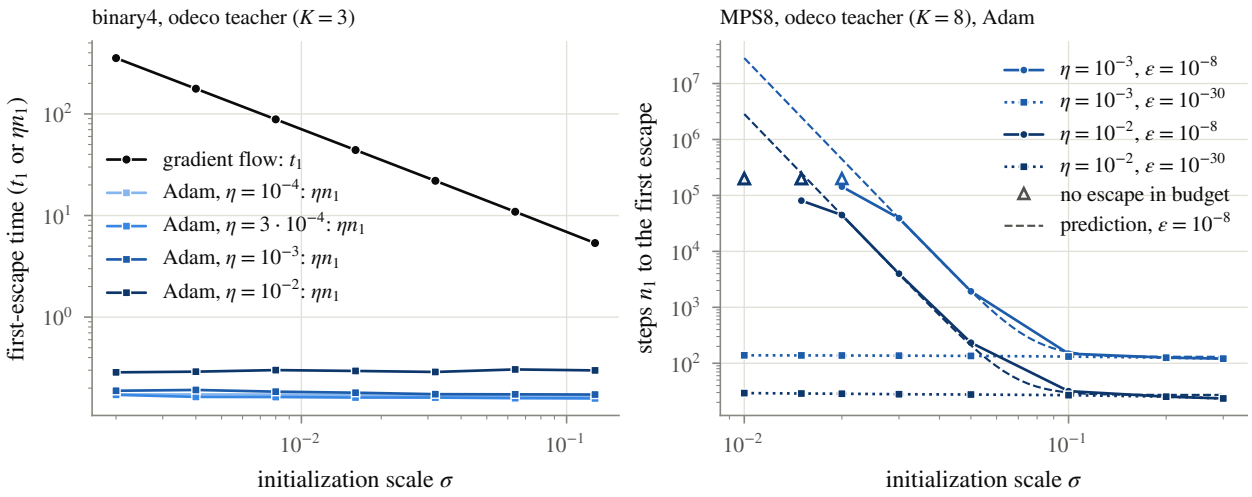


Figure 24: Adam's first escape. Left: binary tree, $N = 4$ ($K = 3$), $d = 3$, $\chi = 4$, odeco teacher $(3, 2, 1)$, fan-in initialization; medians over six seeds of gradient flow's first-escape time t_1 and of Adam's ηn_1 ($\epsilon = 10^{-8}$). Gradient flow follows σ^{-1}; Adam does not depend on σ, and $\eta n_1 \to 0.17$ as $\eta \to 0$ (0.30, 0.18, 0.16, 0.17 for $\eta = 10^{-2}$ to 10^{-4}). Right: MPS, $N = 8$ ($K = 8$), same teacher, balanced initialization, six seeds; median steps to the first escape over the runs that escaped within $2 \cdot 10^5$ steps, for $\epsilon = 10^{-8}$ (solid) and 10^{-30} (dotted); triangles, drawn at the budget, mark σ with runs that did not escape; dashed: (25) with median inputs. Colour: Adam's step η, light to dark, the same in both panels; black: gradient flow. Without ϵ the escape takes a median of 121–139 steps ($\eta = 10^{-3}$; single runs 111–140) and 23.5–29.3 steps ($\eta = 10^{-2}$; single runs 21.5–30.9) at every σ; with $\epsilon = 10^{-8}$ the law σ^{-6} returns below $\sigma \approx 0.05$–0.1. Takeaway: Adam's first escape does not depend on σ until ϵ brings gradient flow's law back. (source: Section B.8, Table 18)

deep tree they must rotate and learning is sequential again, with σ-free event times (exponents at most 0.02 in size; on the MPS with $N = 8$ the times at $\sigma = 0.3$ and 0.8 differ by 27–44%). Gram preconditioning moves each core towards its own least-squares solution, of size $\sim \sigma^{-(K-1)}$ (σ^{-K} times the core), so its first step overshoots the loss by $\sim \sigma^{-2K(K-1)}$ (derived in the aligned scalar model; medians up to $4 \cdot 10^{34} L_0$ on binary $N = 4$, single runs up to $5 \cdot 10^{36}$, deep trees 10^{52}); it has no flat plateau, but its time depends on σ (Table 6). In aligned deep matrix factorization, undamped Gauss–Newton blocks likewise remove the escape barrier and damping restores the gradient-descent law known (Sabanayagam and Lucey, 2026, Prop. 9).

Finite samples under Adam. empirical With $n = 45$ and 60 samples on binary $N = 4$ (four data draws, six initializations, $\sigma \in \{0.01, 0.1\}$, paired comparisons), gradient flow and Adam show no consistent difference in population loss: at $n = 60$ every run that met the training tolerance recovered the teacher; at $n = 45$ none did, and the paired differences range from -5.1 to $+4.0$ ($L_0 = 7$; Adam better in 8–13 of 21–23 pairs per setting). Only a threshold count differs (a fourth bond singular value above $0.1\, s^{e,*}_{\min}$: Adam with $\eta = 10^{-2}$ in 35 of 44 runs, gradient flow in 5); the entropy rank does not.

Which invariance removes which feature. Two properties of an update decide what survives (Table 6, second column). The first is how its step scales with the gradient, which near the origin is $O(\sigma^{K-1})$ while the parameters must move by their own size: a step proportional to the gradient needs a time $\sigma^{-(K-2)}$, a step invariant under its rescaling (Adam) does not, and an absolute scale in the preconditioner, such as Adam's ϵ, brings the law back derived; Adam still follows the gradient's sign, which one component dominates, so learning stays sequential and its first component

is gradient flow's (conjecture). The second is the metric. Under gradient flow W moves through $JJ^\top$ ($J = \partial W/\partial\theta$), which vanishes like $\sigma^{2(K-1)}$ at the origin, whereas Riemannian GD (in continuous time the Gauss–Newton flow $\dot\theta = J^+(W^* - W)$) and the exact block minimizations of ALS and DMRG move W at a rate of order one. These methods are equivariant under the whole gauge group (14), including the rescaling of a core against its neighbour, and see W_0 only through its frames, so they keep no σ (proven for ALS and DMRG, exact for Riemannian GD at $\eta = 1$); because they move W at a rate of order one they keep no plateaus (derived); re-gauged GD shares the equivariance and the σ-independence but not the rate. Plateaus at singular points of a parametrization, and the better behaviour of the natural gradient there, are known for multilayer perceptrons known (Amari et al., 2006). Equivariance alone is not enough: Gram preconditioning is gauge-equivariant up to damping derived but moves all cores at once from their current values. Balance and the flattest minimum (Proposition A.43) rest on the conserved Q_e, the charges of the gauge symmetry under Euclidean gradient flow (Proposition 2.5). Their relative drift is about 10^{-11} under gradient flow and, growing with σ, $4\cdot10^{-5}$–$2\cdot10^{-3}$ under GD and 10^{-3}–$5\cdot10^{-2}$ under heavy ball on binary $N = 4$ (momentum flows conserve fewer quantities than gradient flow, with time- and velocity-dependent laws, Marcotte et al., 2024), while Adam, equivariant only under coordinate permutations and sign flips, breaks them at order one empirical. A law for Adam's later plateaus is open (Section 7.2).

Provenance and corrections after review. The results on steps, noise and samples come from the experiment `experiments/optim_samples/`, with findings in `notes/optim_samples.md` (fix round 1) and theory in `notes/theory_optimizer.md` (v2) and Section A.7; every number is an entry of the committed tables in its `results/` folder (for example `sgd_summary.csv`, `sgd_runs.csv`, `p9_sgd.csv`, `p9_fit.csv`, `gd_sweep.csv` and `samples_summary.csv`). The optimizer results come from the experiment `experiments/optimizers/`, with findings in `notes/optimizers.md` (v2; its section 5 lists the claims withdrawn after review round 1) and tables in its `results/` folder; the binary $N = 8$ gradient-flow exponent 4.991 ± 0.002 is from `later_b8_summary.csv`, the order agreement from `order_agreement_with_gf.csv`. Reviews: `notes/reviews/optim_samples_round1.md`, `optim_samples_round2_verification.md` (errata 1–7 applied), `optimizers_round1.md` and `final_review_claims_B.md`. Corrections after review: SGD is compared with gradient flow on means, since single runs scatter to both sides; the trapped aligned-route runs are counted separately from those that also forgot; the consequences of the SGD drift law carry its conjecture tag; the GD order check on the deep trees compares only the components that gradient flow learns within the budget; Adam is credited with gradient flow's first component, not its full order; and the small-σ statements of the synthesis are scoped to the trees and batch sizes tested. The fix round of the optimizer experiment (exact gradient-flow events, Gram re-runs, four-draw finite samples, route-end check, GD delay slope) was recomputed from the committed tables in the final review but not re-run independently.

D.4 Covariance and teacher structure: supplementary results

Material moved from Section 6: per-teacher values and secondary checks. Settings are in Sections B.6 and B.9; notation as in Section 6.

Aligned covariance (Section 6.1). empirical $N = 4$, $d = \chi = 4$, $\beta = 0.25$, so the high-variance eigenvector e_1 has twice the Λ of e_2. Teachers T1 $(1, 0.8)$ and T2 $(1, 0.6)$ put the smaller component on e_1. Forty seeds, binary4 at $\sigma = 0.02$–0.1, MPS4 at 0.05–0.2, four variants from the same initialization: (Σ, s), $(I, \Lambda s)$, $(I, \Lambda^{1/2}s)$, and the last from the mapped initialization (same landscape and starting point as (Σ, s), whitened flow).

- *Clocks.* From aligned initialization half-times match the clock $\Lambda_r t$ to 10^{-9}; the naive substitution $s_r \to \Lambda_r s_r$ misses the widths by up to 47%.
- *Race.* Paired (Σ, s) and $(I, \Lambda s)$ runs learn the same component first in all MPS4 and 95–100% of the binary4 pairs, with first-escape times within 1.3%. The covariance reverses the order: in T1 the 0.8 component comes first in 90–98% (binary4) and 78% (MPS4) of the runs. The mapped-initialization control picks a different first component in 10–15% (MPS4) and 12–53% (binary4) of the seeds and escapes 11–23% later.
- *Exponents.* $t_1 \propto \sigma^{-1.010\pm0.002}$ (binary4) and $\sigma^{-1.999\pm0.001}$ (MPS4); $t_1 \propto (\Lambda s)^{-1}$ with per-seed exponents -1.000 to -1.003 within each eigenvector direction ($\Lambda = 0.008$–33, ten seeds).
- *Landscape.* On binary4 no swap follows a landscape-correct first component in any of the 24 settings, while in T2 the probability of a swap after the 0.6 component came first is 0.33, 0.68, 0.94 at $\sigma = 0.1$, 0.05, 0.02. The Hessian at the saddle equals the formula to 10^{-15} in the 6 strict binary4 cases; MPS4 has no negative eigenvalue (12 of 12) and no swap in 960 runs. Plateau levels are $\frac{1}{2}\sum_{r\notin S}\Lambda_r s_r^2$ (median relative error 10^{-8}–10^{-3}; binary4 runs of T2 on their way to a swap sit 1–3% from it).

Generic teachers (Section 6.2). Table 27 lists the thirteen teachers and the escape from the exact saddle. Where the realized edge is not the fastest (two random teachers with $N = 8$), the slope follows the realized edge, not $1/|\lambda_{\min}|$. Changing the onset threshold from 3% to 10% of the teacher's singular value moves the MPS slopes by at most 0.4%; fits over all ε, not only the five smallest, deviate by up to 11% (pre-asymptotic points). The misaligned teacher with $N = 8$ at 20°, timed at its intermediate level, gives exponents 1.07 and 1.05 (binary, HT) over 1.3 decades of ε, and 1.03 and 1.02 over its three smallest ε.

- *Random initialization, MPS4.* Slopes over the three smallest σ against h_e: 33.95 ± 0.18 (34.14), 5.674 ± 0.059 (5.682), 12.005 ± 0.071 (12.014), 1.038 ± 0.005 (1.051; 2.78 without the head start), 1.401 ± 0.013 (1.386). The misaligned teacher, which has no prediction under the preconditioned flow, gives 52.46 ± 0.10, below the bound $1/|\lambda_{\min}| = 52.94$. The curvature at the plateau is -0.1510 against -0.1509 at the exact saddle (overlap 0.45). The first escape keeps its $\sigma^{-(K-2)}$ law: per-seed exponents 1.998–2.014 (MPS4) and 1.002–1.017 (binary4) over all nine σ, 2.0000 and 1.0003 over the three smallest.
- *A failure mode of onset events* empirical new (methodological). The misaligned teacher with $N = 8$ at 20° escapes at the predicted rate to an intermediate saddle with second singular values 3.7–$3.9 \cdot 10^{-4}$ on every bond, nine times below the onset threshold, so the threshold rule times the next escape (a slope 24 times the prediction). Timed at the intermediate level, the slope is 1979.6 ± 1.8 against $1/\lambda_e = 1974.8$. A threshold relative to the teacher's spectrum misses escapes that end below it.

Non-orthogonal components (Section 6.3). empirical Single-σ runs: $\sigma = 0.05$ on binary4 and 0.1 on MPS4, 20 seeds.

- *Overlapping CP components* ($d = \chi = 4$, weights $(1, s_2)$, $s_2 \in \{0.7, 1\}$, overlaps 0–0.8; an MPS4 sweep with $s_2 = 0.7$, overlaps 0, 0.05, 0.2, 0.45, nine σ from $3 \cdot 10^{-4}$ to 0.3, ten seeds). MPS4 has two rank-one minima for overlaps up to 0.45 and one from 0.55 ($s_2 = 1$), two up to 0.35 and one from 0.45 ($s_2 = 0.7$); binary4 one. Median relative level errors of the first plateaus: $5 \cdot 10^{-5}$–$2 \cdot 10^{-3}$ on binary4, up to 3% on barely resolved MPS4 plateaus; some binary4 runs with $s_2 = 1$ sit at the second singular pair, a strict saddle.

Table 27: Escape from the exact bond-rank-one saddle, 13 generic teachers ($d = 2$, $\chi = 4$, five seeds, asymptotic per-seed fits, mean ± s.e.). MPS: $\lambda_{\min}$ at the saddle, $1/\lambda_e$ of the realized edge, and the slope A of t_{esc} against $\ln(1/\varepsilon)$, $\varepsilon = 10^{-8}$–10^{-4} (its deviation from $1/\lambda_e$). Binary, HT: exponent of $t_{\mathrm{esc}} \propto \varepsilon^{-\mathrm{exponent}}$, $\varepsilon = 10^{-5}$–10^{-3}. †: intermediate-level event, $\varepsilon = 10^{-6}$–$2 \cdot 10^{-3}$ (MPS), 10^{-3}–$2 \cdot 10^{-2}$ (binary, HT). ‡: up to $\varepsilon = 2 \cdot 10^{-2}$. §: two realized edges, $1/\lambda_e = 5.47$, 5.27. Random teachers are named by their seed.

N	teacher	$\lambda_{\min}$	$1/\lambda_e$	A (MPS)	exponent (binary)	exponent (HT)
4	CP, overlap 0.2	−0.0285	35.14	35.156 ± 0.013 (+0.05%)	1.0001 ± 0.0001	1.0005 ± 0.0003
4	CP, overlap 0.45	−0.151	6.627	6.642 ± 0.005 (+0.22%)	1.0001 ± 0.0001	1.0005 ± 0.0001
4	misaligned, 20°	−0.0189	52.94	53.04 ± 0.06 (+0.18%)	1.0001 ± 0.0001	1.0002 ± 0.0004
4	random binary (302)	−0.0764	13.09	13.12 ± 0.01 (+0.18%)	1.0002 ± 0.0001	1.0001 ± 0.0002
4	random binary (303)	−0.360	2.779	2.786 ± 0.002 (+0.27%)	1.0001 ± 0.0001	1.0001 ± 0.0001
4	random MPS (301)	−0.376	2.658	2.662 ± 0.001 (+0.15%)	1.0000 ± 0.0001	1.0002 ± 0.0001
8	CP, overlap 0.45	−0.0048	209.70	209.60 ± 0.05 (−0.05%)	1.0001 ± 0.0001	$1.0036 \pm 0.0006^{\ddagger}$
8	CP, overlap 0.7	−0.0574	17.41	17.45 ± 0.02 (+0.24%)	1.0001 ± 0.0001	0.9998 ± 0.0001
8	misaligned, 45°	−0.0174	57.39	57.49 ± 0.03 (+0.16%)	1.0003 ± 0.00004	0.9999 ± 0.0001
8	misaligned, 20°	−0.00051	1974.8	1979.6 ± 1.8 (+0.24%)†	$1.071 \pm 0.004^{\dagger}$	$1.048 \pm 0.012^{\dagger}$
8	random binary (302)	−0.371	2.698	2.708 ± 0.002 (+0.36%)	0.9999 ± 0.0001	0.9999 ± 0.0002
8	random binary (303)	−0.190	5.42§	5.43 ± 0.04 (+0.08%)	1.0001 ± 0.0001	1.0001 ± 0.0002
8	random MPS (303)	−0.164	6.181	6.202 ± 0.007 (+0.34%)	1.0003 ± 0.0001	0.9999 ± 0.0001

- *Misaligned covariance* ($d = \chi = 4$, weights $(1, 0.8)$, $\beta \in \{0.5, 1\}$, angles 0–45°; an MPS4 sweep with $\beta = 1$, angles 0, 5°, 20°, nine σ from $3 \cdot 10^{-4}$ to 0.3, ten seeds). For angles up to 30° the median $|\cos|$ between the first plateau and the critical tensor is at least 0.998, over the resolved plateaus (20–50% of the MPS4 runs per angle have none at $\sigma = 0.1$). At 45°, 65% ($\beta = 0.5$) and 90% ($\beta = 1$) of the MPS4 runs have no resolved plateau; for the two resolved ones at $\beta = 1$ the medians are a level error of 5.8% and $|\cos| = 0.53$. Binary4 resolves every plateau at every angle.
- *Correlated sites.* On binary4 ($d = \chi = 4$, weights $(1, 0.7)$) the first plateau is at 0.33, 0.29, 0.20, 0.14, 0.10 $L(0)$ for $c = 0$, 0.2, 0.3, 0.4, 0.5 (median relative error $\leq 5 \cdot 10^{-4}$). The σ sweep ($d = 3$, weights $(1, 0.6)$, six σ from 0.0016 to 0.05, five seeds) dates events on the singular values across 01|23; seed 2 of MPS4 sits on the minor saddle at every c and is excluded. Online SGD (binary4, $c = 0.3$, $\sigma = 0.05$, three seeds) crosses the same loss levels within 2% of gradient flow at batch size 4096 and within 7% at batch size 256 (event times on a grid of 0.2).

Random TTN teachers (Section 6.4). empirical Per teacher, the distance of the rank-one plateau to the bond-rank-one critical point falls as $\sigma^{0.72}$, $\sigma^{0.83}$, $\sigma^{0.41}$ (one seed), $\sigma^{0.93}$, $\sigma^{0.80}$ (±0.01–0.03), against the heuristic exponents 0.64, 0.68, 0.34, 0.80, 0.72. For the three separated teachers, at $\sigma = 10^{-3}$ the plateau lies 10^{-4}–$2.5 \cdot 10^{-3}$ from the critical point and $1.5 \cdot 10^{-3}$–0.028 from the nearest SVD-type truncation (relative to $\|W^*\|_F$); the SVD-type truncations compared are the hierarchical SVD and the TT-SVD in both directions. Example of a single-route slope: 5.188±0.008 against $h_{e^*} = 5.175$.

Teacher and student on different trees (Section 6.5). empirical An MPS8 student of binary8 teachers at $\sigma = 0.45$ grows the predicted bond first in 2 of 24 runs with onset events at 3% or 5% of the teacher's singular value (1 of 24 at 10%; a rule on the bond spectra gives 6 of 24); with $\ln(1/\sigma) = 0.8$ this is no test of the asymptotic race of Proposition A.38.

Star teachers (Section 6.6).

- *Hessian.* At $s_r = 1.6$, $\lambda_{\min} = -0.3716, -0.3310, -0.4311, -0.4311$ on MPS4, binary4, HT4 and the binary8 analogue, as the formula gives.
- *Near the threshold* (MPS4, $s_r = 1.42, 1.44$) the blended rank-one point keeps $a_q = 0.98$, 0.93; with λ_b its rate ($-\lambda_{\min}$ there), the measured slopes 130.0 ± 0.2 and 29.39 ± 0.02 equal $1/\lambda_q + 1/\lambda_b = 129.9$ and 29.3 (one stage would give 86.4 and 19.4). On HT4 leaving the blended point takes a time $\propto \varepsilon^{-1}$ (fitted exponents 0.995, 0.998); at $s_r = 1.6$, a_q falls below $\frac{1}{2}$ after $3.4 \cdot 10^3$, $3.3 \cdot 10^4$ and $3.3 \cdot 10^6$ time units for $\varepsilon = 10^{-3}$, 10^{-4}, 10^{-6}.
- *Random initialization* ($s_r = 1.6$; $\sigma = 0.01$–0.1 on MPS4, 0.005–0.05 on binary4, ten seeds). The flow spends at most 0.13 time units at the loss level of θ_q (one seed 1.7), and up to 538 at "q forgotten". Classes, set by the seed: q learned last (binary4 7 of 10 seeds, MPS4 1 of 10); a_q rising to 0.78–1.00 (binary4 3, MPS4 4 of 10), then decaying except on MPS4 for seed 6 at $\sigma \geq 0.025$ and seeds 3, 5 and 8 at $\sigma = 0.1$, which keep q ($\min a_q > 0.5$); a blended start with $a_q \leq 0.68$ (MPS4, 5 of 10). Sub-critical stars ($s_r = 1.2$): the smallest $\min a_q$ over all runs is 0.9918 (MPS4, $\sigma = 0.1$).

Real data (Section 6.7).

- *Route couplings.* The largest sibling coupling rises first in 22 of the 26 critical-point units with our grouping of states, and in 19–20 of 23–24 with other groupings; 10 of the 11 independent runs use new orders or seeds.
- *Robustness of the $k = 3$ gap law.* The verdicts of Figure 17 also hold on the full sweep of 124 binary runs, each point attributed to the drop that contains the route's rank crossing (91 points kept, 33 excluded with their reason; ten seed series). The one whitened series that meets the selection rule has $t_0 = +10.8$, so $t_0 < 0$ holds for raw features only.
- *Tree shape on the two clocks* empirical. On the clock τ started at the first escape the snake MPS is fastest (MNIST, $\sigma = 0.1$: $L \leq 0.2$ after 56 time units, the quadtree after 333). On the absolute clock the first escape dominates ($t_a \geq 10^5$), $t_a \propto \sigma^{-14}$ on MPS16 against σ^{-13} on binary16 proven, and the MPS escapes later than the quadtree in 4 of 5 seeds at $\sigma = 0.1$; on Fashion-MNIST it is the slowest tree to $L \leq 0.25$ at $\sigma \leq 0.14$ but the first to reach $L \leq 0.12$. Only snake chains were swept (three chains in other orders, one seed, are independent runs), so tree shape and pixel order are confounded; that the logarithmic routes make the MPS fast on the τ clock is a conjecture.
- *Other task and bond dimension* empirical. A sibling pair rises first at $\chi = 8$ (14 of 15 runs per binary order) and on Fashion-MNIST (sneaker against ankle boot; first two rising edges siblings in 15 and 14 of 15 runs, rising together in only 3 and 2). There a strict saddle ($\lambda_{\min} \propto \sigma^{0.03 \pm 0.09}$, error of the pooled fit) separates two rank-one states, and the quadtree has no advantage.
- *Large initialization* empirical. The centred sweep ($\sigma = 0.5$–4, quadtree) gives no regime map: a rich label would rest on $\sigma = 1.4$ alone, the boundary of the saddle-to-saddle label moves over $\sigma = 0.07$–1 with the separation threshold, and the lazy runs are unconverged at their horizon; the small-initialization runs end below the floor of the linearized model for four of five seeds. From $\sigma = 2.8$ on the runs follow the linearized model to $5 \cdot 10^{-5}$ (at $\sigma = 2$ up to 1.5%, one run labelled rich); its tangent kernel has rank $304 = \sum_v F_v \chi_v - \sum_e \chi_e^2$ of 528 parameters (Proposition A.46(c)).

- *Procedural points of the order test.* The run configuration and the analysis script were committed four minutes after the runs started (before any outcome), and an interim summary of 72 of the 81 runs (all 60 random-order runs) was written, and probably analysed, before the last runs finished. The registration (hypotheses, orders, outcomes, analysis) and the predictor values of all 27 orders were committed before the runs.

Provenance and corrections after review. Both experiments were checked by independent verifications that recomputed the numbers from the committed outputs (Hessians at 12 stored saddles with independent code; the saddle-start escape times rerun on another platform). The two-stage law of the correlated-input escape and the star Hessian formula were checked numerically when they were derived, in the same settings, so their experiments are replications with more seeds rather than independent tests (on binary4 the star formula is the known deep-linear-network result). After review, the early plateau durations at a single σ were replaced by σ sweeps; claims about later plateaus were narrowed to the escape from the rank-one saddle, the only one timed; the binary/HT contrast was limited to the settings run (no correlated inputs on HT, no random initialization on HT); a fit over all ε that suggested a slow crossover was traced to pre-asymptotic points; a teacher first reported as degenerate turned out to be the event-rule failure above; the bottom-up count was corrected after de-duplication of runs; and power laws for the forgetting depth and the relearning time of stars from random initialization were withdrawn (they pooled two classes of runs and describe runs that never visit θ_q). For the real data, an early first-route exponent came from an attribution bug; counts per run across σ were replaced by counts per (order, seed) or critical-point unit; statements about the tree shape on every clock, about locality as a small-initialization effect and about later routes of higher degree, and the exponents of the Fashion-MNIST and $\chi = 8$ sweeps, which are in no committed output, were withdrawn (the $\chi = 8$ comparison at $\sigma = 0.5$ is also confounded, since all five quadtree runs there stopped at the wall-clock limit; 23 of the 45 runs did); and the two procedural points of the order test were found by the verification, not listed by the experiment. The order test was registered in the commit 2f4d99e, the predictor values in 44feb2f, and the results are in 01144a5.